Readability of Online Patient-Education Materials for Pre-Procedure Cardiac CT

Clinical Accuracy Review Instrument

Standard (Labeled) Instrument

Document	Review instrument as administered
Variant	Standard (Labeled) Instrument
Items	77 paired passages
Scales	3 ordinal scales, 1–5
Prepared	15 August 2026
Participant identifier	ID only; no names recorded

This variant identifies which passage is the original page and which is the AI-generated rewrite. Reviewers remained blinded to which of the three language models produced each rewrite.

Instructions to reviewers

- You will see a series of paired passages about a cardiac CT procedure. For each pair, read both passages and score the second one on the three scales below.
- Score each passage on its own merits. Do not compare passages across items, and do not revise an earlier score after seeing a later item.
- If a passage omits something you would expect but that is also absent from the reference passage, this is not an omission by the passage under review.
- Record your participant ID at the top of the response sheet. Do not write your name on any page of this instrument.

Scoring scales

Scale	Range	Definition
Factual accuracy	1–5	Is every clinical statement in the passage correct? 5 = no factual errors; 1 = serious factual error that could mislead a patient.
Completeness	1–5	Does the passage retain the clinically important content of the reference passage? 5 = nothing important lost; 1 = major clinically important omission.
Added errors	1–5	Does the passage introduce assertions not supported by the reference passage or inconsistent with standard of care? 1 = none added; 5 = substantial invented content. Lower is better on this scale.

Items

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Original page

Coronary Computed Tomography Angiography (CCTA)

Coronary computed tomography angiography (CCTA) is a noninvasive 3D imaging test that identifies plaque and blockages or narrowing (stenosis) of the coronary arteries. During the test, a dye is injected through an intravenous (IV) line in the hand or arm, and computed tomography (CT), a combination of X-rays and computer technology, is used to produce images from inside the body.

Johns Hopkins cardiologist and director of the Cardiac Computed Tomography Program, Dr. Armin Zadeh, explains why a CCTA may be needed and what to expect.

What is a CCTA?

A CCTA is a diagnostic test that produces detailed 3D images of the arteries in your heart to detect abnormalities in how blood flows through your heart and to diagnose cardiovascular disease. It is sometimes used to determine overall function of the heart. An abnormality of the arteries may include plaque buildup, which may consist of calcium, cholesterol or fat that can lead to cardiovascular diseases like coronary artery disease or heart failure.

Patients undergoing a CCTA are injected with a nontoxic contrast dye that makes blockages and other abnormalities more visible.

A CT scan is an X-ray test that creates cross-sectional images of your body. A CCTA produces images of the heart when contrast dye is injected to highlight diseased areas.

Why would I need a CCTA?

You may need a CCTA if you have symptoms of cardiovascular disease or if you are diagnosed and your doctor needs more information about your condition. Reasons for a CCTA may include:

Abnormal test results (stress testing, echocardiogram, electrocardiogram)

Abnormal coronary artery structure

Risk for developing coronary artery disease

New or worsening coronary artery disease symptoms

You have undergone coronary artery bypass graft (CABG) surgery

A CCTA can also determine any type of heart disease, including heart structure or aortic abnormalities.

How do I prepare for a CCTA?

Your doctor may ask you to not eat or drink anything several hours before your CCTA. You may be asked not to exercise or consume caffeine before the test. Let your doctor know if you have a history of heart disease, allergies, recent illness or other medical conditions.

You should let your doctor know if you are pregnant, may be pregnant or are breastfeeding before scheduling a CCTA. Please tell your doctor about medications you are taking.

Remove any glasses, jewelry, piercings or metal objects, as they may interfere with the images produced by the CT scan.

How is a CCTA performed?

During a CCTA, your nurse or technologist will direct you to lay down on the table of the CT machine and will then place electrodes (small, plastic disks) on your chest that are connected to an electrocardiograph (ECG) monitor, which records your heart's electrical signals at rest.

Your nurse or technologist will insert an IV line containing the contrast material through your arm or hand. As the table moves through the scanner it will take several images at different angles for precise imaging. Your nurse or technologist may ask you to hold your breath and remain still, as slight body movements may impact the quality of the imaging produced by the CT machine.

To minimize the motion of the heart during imaging, you may be asked to take medications to lower your heart rate prior to the procedure. These are commonly prescribed medications, such as beta blockers. You may also be asked to take nitroglycerin just prior to scanning to make your arteries more visible.

After a CCTA, your nurse or technologist will remove the IV and you can return to normal daily activities. Your results will be examined by a radiologist, a physician who interprets radiology exams and will discuss the findings with your doctor.

What are the risks of a CCTA?

There is a slight risk for cancer development years after exposure to radiation, but the amount of radiation from CT imaging is very small.

Although very unlikely, the contrast material may worsen kidney function, particularly if you have poor kidney function to begin with, or contrast material may leak from the injection site, causing skin irritation or damage to the blood vessels and nerves close to where the contrast material was injected.

Like most medications or media, contrast material can trigger an allergic reaction, which may manifest through a skin rash or itching, and very rarely may result in anaphylaxis, a life-threatening condition.

There is a potential risk to unborn babies if patients are pregnant and undergo a CCTA or any other type of X-ray. You will be asked to undergo a pregnancy test if you are of childbearing age.

Medications given prior to the scan for lowering your heart rate or widening your arteries sometimes cause a drop in blood pressure, which may result in lightheadedness or headaches. Very rarely, these medications cause shortness of breath and worsen preexisting heart failure.

You will be closely monitored throughout your visit for any side effects.

AI rewrite

A Coronary Computed Tomography Angiography is often called a CCTA. It is a test that takes 3D pictures of your heart without surgery. It checks the blood vessels in your heart for narrowing or blockages. These blockages are made of plaque. Plaque is a mix of calcium, cholesterol, and fat. Too much plaque can lead to heart disease or heart failure. For the test, a safe dye goes into your arm or hand through an IV tube. The dye makes your blood vessels and any blockages easier to see. Then, a machine uses X-rays and a computer to take pictures of the inside of your body. Dr. Armin Zadeh, a heart doctor at Johns Hopkins, explains why you might need this test and what to expect.

Your doctor might order a CCTA if you have signs of heart disease. You might also need this test if other heart tests showed a problem. It is used if your blood vessels are not shaped normally or if you have a high chance of getting heart disease. You may need it if your heart symptoms are new or getting worse, or if you have had heart bypass surgery in the past. This test helps your doctor see how well your heart works. It can also find problems with the shape of your heart or your aorta, which is a large blood vessel.

Before your test, your doctor may tell you not to eat or drink for a few hours. You may also need to skip exercise and caffeine that day. Tell your doctor about all the medicines you take. Also, tell them if you have allergies, heart problems, a recent illness, or other medical conditions. You must tell your doctor if you are pregnant, might be pregnant, or are breastfeeding. Before the test starts, you need to take off glasses, jewelry, piercings, and any metal objects. Metal can ruin the pictures from the scanner.

During the test, you will lie down on the scanner table. A worker will put small, plastic patches on your chest. These patches connect to a monitor. The monitor records your heart's electrical signals while you rest. A nurse will put an IV tube into a vein in your arm or hand. The special dye will go through this tube. The table will slowly move through the scanner to take pictures from different angles. You will need to hold your breath and stay very still. Moving your body can make the pictures blurry. You might get medicine, like beta blockers, to slow down your heart rate. This helps the machine take clear pictures. You might also get a medicine called nitroglycerin. This makes your blood vessels wider and easier to see. After the test, the nurse will take out your IV. You can go back to your normal daily activities right away. A special doctor called a radiologist will look at your pictures and share the results with your doctor.

There are a few risks with a CCTA. The machine uses a very small amount of radiation. This causes a tiny risk of getting cancer years later. The special dye can sometimes harm your kidneys. This is more likely if you already have poor kidney function. The dye could also leak out of your vein. This can hurt your skin, blood vessels, or nerves near the IV. Some people are allergic to the dye. An allergy might cause a skin rash or itching. In very rare cases, it can cause a severe, life-threatening reaction. The X-rays can harm an unborn baby. If you are able to get pregnant, you will need to take a pregnancy test first. The medicines used to slow your heart or open your blood vessels can lower your blood pressure. This might give you a headache or make you feel dizzy. Very rarely, these medicines can make it hard to breathe or make heart failure worse. Your care team will watch you closely the whole time to keep you safe.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

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Original page

Coronary Computed Tomography Angiography (CCTA)

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Johns Hopkins cardiologist and director of the Cardiac Computed Tomography Program, Dr. Armin Zadeh, explains why a CCTA may be needed and what to expect.

What is a CCTA?

A CCTA is a diagnostic test that produces detailed 3D images of the arteries in your heart to detect abnormalities in how blood flows through your heart and to diagnose cardiovascular disease. It is sometimes used to determine overall function of the heart. An abnormality of the arteries may include plaque buildup, which may consist of calcium, cholesterol or fat that can lead to cardiovascular diseases like coronary artery disease or heart failure.

Patients undergoing a CCTA are injected with a nontoxic contrast dye that makes blockages and other abnormalities more visible.

A CT scan is an X-ray test that creates cross-sectional images of your body. A CCTA produces images of the heart when contrast dye is injected to highlight diseased areas.

Why would I need a CCTA?

You may need a CCTA if you have symptoms of cardiovascular disease or if you are diagnosed and your doctor needs more information about your condition. Reasons for a CCTA may include:

Abnormal test results (stress testing, echocardiogram, electrocardiogram)

Abnormal coronary artery structure

Risk for developing coronary artery disease

New or worsening coronary artery disease symptoms

You have undergone coronary artery bypass graft (CABG) surgery

A CCTA can also determine any type of heart disease, including heart structure or aortic abnormalities.

How do I prepare for a CCTA?

Your doctor may ask you to not eat or drink anything several hours before your CCTA. You may be asked not to exercise or consume caffeine before the test. Let your doctor know if you have a history of heart disease, allergies, recent illness or other medical conditions.

You should let your doctor know if you are pregnant, may be pregnant or are breastfeeding before scheduling a CCTA. Please tell your doctor about medications you are taking.

Remove any glasses, jewelry, piercings or metal objects, as they may interfere with the images produced by the CT scan.

How is a CCTA performed?

During a CCTA, your nurse or technologist will direct you to lay down on the table of the CT machine and will then place electrodes (small, plastic disks) on your chest that are connected to an electrocardiograph (ECG) monitor, which records your heart's electrical signals at rest.

Your nurse or technologist will insert an IV line containing the contrast material through your arm or hand. As the table moves through the scanner it will take several images at different angles for precise imaging. Your nurse or technologist may ask you to hold your breath and remain still, as slight body movements may impact the quality of the imaging produced by the CT machine.

To minimize the motion of the heart during imaging, you may be asked to take medications to lower your heart rate prior to the procedure. These are commonly prescribed medications, such as beta blockers. You may also be asked to take nitroglycerin just prior to scanning to make your arteries more visible.

After a CCTA, your nurse or technologist will remove the IV and you can return to normal daily activities. Your results will be examined by a radiologist, a physician who interprets radiology exams and will discuss the findings with your doctor.

What are the risks of a CCTA?

There is a slight risk for cancer development years after exposure to radiation, but the amount of radiation from CT imaging is very small.

Although very unlikely, the contrast material may worsen kidney function, particularly if you have poor kidney function to begin with, or contrast material may leak from the injection site, causing skin irritation or damage to the blood vessels and nerves close to where the contrast material was injected.

Like most medications or media, contrast material can trigger an allergic reaction, which may manifest through a skin rash or itching, and very rarely may result in anaphylaxis, a life-threatening condition.

There is a potential risk to unborn babies if patients are pregnant and undergo a CCTA or any other type of X-ray. You will be asked to undergo a pregnancy test if you are of childbearing age.

Medications given prior to the scan for lowering your heart rate or widening your arteries sometimes cause a drop in blood pressure, which may result in lightheadedness or headaches. Very rarely, these medications cause shortness of breath and worsen preexisting heart failure.

You will be closely monitored throughout your visit for any side effects.

AI rewrite

Coronary Computed Tomography Angiography (CCTA)

Coronary computed tomography angiography is also called CCTA. It is a test that takes 3D pictures of the blood vessels in your heart. These blood vessels are called coronary arteries.

A CCTA can show plaque, blockages, or narrow areas in these arteries. During the test, a dye is put into a vein in your hand or arm through an IV. This dye helps the pictures show the blood vessels more clearly. The test uses CT, which is a type of X-ray test that uses a computer to make pictures from inside the body.

What is a CCTA?

A CCTA is a test that helps doctors see the arteries in your heart. It makes detailed 3D pictures. It can show problems with how blood flows through your heart. It can also help diagnose heart and blood vessel disease. Sometimes it is used to check how well the heart works.

Plaque can build up in the arteries. Plaque may be made of calcium, cholesterol, or fat. Plaque can lead to heart and blood vessel diseases, such as coronary artery disease or heart failure.

For a CCTA, you are given a nontoxic contrast dye. This dye helps blockages and other problems show up better.

A CT scan is an X-ray test. It makes pictures of “slices” of your body. A CCTA takes pictures of your heart after the dye is given. The dye helps show diseased areas.

Why would I need a CCTA?

You may need a CCTA if you have symptoms of heart or blood vessel disease. You may also need one if you already have a diagnosis and your doctor needs more information.

Reasons for a CCTA may include:

Abnormal test results, such as a stress test, echocardiogram, or electrocardiogram

Abnormal structure of the coronary arteries

Risk for coronary artery disease

New or worsening symptoms of coronary artery disease

You have had coronary artery bypass graft surgery, also called CABG

A CCTA can also help find other types of heart disease. This includes problems with the heart’s structure or problems with the aorta.

How do I prepare for a CCTA?

Your doctor may ask you not to eat or drink for several hours before the test. You may be asked not to exercise or have caffeine before the test.

Tell your doctor if you have a history of heart disease. Also tell your doctor if you have allergies, a recent illness, or other medical conditions.

Tell your doctor before scheduling the test if you are pregnant, may be pregnant, or are breastfeeding.

Tell your doctor about all medicines you take.

Remove glasses, jewelry, piercings, or metal objects. These can affect the CT pictures.

How is a CCTA done?

During a CCTA, your nurse or technologist will help you lie down on the CT table. They will place small plastic disks on your chest. These are called electrodes. They are connected to an electrocardiograph monitor, also called an ECG. The ECG records the heart's electrical signals while you rest.

Your nurse or technologist will put an IV into a vein in your hand or arm. The IV gives you the contrast dye.

The table will move through the scanner. The scanner will take many pictures from different angles. Your nurse or technologist may ask you to hold your breath and stay still. Even small movements can make the pictures less clear.

You may be asked to take medicine before the test to slow your heart rate. This helps reduce heart movement during the scan. These medicines are often common medicines, such as beta blockers. You may also be asked to take nitroglycerin right before the scan. This helps make your arteries easier to see.

After the CCTA, your nurse or technologist will remove the IV. You can return to your normal daily activities. A radiologist will look at your results. A radiologist is a doctor who reads imaging tests. The radiologist will discuss the results with your doctor.

What are the risks of a CCTA?

There is a small risk of cancer years after being exposed to radiation. But the amount of radiation from CT imaging is very small.

The contrast dye may worsen kidney function. This is very unlikely. It is more likely if you already have poor kidney function.

The contrast dye may leak from the injection site. This can cause skin irritation or damage to nearby blood vessels and nerves.

Like many medicines or dyes, contrast dye can cause an allergic reaction. This may cause a skin rash or itching. Very rarely, it can cause anaphylaxis. Anaphylaxis is a life-threatening reaction.

There is a possible risk to unborn babies if a patient is pregnant and has a CCTA or any other X-ray test. If you are of childbearing age, you will be asked to take a pregnancy test.

Medicines given before the scan may slow your heart rate or widen your arteries. These medicines can sometimes cause a drop in blood pressure. This may make you feel lightheaded or cause a headache. Very rarely, these medicines can cause shortness of breath or make existing heart failure worse.

You will be closely watched during your visit for any side effects.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

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Original page

Coronary Computed Tomography Angiography (CCTA)

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Johns Hopkins cardiologist and director of the Cardiac Computed Tomography Program, Dr. Armin Zadeh, explains why a CCTA may be needed and what to expect.

What is a CCTA?

A CCTA is a diagnostic test that produces detailed 3D images of the arteries in your heart to detect abnormalities in how blood flows through your heart and to diagnose cardiovascular disease. It is sometimes used to determine overall function of the heart. An abnormality of the arteries may include plaque buildup, which may consist of calcium, cholesterol or fat that can lead to cardiovascular diseases like coronary artery disease or heart failure.

Patients undergoing a CCTA are injected with a nontoxic contrast dye that makes blockages and other abnormalities more visible.

A CT scan is an X-ray test that creates cross-sectional images of your body. A CCTA produces images of the heart when contrast dye is injected to highlight diseased areas.

Why would I need a CCTA?

You may need a CCTA if you have symptoms of cardiovascular disease or if you are diagnosed and your doctor needs more information about your condition. Reasons for a CCTA may include:

Abnormal test results (stress testing, echocardiogram, electrocardiogram)

Abnormal coronary artery structure

Risk for developing coronary artery disease

New or worsening coronary artery disease symptoms

You have undergone coronary artery bypass graft (CABG) surgery

A CCTA can also determine any type of heart disease, including heart structure or aortic abnormalities.

How do I prepare for a CCTA?

Your doctor may ask you to not eat or drink anything several hours before your CCTA. You may be asked not to exercise or consume caffeine before the test. Let your doctor know if you have a history of heart disease, allergies, recent illness or other medical conditions.

You should let your doctor know if you are pregnant, may be pregnant or are breastfeeding before scheduling a CCTA. Please tell your doctor about medications you are taking.

Remove any glasses, jewelry, piercings or metal objects, as they may interfere with the images produced by the CT scan.

How is a CCTA performed?

During a CCTA, your nurse or technologist will direct you to lay down on the table of the CT machine and will then place electrodes (small, plastic disks) on your chest that are connected to an electrocardiograph (ECG) monitor, which records your heart's electrical signals at rest.

Your nurse or technologist will insert an IV line containing the contrast material through your arm or hand. As the table moves through the scanner it will take several images at different angles for precise imaging. Your nurse or technologist may ask you to hold your breath and remain still, as slight body movements may impact the quality of the imaging produced by the CT machine.

To minimize the motion of the heart during imaging, you may be asked to take medications to lower your heart rate prior to the procedure. These are commonly prescribed medications, such as beta blockers. You may also be asked to take nitroglycerin just prior to scanning to make your arteries more visible.

After a CCTA, your nurse or technologist will remove the IV and you can return to normal daily activities. Your results will be examined by a radiologist, a physician who interprets radiology exams and will discuss the findings with your doctor.

What are the risks of a CCTA?

There is a slight risk for cancer development years after exposure to radiation, but the amount of radiation from CT imaging is very small.

Although very unlikely, the contrast material may worsen kidney function, particularly if you have poor kidney function to begin with, or contrast material may leak from the injection site, causing skin irritation or damage to the blood vessels and nerves close to where the contrast material was injected.

Like most medications or media, contrast material can trigger an allergic reaction, which may manifest through a skin rash or itching, and very rarely may result in anaphylaxis, a life-threatening condition.

There is a potential risk to unborn babies if patients are pregnant and undergo a CCTA or any other type of X-ray. You will be asked to undergo a pregnancy test if you are of childbearing age.

Medications given prior to the scan for lowering your heart rate or widening your arteries sometimes cause a drop in blood pressure, which may result in lightheadedness or headaches. Very rarely, these medications cause shortness of breath and worsen preexisting heart failure.

You will be closely monitored throughout your visit for any side effects.

AI rewrite

Coronary Computed Tomography Angiography (CCTA)

A CCTA is a test that takes 3D pictures of the blood vessels in your heart. These blood vessels are called coronary arteries. The test does not need surgery. It can show plaque, blockages, or narrow spots in these arteries.

During the test, a dye goes into your body through a small tube called an IV. The IV goes in your hand or arm. Then a CT scanner takes pictures. A CT scanner uses X-rays and a computer to make pictures of the inside of your body.

Dr. Armin Zadeh is a heart doctor at Johns Hopkins. He leads the Cardiac Computed Tomography Program. He explains why you might need a CCTA and what to expect.

What is a CCTA?

A CCTA makes clear 3D pictures of the arteries in your heart. It helps doctors see how blood moves through your heart. It can show heart and blood vessel disease. Sometimes it shows how well your whole heart works.

The test can find problems in your arteries. One problem is plaque. Plaque is a buildup of calcium, cholesterol, or fat. Plaque can lead to heart disease.

For this test, you get a dye called contrast. The dye is not poisonous. It makes blockages and other problems easier to see.

A CT scan is an X-ray test. It makes pictures of slices of your body. With the dye, a CCTA shows the sick areas of your heart more clearly.

Why would I need a CCTA?

You may need a CCTA if you have signs of heart disease. You may also need it if you already have heart disease and your doctor needs to learn more. You may need a CCTA if:

- Other tests were not normal (such as a stress test, echocardiogram, or electrocardiogram)

- Your arteries have an unusual shape

- You are at risk for heart artery disease

- You have new or worse signs of heart artery disease

- You had heart bypass surgery (CABG)

A CCTA can also find other kinds of heart problems. This includes problems with the shape of your heart or your aorta.

How do I prepare for a CCTA?

Your doctor may tell you not to eat or drink for a few hours before the test. You may be told not to exercise. You may be told not to have caffeine before the test.

Tell your doctor if you have heart disease, allergies, a recent illness, or other health problems.

Tell your doctor if you are pregnant, might be pregnant, or are breastfeeding. Tell your doctor before you set up the test. Also tell your doctor about all the medicines you take.

Take off your glasses, jewelry, piercings, and any metal. These things can mess up the pictures.

How is a CCTA performed?

A nurse or technologist will help you lie down on the table of the CT machine. They will put small plastic disks, called electrodes, on your chest. These connect to a heart monitor (ECG). The monitor records your heart's signals while you rest.

The nurse or technologist will put an IV in your arm or hand. The dye goes through this IV. The table will move through the scanner. The machine takes many pictures from different angles. You may be asked to hold your breath and stay very still. Even small moves can blur the pictures.

To keep your heart still during the test, you may be asked to take medicine to slow your heart rate. One common medicine for this is a beta blocker. You may also be asked to take nitroglycerin just before the scan. This medicine makes your arteries easier to see.

After the test, the nurse or technologist will take out the IV. You can go back to your normal activities. A radiologist will look at your pictures. A radiologist is a doctor who reads these kinds of tests. This doctor will talk about the results with your doctor.

What are the risks of a CCTA?

The test uses radiation. There is a small chance that radiation could lead to cancer years later. But the amount of radiation is very small.

The dye can cause problems, but this is rare. The dye may make kidney problems worse. This is more likely if your kidneys are already weak. The dye may also leak where the IV went in. This can hurt your skin, blood vessels, or nerves nearby.

Like many medicines, the dye can cause an allergic reaction. This may be a skin rash or itching. Very rarely, it can cause a serious reaction called anaphylaxis. This reaction can be life-threatening.

X-rays can be risky for an unborn baby. If you could become pregnant, you will be asked to take a pregnancy test first.

The medicines that slow your heart or widen your arteries can lower your blood pressure. This may make you feel dizzy or give you a headache. Very rarely, these medicines cause trouble breathing or make heart failure worse.

You will be watched closely during your whole visit for any side effects.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

CT coronary angiogram

Overview

A computerized tomography (CT) coronary angiogram is an imaging test that looks at the arteries that supply blood to the heart. A CT coronary angiogram uses a powerful X-ray machine to make images of the heart and its blood vessels. The test is used to diagnose many different heart conditions.

A CT coronary angiogram involves no surgical cuts in the body. And it doesn't require recovery time.

Why it's done

A CT coronary angiogram mainly is done to check for narrowed or blocked arteries in the heart. It may be done if you have symptoms of coronary artery disease. But the test can look for other heart conditions too.

A CT coronary angiogram is different from a standard coronary angiogram. With a standard coronary angiogram, a healthcare professional makes a small cut in the groin or wrist. A flexible tube called a catheter is threaded through the artery in the groin or wrist to the heart arteries. For those with known coronary artery disease, this approach also can be used as treatment.

A CT coronary angiogram also is different from a test called a CT coronary calcium scan. A CT coronary angiogram looks for buildups of plaque and other substances in the coronary artery walls. A CT coronary calcium scan looks only at how much calcium is in the artery walls.

Risks

A CT coronary angiogram exposes you to radiation. The amount varies depending on the type of machine used. If you're pregnant or you may be pregnant, you ideally should not have a CT angiogram. There's a risk that the radiation may harm an unborn child. But some people with serious health conditions may need CT imaging during pregnancy. For these people, steps are taken to minimize any possible radiation exposure to the unborn children.

A CT coronary angiogram is done using dye called contrast. Tell your healthcare professional if you're breastfeeding, because the dye can get into breast milk. Also, be aware that some people have an allergic reaction to the contrast dye. Talk to your healthcare professional if you're concerned about having an allergic reaction.

If you have a contrast dye allergy, you may be asked to take steroid medicine 12 hours before a CT coronary angiogram. This lowers the risk of a reaction. Rarely, contrast dye also may damage the kidneys, especially in people with chronic kidney conditions.

How you prepare

A healthcare professional tells you how to prepare for a CT coronary angiogram. Driving yourself to and from the test should be OK.

Food and medicines

Usually, a CT coronary angiogram requires not eating anything for at least 8 hours before the test. Drinking water is OK. Don't have drinks that contain caffeine 12 hours before. Caffeine can raise the heart rate, making it hard to get clear pictures of the heart.

Tell your healthcare team about the medicines that you take. You may be asked not to take a certain medicine before the test.

Clothing and personal items

The procedure requires removing jewelry, glasses and clothing above the waist. You'll be asked to change into a hospital gown.

What you can expect

A CT coronary angiogram usually is done in the radiology department of a hospital or an outpatient imaging facility.

Before the procedure

Before a CT coronary angiogram, a healthcare professional may give you medicine called a beta blocker. This slows your heart rate to help the CT scanner make clearer images. Let your healthcare professional know if you've had side effects from beta blockers in the past.

You also may be given nitroglycerin to widen your coronary arteries. This helps your healthcare professional see the arteries more easily in the images.

During the procedure

A healthcare professional places a thin tube called an IV into the hand or arm. Dye, called contrast, flows through this IV. The dye helps blood vessels show up better on the images taken during the test. Sticky patches called electrodes also are attached to your chest to record your heart rate.

You lie on a long table that slides through a short, tunnel-like machine called a CT scanner. If you're not comfortable in closed spaces, ask your healthcare professional about medicine to help you relax. Do this before the day of your test.

During the scan you need to stay still and hold your breath as directed. Movement can cause blurry images.

A healthcare professional controls the CT machine from a room that's separated from your exam room by a glass window. An intercom system lets you and the healthcare professional talk to each other.

The scanning parts of the test take as few as five seconds. But the whole process may take up to an hour if you're given medicines such as beta blockers or nitroglycerin.

After the procedure

After your CT coronary angiogram is done, you usually can return to your daily activities. You should be able to drive yourself home or to work as long as the CT scan causes no complications. Drink plenty of water to help flush the dye from your body.

Results

The images from your CT coronary angiogram should be ready soon after your test. The healthcare professional who asked you to have the test gives you the results.

If your test suggests that you have or are at risk of heart disease, you and your healthcare professional can talk about treatment options.

Regardless of the test results, it's always a good idea to make lifestyle changes to help protect the heart. Try these heart-healthy habits:

Get regular exercise. Exercise helps manage weight and control diabetes, high cholesterol and high blood pressure - all risk factors for coronary artery disease. Get at least 150 minutes of moderate aerobic activity or 75 minutes of vigorous aerobic activity a week, or a combination of moderate and vigorous activity.

Eat healthy foods. Eat plenty of fruits, vegetables, whole grains, legumes and nuts. Avoid saturated fats and trans fats. Reduce salt and sugar. Eating one or two servings of fish a week also may help keep the heart healthy.

Lose extra weight. Reaching a healthy weight and staying at it is good for your heart. Losing even a small amount of weight can help reduce risk factors for coronary artery disease. You can ask your healthcare professional to set a goal weight for you.

Don't smoke or use tobacco. Smoking is a major risk factor for coronary artery disease. Nicotine tightens blood vessels and forces the heart to work harder. Not smoking is one of the best ways to lower the risk of a heart attack. If you need help quitting, talk with your healthcare professional.

Manage health conditions. For high blood pressure, high cholesterol or diabetes, take medicines as directed. Ask your healthcare professional how often you need health checkups.

Lower stress. Stress can cause blood vessels to tighten. This raises the risk of a heart attack. Some ways to ease stress are to get more exercise, practice mindfulness and connect with others in support groups.

Get enough sleep. Adults should try to get 7 to 9 hours of sleep a night.

AI rewrite

CT Coronary Angiogram

Overview

A CT coronary angiogram is a test that looks at the blood vessels going to your heart. It uses a strong X-ray machine to take pictures of your heart and blood vessels. Doctors use this test to find many heart problems. This test does not need any cuts on your body. You do not need time to recover.

Why it is done

This test checks for narrow or blocked blood vessels in your heart. You might get this test if you have signs of heart disease. It can also look for other heart problems.

This test is not the same as a standard angiogram. In a standard angiogram, a doctor makes a small cut in your wrist or groin. Then, they put a soft tube called a catheter into your blood vessel. They move it up to your heart. For people who already have heart disease, that standard test can also be used to treat the problem.

This test is also different from a CT calcium scan. A CT angiogram looks for plaque and other buildup inside the blood vessel walls. A calcium scan only looks at how much calcium is in those walls.

Risks

This test uses radiation. The amount depends on the machine. If you are pregnant or might be pregnant, you should try not to have this test. The radiation could harm an unborn baby. But some pregnant people with serious health problems may still need the test. If so, the doctor will take steps to protect the baby from radiation.

The test uses a special dye called contrast. Tell your doctor if you are breastfeeding. The dye can get into breast milk. Also, some people are allergic to this dye. Talk to your doctor if you are worried about an allergic reaction.

If you are allergic to the dye, you might need to take steroid medicine 12 hours before the test. This lowers the chance of a bad reaction. In rare cases, the dye can hurt the kidneys. This is more likely if you already have kidney problems.

How you prepare

Your care team will tell you how to get ready for the test. It is usually safe to drive yourself to and from the test.

Food and medicines

You usually cannot eat anything for at least 8 hours before the test. You can drink water. Do not have drinks with caffeine for 12 hours before the test. Caffeine can make your heart beat faster. This makes it hard to get clear pictures of your heart.

Tell your care team about all the medicines you take. They might ask you to stop taking certain medicines before the test.

Clothing and personal items

You will need to take off your jewelry, glasses, and clothing above your waist. You will wear a hospital gown.

What you can expect

You will likely have the test in a hospital X-ray department or an imaging center.

Before the procedure

Before the test, a doctor or nurse may give you a medicine called a beta blocker. This slows down your heart rate so the machine can take clearer pictures. Tell your doctor if you have ever had bad side effects from beta blockers.

You might also get a medicine called nitroglycerin. This opens up your blood vessels. It helps the doctor see them better in the pictures.

During the procedure

A nurse or doctor will put a thin tube called an IV into your hand or arm. The special dye goes through this IV. The dye helps your blood vessels show up clearly in the pictures. Sticky patches called electrodes will be put on your chest. These track your heart rate.

You will lie on a long table. The table slides into a short, tunnel-like machine called a CT scanner. If you are afraid of small spaces, ask your doctor for medicine to help you relax. You should ask for this before the day of your test.

During the scan, you must stay very still. You will be asked to hold your breath at certain times. Moving can make the pictures blurry.

The person running the machine will be in another room behind a glass window. You can talk to each other through a speaker system.

The actual scan takes only about five seconds. But the whole visit can take up to an hour if you need to take the medicines first.

After the procedure

When the test is done, you can usually go back to your normal day. You can drive yourself home or to work if there were no problems. Drink plenty of water to help wash the dye out of your body.

Results

The pictures from your test should be ready soon. The doctor who ordered the test will give you the results.

If the test shows you have heart disease or are at risk for it, you and your doctor will talk about how to treat it.

No matter what your results are, it is always smart to protect your heart. Try these healthy habits:

Get regular exercise. Exercise helps you control your weight, diabetes, cholesterol, and blood pressure. These are all things that can cause heart disease. Try to get at least 150 minutes of moderate exercise a week. This means activities like brisk walking. Or you can do 75 minutes of hard exercise, like running. You can also do a mix of both.

Eat healthy foods. Eat lots of fruits, vegetables, whole grains, beans, and nuts. Avoid unhealthy fats like saturated and trans fats. Eat less salt and sugar. Eating fish once or twice a week can also help your heart.

Lose extra weight. Getting to a healthy weight is good for your heart. Losing even a little bit of weight can lower your risk of heart disease. Ask your doctor what a good weight is for you.

Do not smoke or use tobacco. Smoking is very bad for your heart. Nicotine makes your blood vessels tight. This makes your heart work harder. Quitting smoking is one of the best ways to prevent a heart attack. If you need help quitting, talk to your doctor.

Manage health conditions. If you have high blood pressure, high cholesterol, or diabetes, take your medicines exactly as told. Ask your doctor how often you need checkups.

Lower stress. Stress can make your blood vessels tight. This raises your risk of a heart attack. To lower stress, you can exercise more, practice mindfulness, or join a support group.

Get enough sleep. Adults should try to sleep 7 to 9 hours each night.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 5 of 77 · Item ID B-GFYGWWQC38

Original page

CT coronary angiogram

Overview

A computerized tomography (CT) coronary angiogram is an imaging test that looks at the arteries that supply blood to the heart. A CT coronary angiogram uses a powerful X-ray machine to make images of the heart and its blood vessels. The test is used to diagnose many different heart conditions.

A CT coronary angiogram involves no surgical cuts in the body. And it doesn't require recovery time.

Why it's done

A CT coronary angiogram mainly is done to check for narrowed or blocked arteries in the heart. It may be done if you have symptoms of coronary artery disease. But the test can look for other heart conditions too.

A CT coronary angiogram is different from a standard coronary angiogram. With a standard coronary angiogram, a healthcare professional makes a small cut in the groin or wrist. A flexible tube called a catheter is threaded through the artery in the groin or wrist to the heart arteries. For those with known coronary artery disease, this approach also can be used as treatment.

A CT coronary angiogram also is different from a test called a CT coronary calcium scan. A CT coronary angiogram looks for buildups of plaque and other substances in the coronary artery walls. A CT coronary calcium scan looks only at how much calcium is in the artery walls.

Risks

A CT coronary angiogram exposes you to radiation. The amount varies depending on the type of machine used. If you're pregnant or you may be pregnant, you ideally should not have a CT angiogram. There's a risk that the radiation may harm an unborn child. But some people with serious health conditions may need CT imaging during pregnancy. For these people, steps are taken to minimize any possible radiation exposure to the unborn children.

A CT coronary angiogram is done using dye called contrast. Tell your healthcare professional if you're breastfeeding, because the dye can get into breast milk. Also, be aware that some people have an allergic reaction to the contrast dye. Talk to your healthcare professional if you're concerned about having an allergic reaction.

If you have a contrast dye allergy, you may be asked to take steroid medicine 12 hours before a CT coronary angiogram. This lowers the risk of a reaction. Rarely, contrast dye also may damage the kidneys, especially in people with chronic kidney conditions.

How you prepare

A healthcare professional tells you how to prepare for a CT coronary angiogram. Driving yourself to and from the test should be OK.

Food and medicines

Usually, a CT coronary angiogram requires not eating anything for at least 8 hours before the test. Drinking water is OK. Don't have drinks that contain caffeine 12 hours before. Caffeine can raise the heart rate, making it hard to get clear pictures of the heart.

Tell your healthcare team about the medicines that you take. You may be asked not to take a certain medicine before the test.

Clothing and personal items

The procedure requires removing jewelry, glasses and clothing above the waist. You'll be asked to change into a hospital gown.

What you can expect

A CT coronary angiogram usually is done in the radiology department of a hospital or an outpatient imaging facility.

Before the procedure

Before a CT coronary angiogram, a healthcare professional may give you medicine called a beta blocker. This slows your heart rate to help the CT scanner make clearer images. Let your healthcare professional know if you've had side effects from beta blockers in the past.

You also may be given nitroglycerin to widen your coronary arteries. This helps your healthcare professional see the arteries more easily in the images.

During the procedure

A healthcare professional places a thin tube called an IV into the hand or arm. Dye, called contrast, flows through this IV. The dye helps blood vessels show up better on the images taken during the test. Sticky patches called electrodes also are attached to your chest to record your heart rate.

You lie on a long table that slides through a short, tunnel-like machine called a CT scanner. If you're not comfortable in closed spaces, ask your healthcare professional about medicine to help you relax. Do this before the day of your test.

During the scan you need to stay still and hold your breath as directed. Movement can cause blurry images.

A healthcare professional controls the CT machine from a room that's separated from your exam room by a glass window. An intercom system lets you and the healthcare professional talk to each other.

The scanning parts of the test take as few as five seconds. But the whole process may take up to an hour if you're given medicines such as beta blockers or nitroglycerin.

After the procedure

After your CT coronary angiogram is done, you usually can return to your daily activities. You should be able to drive yourself home or to work as long as the CT scan causes no complications. Drink plenty of water to help flush the dye from your body.

Results

The images from your CT coronary angiogram should be ready soon after your test. The healthcare professional who asked you to have the test gives you the results.

If your test suggests that you have or are at risk of heart disease, you and your healthcare professional can talk about treatment options.

Regardless of the test results, it's always a good idea to make lifestyle changes to help protect the heart. Try these heart-healthy habits:

Get regular exercise. Exercise helps manage weight and control diabetes, high cholesterol and high blood pressure - all risk factors for coronary artery disease. Get at least 150 minutes of moderate aerobic activity or 75 minutes of vigorous aerobic activity a week, or a combination of moderate and vigorous activity.

Eat healthy foods. Eat plenty of fruits, vegetables, whole grains, legumes and nuts. Avoid saturated fats and trans fats. Reduce salt and sugar. Eating one or two servings of fish a week also may help keep the heart healthy.

Lose extra weight. Reaching a healthy weight and staying at it is good for your heart. Losing even a small amount of weight can help reduce risk factors for coronary artery disease. You can ask your healthcare professional to set a goal weight for you.

Don't smoke or use tobacco. Smoking is a major risk factor for coronary artery disease. Nicotine tightens blood vessels and forces the heart to work harder. Not smoking is one of the best ways to lower the risk of a heart attack. If you need help quitting, talk with your healthcare professional.

Manage health conditions. For high blood pressure, high cholesterol or diabetes, take medicines as directed. Ask your healthcare professional how often you need health checkups.

Lower stress. Stress can cause blood vessels to tighten. This raises the risk of a heart attack. Some ways to ease stress are to get more exercise, practice mindfulness and connect with others in support groups.

Get enough sleep. Adults should try to get 7 to 9 hours of sleep a night.

AI rewrite

CT Coronary Angiogram

Overview

A CT coronary angiogram is a test that takes pictures of your heart. It looks at the arteries that bring blood to your heart. The test uses a strong X-ray machine to make these pictures. Doctors use this test to find many heart problems.

This test does not cut into your body. You do not need time to heal after it.

Why It's Done

A CT coronary angiogram mostly checks for arteries in the heart that are narrow or blocked. You may get this test if you have signs of heart disease. But the test can find other heart problems too.

This test is not the same as a standard coronary angiogram. With a standard one, a doctor makes a small cut in your groin or wrist. Then they push a thin, bendy tube into the artery. The tube goes all the way to your heart arteries. For people who already have heart disease, this way can also treat the problem.

This test is also not the same as a CT coronary calcium scan. A CT coronary angiogram looks for plaque and other buildup in the artery walls. A calcium scan only checks how much calcium is in the walls.

Risks

This test gives off radiation. The amount depends on the machine. If you are pregnant or might be pregnant, you should not have this test. The radiation may hurt the baby. But some people with serious health problems may need this test while pregnant. If so, the team will take steps to keep the baby safe from as much radiation as possible.

This test uses a special dye called contrast. Tell your doctor if you are breastfeeding. The dye can get into breast milk. Also, some people have an allergic reaction to the dye. Talk to your doctor if you are worried about a reaction.

If you are allergic to the dye, you may need to take a steroid medicine 12 hours before the test. This lowers the chance of a reaction. In rare cases, the dye can harm the kidneys. This is more likely in people who already have kidney problems.

How to Prepare

Your doctor will tell you how to get ready for the test. You can drive yourself to and from the test.

Food and Medicine

Most of the time, you should not eat for at least 8 hours before the test. You can drink water. Do not have anything with caffeine for 12 hours before the test. Caffeine can speed up your heart. This makes it hard to get clear pictures.

Tell your care team about all the medicines you take. You may be told not to take a certain medicine before the test.

Clothes and Personal Items

You will need to take off jewelry, glasses, and clothing above your waist. You will put on a hospital gown.

What to Expect

This test is usually done in the radiology part of a hospital or at an imaging center.

Before the Test

Before the test, your doctor may give you a medicine called a beta blocker. This slows your heart rate. A slower heart helps the machine make clearer pictures. Tell your doctor if you have had bad effects from beta blockers before.

You may also get a medicine called nitroglycerin. It makes your heart arteries wider. This helps the doctor see them better in the pictures.

During the Test

A care team member puts a thin tube called an IV into your hand or arm. The dye flows in through this IV. The dye helps your blood vessels show up better in the pictures. Sticky patches called electrodes are put on your chest. They record your heart rate.

You lie on a long table. The table slides through a short machine shaped like a tunnel. This machine is the CT scanner. If small or closed spaces make you uneasy, ask your doctor about medicine to help you relax. Ask before the day of the test.

During the scan, you must stay still. You will also need to hold your breath when told. Moving can make the pictures blurry.

A care team member runs the machine from another room. A glass window separates the two rooms. You can talk to each other through an intercom.

The scan part can take as little as five seconds. But the whole test may take up to an hour if you get medicines like beta blockers or nitroglycerin.

After the Test

After the test, you can usually go back to your normal day. You should be able to drive yourself home or to work, as long as there are no problems from the scan. Drink lots of water. This helps flush the dye out of your body.

Results

The pictures should be ready soon after the test. The doctor who ordered the test will give you the results.

If the test shows you have heart disease or may be at risk, you and your doctor can talk about ways to treat it.

No matter what the results show, it is always good to make healthy changes for your heart. Try these healthy habits:

Get regular exercise. Exercise helps you stay at a healthy weight. It also helps control diabetes, high cholesterol, and high blood pressure. These are all risk factors for heart disease. Try to get at least 150 minutes of moderate exercise each week. Or get 75 minutes of hard exercise. You can also mix the two.

Eat healthy foods. Eat lots of fruits, vegetables, whole grains, beans, and nuts. Stay away from saturated fats and trans fats. Cut back on salt and sugar. Eating one or two servings of fish a week may also help your heart.

Lose extra weight. Reaching a healthy weight and staying there is good for your heart. Even losing a little weight can help lower your risk of heart disease. You can ask your doctor to set a goal weight for you.

Do not smoke or use tobacco. Smoking is a big risk factor for heart disease. Nicotine makes your blood vessels tighter. This makes your heart work harder. Not smoking is one of the best ways to lower your chance of a heart attack. If you need help to quit, talk with your doctor.

Take care of health problems. If you have high blood pressure, high cholesterol, or diabetes, take your medicines as told. Ask your doctor how often you need checkups.

Lower your stress. Stress can make your blood vessels tighter. This raises your chance of a heart attack. To ease stress, you can exercise more, practice mindfulness, or join a support group.

Get enough sleep. Adults should try to get 7 to 9 hours of sleep each night.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

CT coronary angiogram

Overview

A computerized tomography (CT) coronary angiogram is an imaging test that looks at the arteries that supply blood to the heart. A CT coronary angiogram uses a powerful X-ray machine to make images of the heart and its blood vessels. The test is used to diagnose many different heart conditions.

A CT coronary angiogram involves no surgical cuts in the body. And it doesn't require recovery time.

Why it's done

A CT coronary angiogram mainly is done to check for narrowed or blocked arteries in the heart. It may be done if you have symptoms of coronary artery disease. But the test can look for other heart conditions too.

A CT coronary angiogram is different from a standard coronary angiogram. With a standard coronary angiogram, a healthcare professional makes a small cut in the groin or wrist. A flexible tube called a catheter is threaded through the artery in the groin or wrist to the heart arteries. For those with known coronary artery disease, this approach also can be used as treatment.

A CT coronary angiogram also is different from a test called a CT coronary calcium scan. A CT coronary angiogram looks for buildups of plaque and other substances in the coronary artery walls. A CT coronary calcium scan looks only at how much calcium is in the artery walls.

Risks

A CT coronary angiogram exposes you to radiation. The amount varies depending on the type of machine used. If you're pregnant or you may be pregnant, you ideally should not have a CT angiogram. There's a risk that the radiation may harm an unborn child. But some people with serious health conditions may need CT imaging during pregnancy. For these people, steps are taken to minimize any possible radiation exposure to the unborn children.

A CT coronary angiogram is done using dye called contrast. Tell your healthcare professional if you're breastfeeding, because the dye can get into breast milk. Also, be aware that some people have an allergic reaction to the contrast dye. Talk to your healthcare professional if you're concerned about having an allergic reaction.

If you have a contrast dye allergy, you may be asked to take steroid medicine 12 hours before a CT coronary angiogram. This lowers the risk of a reaction. Rarely, contrast dye also may damage the kidneys, especially in people with chronic kidney conditions.

How you prepare

A healthcare professional tells you how to prepare for a CT coronary angiogram. Driving yourself to and from the test should be OK.

Food and medicines

Usually, a CT coronary angiogram requires not eating anything for at least 8 hours before the test. Drinking water is OK. Don't have drinks that contain caffeine 12 hours before. Caffeine can raise the heart rate, making it hard to get clear pictures of the heart.

Tell your healthcare team about the medicines that you take. You may be asked not to take a certain medicine before the test.

Clothing and personal items

The procedure requires removing jewelry, glasses and clothing above the waist. You'll be asked to change into a hospital gown.

What you can expect

A CT coronary angiogram usually is done in the radiology department of a hospital or an outpatient imaging facility.

Before the procedure

Before a CT coronary angiogram, a healthcare professional may give you medicine called a beta blocker. This slows your heart rate to help the CT scanner make clearer images. Let your healthcare professional know if you've had side effects from beta blockers in the past.

You also may be given nitroglycerin to widen your coronary arteries. This helps your healthcare professional see the arteries more easily in the images.

During the procedure

A healthcare professional places a thin tube called an IV into the hand or arm. Dye, called contrast, flows through this IV. The dye helps blood vessels show up better on the images taken during the test. Sticky patches called electrodes also are attached to your chest to record your heart rate.

You lie on a long table that slides through a short, tunnel-like machine called a CT scanner. If you're not comfortable in closed spaces, ask your healthcare professional about medicine to help you relax. Do this before the day of your test.

During the scan you need to stay still and hold your breath as directed. Movement can cause blurry images.

A healthcare professional controls the CT machine from a room that's separated from your exam room by a glass window. An intercom system lets you and the healthcare professional talk to each other.

The scanning parts of the test take as few as five seconds. But the whole process may take up to an hour if you're given medicines such as beta blockers or nitroglycerin.

After the procedure

After your CT coronary angiogram is done, you usually can return to your daily activities. You should be able to drive yourself home or to work as long as the CT scan causes no complications. Drink plenty of water to help flush the dye from your body.

Results

The images from your CT coronary angiogram should be ready soon after your test. The healthcare professional who asked you to have the test gives you the results.

If your test suggests that you have or are at risk of heart disease, you and your healthcare professional can talk about treatment options.

Regardless of the test results, it's always a good idea to make lifestyle changes to help protect the heart. Try these heart-healthy habits:

Get regular exercise. Exercise helps manage weight and control diabetes, high cholesterol and high blood pressure - all risk factors for coronary artery disease. Get at least 150 minutes of moderate aerobic activity or 75 minutes of vigorous aerobic activity a week, or a combination of moderate and vigorous activity.

Eat healthy foods. Eat plenty of fruits, vegetables, whole grains, legumes and nuts. Avoid saturated fats and trans fats. Reduce salt and sugar. Eating one or two servings of fish a week also may help keep the heart healthy.

Lose extra weight. Reaching a healthy weight and staying at it is good for your heart. Losing even a small amount of weight can help reduce risk factors for coronary artery disease. You can ask your healthcare professional to set a goal weight for you.

Don't smoke or use tobacco. Smoking is a major risk factor for coronary artery disease. Nicotine tightens blood vessels and forces the heart to work harder. Not smoking is one of the best ways to lower the risk of a heart attack. If you need help quitting, talk with your healthcare professional.

Manage health conditions. For high blood pressure, high cholesterol or diabetes, take medicines as directed. Ask your healthcare professional how often you need health checkups.

Lower stress. Stress can cause blood vessels to tighten. This raises the risk of a heart attack. Some ways to ease stress are to get more exercise, practice mindfulness and connect with others in support groups.

Get enough sleep. Adults should try to get 7 to 9 hours of sleep a night.

AI rewrite

CT coronary angiogram

Overview

A CT coronary angiogram is a test that takes pictures of the arteries that bring blood to your heart. CT stands for computerized tomography.

The test uses a strong X-ray machine to take pictures of your heart and blood vessels. It can help find many heart problems.

This test does not need any cuts in your body. You do not need recovery time after it.

Why it is done

A CT coronary angiogram is mainly done to check for narrowed or blocked heart arteries. It may be done if you have symptoms of coronary artery disease. The test also can look for other heart problems.

A CT coronary angiogram is different from a standard coronary angiogram. In a standard coronary angiogram, a healthcare professional makes a small cut in the groin or wrist. A thin, bendable tube called a catheter is put into an artery. It is then moved to the heart arteries. For people who already have coronary artery disease, this test also can be used to treat the problem.

A CT coronary angiogram is also different from a CT coronary calcium scan. A CT coronary angiogram looks for plaque and other buildup in the walls of the heart arteries. A CT coronary calcium scan looks only at how much calcium is in the artery walls.

Risks

A CT coronary angiogram uses radiation. The amount of radiation depends on the type of machine used.

If you are pregnant or may be pregnant, you should try not to have a CT angiogram. Radiation may harm an unborn baby. But some people with serious health problems may need CT imaging during pregnancy. If this happens, steps are taken to lower the radiation to the unborn baby as much as possible.

A CT coronary angiogram uses a dye called contrast. Tell your healthcare professional if you are breastfeeding. The dye can get into breast milk.

Some people have an allergic reaction to the contrast dye. Talk with your healthcare professional if you are worried about this.

If you are allergic to contrast dye, you may need to take steroid medicine 12 hours before the test. This lowers the chance of a reaction.

Rarely, contrast dye may harm the kidneys. This is more likely in people with long-term kidney problems.

How to prepare

A healthcare professional will tell you how to get ready for your CT coronary angiogram. It is usually OK to drive yourself to and from the test.

Food and medicines

Usually, you should not eat anything for at least 8 hours before the test. Drinking water is OK.

Do not have drinks with caffeine for 12 hours before the test. Caffeine can make your heart beat faster. This can make it harder to get clear pictures of your heart.

Tell your healthcare team about all the medicines you take. You may be asked not to take a certain medicine before the test.

Clothing and personal items

You will need to remove jewelry, glasses, and clothing above the waist. You will change into a hospital gown.

What to expect

A CT coronary angiogram is usually done in the radiology department of a hospital or at an outpatient imaging center.

Before the test

Before the test, a healthcare professional may give you medicine called a beta blocker. This slows your heart rate. A slower heart rate helps the CT scanner take clearer pictures.

Tell your healthcare professional if you have had side effects from beta blockers before.

You also may be given nitroglycerin. This medicine widens your heart arteries. It helps your healthcare professional see the arteries more clearly in the pictures.

During the test

A healthcare professional puts a thin tube called an IV into your hand or arm. Contrast dye flows through the IV. The dye helps blood vessels show up better in the pictures.

Sticky patches called electrodes are put on your chest. They record your heart rate.

You lie on a long table. The table slides through a short, tunnel-like machine called a CT scanner.

If you are not comfortable in small or closed spaces, ask your healthcare professional about medicine to help you relax. Ask before the day of your test.

During the scan, you need to stay still. You also need to hold your breath when you are told to. Moving can make the pictures blurry.

A healthcare professional controls the CT machine from another room. A glass window separates the rooms. You can talk to each other through an intercom.

The scanning part of the test may take as little as 5 seconds. The whole process may take up to an hour if you are given medicines such as beta blockers or nitroglycerin.

After the test

After the CT coronary angiogram, you can usually go back to your daily activities. You should be able to drive yourself home or to work if the CT scan causes no problems.

Drink plenty of water. This helps flush the dye from your body.

Results

The pictures from your CT coronary angiogram should be ready soon after your test. The healthcare professional who ordered the test will give you the results.

If your test shows that you have heart disease or may be at risk for it, you and your healthcare professional can talk about treatment choices.

No matter what your results are, it is always a good idea to make healthy changes to help protect your heart. Try these heart-healthy habits:

Get regular exercise. Exercise helps manage weight. It also helps control diabetes, high cholesterol, and high blood pressure. These are all risk factors for coronary artery disease. Try to get at least 150 minutes of moderate aerobic activity each week. Or get 75 minutes of vigorous aerobic activity each week. You also can do a mix of both.

Eat healthy foods. Eat plenty of fruits, vegetables, whole grains, beans, peas, lentils, and nuts. Avoid saturated fats and trans fats. Cut back on salt and sugar. Eating one or two servings of fish a week also may help keep the heart healthy.

Lose extra weight. Reaching and staying at a healthy weight is good for your heart. Losing even a small amount of weight can help lower risk factors for coronary artery disease. You can ask your healthcare professional to help set a goal weight for you.

Do not smoke or use tobacco. Smoking is a major risk factor for coronary artery disease. Nicotine tightens blood vessels. This makes the heart work harder. Not smoking is one of the best ways to lower the risk of a heart attack. If you need help quitting, talk with your healthcare professional.

Manage health conditions. If you have high blood pressure, high cholesterol, or diabetes, take your medicines as directed. Ask your healthcare professional how often you need checkups.

Lower stress. Stress can make blood vessels tighten. This raises the risk of a heart attack. To ease stress, try to exercise more, practice mindfulness, and connect with others in support groups.

Get enough sleep. Adults should try to get 7 to 9 hours of sleep each night.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

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Original page

Transcatheter aortic valve replacement (TAVR)

Overview

Transcatheter aortic valve replacement, also called TAVR, is a treatment to replace a narrowed aortic valve that doesn't open fully. The aortic valve is between the lower left heart chamber and the body's main artery. Narrowing of the aortic valve is called aortic valve stenosis. The valve condition blocks or slows blood flow from the heart to the body.

TAVR is minimally invasive. That means it uses smaller surgical cuts than open-heart valve surgery. TAVR may be an option for people who can't have open-heart surgery to replace the aortic valve. TAVR can help ease chest pain, shortness of breath and other symptoms of aortic valve stenosis.

The decision to have TAVR is made after talking with a team of doctors and surgeons who are focused on the heart. The team works together to figure out the best treatment for you.

Transcatheter aortic valve replacement also may be called transcatheter aortic valve implantation, known as TAVI.

What is TAVR?

Transcatheter aortic valve replacement, also called TAVR, is a minimally invasive treatment for aortic valve stenosis. This animation shows how TAVR is performed.

Here's how TAVR works:

A thin flexible tube called a balloon catheter delivers a collapsible replacement valve to the heart.

The new valve is placed inside the diseased valve and expanded - pushing the old valve leaflets out of the way.

The new valve works like a healthy one, restoring normal blood flow from the ventricle.

Why it's done

Transcatheter aortic valve replacement, also called TAVR, is a treatment for aortic valve stenosis. In this condition, also called aortic stenosis, the heart's aortic valve thickens and becomes stiff and narrow. As a result, the valve can't fully open and less blood flows to the body.

TAVR is an alternative to open-heart aortic valve replacement surgery. People who have TAVR often have a shorter hospital stay than those who have open-heart surgery.

Your healthcare professional may recommend TAVR if you have:

Severe aortic stenosis that causes symptoms such as chest pain and shortness of breath.

A replacement valve made from animal or human heart tissue that isn't working as well as it should.

Another health condition, such as lung or kidney disease, that makes open-heart surgery too risky.

Risks

All surgeries and heart valve treatments come with some type of risk. Possible risks of transcatheter aortic valve replacement, also called TAVR, can include:

Bleeding.

Blood vessel injuries.

Problems with the replacement valve, such as the valve slipping out of place or leaking.

Stroke.

Heart rhythm conditions and the need for a pacemaker.

Kidney disease.

Heart attack.

Infection.

Death.

Studies have found that the risks of disabling stroke and death are similar among those who have TAVR and aortic valve replacement surgery.

How you prepare

Your healthcare team gives you instructions on how to prepare for transcatheter aortic valve replacement, also called TAVR. Talk with your healthcare if you have any questions about TAVR.

Food and medications

Tell your healthcare team about all the medicines you take. Include medicines and supplements bought without a prescription. Also tell the team if you have allergies to any medicines.

Before having TAVR, ask your healthcare team if and when you can take your regular medicines.

Most people are told not to drink or eat for a certain amount of time before TAVR. Ask your healthcare team when you need to stop eating or drinking.

Clothing and personal items

You may be asked to bring the following items to the hospital:

Eyeglasses, hearing aids or dentures.

Personal care items, such as a brush or comb, toothbrush, and shaving equipment.

Loose-fitting, comfortable clothing.

Items that may help you relax, such as portable music players or books.

During your treatment, do not wear:

Contact lenses.

Dentures.

Eyeglasses.

Jewelry.

Nail polish.

What you can expect

Before

A healthcare professional places an IV into your forearm or hand. Medicine called a sedative goes through the IV. The medicine helps you feel relaxed, calm or sleepy. Medicines to prevent blood clots and infection also may be given through the IV.

Your treatment team gives you medicine called an anesthetic to prevent pain. You may be given a local anesthetic that prevents pain in the treatment area. This is given along with a sedative. Or you may receive a combination of medicines that put you in a sleeplike state.

A member of your healthcare team may shave any hair from the area on your body where the treatment will take place.

During

During transcatheter aortic valve replacement, a doctor replaces a damaged aortic valve with one made from cow or pig heart tissue. The cow or pig valve is called a biological tissue valve. Sometimes, doctors place a biological tissue valve into an existing valve that no longer works.

TAVR uses small cuts and a flexible tube called a catheter to reach the heart. It's different from open-heart surgery to replace the aortic valve. That surgery requires a long cut down the chest.

In TAVR, a doctor inserts a catheter into a blood vessel, usually in the groin or chest area. The doctor guides the catheter to the site of the aortic valve in the heart using X-ray or other imaging tools as a guide.

The doctor sends a biological tissue valve through the catheter and places it near the aortic valve. A balloon on the catheter tip expands to press the new aortic valve into place. Some replacement valves expand without the use of a balloon.

The doctor removes the catheter once the new valve is securely in place.

During TAVR, your healthcare team watches you carefully. Your blood pressure, heart rate and rhythm, and breathing are checked constantly.

After

You may spend the night in a hospital's intensive care unit. That way, you can be watched carefully after your treatment. How long you stay in the hospital depends on many things. Some people who have TAVR might go home the next day. Less often, some go home the same day.

Before you leave the hospital, your treatment team tells you how to care for any wounds. Your team also tells you how to watch for signs of infection. These include fever, greater pain, swelling, and draining or oozing at the catheter site.

Some medicines may be prescribed after TAVR, including:

Blood thinners, also called anticoagulants. This medicine helps prevent blood clots. Your healthcare team tells you how long you may need to take this medicine. Always take medicines as directed.

Antibiotics. These medicines treat and prevent infections caused by bacteria. Germs can stick to or infect an artificial heart valve. Most bacteria that cause heart valve infections come from the bacteria in the mouth. Taking good care of your teeth and mouth can help prevent these infections. Get regular dental checkups. Antibiotics may be prescribed for use before certain dental treatments.

You need regular healthcare checkups and imaging tests after TAVR. These checkups help your healthcare team make sure that the new valve is working properly. Let a member of your care team know if you have any new symptoms or symptoms that get worse, including:

Dizziness or lightheadedness.

Swelling of the ankles.

Extreme tiredness with activity.

Swelling, change in skin color, tenderness or other infection symptoms at the catheter site.

Seek emergency medical help if you have:

Chest pain, pressure or tightness.

Severe, sudden shortness of breath.

Fainting.

Results

Transcatheter aortic valve replacement, also called TAVR, may ease symptoms of aortic valve stenosis. Having fewer symptoms may help improve quality of life.

It's important to follow a heart-healthy lifestyle as you recover from TAVR. Such lifestyle habits also can help prevent other heart conditions. After TAVR:

If you smoke, quit. You can ask your healthcare professional for help with this.

Eat a heathy diet rich in fruits and vegetables and low in salt, saturated fats and trans fats.

Get regular exercise. But talk with your healthcare professional before starting a new exercise routine.

Stay at a healthy weight. Ask your healthcare team what a healthy weight is for you.

AI rewrite

Heart Valve Replacement Through a Tube (TAVR)

Overview

TAVR is a way to fix a heart valve that is too tight. The full name is transcatheter aortic valve replacement. The aortic valve is between the lower left part of your heart and the body's main blood tube. When this valve is too narrow, it is called aortic valve stenosis. A tight valve blocks or slows blood as it leaves the heart and goes to the body.

TAVR uses smaller cuts than open-heart surgery. This may be a good choice for people who cannot have open-heart surgery. TAVR can help with chest pain, trouble breathing, and other signs of a tight valve.

You and a team of heart doctors and surgeons will decide together if TAVR is right for you.

TAVR is sometimes also called TAVI. That stands for transcatheter aortic valve implantation.

What is TAVR?

TAVR is a way to fix a tight aortic valve using small cuts. Here is how it works:

- A thin, bendy tube called a balloon catheter carries a new valve to your heart. The new valve is folded up small.

- The doctor puts the new valve inside the old valve. Then the new valve opens up. This pushes the old valve flaps out of the way.

- The new valve works like a healthy one. Blood can flow the right way again.

Why It's Done

TAVR treats a tight aortic valve. With this problem, the valve gets thick, stiff, and narrow. It cannot open all the way. So less blood flows to the body.

TAVR is another choice instead of open-heart surgery. People who have TAVR often go home from the hospital sooner than people who have open-heart surgery.

Your doctor may suggest TAVR if you have:

- A very tight valve that gives you chest pain or trouble breathing.

- A past valve replacement, made from animal or human heart tissue, that is no longer working well.

- Another health problem, like lung or kidney disease, that makes open-heart surgery too risky.

Risks

All surgeries and valve treatments have some risk. Risks of TAVR can include:

- Bleeding.

- Harm to blood vessels.

- Problems with the new valve, like it slipping out of place or leaking.

- Stroke.

- Heart beat problems and the need for a pacemaker.

- Kidney disease.

- Heart attack.

- Infection.

- Death.

Studies show that the risk of a serious stroke or death is about the same for TAVR and open-heart surgery.

How You Prepare

Your care team will tell you how to get ready for TAVR. Ask them if you have any questions.

Food and Medicines

Tell your care team about all the medicines you take. This includes medicines and vitamins you buy without a prescription. Also tell them if you are allergic to any medicines.

Before TAVR, ask your care team if and when you can take your normal medicines.

Most people are told not to eat or drink for some time before TAVR. Ask your care team when you need to stop eating and drinking.

Clothes and Personal Items

You may be asked to bring these things to the hospital:

- Eyeglasses, hearing aids, or false teeth.

- Personal care items, like a brush or comb, toothbrush, and shaving tools.

- Loose, comfy clothes.

- Things to help you relax, like music players or books.

During your treatment, do not wear:

- Contact lenses.

- False teeth.

- Eyeglasses.

- Jewelry.

- Nail polish.

What You Can Expect

Before

A care team member puts an IV into your arm or hand. Medicine to help you relax goes in through the IV. This medicine helps you feel calm or sleepy. You may also get medicine to stop blood clots and infection through the IV.

Your team gives you medicine to stop pain. You may get medicine that numbs just the treatment area. You also get medicine to help you relax. Or you may get medicine that puts you in a deep, sleep-like state.

A team member may shave hair from the spot where the treatment will happen.

During

In TAVR, a doctor replaces your bad aortic valve with a new one. The new valve is made from cow or pig heart tissue. This is called a biological tissue valve. Sometimes the doctor puts the new valve inside an old valve that no longer works.

TAVR uses small cuts and a bendy tube called a catheter to reach the heart. This is different from open-heart surgery. That surgery needs a long cut down the chest.

In TAVR, the doctor puts a catheter into a blood vessel. This is often in the groin or chest. The doctor guides the catheter to the aortic valve. They use X-ray or other pictures to help guide it.

The doctor sends the new valve through the catheter and places it near the aortic valve. A balloon on the tip of the catheter blows up. This presses the new valve into place. Some new valves open without a balloon.

The doctor takes out the catheter once the new valve is in place.

During TAVR, your care team watches you closely. They check your blood pressure, heart beat, and breathing the whole time.

After

You may spend the night in the hospital's intensive care unit. This lets the team watch you closely after your treatment. How long you stay depends on many things. Some people go home the next day. Less often, some go home the same day.

Before you leave, your team tells you how to care for any wounds. They also tell you how to watch for signs of infection. These signs are fever, more pain, swelling, and fluid leaking from the cut.

You may get some medicines after TAVR, like:

- Blood thinners. These help stop blood clots. Your care team tells you how long to take them. Always take medicine the way you are told.

- Antibiotics. These treat and stop infections caused by germs. Germs can stick to or infect a new heart valve. Most germs that cause valve infections come from the mouth. Taking good care of your teeth and mouth can help stop these infections. Get regular dental checkups. You may need antibiotics before some dental work.

You need regular checkups and picture tests after TAVR. These help your care team make sure the new valve works well. Tell your care team if you have any new signs, or signs that get worse, like:

- Feeling dizzy or like you might faint.

- Swollen ankles.

- Feeling very tired when you move around.

- Swelling, skin color changes, soreness, or other signs of infection at the cut site.

Get emergency help right away if you have:

- Chest pain, pressure, or tightness.

- Very bad, sudden trouble breathing.

- Fainting.

Results

TAVR may help with the signs of a tight aortic valve. Fewer signs may help you feel better and live a fuller life.

It is important to live in a heart-healthy way as you heal. These habits can also help stop other heart problems. After TAVR:

- If you smoke, quit. Your care team can help you.
- Eat healthy foods with lots of fruits and vegetables. Eat less salt and less bad fat.
- Get regular exercise. But talk with your care team before you start a new exercise plan.
- Stay at a healthy weight. Ask your care team what a healthy weight is for you.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Transcatheter aortic valve replacement (TAVR)

Overview

Transcatheter aortic valve replacement, also called TAVR, is a treatment to replace a narrowed aortic valve that doesn't open fully. The aortic valve is between the lower left heart chamber and the body's main artery. Narrowing of the aortic valve is called aortic valve stenosis. The valve condition blocks or slows blood flow from the heart to the body.

TAVR is minimally invasive. That means it uses smaller surgical cuts than open-heart valve surgery. TAVR may be an option for people who can't have open-heart surgery to replace the aortic valve. TAVR can help ease chest pain, shortness of breath and other symptoms of aortic valve stenosis.

The decision to have TAVR is made after talking with a team of doctors and surgeons who are focused on the heart. The team works together to figure out the best treatment for you.

Transcatheter aortic valve replacement also may be called transcatheter aortic valve implantation, known as TAVI.

What is TAVR?

Transcatheter aortic valve replacement, also called TAVR, is a minimally invasive treatment for aortic valve stenosis. This animation shows how TAVR is performed.

Here's how TAVR works:

A thin flexible tube called a balloon catheter delivers a collapsible replacement valve to the heart.

The new valve is placed inside the diseased valve and expanded - pushing the old valve leaflets out of the way.

The new valve works like a healthy one, restoring normal blood flow from the ventricle.

Why it's done

Transcatheter aortic valve replacement, also called TAVR, is a treatment for aortic valve stenosis. In this condition, also called aortic stenosis, the heart's aortic valve thickens and becomes stiff and narrow. As a result, the valve can't fully open and less blood flows to the body.

TAVR is an alternative to open-heart aortic valve replacement surgery. People who have TAVR often have a shorter hospital stay than those who have open-heart surgery.

Your healthcare professional may recommend TAVR if you have:

Severe aortic stenosis that causes symptoms such as chest pain and shortness of breath.

A replacement valve made from animal or human heart tissue that isn't working as well as it should.

Another health condition, such as lung or kidney disease, that makes open-heart surgery too risky.

Risks

All surgeries and heart valve treatments come with some type of risk. Possible risks of transcatheter aortic valve replacement, also called TAVR, can include:

Bleeding.

Blood vessel injuries.

Problems with the replacement valve, such as the valve slipping out of place or leaking.

Stroke.

Heart rhythm conditions and the need for a pacemaker.

Kidney disease.

Heart attack.

Infection.

Death.

Studies have found that the risks of disabling stroke and death are similar among those who have TAVR and aortic valve replacement surgery.

How you prepare

Your healthcare team gives you instructions on how to prepare for transcatheter aortic valve replacement, also called TAVR. Talk with your healthcare if you have any questions about TAVR.

Food and medications

Tell your healthcare team about all the medicines you take. Include medicines and supplements bought without a prescription. Also tell the team if you have allergies to any medicines.

Before having TAVR, ask your healthcare team if and when you can take your regular medicines.

Most people are told not to drink or eat for a certain amount of time before TAVR. Ask your healthcare team when you need to stop eating or drinking.

Clothing and personal items

You may be asked to bring the following items to the hospital:

Eyeglasses, hearing aids or dentures.

Personal care items, such as a brush or comb, toothbrush, and shaving equipment.

Loose-fitting, comfortable clothing.

Items that may help you relax, such as portable music players or books.

During your treatment, do not wear:

Contact lenses.

Dentures.

Eyeglasses.

Jewelry.

Nail polish.

What you can expect

Before

A healthcare professional places an IV into your forearm or hand. Medicine called a sedative goes through the IV. The medicine helps you feel relaxed, calm or sleepy. Medicines to prevent blood clots and infection also may be given through the IV.

Your treatment team gives you medicine called an anesthetic to prevent pain. You may be given a local anesthetic that prevents pain in the treatment area. This is given along with a sedative. Or you may receive a combination of medicines that put you in a sleeplike state.

A member of your healthcare team may shave any hair from the area on your body where the treatment will take place.

During

During transcatheter aortic valve replacement, a doctor replaces a damaged aortic valve with one made from cow or pig heart tissue. The cow or pig valve is called a biological tissue valve. Sometimes, doctors place a biological tissue valve into an existing valve that no longer works.

TAVR uses small cuts and a flexible tube called a catheter to reach the heart. It's different from open-heart surgery to replace the aortic valve. That surgery requires a long cut down the chest.

In TAVR, a doctor inserts a catheter into a blood vessel, usually in the groin or chest area. The doctor guides the catheter to the site of the aortic valve in the heart using X-ray or other imaging tools as a guide.

The doctor sends a biological tissue valve through the catheter and places it near the aortic valve. A balloon on the catheter tip expands to press the new aortic valve into place. Some replacement valves expand without the use of a balloon.

The doctor removes the catheter once the new valve is securely in place.

During TAVR, your healthcare team watches you carefully. Your blood pressure, heart rate and rhythm, and breathing are checked constantly.

After

You may spend the night in a hospital's intensive care unit. That way, you can be watched carefully after your treatment. How long you stay in the hospital depends on many things. Some people who have TAVR might go home the next day. Less often, some go home the same day.

Before you leave the hospital, your treatment team tells you how to care for any wounds. Your team also tells you how to watch for signs of infection. These include fever, greater pain, swelling, and draining or oozing at the catheter site.

Some medicines may be prescribed after TAVR, including:

Blood thinners, also called anticoagulants. This medicine helps prevent blood clots. Your healthcare team tells you how long you may need to take this medicine. Always take medicines as directed.

Antibiotics. These medicines treat and prevent infections caused by bacteria. Germs can stick to or infect an artificial heart valve. Most bacteria that cause heart valve infections come from the bacteria in the mouth. Taking good care of your teeth and mouth can help prevent these infections. Get regular dental checkups. Antibiotics may be prescribed for use before certain dental treatments.

You need regular healthcare checkups and imaging tests after TAVR. These checkups help your healthcare team make sure that the new valve is working properly. Let a member of your care team know if you have any new symptoms or symptoms that get worse, including:

Dizziness or lightheadedness.

Swelling of the ankles.

Extreme tiredness with activity.

Swelling, change in skin color, tenderness or other infection symptoms at the catheter site.

Seek emergency medical help if you have:

Chest pain, pressure or tightness.

Severe, sudden shortness of breath.

Fainting.

Results

Transcatheter aortic valve replacement, also called TAVR, may ease symptoms of aortic valve stenosis. Having fewer symptoms may help improve quality of life.

It's important to follow a heart-healthy lifestyle as you recover from TAVR. Such lifestyle habits also can help prevent other heart conditions. After TAVR:

If you smoke, quit. You can ask your healthcare professional for help with this.

Eat a heathy diet rich in fruits and vegetables and low in salt, saturated fats and trans fats.

Get regular exercise. But talk with your healthcare professional before starting a new exercise routine.

Stay at a healthy weight. Ask your healthcare team what a healthy weight is for you.

AI rewrite

Transcatheter aortic valve replacement (TAVR)

Overview

Transcatheter aortic valve replacement is also called TAVR. It is a treatment to replace a narrowed aortic valve that does not open all the way.

The aortic valve is between the lower left chamber of the heart and the body’s main artery. When this valve gets narrow, it is called aortic valve stenosis. This can block or slow blood flow from the heart to the body.

TAVR is minimally invasive. This means it uses smaller cuts than open-heart valve surgery.

TAVR may be an option for people who cannot have open-heart surgery to replace the aortic valve. TAVR can help ease chest pain, shortness of breath, and other symptoms of aortic valve stenosis.

The choice to have TAVR is made after you talk with a team of heart doctors and surgeons. The team works together to choose the best treatment for you.

TAVR may also be called transcatheter aortic valve implantation, or TAVI.

What is TAVR?

TAVR is a minimally invasive treatment for aortic valve stenosis.

Here is how TAVR works:

A thin, flexible tube called a balloon catheter carries a folded replacement valve to the heart.

The new valve is placed inside the damaged valve and opened. This pushes the old valve parts out of the way.

The new valve works like a healthy valve. It helps blood flow normally from the heart chamber to the body.

Why it is done

TAVR is a treatment for aortic valve stenosis. This is also called aortic stenosis. In this condition, the aortic valve becomes thick, stiff, and narrow. The valve cannot open all the way. Less blood flows to the body.

TAVR is an option instead of open-heart surgery to replace the aortic valve. People who have TAVR often stay in the hospital for less time than people who have open-heart surgery.

Your healthcare professional may suggest TAVR if you have:

Severe aortic stenosis that causes symptoms, such as chest pain and shortness of breath.

A replacement valve made from animal or human heart tissue that is not working as well as it should.

Another health problem, such as lung or kidney disease, that makes open-heart surgery too risky.

Risks

All surgeries and heart valve treatments have some risks. Possible risks of TAVR include:

Bleeding.

Injury to blood vessels.

Problems with the replacement valve, such as the valve moving out of place or leaking.

Stroke.

Heart rhythm problems and the need for a pacemaker.

Kidney disease.

Heart attack.

Infection.

Death.

Studies have found that the risks of disabling stroke and death are similar for people who have TAVR and people who have aortic valve replacement surgery.

How you prepare

Your healthcare team will tell you how to prepare for TAVR. Talk with your healthcare team if you have questions about TAVR.

Food and medicines

Tell your healthcare team about all medicines you take. Include medicines and supplements you buy without a prescription. Also tell the team if you are allergic to any medicines.

Before TAVR, ask your healthcare team if you can take your regular medicines and when to take them.

Most people are told not to eat or drink for a certain amount of time before TAVR. Ask your healthcare team when you need to stop eating or drinking.

Clothing and personal items

You may be asked to bring these items to the hospital:

Eyeglasses, hearing aids, or dentures.

Personal care items, such as a brush or comb, toothbrush, and shaving items.

Loose, comfortable clothes.

Items that may help you relax, such as a music player or books.

During your treatment, do not wear:

Contact lenses.

Dentures.

Eyeglasses.

Jewelry.

Nail polish.

What you can expect

Before

A healthcare professional puts an IV into your forearm or hand. Medicine called a sedative goes through the IV. This medicine helps you feel relaxed, calm, or sleepy. Medicines to prevent blood clots and infection also may be given through the IV.

Your treatment team gives you medicine called an anesthetic to prevent pain. You may get a local anesthetic. This prevents pain in the treatment area. It is given with a sedative. Or you may get medicines that put you in a sleeplike state.

A member of your healthcare team may shave hair from the area where the treatment will happen.

During

During TAVR, a doctor replaces a damaged aortic valve with one made from cow or pig heart tissue. The cow or pig valve is called a biological tissue valve. Sometimes, doctors put a biological tissue valve into an existing valve that no longer works.

TAVR uses small cuts and a flexible tube called a catheter to reach the heart. It is different from open-heart surgery to replace the aortic valve. Open-heart surgery needs a long cut down the chest.

In TAVR, a doctor puts a catheter into a blood vessel. This is usually in the groin or chest area. The doctor guides the catheter to the aortic valve in the heart. X-ray or other imaging tools help guide it.

The doctor sends a biological tissue valve through the catheter and places it near the aortic valve. A balloon on the tip of the catheter opens to press the new aortic valve into place. Some replacement valves open without a balloon.

The doctor removes the catheter after the new valve is firmly in place.

During TAVR, your healthcare team watches you closely. Your blood pressure, heart rate and rhythm, and breathing are checked all the time.

After

You may spend the night in a hospital intensive care unit. This lets your team watch you closely after treatment. How long you stay in the hospital depends on many things. Some people who have TAVR may go home the next day. Less often, some go home the same day.

Before you leave the hospital, your treatment team tells you how to care for any wounds. Your team also tells you how to watch for signs of infection. These include fever, more pain, swelling, and fluid or pus coming from the catheter site.

Some medicines may be prescribed after TAVR, including:

Blood thinners, also called anticoagulants. This medicine helps prevent blood clots. Your healthcare team tells you how long you may need to take it. Always take medicines as directed.

Antibiotics. These medicines treat and prevent infections caused by bacteria. Germs can stick to or infect an artificial heart valve. Most bacteria that cause heart valve infections come from the mouth. Taking good care of your teeth and mouth can help prevent these infections. Get regular dental checkups. Antibiotics may be prescribed before some dental treatments.

You need regular healthcare checkups and imaging tests after TAVR. These checkups help your healthcare team make sure the new valve is working well.

Tell a member of your care team if you have any new symptoms or symptoms that get worse, including:

Dizziness or lightheadedness.

Swelling of the ankles.

Feeling very tired during activity.

Swelling, skin color change, tenderness, or other signs of infection at the catheter site.

Get emergency medical help if you have:

Chest pain, pressure, or tightness.

Severe, sudden shortness of breath.

Fainting.

Results

TAVR may ease symptoms of aortic valve stenosis. Having fewer symptoms may help improve quality of life.

It is important to follow a heart-healthy lifestyle as you recover from TAVR. These habits also can help prevent other heart conditions. After TAVR:

If you smoke, quit. You can ask your healthcare professional for help.

Eat a healthy diet with lots of fruits and vegetables. Eat foods low in salt, saturated fats, and trans fats.

Get regular exercise. Talk with your healthcare professional before starting a new exercise routine.

Stay at a healthy weight. Ask your healthcare team what a healthy weight is for you.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

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Original page

Transcatheter aortic valve replacement (TAVR)

Overview

Transcatheter aortic valve replacement, also called TAVR, is a treatment to replace a narrowed aortic valve that doesn't open fully. The aortic valve is between the lower left heart chamber and the body's main artery. Narrowing of the aortic valve is called aortic valve stenosis. The valve condition blocks or slows blood flow from the heart to the body.

TAVR is minimally invasive. That means it uses smaller surgical cuts than open-heart valve surgery. TAVR may be an option for people who can't have open-heart surgery to replace the aortic valve. TAVR can help ease chest pain, shortness of breath and other symptoms of aortic valve stenosis.

The decision to have TAVR is made after talking with a team of doctors and surgeons who are focused on the heart. The team works together to figure out the best treatment for you.

Transcatheter aortic valve replacement also may be called transcatheter aortic valve implantation, known as TAVI.

What is TAVR?

Transcatheter aortic valve replacement, also called TAVR, is a minimally invasive treatment for aortic valve stenosis. This animation shows how TAVR is performed.

Here's how TAVR works:

A thin flexible tube called a balloon catheter delivers a collapsible replacement valve to the heart.

The new valve is placed inside the diseased valve and expanded - pushing the old valve leaflets out of the way.

The new valve works like a healthy one, restoring normal blood flow from the ventricle.

Why it's done

Transcatheter aortic valve replacement, also called TAVR, is a treatment for aortic valve stenosis. In this condition, also called aortic stenosis, the heart's aortic valve thickens and becomes stiff and narrow. As a result, the valve can't fully open and less blood flows to the body.

TAVR is an alternative to open-heart aortic valve replacement surgery. People who have TAVR often have a shorter hospital stay than those who have open-heart surgery.

Your healthcare professional may recommend TAVR if you have:

Severe aortic stenosis that causes symptoms such as chest pain and shortness of breath.

A replacement valve made from animal or human heart tissue that isn't working as well as it should.

Another health condition, such as lung or kidney disease, that makes open-heart surgery too risky.

Risks

All surgeries and heart valve treatments come with some type of risk. Possible risks of transcatheter aortic valve replacement, also called TAVR, can include:

Bleeding.

Blood vessel injuries.

Problems with the replacement valve, such as the valve slipping out of place or leaking.

Stroke.

Heart rhythm conditions and the need for a pacemaker.

Kidney disease.

Heart attack.

Infection.

Death.

Studies have found that the risks of disabling stroke and death are similar among those who have TAVR and aortic valve replacement surgery.

How you prepare

Your healthcare team gives you instructions on how to prepare for transcatheter aortic valve replacement, also called TAVR. Talk with your healthcare if you have any questions about TAVR.

Food and medications

Tell your healthcare team about all the medicines you take. Include medicines and supplements bought without a prescription. Also tell the team if you have allergies to any medicines.

Before having TAVR, ask your healthcare team if and when you can take your regular medicines.

Most people are told not to drink or eat for a certain amount of time before TAVR. Ask your healthcare team when you need to stop eating or drinking.

Clothing and personal items

You may be asked to bring the following items to the hospital:

Eyeglasses, hearing aids or dentures.

Personal care items, such as a brush or comb, toothbrush, and shaving equipment.

Loose-fitting, comfortable clothing.

Items that may help you relax, such as portable music players or books.

During your treatment, do not wear:

Contact lenses.

Dentures.

Eyeglasses.

Jewelry.

Nail polish.

What you can expect

Before

A healthcare professional places an IV into your forearm or hand. Medicine called a sedative goes through the IV. The medicine helps you feel relaxed, calm or sleepy. Medicines to prevent blood clots and infection also may be given through the IV.

Your treatment team gives you medicine called an anesthetic to prevent pain. You may be given a local anesthetic that prevents pain in the treatment area. This is given along with a sedative. Or you may receive a combination of medicines that put you in a sleeplike state.

A member of your healthcare team may shave any hair from the area on your body where the treatment will take place.

During

During transcatheter aortic valve replacement, a doctor replaces a damaged aortic valve with one made from cow or pig heart tissue. The cow or pig valve is called a biological tissue valve. Sometimes, doctors place a biological tissue valve into an existing valve that no longer works.

TAVR uses small cuts and a flexible tube called a catheter to reach the heart. It's different from open-heart surgery to replace the aortic valve. That surgery requires a long cut down the chest.

In TAVR, a doctor inserts a catheter into a blood vessel, usually in the groin or chest area. The doctor guides the catheter to the site of the aortic valve in the heart using X-ray or other imaging tools as a guide.

The doctor sends a biological tissue valve through the catheter and places it near the aortic valve. A balloon on the catheter tip expands to press the new aortic valve into place. Some replacement valves expand without the use of a balloon.

The doctor removes the catheter once the new valve is securely in place.

During TAVR, your healthcare team watches you carefully. Your blood pressure, heart rate and rhythm, and breathing are checked constantly.

After

You may spend the night in a hospital's intensive care unit. That way, you can be watched carefully after your treatment. How long you stay in the hospital depends on many things. Some people who have TAVR might go home the next day. Less often, some go home the same day.

Before you leave the hospital, your treatment team tells you how to care for any wounds. Your team also tells you how to watch for signs of infection. These include fever, greater pain, swelling, and draining or oozing at the catheter site.

Some medicines may be prescribed after TAVR, including:

Blood thinners, also called anticoagulants. This medicine helps prevent blood clots. Your healthcare team tells you how long you may need to take this medicine. Always take medicines as directed.

Antibiotics. These medicines treat and prevent infections caused by bacteria. Germs can stick to or infect an artificial heart valve. Most bacteria that cause heart valve infections come from the bacteria in the mouth. Taking good care of your teeth and mouth can help prevent these infections. Get regular dental checkups. Antibiotics may be prescribed for use before certain dental treatments.

You need regular healthcare checkups and imaging tests after TAVR. These checkups help your healthcare team make sure that the new valve is working properly. Let a member of your care team know if you have any new symptoms or symptoms that get worse, including:

Dizziness or lightheadedness.

Swelling of the ankles.

Extreme tiredness with activity.

Swelling, change in skin color, tenderness or other infection symptoms at the catheter site.

Seek emergency medical help if you have:

Chest pain, pressure or tightness.

Severe, sudden shortness of breath.

Fainting.

Results

Transcatheter aortic valve replacement, also called TAVR, may ease symptoms of aortic valve stenosis. Having fewer symptoms may help improve quality of life.

It's important to follow a heart-healthy lifestyle as you recover from TAVR. Such lifestyle habits also can help prevent other heart conditions. After TAVR:

If you smoke, quit. You can ask your healthcare professional for help with this.

Eat a heathy diet rich in fruits and vegetables and low in salt, saturated fats and trans fats.

Get regular exercise. But talk with your healthcare professional before starting a new exercise routine.

Stay at a healthy weight. Ask your healthcare team what a healthy weight is for you.

AI rewrite

Transcatheter aortic valve replacement (TAVR)

Overview

Transcatheter aortic valve replacement is also called TAVR. It is a treatment to replace a narrowed heart valve that does not open all the way. This valve is called the aortic valve. It sits between the lower left part of your heart and your body's main blood vessel. When this valve gets too narrow, it is called aortic valve stenosis. This problem blocks or slows down blood flowing from your heart to your body.

TAVR uses very small cuts. It does not need a large cut like open-heart surgery. TAVR might be a good choice for people who cannot have open-heart surgery. It can help stop chest pain, trouble breathing, and other symptoms.

You and a team of heart doctors will decide if TAVR is right for you. The team works together to find your best treatment. TAVR is sometimes called TAVI.

What is TAVR?

TAVR is a treatment for a narrowed heart valve. Here is how it works:

* A doctor uses a thin, bendable tube called a catheter. This tube carries a folded-up new valve to your heart.

* The new valve is placed inside your old, sick valve.

* The new valve is opened up. This pushes the old valve out of the way.

* The new valve works like a healthy one. It lets blood flow normally again.

Why it's done

TAVR treats a narrowed aortic valve. When you have this problem, your heart valve gets thick, stiff, and narrow. The valve cannot open all the way. This means less blood flows to your body.

TAVR is a choice instead of open-heart surgery. People who have TAVR often go home from the hospital sooner. Your doctor might suggest TAVR if you have:

* Severe narrowing that causes chest pain and trouble breathing.

* An older replacement valve made from animal or human tissue that is not working well anymore.

* Other health problems, like lung or kidney disease, that make open-heart surgery too dangerous.

Risks

All surgeries and heart treatments have some risks. The risks of TAVR include:

* Bleeding.

* Hurt blood vessels.

* Problems with the new valve, like slipping or leaking.

* Stroke.

* Heartbeat problems that might need a pacemaker.

* Kidney disease.

* Heart attack.

* Infection.

* Death.

Studies show that the risks of a bad stroke or death are about the same for TAVR and open-heart surgery.

How you prepare

Your care team will tell you how to get ready for TAVR. Ask them if you have any questions.

Food and medications

Tell your team about all the medicines and vitamins you take. This includes things you buy without a prescription. Tell them if you are allergic to any medicines. Ask your team if you should take your normal medicines before your TAVR.

Most people must stop eating and drinking for a certain amount of time before the treatment. Ask your team when you need to stop.

Clothing and personal items

You may need to bring these things to the hospital:

* Glasses, hearing aids, or false teeth.

* A brush, comb, toothbrush, and shaving items.

* Loose, comfortable clothes.

* Things to help you relax, like a book or music player.

During your treatment, do not wear:

* Contact lenses.

* False teeth.

* Glasses.

* Jewelry.

* Nail polish.

What you can expect

Before

A nurse will put an IV tube into a vein in your hand or arm. Medicine will go through the IV to make you feel calm and sleepy. You might also get medicines to stop blood clots and infections.

Your team will give you medicine to stop pain. You might get medicine to numb just the treatment area while you are sleepy. Or, you might get medicine that puts you completely to sleep.

A team member may shave hair from the spot on your body where the doctor will work.

During

The doctor will replace your bad valve with a new one. The new valve is made from cow or pig heart tissue. Sometimes, the new valve is placed right inside an old replacement valve that stopped working.

The doctor makes a small cut, usually in your groin or chest. They put a thin tube called a catheter into a blood vessel. They use X-ray pictures to guide the tube up to your heart.

The doctor sends the new valve through the tube. A small balloon at the end of the tube may blow up to press the new valve into place. Some valves open up without a balloon. Once the new valve is in place, the doctor takes the tube out.

Your care team will watch you very closely the whole time. They will check your blood pressure, heartbeat, and breathing.

After

You might spend the night in the intensive care unit (ICU). This lets the team watch you closely. How long you stay in the hospital depends on how you feel. Some people go home the next day. A few people go home the same day.

Before you leave, your team will teach you how to care for your cuts. They will tell you how to spot signs of an infection. Signs of infection include a fever, more pain, swelling, or fluid leaking from your cut.

You may need to take new medicines after TAVR:

* Blood thinners: These stop blood clots from forming. Your team will tell you how long to take them. Always take them exactly as told.

* Antibiotics: These stop infections caused by germs. Germs can stick to your new heart valve. Most germs that infect heart valves come from your mouth. Brushing your teeth and seeing a dentist regularly helps stop these infections. You might need to take antibiotics before you go to the dentist.

You will need regular doctor visits and tests after TAVR. This helps your team make sure the new valve is working right. Tell your doctor if you have new symptoms or if these symptoms get worse:

* Feeling dizzy or lightheaded.

* Swollen ankles.

* Feeling very tired when you move around.

* Swelling, pain, color changes, or other signs of infection where the cut was made.

Get emergency medical help right away if you have:

* Chest pain, pressure, or tightness.

* Sudden and severe trouble breathing.

* Fainting.

Results

TAVR can help stop the symptoms of a narrow heart valve. Feeling better can help you enjoy life more.

It is important to live a heart-healthy life as you heal. This can also stop other heart problems. After TAVR:

* If you smoke, you should quit. Ask your doctor for help.

* Eat healthy foods. Eat lots of fruits and vegetables. Eat less salt, saturated fats, and trans fats.

* Exercise regularly. Talk to your doctor before starting a new exercise plan.

* Stay at a healthy weight. Ask your team what a healthy weight is for you.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Transcatheter Aortic Valve Replacement (TAVR)

Cardiovascular Aortic Valve Treatment

What is transcatheter aortic valve replacement?

Transcatheter aortic valve replacement (TAVR) is a procedure that replaces a diseased aortic valve with a man-made valve. Aortic valve replacement can also be performed with open-heart surgery; this procedure is surgical aortic valve replacement (SAVR).

Your aortic valve controls blood flow from your heart to your body. If your valve becomes stiff, you have a condition called aortic stenosis. Your heart may have to work too hard to pump blood through the small valve opening to the rest of your body. This may lead to increasing heart failure.

The Food and Drug Administration approved TAVR for use in a broad spectrum of patients following multiple research studies comparing TAVR to SAVR. Whether TAVR or SAVR is more appropriate for a given individual depends on multiple factors and is discussed with each patient by both an interventional cardiologist and a cardiac surgeon. During TAVR, your doctor inserts a catheter through a blood vessel in your leg to deliver and implant the artificial valve into your heart. Significant research is exploring how to both advance the use of this technique and improve the devices that are used for TAVR.

Why might I need TAVR or SAVR?

You may need replacement of your aortic valve if you have progressive heart failure due to aortic valve stenosis. Factors making TAVR more likely to be recommended than SAVR include conditions that would increase the risk of traditional open-heart aortic valve replacement surgery such as:

Older age.

Frailty.

Weaker heart.

Previous heart surgery.

History of stroke.

Chronic obstructive lung disease (COPD).

Liver disease.

Kidney disease.

Diabetes.

Previous radiation treatment to your chest.

Large calcium deposits in the blood vessel that carries blood away from your heart (ascending aorta), called porcelain aorta.

What are the risks of TAVR?

Johns Hopkins has been performing TAVRs for more than a decade and has performed nearly 2,000 procedures. However, TAVR is still a major procedure that has risks. Most TAVR procedures are performed with sedation without the need for general anesthesia. The most common risks associated with TAVR include:

Damage to your blood vessels.

Bleeding.

Decreased blood supply to your brain, causing a stroke.

Heart attack.

Kidney failure.

Collection of fluid around your heart.

Leaking of the replacement valve.

Severe heart failure.

Death.

There may be other risks, depending on your specific medical condition. Be sure to discuss any concerns with your doctor before the procedure.

How do I prepare for TAVR?

Before surgery, your medical and surgical team will evaluate your overall health. This may include X-rays, CT scans, blood tests and other tests to check the health of your lungs and heart. Your medical team will also give you an echocardiogram to evaluate your aortic valve. This test uses sound waves to create images of your heart. Your medical team may also do a cardiac catheterization to evaluate the arteries that supply blood to your heart. You will also need to:

Tell your healthcare provider about any drugs you are taking, including over-the-counter medicines.

Quit smoking if you still smoke, because continuing to smoke increases the risk of a procedure-related problem with your lungs.

Stop certain medicines if your health care provider instructs you to.

Stop eating and drinking, usually at midnight before surgery.

What happens during TAVR?

Methods may vary, depending on your condition and your doctor's practices. Talk with your doctor about what to expect. Generally, a TAVR procedure follows this process:

You will remove any jewelry or other objects that may interfere with the procedure.

You will remove your clothing and will be given a gown to wear.

You will empty your bladder prior to the procedure.

An IV line will be started in your arm or hand. Additional catheters will be inserted in your wrist to help guide the procedure. Alternate sites for the additional catheters may include the neck and the groin.

You will be positioned on the operating table, lying on your back.

The anesthesiologist will continuously monitor your heart rate, blood pressure, breathing, and blood oxygen level during the surgery. Sedation will be given to make you comfortable. Rarely will general anesthesia be required, but then, once you are sedated, a breathing tube will be inserted into your throat and into your trachea to provide oxygen to your lungs. You will be connected to a ventilator, which will breathe for you during the surgery.

The skin over the surgical site in your groin will be cleansed with an antiseptic solution.

The surgeon will place a catheter into the leg artery (femoral artery) and thread it to your heart and through your aorta, to reach your aortic valve.

Your surgeon will place other catheters in your heart to take measurements and X-ray pictures during the procedure.

Your surgeon will guide your replacement valve, either a self-expanding valve or a balloon expandable valve, up the femoral artery catheter and through your old aortic valve.

Once the new valve is properly positioned, your surgeon will implant the new valve to replace the old one.

Your doctor will take measurements and images to make sure your new valve works properly before removing the catheters.

Your doctor will close your femoral artery with a suture device that does not require any incision.

Transthoracic echocardiography is then performed to assess the new valve's function.

What happens after TAVR?

You will be moved to the cardiac recovery floor of the hospital, so you can be monitored closely during your recovery. You will soon be able to get up and walk and return to a normal diet. You will stay overnight in the hospital. Most patients go home the next day and are allowed to resume their normal activities.

After your doctors feel you have recovered enough to go home, follow all instructions for medicines, pain control, diet, activity and wound care. Make sure to keep all your follow-up appointments.

Complete recovery from deconditioning caused by the aortic stenosis may take several weeks. Here are some helpful guidelines to follow as you heal:

Your doctor may give you medication to take for several months after TAVR.

Walk around as much as possible.

Gradually resume normal activities, but avoid any heavy lifting.

Participating in a cardiac rehabilitation program is highly recommended and will be arranged for you by your doctors.

Ask your doctors when you can resume driving, work and sexual activity.

Watch your groin for any sign of swelling, redness, bleeding or discharge.

Let your doctor know if you feel any pain or have fever, bleeding or shortness of breath.

Eat a heart-healthy diet and maintain a healthy weight.

Don't smoke.

Next steps

Before you agree to the procedure, make sure you know:

The name of the procedure.

The reason you are having the procedure.

What results to expect and what they mean.

The risks and benefits of the procedure.

What the possible side effects or complications are.

When and where you are to have the procedure.

Who will do the procedure and what that person's qualifications are.

What would happen if you did not have the procedure.

Any alternative procedures to think about.

Who to call after the procedure if you have questions or problems.

How much you will pay for the procedure.

AI rewrite

Transcatheter Aortic Valve Replacement (TAVR)

Heart Valve Treatment

What is TAVR?

TAVR is a procedure to replace a sick heart valve with a man-made one. Doctors can also replace the valve using open-heart surgery. That is called SAVR.

Your aortic valve controls blood flow from your heart to your body. Sometimes this valve gets stiff. This is called aortic stenosis. Your heart has to work too hard to pump blood through the small opening. This can lead to heart failure.

TAVR is approved for many patients. Your heart doctors will talk with you to decide if TAVR or SAVR is best for you. During TAVR, your doctor puts a thin tube called a catheter into a blood vessel in your leg. They use this tube to place the new valve into your heart. Doctors are always studying ways to make this procedure even better.

Why might I need TAVR or SAVR?

You may need a new valve if your stiff valve is causing heart failure. Doctors may choose TAVR instead of open-heart surgery if open-heart surgery is too risky for you. TAVR might be better if you have:

* Older age.

* Weakness or frailty.

* A weak heart.

* Past heart surgery.

* Past stroke.

* Lung disease (COPD).

* Liver disease.

* Kidney disease.

* Diabetes.

* Past radiation treatment to your chest.

* A lot of calcium buildup in your main blood vessel (porcelain aorta).

What are the risks of TAVR?

Johns Hopkins has been doing TAVRs for more than 10 years and has done nearly 2,000 of them. Still, TAVR is a major procedure with some risks. Most of the time, you will get medicine to make you sleepy and relaxed. You will not need to be fully put to sleep. The most common risks of TAVR include:

* Damage to your blood vessels.

* Bleeding.

* Less blood flow to your brain, which can cause a stroke.

* Heart attack.

* Kidney failure.

* Fluid building up around your heart.

* A leak in the new valve.

* Severe heart failure.

* Death.

You may have other risks based on your health. Talk to your doctor about any worries before the procedure.

How do I prepare for TAVR?

Before the procedure, your care team will check your health. You may have X-rays, CT scans, and blood tests. These check your heart and lungs. You will also have an echocardiogram. This test uses sound waves to take pictures of your heart and valve. Your team may also check the blood vessels in your heart with a thin tube.

You will also need to:

* Tell your doctor about all the medicines you take. This includes medicines you buy without a prescription.

* Stop smoking. Smoking raises the chance of lung problems during the procedure.

* Stop taking certain medicines if your doctor tells you to.

* Stop eating and drinking. This usually starts at midnight before your procedure.

What happens during TAVR?

The steps may change based on your health. Talk with your doctor about what to expect. Most of the time, TAVR goes like this:

* You will take off any jewelry or objects that could get in the way.

* You will take off your clothes and put on a hospital gown.

* You will empty your bladder before the procedure.

* A nurse will put an IV line into your arm or hand. The doctor will put extra tubes in your wrist, neck, or groin to help guide the procedure.

* You will lie on your back on the operating table.

* A doctor will watch your heart rate, blood pressure, and breathing. You will get medicine to make you sleepy and comfortable. In rare cases, you may be fully put to sleep. If so, a breathing tube will be put down your throat to help you breathe.

* The skin on your groin will be cleaned to prevent infection.

* The doctor will put a thin tube into a blood vessel in your leg. They will guide it up to your heart to reach your old valve.

* The doctor will put other tubes in your heart to measure things and take X-ray pictures.

* The doctor will push the new valve through the tube in your leg and into your old valve.

* When the new valve is in the right spot, the doctor will lock it in place.

* The doctor will check that the new valve works right before taking out the tubes.

* The doctor will close the blood vessel in your leg with a special stitch. This does not need a large cut.

* You will have an ultrasound test on your chest to check how the new valve is working.

What happens after TAVR?

You will go to a heart recovery room in the hospital. Nurses will watch you closely. Soon, you will be able to get up, walk, and eat normal food. You will stay in the hospital overnight. Most people go home the next day and can go back to normal activities.

When you go home, follow all directions from your doctor. This includes rules for medicines, pain, diet, activity, and caring for your wound. Make sure to go to all of your follow-up visits.

It may take a few weeks to fully get your strength back. Here are some tips to help you heal:

* Take the medicines your doctor gives you. You may need them for a few months.

* Walk around as much as you can.

* Slowly go back to your normal activities. Do not lift heavy things.

* Your doctor will set up a heart rehab program for you. It is highly recommended that you go.

* Ask your doctor when it is safe to drive, work, and have sex.

* Check your groin for swelling, redness, bleeding, or fluid leaking out.

* Call your doctor if you have pain, a fever, bleeding, or trouble breathing.

* Eat healthy foods and keep a healthy weight.

* Do not smoke.

Next steps

Before you agree to the procedure, make sure you know:

* The name of the procedure.

* The reason you are having it.

* What results to expect and what they mean.

* The risks and benefits.

* The possible side effects or problems.

* When and where you will have it.

* Who will do it and what their training is.

* What would happen if you do not have it.

* Other choices you might have.

* Who to call if you have questions or problems later.

* How much you will have to pay.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Transcatheter Aortic Valve Replacement (TAVR)

Cardiovascular Aortic Valve Treatment

What is transcatheter aortic valve replacement?

Transcatheter aortic valve replacement (TAVR) is a procedure that replaces a diseased aortic valve with a man-made valve. Aortic valve replacement can also be performed with open-heart surgery; this procedure is surgical aortic valve replacement (SAVR).

Your aortic valve controls blood flow from your heart to your body. If your valve becomes stiff, you have a condition called aortic stenosis. Your heart may have to work too hard to pump blood through the small valve opening to the rest of your body. This may lead to increasing heart failure.

The Food and Drug Administration approved TAVR for use in a broad spectrum of patients following multiple research studies comparing TAVR to SAVR. Whether TAVR or SAVR is more appropriate for a given individual depends on multiple factors and is discussed with each patient by both an interventional cardiologist and a cardiac surgeon. During TAVR, your doctor inserts a catheter through a blood vessel in your leg to deliver and implant the artificial valve into your heart. Significant research is exploring how to both advance the use of this technique and improve the devices that are used for TAVR.

Why might I need TAVR or SAVR?

You may need replacement of your aortic valve if you have progressive heart failure due to aortic valve stenosis. Factors making TAVR more likely to be recommended than SAVR include conditions that would increase the risk of traditional open-heart aortic valve replacement surgery such as:

Older age.

Frailty.

Weaker heart.

Previous heart surgery.

History of stroke.

Chronic obstructive lung disease (COPD).

Liver disease.

Kidney disease.

Diabetes.

Previous radiation treatment to your chest.

Large calcium deposits in the blood vessel that carries blood away from your heart (ascending aorta), called porcelain aorta.

What are the risks of TAVR?

Johns Hopkins has been performing TAVRs for more than a decade and has performed nearly 2,000 procedures. However, TAVR is still a major procedure that has risks. Most TAVR procedures are performed with sedation without the need for general anesthesia. The most common risks associated with TAVR include:

Damage to your blood vessels.

Bleeding.

Decreased blood supply to your brain, causing a stroke.

Heart attack.

Kidney failure.

Collection of fluid around your heart.

Leaking of the replacement valve.

Severe heart failure.

Death.

There may be other risks, depending on your specific medical condition. Be sure to discuss any concerns with your doctor before the procedure.

How do I prepare for TAVR?

Before surgery, your medical and surgical team will evaluate your overall health. This may include X-rays, CT scans, blood tests and other tests to check the health of your lungs and heart. Your medical team will also give you an echocardiogram to evaluate your aortic valve. This test uses sound waves to create images of your heart. Your medical team may also do a cardiac catheterization to evaluate the arteries that supply blood to your heart. You will also need to:

Tell your healthcare provider about any drugs you are taking, including over-the-counter medicines.

Quit smoking if you still smoke, because continuing to smoke increases the risk of a procedure-related problem with your lungs.

Stop certain medicines if your health care provider instructs you to.

Stop eating and drinking, usually at midnight before surgery.

What happens during TAVR?

Methods may vary, depending on your condition and your doctor's practices. Talk with your doctor about what to expect. Generally, a TAVR procedure follows this process:

You will remove any jewelry or other objects that may interfere with the procedure.

You will remove your clothing and will be given a gown to wear.

You will empty your bladder prior to the procedure.

An IV line will be started in your arm or hand. Additional catheters will be inserted in your wrist to help guide the procedure. Alternate sites for the additional catheters may include the neck and the groin.

You will be positioned on the operating table, lying on your back.

The anesthesiologist will continuously monitor your heart rate, blood pressure, breathing, and blood oxygen level during the surgery. Sedation will be given to make you comfortable. Rarely will general anesthesia be required, but then, once you are sedated, a breathing tube will be inserted into your throat and into your trachea to provide oxygen to your lungs. You will be connected to a ventilator, which will breathe for you during the surgery.

The skin over the surgical site in your groin will be cleansed with an antiseptic solution.

The surgeon will place a catheter into the leg artery (femoral artery) and thread it to your heart and through your aorta, to reach your aortic valve.

Your surgeon will place other catheters in your heart to take measurements and X-ray pictures during the procedure.

Your surgeon will guide your replacement valve, either a self-expanding valve or a balloon expandable valve, up the femoral artery catheter and through your old aortic valve.

Once the new valve is properly positioned, your surgeon will implant the new valve to replace the old one.

Your doctor will take measurements and images to make sure your new valve works properly before removing the catheters.

Your doctor will close your femoral artery with a suture device that does not require any incision.

Transthoracic echocardiography is then performed to assess the new valve's function.

What happens after TAVR?

You will be moved to the cardiac recovery floor of the hospital, so you can be monitored closely during your recovery. You will soon be able to get up and walk and return to a normal diet. You will stay overnight in the hospital. Most patients go home the next day and are allowed to resume their normal activities.

After your doctors feel you have recovered enough to go home, follow all instructions for medicines, pain control, diet, activity and wound care. Make sure to keep all your follow-up appointments.

Complete recovery from deconditioning caused by the aortic stenosis may take several weeks. Here are some helpful guidelines to follow as you heal:

Your doctor may give you medication to take for several months after TAVR.

Walk around as much as possible.

Gradually resume normal activities, but avoid any heavy lifting.

Participating in a cardiac rehabilitation program is highly recommended and will be arranged for you by your doctors.

Ask your doctors when you can resume driving, work and sexual activity.

Watch your groin for any sign of swelling, redness, bleeding or discharge.

Let your doctor know if you feel any pain or have fever, bleeding or shortness of breath.

Eat a heart-healthy diet and maintain a healthy weight.

Don't smoke.

Next steps

Before you agree to the procedure, make sure you know:

The name of the procedure.

The reason you are having the procedure.

What results to expect and what they mean.

The risks and benefits of the procedure.

What the possible side effects or complications are.

When and where you are to have the procedure.

Who will do the procedure and what that person's qualifications are.

What would happen if you did not have the procedure.

Any alternative procedures to think about.

Who to call after the procedure if you have questions or problems.

How much you will pay for the procedure.

AI rewrite

Transcatheter Aortic Valve Replacement (TAVR)

Heart and Aortic Valve Treatment

What is transcatheter aortic valve replacement?

Transcatheter aortic valve replacement is also called TAVR. It is a procedure to replace a diseased aortic valve with a man-made valve.

The aortic valve can also be replaced with open-heart surgery. This is called surgical aortic valve replacement, or SAVR.

Your aortic valve controls blood flow from your heart to the rest of your body. If the valve gets stiff, you have a condition called aortic stenosis. Your heart may have to work too hard to push blood through the small valve opening. This can lead to worsening heart failure.

The Food and Drug Administration approved TAVR for many types of patients. This happened after many studies compared TAVR with SAVR.

Whether TAVR or SAVR is better for you depends on many things. An interventional cardiologist and a heart surgeon will talk with you about this.

During TAVR, your doctor puts a thin tube called a catheter into a blood vessel in your leg. The catheter carries the new valve to your heart. The doctor places the new valve inside your heart.

Researchers are still studying how to improve TAVR and the valves used for it.

Why might I need TAVR or SAVR?

You may need your aortic valve replaced if you have worsening heart failure caused by aortic stenosis.

TAVR may be more likely than SAVR if open-heart surgery would be higher risk for you. These risks may include:

Older age.

Frailty.

A weaker heart.

Past heart surgery.

History of stroke.

Chronic obstructive lung disease, or COPD.

Liver disease.

Kidney disease.

Diabetes.

Past radiation treatment to your chest.

Large calcium deposits in the blood vessel that carries blood away from your heart. This blood vessel is called the ascending aorta. This problem is called porcelain aorta.

What are the risks of TAVR?

Johns Hopkins has been doing TAVR for more than 10 years. It has done nearly 2,000 TAVR procedures. But TAVR is still a major procedure. It has risks.

Most TAVR procedures are done with sedation. This helps you relax and feel comfortable. General anesthesia is usually not needed.

The most common risks of TAVR include:

Damage to your blood vessels.

Bleeding.

Less blood flow to your brain, which can cause a stroke.

Heart attack.

Kidney failure.

Fluid around your heart.

Leaking of the new valve.

Severe heart failure.

Death.

You may have other risks based on your health. Talk with your doctor about any concerns before the procedure.

How do I prepare for TAVR?

Before the procedure, your medical and surgery team will check your overall health. You may have X-rays, CT scans, blood tests, and other tests. These tests check your lungs and heart.

You will also have an echocardiogram. This test uses sound waves to make pictures of your heart. It helps check your aortic valve.

Your medical team may also do a cardiac catheterization. This checks the arteries that bring blood to your heart.

You will also need to:

Tell your health care provider about all medicines you take. This includes medicines you buy without a prescription.

Quit smoking if you still smoke. Smoking raises the risk of lung problems related to the procedure.

Stop taking certain medicines if your health care provider tells you to.

Stop eating and drinking, usually at midnight before the procedure.

What happens during TAVR?

Methods may be different based on your health and your doctor’s way of doing the procedure. Ask your doctor what to expect.

In general, TAVR follows these steps:

You will take off jewelry or other items that may get in the way.

You will take off your clothes and put on a hospital gown.

You will empty your bladder before the procedure.

An IV line will be placed in your arm or hand.

More catheters will be placed in your wrist to help guide the procedure. Other places for these catheters may include your neck or groin.

You will lie on your back on the operating table.

The anesthesia doctor will watch your heart rate, blood pressure, breathing, and blood oxygen level during the procedure.

You will be given sedation to help you feel comfortable.

General anesthesia is rarely needed. If it is needed, after you are asleep, a breathing tube will be placed in your throat and windpipe. This gives oxygen to your lungs. You will be connected to a ventilator. The ventilator will breathe for you during the procedure.

The skin over the procedure site in your groin will be cleaned with a germ-killing solution.

The surgeon will place a catheter into the leg artery, called the femoral artery. The surgeon will move it up to your heart and through your aorta to reach your aortic valve.

Your surgeon will place other catheters in your heart. These are used to take measurements and X-ray pictures during the procedure.

Your surgeon will move the new valve through the catheter in your femoral artery. The valve may open on its own or may be opened with a balloon. It will be guided through your old aortic valve.

When the new valve is in the right place, your surgeon will place it to replace the old valve.

Your doctor will take measurements and pictures to make sure the new valve works well before removing the catheters.

Your doctor will close your femoral artery with a stitch device. This does not need a cut.

Then an echocardiogram through the chest is done to check how the new valve works.

What happens after TAVR?

You will be moved to the heart recovery floor in the hospital. Your care team will watch you closely as you recover.

Soon, you will be able to get up and walk. You will also be able to return to a normal diet.

You will stay in the hospital overnight. Most people go home the next day. Most people are allowed to return to normal activities.

When your doctors think you are ready to go home, follow all instructions. These may include instructions for medicines, pain control, diet, activity, and wound care.

Keep all follow-up appointments.

Full recovery from weakness caused by aortic stenosis may take several weeks.

Follow these guidelines as you heal:

Your doctor may give you medicine to take for several months after TAVR.

Walk as much as you can.

Slowly return to normal activities. Do not lift heavy things.

A cardiac rehabilitation program is highly recommended. Your doctors will arrange this for you.

Ask your doctors when you can drive, work, and have sex again.

Watch your groin for swelling, redness, bleeding, or drainage.

Tell your doctor if you have pain, fever, bleeding, or shortness of breath.

Eat a heart-healthy diet and keep a healthy weight.

Do not smoke.

Next steps

Before you agree to the procedure, make sure you know:

The name of the procedure.

Why you are having the procedure.

What results to expect and what they mean.

The risks and benefits of the procedure.

Possible side effects or problems.

When and where you will have the procedure.

Who will do the procedure and that person’s training and experience.

What could happen if you do not have the procedure.

Other procedures you could think about.

Who to call after the procedure if you have questions or problems.

How much you will pay for the procedure.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Transcatheter Aortic Valve Replacement (TAVR)

Cardiovascular Aortic Valve Treatment

What is transcatheter aortic valve replacement?

Transcatheter aortic valve replacement (TAVR) is a procedure that replaces a diseased aortic valve with a man-made valve. Aortic valve replacement can also be performed with open-heart surgery; this procedure is surgical aortic valve replacement (SAVR).

Your aortic valve controls blood flow from your heart to your body. If your valve becomes stiff, you have a condition called aortic stenosis. Your heart may have to work too hard to pump blood through the small valve opening to the rest of your body. This may lead to increasing heart failure.

The Food and Drug Administration approved TAVR for use in a broad spectrum of patients following multiple research studies comparing TAVR to SAVR. Whether TAVR or SAVR is more appropriate for a given individual depends on multiple factors and is discussed with each patient by both an interventional cardiologist and a cardiac surgeon. During TAVR, your doctor inserts a catheter through a blood vessel in your leg to deliver and implant the artificial valve into your heart. Significant research is exploring how to both advance the use of this technique and improve the devices that are used for TAVR.

Why might I need TAVR or SAVR?

You may need replacement of your aortic valve if you have progressive heart failure due to aortic valve stenosis. Factors making TAVR more likely to be recommended than SAVR include conditions that would increase the risk of traditional open-heart aortic valve replacement surgery such as:

Older age.

Frailty.

Weaker heart.

Previous heart surgery.

History of stroke.

Chronic obstructive lung disease (COPD).

Liver disease.

Kidney disease.

Diabetes.

Previous radiation treatment to your chest.

Large calcium deposits in the blood vessel that carries blood away from your heart (ascending aorta), called porcelain aorta.

What are the risks of TAVR?

Johns Hopkins has been performing TAVRs for more than a decade and has performed nearly 2,000 procedures. However, TAVR is still a major procedure that has risks. Most TAVR procedures are performed with sedation without the need for general anesthesia. The most common risks associated with TAVR include:

Damage to your blood vessels.

Bleeding.

Decreased blood supply to your brain, causing a stroke.

Heart attack.

Kidney failure.

Collection of fluid around your heart.

Leaking of the replacement valve.

Severe heart failure.

Death.

There may be other risks, depending on your specific medical condition. Be sure to discuss any concerns with your doctor before the procedure.

How do I prepare for TAVR?

Before surgery, your medical and surgical team will evaluate your overall health. This may include X-rays, CT scans, blood tests and other tests to check the health of your lungs and heart. Your medical team will also give you an echocardiogram to evaluate your aortic valve. This test uses sound waves to create images of your heart. Your medical team may also do a cardiac catheterization to evaluate the arteries that supply blood to your heart. You will also need to:

Tell your healthcare provider about any drugs you are taking, including over-the-counter medicines.

Quit smoking if you still smoke, because continuing to smoke increases the risk of a procedure-related problem with your lungs.

Stop certain medicines if your health care provider instructs you to.

Stop eating and drinking, usually at midnight before surgery.

What happens during TAVR?

Methods may vary, depending on your condition and your doctor's practices. Talk with your doctor about what to expect. Generally, a TAVR procedure follows this process:

You will remove any jewelry or other objects that may interfere with the procedure.

You will remove your clothing and will be given a gown to wear.

You will empty your bladder prior to the procedure.

An IV line will be started in your arm or hand. Additional catheters will be inserted in your wrist to help guide the procedure. Alternate sites for the additional catheters may include the neck and the groin.

You will be positioned on the operating table, lying on your back.

The anesthesiologist will continuously monitor your heart rate, blood pressure, breathing, and blood oxygen level during the surgery. Sedation will be given to make you comfortable. Rarely will general anesthesia be required, but then, once you are sedated, a breathing tube will be inserted into your throat and into your trachea to provide oxygen to your lungs. You will be connected to a ventilator, which will breathe for you during the surgery.

The skin over the surgical site in your groin will be cleansed with an antiseptic solution.

The surgeon will place a catheter into the leg artery (femoral artery) and thread it to your heart and through your aorta, to reach your aortic valve.

Your surgeon will place other catheters in your heart to take measurements and X-ray pictures during the procedure.

Your surgeon will guide your replacement valve, either a self-expanding valve or a balloon expandable valve, up the femoral artery catheter and through your old aortic valve.

Once the new valve is properly positioned, your surgeon will implant the new valve to replace the old one.

Your doctor will take measurements and images to make sure your new valve works properly before removing the catheters.

Your doctor will close your femoral artery with a suture device that does not require any incision.

Transthoracic echocardiography is then performed to assess the new valve's function.

What happens after TAVR?

You will be moved to the cardiac recovery floor of the hospital, so you can be monitored closely during your recovery. You will soon be able to get up and walk and return to a normal diet. You will stay overnight in the hospital. Most patients go home the next day and are allowed to resume their normal activities.

After your doctors feel you have recovered enough to go home, follow all instructions for medicines, pain control, diet, activity and wound care. Make sure to keep all your follow-up appointments.

Complete recovery from deconditioning caused by the aortic stenosis may take several weeks. Here are some helpful guidelines to follow as you heal:

Your doctor may give you medication to take for several months after TAVR.

Walk around as much as possible.

Gradually resume normal activities, but avoid any heavy lifting.

Participating in a cardiac rehabilitation program is highly recommended and will be arranged for you by your doctors.

Ask your doctors when you can resume driving, work and sexual activity.

Watch your groin for any sign of swelling, redness, bleeding or discharge.

Let your doctor know if you feel any pain or have fever, bleeding or shortness of breath.

Eat a heart-healthy diet and maintain a healthy weight.

Don't smoke.

Next steps

Before you agree to the procedure, make sure you know:

The name of the procedure.

The reason you are having the procedure.

What results to expect and what they mean.

The risks and benefits of the procedure.

What the possible side effects or complications are.

When and where you are to have the procedure.

Who will do the procedure and what that person's qualifications are.

What would happen if you did not have the procedure.

Any alternative procedures to think about.

Who to call after the procedure if you have questions or problems.

How much you will pay for the procedure.

AI rewrite

Transcatheter Aortic Valve Replacement (TAVR)

Heart Aortic Valve Treatment

What is transcatheter aortic valve replacement?

TAVR is a way to replace a sick heart valve with a man-made valve. The valve that gets replaced is called the aortic valve. There is also another way to replace this valve. That way uses open-heart surgery. It is called surgical aortic valve replacement, or SAVR.

Your aortic valve controls how blood flows from your heart to your body. Sometimes this valve gets stiff. This is called aortic stenosis. When the valve gets stiff, the opening gets small. Your heart has to work too hard to push blood through it. Over time, this can cause heart failure.

Many studies compared TAVR to SAVR. After these studies, the Food and Drug Administration (FDA) approved TAVR for many kinds of patients. Which one is best for you depends on many things. A heart doctor (interventional cardiologist) and a heart surgeon will talk with you about which is best.

During TAVR, your doctor puts a thin tube (catheter) into a blood vessel in your leg. The doctor uses this tube to put the new valve into your heart. Doctors are still doing research to make TAVR better and to improve the valves they use.

Why might I need TAVR or SAVR?

You may need a new aortic valve if you have heart failure that is getting worse because of aortic stenosis. Some health problems make open-heart surgery riskier. If you have these problems, your doctor may suggest TAVR instead of SAVR. These include:

- Older age.

- Being frail (weak).

- A weaker heart.

- Past heart surgery.

- A history of stroke.

- A lung disease called COPD (chronic obstructive lung disease).

- Liver disease.

- Kidney disease.

- Diabetes.

- Past radiation treatment to your chest.

- Large calcium buildup in the big blood vessel that carries blood away from your heart (the ascending aorta). This is called porcelain aorta.

What are the risks of TAVR?

Johns Hopkins has done TAVR for more than 10 years. We have done almost 2,000 of these procedures. But TAVR is still a major procedure, and it has risks. Most of the time, you will get medicine to make you sleepy and calm. You usually will not need to be put fully to sleep (general anesthesia). The most common risks of TAVR are:

- Damage to your blood vessels.

- Bleeding.

- Less blood going to your brain, which can cause a stroke.

- Heart attack.

- Kidney failure.

- Fluid building up around your heart.

- A leak in the new valve.

- Bad heart failure.

- Death.

There may be other risks too, based on your own health. Talk with your doctor about any worries before the procedure.

How do I prepare for TAVR?

Before the procedure, your team will check your health. They may do X-rays, CT scans, and blood tests. These tests check your heart and lungs. You will also get a test called an echocardiogram to check your aortic valve. This test uses sound waves to make pictures of your heart. Your team may also do a cardiac catheterization. This checks the blood vessels that feed your heart.

You will also need to:

- Tell your doctor about all the medicines you take. This includes medicines you buy without a prescription.

- Quit smoking if you smoke. Smoking raises the chance of lung problems from the procedure.

- Stop taking some medicines if your doctor tells you to.

- Stop eating and drinking. This is usually at midnight before the procedure.

What happens during TAVR?

Each doctor may do things a little differently. It also depends on your health. Talk with your doctor about what to expect. In general, TAVR works like this:

- You will take off any jewelry or other things that could get in the way.

- You will take off your clothes and put on a gown.

- You will empty your bladder before the procedure.

- A nurse will start an IV line in your arm or hand. The team will put more tubes in your wrist to help guide the procedure. They may use your neck or groin instead.

- You will lie on your back on the operating table.

- The anesthesiologist will watch your heart rate, blood pressure, breathing, and oxygen the whole time. You will get medicine to make you comfortable. You usually will not be put fully to sleep. But if you are, the team will put a tube in your throat and windpipe to give your lungs oxygen. This tube connects to a machine that breathes for you during the procedure.

- The team will clean the skin in your groin with a germ-killing liquid.

- The surgeon will put a tube into the artery in your leg (femoral artery). They will move it up to your heart and through your aorta to reach your aortic valve.

- The surgeon will put other tubes in your heart to take measurements and X-ray pictures.

- The surgeon will guide your new valve up the leg tube and through your old valve. The new valve may open by itself or open with a balloon.

- When the new valve is in the right spot, the surgeon will put it in place of the old one.

- The doctor will take measurements and pictures to make sure the new valve works well. Then they will take out the tubes.

- The doctor will close the leg artery with a special device. This does not need a cut.

- Last, the team does an echocardiogram to check that the new valve is working.

What happens after TAVR?

You will go to the heart recovery floor of the hospital. The team will watch you closely. Soon you will be able to get up, walk, and eat normal food. You will stay in the hospital overnight. Most patients go home the next day. Most can go back to normal activities.

When your doctors say you are ready to go home, follow all of their instructions. This includes instructions for your medicines, pain, food, activity, and wound care. Keep all your follow-up appointments.

It may take a few weeks to fully recover. The aortic stenosis may have made you weak, and it takes time to get strong again. Here are some tips to help you heal:

- Your doctor may give you medicine to take for a few months after TAVR.

- Walk around as much as you can.

- Slowly go back to your normal activities. But do not lift heavy things.

- It is a good idea to join a heart rehab program. Your doctors will set this up for you.

- Ask your doctors when you can drive, work, and have sex again.

- Watch your groin for swelling, redness, bleeding, or fluid.

- Tell your doctor if you have pain, fever, bleeding, or trouble breathing.

- Eat a heart-healthy diet. Try to stay at a healthy weight.

- Do not smoke.

Next steps

Before you agree to the procedure, make sure you know:

- The name of the procedure.

- Why you are having it.

- What results to expect and what they mean.

- The risks and benefits.

- The possible side effects or problems.

- When and where you will have it.

- Who will do it and what their training is.

- What would happen if you did not have it.

- Other choices you could think about.

- Who to call after the procedure if you have questions or problems.

- How much you will pay for it.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Left Atrial Appendage Closure Procedures

Cardiovascular

Stroke is a serious risk for patients with atrial fibrillation, which is caused by a disruption of electrical signals in the muscles of the upper chambers of the heart. When the heart pumps in an irregular, uncoordinated way, there is a decreased flow of blood to the body, which is typically slow moving and likely to cause a blood clot. In the majority of cases, the clots form in the left atrial appendage, a small, pouchlike sac in the top left chamber of the heart. If the clots travel through the arteries in the heart, they can cause a stroke.

If you are a patient with atrial fibrillation who is at increased stroke risk, your doctor will recommend that you be placed on a blood thinner. If you are unable to take a blood thinner because of risk of bleeding or falls, your doctor may recommend a procedure to occlude your left atrial appendage.

Catheter-Inserted Closure Devices

WATCHMANTM Device

One procedure used to close the left atrial appendage is a self-expanding, parachute-shaped device with an attached woven plastic cap called the WATCHMANTM device. This device is approved by the FDA to reduce the risk of thromboembolism from the left atrial appendage in patients with nonvalvular atrial fibrillation who are not good candidates for long-term anti-coagulation. Patients will be required to take low-dose anticoagulant therapy before the Watchman implantation and to remain on it for at least 45 days after the procedure.

The Procedure

The procedure is performed under general anesthesia. The physician inserts the delivery catheter into the body through a vein in the leg. The catheter is advanced through the bloodstream until it reaches the upper right chamber of the heart. The physician makes a small hole through the wall between the two upper chambers of the heart so that the catheter reaches the left atrium. The physician then pushes the device through the delivery catheter into the left atrial appendage, where it opens up like an umbrella and is permanently implanted. Once the device is in place, a thin layer of tissue grows over it in about 45 days. This keeps blood clots in the left atrial appendage from entering the bloodstream.

What are the risks?

Although the device is a permanent solution, there is a small incidence of complications associated with the implantation procedure. Your doctor can go over the risks and benefits of this procedure with you in detail.

What happens after the procedure?

You will have a follow-up transesophageal echocardiogram, which requires sedation, 45 days after the procedure. At that point, your anticoagulant medication will be stopped or adjusted if complete occlusion of the left atrial appendage is demonstrated.

You will have a follow-up clinic visit with your provider six months after the procedure. At that point, you will remain on low-dose aspirin indefinitely.

AI rewrite

Left Atrial Appendage Closure Procedures

Cardiovascular

Stroke is a serious risk for people with atrial fibrillation. Atrial fibrillation happens when the electrical signals in the upper chambers of the heart do not work the right way.

When the heart beats in an uneven way, blood may not move well through the body. Slow-moving blood can form a blood clot. Most of these clots form in the left atrial appendage. This is a small pouch in the upper left chamber of the heart. If a clot travels through the arteries, it can cause a stroke.

If you have atrial fibrillation and a higher risk of stroke, your doctor will likely recommend a blood thinner. If you cannot take a blood thinner because of bleeding risk or fall risk, your doctor may recommend a procedure to close your left atrial appendage.

Catheter-Inserted Closure Devices

WATCHMAN™ Device

One device used to close the left atrial appendage is called the WATCHMAN™ device. It opens on its own and is shaped like a parachute. It has a woven plastic cap attached to it.

The WATCHMAN™ device is approved by the FDA. It lowers the risk of blood clots from the left atrial appendage in people with nonvalvular atrial fibrillation who are not good candidates for long-term blood thinner treatment.

You will need to take a low-dose blood thinner before the WATCHMAN™ device is placed. You will need to keep taking it for at least 45 days after the procedure.

The Procedure

The procedure is done while you are under general anesthesia. This means you are asleep and do not feel pain.

The doctor puts a delivery catheter into a vein in your leg. A catheter is a thin, flexible tube. The doctor moves the catheter through your blood vessels until it reaches the upper right chamber of your heart.

The doctor makes a small hole in the wall between the two upper chambers of the heart. This lets the catheter reach the left atrium. The doctor then moves the device through the catheter into the left atrial appendage.

The device opens like an umbrella and stays there permanently. After the device is in place, a thin layer of tissue grows over it in about 45 days. This helps keep blood clots in the left atrial appendage from entering the bloodstream.

What are the risks?

The device is meant to be permanent. But there is a small chance of problems from the procedure to place it. Your doctor can talk with you in detail about the risks and benefits.

What happens after the procedure?

You will have a follow-up transesophageal echocardiogram 45 days after the procedure. This test uses sedation. At that time, your blood thinner medicine will be stopped or changed if the left atrial appendage is fully closed.

You will have a follow-up clinic visit with your provider 6 months after the procedure. At that time, you will stay on low-dose aspirin for the rest of your life.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

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Original page

Left Atrial Appendage Closure Procedures

Cardiovascular

Stroke is a serious risk for patients with atrial fibrillation, which is caused by a disruption of electrical signals in the muscles of the upper chambers of the heart. When the heart pumps in an irregular, uncoordinated way, there is a decreased flow of blood to the body, which is typically slow moving and likely to cause a blood clot. In the majority of cases, the clots form in the left atrial appendage, a small, pouchlike sac in the top left chamber of the heart. If the clots travel through the arteries in the heart, they can cause a stroke.

If you are a patient with atrial fibrillation who is at increased stroke risk, your doctor will recommend that you be placed on a blood thinner. If you are unable to take a blood thinner because of risk of bleeding or falls, your doctor may recommend a procedure to occlude your left atrial appendage.

Catheter-Inserted Closure Devices

WATCHMANTM Device

One procedure used to close the left atrial appendage is a self-expanding, parachute-shaped device with an attached woven plastic cap called the WATCHMANTM device. This device is approved by the FDA to reduce the risk of thromboembolism from the left atrial appendage in patients with nonvalvular atrial fibrillation who are not good candidates for long-term anti-coagulation. Patients will be required to take low-dose anticoagulant therapy before the Watchman implantation and to remain on it for at least 45 days after the procedure.

The Procedure

The procedure is performed under general anesthesia. The physician inserts the delivery catheter into the body through a vein in the leg. The catheter is advanced through the bloodstream until it reaches the upper right chamber of the heart. The physician makes a small hole through the wall between the two upper chambers of the heart so that the catheter reaches the left atrium. The physician then pushes the device through the delivery catheter into the left atrial appendage, where it opens up like an umbrella and is permanently implanted. Once the device is in place, a thin layer of tissue grows over it in about 45 days. This keeps blood clots in the left atrial appendage from entering the bloodstream.

What are the risks?

Although the device is a permanent solution, there is a small incidence of complications associated with the implantation procedure. Your doctor can go over the risks and benefits of this procedure with you in detail.

What happens after the procedure?

You will have a follow-up transesophageal echocardiogram, which requires sedation, 45 days after the procedure. At that point, your anticoagulant medication will be stopped or adjusted if complete occlusion of the left atrial appendage is demonstrated.

You will have a follow-up clinic visit with your provider six months after the procedure. At that point, you will remain on low-dose aspirin indefinitely.

AI rewrite

Closing the Left Atrial Appendage

Heart and Blood Vessels

People with atrial fibrillation have a higher risk of stroke. Atrial fibrillation happens when the electrical signals in the top parts of the heart get mixed up. This makes the heart pump in a fast, uneven way. When this happens, less blood moves through the body. The blood moves slowly, and slow blood can form a clot.

Most of the time, these clots form in the left atrial appendage. This is a small pouch in the top left part of the heart. If a clot leaves this pouch and moves through your blood vessels, it can cause a stroke.

If you have atrial fibrillation and a higher risk of stroke, your doctor will want you to take a blood thinner. But some people cannot take blood thinners. This may be because they bleed too easily or they fall a lot. If you cannot take a blood thinner, your doctor may suggest a procedure to close off the left atrial appendage.

Closure Devices Put in Through a Tube

WATCHMAN™ Device

One way to close the left atrial appendage is with the WATCHMAN™ device. This small device opens up like a parachute. It has a woven plastic cap on it. The FDA has approved this device. It helps lower the risk of clots from the left atrial appendage. It is for people with nonvalvular atrial fibrillation who cannot take blood thinners for a long time.

Before you get this device, you will need to take a low dose of a blood thinner. You will need to keep taking it for at least 45 days after the procedure.

The Procedure

You will be asleep during the procedure. The doctor puts a thin tube, called a catheter, into a vein in your leg. The doctor moves the tube through your blood vessels until it reaches the top right part of your heart. Then the doctor makes a small hole in the wall between the two top parts of the heart. This lets the tube reach the left atrium.

Next, the doctor pushes the device through the tube and into the left atrial appendage. There, it opens up like an umbrella. The device stays in your heart for good. Over about 45 days, a thin layer of tissue grows over the device. This keeps blood clots in the pouch from getting into your blood.

What are the risks?

The device is a permanent fix. But there is a small chance of problems from the procedure. Your doctor can talk with you about the risks and benefits in more detail.

What happens after the procedure?

About 45 days after the procedure, you will have a test called a transesophageal echocardiogram. You will get medicine to help you relax for this test. If the test shows that the left atrial appendage is fully closed, your doctor will stop or change your blood thinner.

Six months after the procedure, you will have a follow-up visit with your provider. After that, you will keep taking low-dose aspirin for the rest of your life.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Left Atrial Appendage Closure Procedures

Cardiovascular

Stroke is a serious risk for patients with atrial fibrillation, which is caused by a disruption of electrical signals in the muscles of the upper chambers of the heart. When the heart pumps in an irregular, uncoordinated way, there is a decreased flow of blood to the body, which is typically slow moving and likely to cause a blood clot. In the majority of cases, the clots form in the left atrial appendage, a small, pouchlike sac in the top left chamber of the heart. If the clots travel through the arteries in the heart, they can cause a stroke.

If you are a patient with atrial fibrillation who is at increased stroke risk, your doctor will recommend that you be placed on a blood thinner. If you are unable to take a blood thinner because of risk of bleeding or falls, your doctor may recommend a procedure to occlude your left atrial appendage.

Catheter-Inserted Closure Devices

WATCHMANTM Device

One procedure used to close the left atrial appendage is a self-expanding, parachute-shaped device with an attached woven plastic cap called the WATCHMANTM device. This device is approved by the FDA to reduce the risk of thromboembolism from the left atrial appendage in patients with nonvalvular atrial fibrillation who are not good candidates for long-term anti-coagulation. Patients will be required to take low-dose anticoagulant therapy before the Watchman implantation and to remain on it for at least 45 days after the procedure.

The Procedure

The procedure is performed under general anesthesia. The physician inserts the delivery catheter into the body through a vein in the leg. The catheter is advanced through the bloodstream until it reaches the upper right chamber of the heart. The physician makes a small hole through the wall between the two upper chambers of the heart so that the catheter reaches the left atrium. The physician then pushes the device through the delivery catheter into the left atrial appendage, where it opens up like an umbrella and is permanently implanted. Once the device is in place, a thin layer of tissue grows over it in about 45 days. This keeps blood clots in the left atrial appendage from entering the bloodstream.

What are the risks?

Although the device is a permanent solution, there is a small incidence of complications associated with the implantation procedure. Your doctor can go over the risks and benefits of this procedure with you in detail.

What happens after the procedure?

You will have a follow-up transesophageal echocardiogram, which requires sedation, 45 days after the procedure. At that point, your anticoagulant medication will be stopped or adjusted if complete occlusion of the left atrial appendage is demonstrated.

You will have a follow-up clinic visit with your provider six months after the procedure. At that point, you will remain on low-dose aspirin indefinitely.

AI rewrite

Atrial fibrillation is a problem with the electrical signals in the top parts of your heart. It causes an irregular heartbeat. This can put you at a high risk for a stroke. When your heart does not beat normally, blood flows too slowly. Slow-moving blood can form clots. Most of these blood clots form in a small pouch in the top left part of your heart. This pouch is called the left atrial appendage. If a blood clot leaves this pouch, it can travel through your blood vessels and cause a stroke.

To lower your risk of a stroke, your doctor will usually give you a blood thinner medicine. But some people cannot take blood thinners because they might bleed too much or fall. If you cannot take blood thinners, your doctor may suggest a procedure to close off the pouch in your heart.

One way to close the pouch is with a WATCHMAN device. This device looks like a tiny parachute with a plastic cap. It is approved by the FDA to lower the risk of blood clots for people who cannot take blood thinners for a long time. You will need to take a low dose of a blood thinner before the device is put in. You must also keep taking it for at least 45 days after the procedure.

During the procedure, you will be fully asleep. The doctor will put a thin tube into a vein in your leg. The doctor will guide this tube up to the top right part of your heart. Then, the doctor makes a tiny hole in the wall between the top two parts of your heart. This lets the tube reach the left side. The doctor pushes the device through the tube and into the pouch. The device opens up like an umbrella and stays in your heart forever. Over the next 45 days, a thin layer of tissue will grow over the device. This seals the pouch so blood clots cannot get out into your bloodstream.

This device is a permanent fix. However, there is a small chance of problems during the procedure. Your doctor will talk with you about all the risks and benefits.

About 45 days after the procedure, you will have a special heart ultrasound called a transesophageal echocardiogram. You will be given medicine to make you sleepy and relaxed for this test. If the test shows the pouch is completely closed, your doctor will stop or change your blood thinner medicine. Six months after the procedure, you will visit your doctor again. From then on, you will need to take a low dose of aspirin for the rest of your life.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Last reviewed on March 11, 2024

Coronary CTA

Coronary computed tomography angiography (also called coronary CT angiography or CCTA) uses an injection of iodine-containing contrast material and CT scanning to examine the arteries that supply blood to the heart and determine whether they have been narrowed. The images generated during a CT scan can be reformatted to create three-dimensional (3D) images that may be viewed on a monitor, printed on film or by a 3D printer, or transferred to electronic media.

Tell your doctor if there's a possibility you are pregnant and discuss any recent illnesses, medical conditions, medications you're taking, and allergies. You will be instructed not to eat or drink anything several hours beforehand and to avoid caffeinated products, Viagra or similar medication. If you have a known allergy to contrast material, your doctor may prescribe medications to reduce the risk of an allergic reaction. These medications must be taken at multiple intervals beginning 13 hours prior to your exam. Leave jewelry at home and wear loose, comfortable clothing. You may be asked to wear a gown. If you are breastfeeding, talk to your doctor about how to proceed.

- What is coronary CTA?

- What are some common uses of the procedure?

- How should I prepare?

- What does the equipment look like?

- How does the procedure work?

- How is the procedure performed?

- What will I experience during and after the procedure?

- Who interprets the results and how do I get them?

- What are the benefits vs. risks?

- What are the limitations of coronary CTA?

What is coronary CTA?

Coronary computed tomography angiography (CCTA) is a heart imaging test that helps determine if plaque buildup has narrowed the coronary arteries, the blood vessels that supply the heart. Plaque is made of various substances such as fat, cholesterol and calcium that deposit along the inner lining of the arteries. Plaque, which builds up over time, can reduce or in some cases completely block blood flow. Patients undergoing a CCTA scan receive an iodine-containing contrast material as an intravenous (IV) injection to ensure the best possible images of the heart blood vessels.

Computed tomography, more commonly known as a CT or CAT scan, is a diagnostic medical imaging test. Like traditional x-rays, it produces multiple images or pictures of the inside of the body.

A CT scan generates images that can be reformatted in multiple planes. It can even generate three-dimensional images. Your doctor can review these images on a computer monitor, print them on film or via a 3D printer, or transfer them to a CD or DVD.

CT images of internal organs, bones, soft tissue, and blood vessels provide greater detail than traditional x-rays. This is especially true for soft tissues and blood vessels.

What are some common uses of the procedure?

Many physicians advocate the careful use of CCTA for patients who have:

- suspected abnormal anatomy of the coronary arteries.

- low or intermediate risk for coronary artery disease, including patients who have chest pain and normal, non-diagnostic or unclear lab and ECG results.

- low to intermediate risk atypical chest pain in the emergency department.

- non-acute chest pain.

- new or worsening symptoms with a previous normal stress test result.

- unclear or inconclusive stress test results.

- new onset heart failure with reduced heart function and low or medium risk for coronary artery disease.

- intermediate risk of coronary artery disease before non-coronary cardiac surgery.

- coronary artery bypass grafts.

For patients meeting the above indications, CCTA can provide important information about the presence and extent of plaque in the coronary arteries. Apart from identifying coronary artery narrowing as the cause of chest discomfort, it can also detect other possible causes of symptoms, such as a collapsed lung, blood clot in the vessels leading to the lungs, or aortic abnormalities. Your primary care physician or cardiac specialist, in consultation with a radiologist who would perform the test, will determine whether CCTA is appropriate for you.

How should I prepare?

Wear comfortable, loose-fitting clothing to your exam. You may need to change into a gown for the procedure.

Metal objects, including jewelry, eyeglasses, dentures, and hairpins, may affect the CT images. Leave them at home or remove them prior to your exam. Some CT exams will require you to remove hearing aids and removable dental work. Women will need to remove bras containing metal underwire. You may need to remove any piercings, if possible.

Your doctor may instruct you to not eat or drink anything for a few hours before your exam if it will use contrast material. Tell your doctor about all medications you are taking and if you have any allergies. If you have a known allergy to contrast material, your doctor may prescribe medications (usually a steroid) to reduce the risk of an allergic reaction. To avoid unnecessary delays, contact your doctor well before the date of your exam.

Also tell your doctor about any recent illnesses or other medical conditions and whether you have a history of heart disease, asthma, diabetes, kidney disease, or thyroid problems. Any of these conditions may increase the risk of an adverse effect.

On the day before and day of your exam, you may be asked to avoid:

- diet pills and caffeinated drinks such as coffee, tea, energy drinks or sodas. These may increase heart rate and limit the ability of the exam to evaluate for plaque in the coronary arteries.

- Viagra or any similar medication. They are not compatible with the medications you will receive during the procedure.

On the night before the procedure, you may be asked to take a beta blocker medication to lower your heart rate to optimize the quality of the exam.

Your child may be asked not to eat or drink anything for several hours beforehand, especially if a sedative or anesthesia will be used in the exam. In general, children who have recently been ill will not be sedated or anesthetized. If this is the case, or if you suspect that your child may be getting sick, you should talk with your physician about rescheduling the CT exam.

You should also inform your physician of any medications your child is taking and if he/she has any allergies, especially to intravenous (IV) or oral contrast materials. The allergy information should also be discussed with the CT technologist or nurse at the time of the CT examination. If your child has a known contrast material allergy, you should inform the doctor and technologist prior to the exam.

Also inform your doctor of any recent illnesses or other medical conditions your child may have, and if there is a history of heart disease, asthma, diabetes, kidney disease or thyroid problems. Any of these conditions may influence the decision on whether contrast material will be given to your child for the CT examination.

Talk to your doctor if you have questions about the instructions you've been given.

Women should always inform their physician and the CT technologist if there is any possibility that they may be pregnant. See the CT Safety During Pregnancy page for more information.

If you are breastfeeding at the time of the exam, ask your doctor how to proceed. It may help to pump breast milk ahead of time. Keep it on hand for use until all contrast material has cleared from your body (about 24 hours after the test). However, the most recent American College of Radiology (ACR) Manual on Contrast Media reports that studies show the amount of contrast absorbed by the infant during breastfeeding is extremely low. For further information please consult the ACR Manual on Contrast Media and its references.

What does the equipment look like?

The CT scanner is typically a large, donut-shaped machine with a short tunnel in the center. You will lie on a narrow table that slides in and out of this short tunnel. Rotating around you, the x-ray tube and electronic x-ray detectors are located opposite each other in a ring, called a gantry. The computer workstation that processes the imaging information is in a separate control room. This is where the technologist operates the scanner and monitors your exam in direct visual contact. The technologist will be able to hear and talk to you using a speaker and microphone.

CCTA is very much like a normal CT scan. The only difference is the speed of the CT scanner and the use of a heart monitor to determine your heart rate.

How does the procedure work?

During the examination, x-rays pass through the body and are picked up by special detectors in the scanner. Typically, higher numbers (especially 64 or more) of these detectors result in clearer final images. For that reason, CCTA often is referred to as multi-detector or multi-slice CT scanning. The information collected during the CCTA examination is used to identify the coronary artery anatomy, plaque, narrowing of the vessel, and, in certain cases, heart function. The radiologist will use the computer to create three-dimensional images and images in various planes to completely evaluate the heart and coronary arteries.

Procedures use contrast material to clearly define the blood vessels being examined by making them appear bright white.

How is the procedure performed?

The nurse will give you a gown to wear during the procedure.

The technologist will clean three small areas of your chest and place electrodes (small, sticky discs) on these areas. Men may require a small area of hair to be shaved on their chest to help the electrodes stick. The electrodes are attached to an electrocardiograph (ECG) monitor, which shows your heart's electrical activity during the test.

A nurse or technologist will insert an intravenous (IV) line into a vein in your arm to administer contrast material during your procedure. While in the CT scanner, you may be given a beta blocker medication through the same IV line or by mouth to help slow your heart rate in order to improve image quality. Nitroglycerin, to dilate and improve visualization of the coronary arteries, may also be administered as a tablet, a spray underneath your tongue or a patch on your skin.

While lying on the scanning table, you may be asked to raise your arms over your head for the duration of the exam. This will help improve image quality.

Next, the table will move quickly through the scanner to determine the correct starting position for the scans. Then, the table will move slowly through the machine for the actual CT scan. Depending on the type of CT scan, the machine may make several passes.

The technologist may ask you to hold your breath during the scanning. Any motion, including breathing and body movements, can lead to artifacts on the images. This loss of image quality can resemble the blurring seen on a photograph taken of a moving object.

Inform your doctor if you have problems holding your breath for 5 to 15 seconds. Breathing during the scan creates blurring on the images and can result in an inconclusive exam.

When the exam is complete, the technologist will ask you to wait until they verify that the images are of high enough quality for accurate interpretation by the radiologist.

The doctor or nurse will remove your IV line before you go home.

Including all preparations, the CCTA scan usually takes about 15 minutes if the heart rate is slow and steady. It may take longer if the baseline heart rate is fast and beta-blocker is given to slow it down. If the beta-blocker is given by mouth it generally will require at least one hour to take effect. If the medication is injected into a vein (intravenously), it may still require multiple doses and up to 20 minutes to reach the slower heart rate.

What will I experience during and after the procedure?

Other than the needle stick when the IV line is placed, most CT exams are fast, easy and painless.

Though the scan is painless, you may have some discomfort from remaining still for several minutes or from placement of an IV. If you have a hard time staying still, are very nervous, anxious, or in pain, you may find a CT exam stressful. The technologist or nurse, under the direction of a doctor, may offer you some medication to help you tolerate the CT exam.

If the exam uses iodinated contrast material, your doctor will screen you for chronic or acute kidney disease. The doctor may administer contrast material intravenously (by vein), so you will feel a pin prick when the nurse inserts the needle into your vein. You may feel warm or flushed as the contrast is injected. You also may have a metallic taste in your mouth. This will pass. You may feel a need to urinate. However, these are only side effects of the contrast injection, and they subside quickly.

The medication given to slow the heart rate has been known to cause some patients to feel dizzy when they stand suddenly due to a lowering of blood pressure. Therefore, you will often be asked to sit up slowly on the table prior to standing. The dizziness is slight and only happens rarely. You may also have your blood pressure taken before the exam, during and following the examination if medications are given. The nitroglycerin medication may also give you a headache; this is not dangerous and will wear off quickly.

When you enter the CT scanner, you may see special light lines projected onto your body. These lines help ensure that you are in the correct position on the exam table. With modern CT scanners, you may hear slight buzzing, clicking and whirring sounds. These occur as the CT scanner's internal parts, not usually visible to you, revolve around you during the imaging process.

You will be alone in the exam room during the CT scan, unless there are special circumstances. For example, sometimes a parent wearing a lead shield may stay in the room with their child. However, the technologist will always be able to see, hear and speak with you through a built-in intercom system.

After a CT exam, the technologist will remove your intravenous line. They will cover the tiny hole made by the needle with a small dressing. You can return to your normal activities immediately.

Who interprets the results and how do I get them?

A radiologist, a doctor specially trained to supervise and interpret radiology exams, will analyze the images. The radiologist will send an official report to the doctor who ordered the exam.

If you are actively having chest pain, your results will be given to the emergency room doctor by the radiologist, and a preliminary result will be reported right away.

You may need a follow-up exam. If so, your doctor will explain why. Sometimes a follow-up exam further evaluates a potential issue with more views or a special imaging technique. It may also see if there has been any change in an issue over time. Follow-up exams are often the best way to see if treatment is working or if a problem needs attention.

What are the benefits vs. risks?

Benefits

- CCTA is not invasive. An alternative test, cardiac catheterization with a coronary angiogram, is invasive, has more complications related to the placement of a long catheter into the groin or wrist arteries extending all the way to the heart, and the movement of the catheter in the blood vessels. Invasive catheterization requires more time for the patient to recover.

- A major advantage of CT is that it is able to view bone, soft tissue and blood vessels all at the same time. It is therefore suited to identify other reasons for your discomfort such as an injury to the aorta or a blood clot in the lungs.

- Unlike conventional x-rays, CT scanning provides very detailed images of many types of tissue.

- CT examinations are fast and simple.

- CT has been shown to be cost-effective for a wide range of medical problems.

- CT is less sensitive to patient movement than MRI.

- Unlike MRI, an implanted medical device of any kind will not prevent you from having a CT scan.

- No radiation remains in a patient's body after a CT exam.

- The x-rays used for CT scanning should have no immediate side effects.

Risks

- In some people with abnormal kidney function, the contrast material used in CT scanning may worsen kidney function.

- If contrast material leaks out from the vessel being injected and spreads under the skin where the IV is placed, skin damage or damage to blood vessels and nerves, though unlikely, can result. If you feel any pain in your arm at the location of the IV during contrast material injection, you should immediately inform the technologist.

- There is always a slight chance of cancer from excessive exposure to radiation. However, the benefit of an accurate diagnosis far outweighs the risk involved with CT scanning.

- The radiation dose for this procedure varies. See the Radiation Dose page for more information.

- Women should always tell their doctor and x-ray or CT technologist if there is any chance they are pregnant. See the Radiation Safety page for more information about pregnancy and x-rays.

- Doctors do not generally recommend CT scanning for pregnant women unless medically necessary because of potential risk to the unborn baby.

- IV contrast manufacturers indicate mothers should not breastfeed their babies for 24-48 hours after contrast material is given. However, the most recent American College of Radiology (ACR) Manual on Contrast Media reports that studies show the amount of contrast absorbed by the infant during breastfeeding is extremely low. For further information please consult the ACR Manual on Contrast Media and its references.

- The risk of serious allergic reaction to contrast materials that contain iodine is rare, and radiology departments are well-equipped to deal with them.

What are the limitations of coronary CTA?

A person who is very large may not fit into the opening of a conventional CT scanner. Or, they may be over the weight limit-usually 450 pounds-for the moving table.

Patients who are extremely overweight or who have abnormal heart rhythms may not be good candidates for this test because image quality may be compromised.

Unlike CCTA, which is only a diagnostic test, invasive coronary angiography can be used for both diagnosis and treatment in a single session. If a narrowing or blockage is found during a CCTA, it cannot be treated at the same time. Patients with a high risk of coronary artery disease and typical symptoms might undergo coronary angiography instead of CCTA because they are more likely to need treatment.

CCTA can be difficult to read if there are many areas of old, calcified (hardened) plaque, which can be the case in older patients.

AI rewrite

A coronary CTA is a special scan of your heart. It checks the blood vessels that bring blood to your heart. The test looks for plaque. Plaque is made of fat, cholesterol, and calcium. Over time, plaque can build up and block blood flow. For this test, you will get a special dye called contrast. It goes into a vein in your arm. The dye helps the doctor see your blood vessels clearly on the scan. The CT machine takes many pictures to make a 3D model of your heart.

Doctors use this test to see if your heart's blood vessels are narrowed. They may order it if you have chest pain. It is also used if other heart tests are not clear. The scan can also find other problems in your chest, like a collapsed lung or a blood clot.

Tell your doctor if you might be pregnant. Also tell them about any illnesses, medical conditions, or allergies. Tell them about all the medicines you take. Let them know if you have heart disease, asthma, diabetes, kidney disease, or thyroid problems. If you are allergic to the special dye, tell your doctor early. You may need to take medicine starting 13 hours before the test to prevent an allergic reaction.

Do not eat or drink for a few hours before the test. The day before and the day of the test, do not have caffeine. This includes coffee, tea, energy drinks, sodas, and diet pills. Caffeine speeds up your heart. Do not take Viagra or similar medicines. They do not mix well with the medicines used in this test. You may need to take a medicine called a beta-blocker the night before. This slows your heart rate for better pictures.

Wear loose, comfortable clothes. Leave jewelry, glasses, hairpins, and metal objects at home. You may need to take off hearing aids, dental work, or bras with metal wires. You might wear a hospital gown. If you are breastfeeding, ask your doctor what to do. The dye is mostly safe, but you can pump breast milk ahead of time just in case. If your child is having this test, tell the doctor about their health and allergies. Sick children usually cannot have medicine to make them sleep for the test.

The CT scanner looks like a large donut with a short tunnel. You will lie on a narrow table that slides into the tunnel. The machine uses X-rays to take pictures. The special dye makes your blood vessels look bright white in the pictures. The person running the machine is called a technologist. They will be in the next room. They can see, hear, and talk to you through a speaker.

A nurse will clean three small spots on your chest. They will put small, sticky patches on these spots to check your heartbeat. Men may need a small amount of chest hair shaved so the patches stick. A nurse will put an IV line into a vein in your arm. The special dye goes through this IV. You may also get medicine to slow your heart rate. You might get another medicine called nitroglycerin. This can be a pill, a spray under your tongue, or a skin patch. It opens your blood vessels so they are easier to see.

You will lie on the table. You may need to hold your arms above your head. The table will move quickly through the scanner to find the right starting spot. Then it will move slowly to take the pictures. You must stay very still. The technologist will ask you to hold your breath for 5 to 15 seconds. If you move or breathe, the pictures will be blurry. Tell your doctor if you cannot hold your breath this long. The test takes about 15 minutes if your heart rate is slow. It may take longer if you need medicine to slow your heart.

The scan does not hurt. You will feel a quick pinch when the IV goes into your arm. When the dye goes in, you may feel warm or flushed. You might taste metal in your mouth. You might feel like you need to pee. These feelings are normal and go away quickly. The medicine used to slow your heart might make you dizzy if you stand up too fast. You should sit up slowly. The nitroglycerin might give you a mild headache, but it will go away soon.

The machine may make buzzing and clicking sounds. This is normal. After the test, the nurse will take out the IV and put a small bandage on your arm. You can go back to your normal activities right away.

A special doctor called a radiologist will look at your pictures. They will send a report to your doctor. If you are in the emergency room with chest pain, the doctor will get the results right away. You might need another test later to check on a problem.

This test has many benefits. It does not require surgery or long tubes put into your heart. It is fast and gives very clear pictures of your blood vessels, bones, and soft tissues. No radiation stays in your body after the test.

There are some risks. The special dye can cause kidney problems for people who already have kidney disease. Rarely, the dye can leak out of the vein and hurt the skin. Tell the technologist right away if your arm hurts during the test. There is a very small chance of getting cancer from the X-rays. But the benefit of finding a heart problem is much greater than this risk. Pregnant women should usually not have this test because X-rays can harm the baby. Allergic reactions to the dye are rare, but doctors are ready to treat you if it happens.

The CT scanner has a weight limit, usually 450 pounds. A very large person might not fit in the machine. The pictures might not be clear if a person is very overweight or has an irregular heartbeat. The test can also be hard to read in older patients who have a lot of hard plaque. Finally, this test only finds problems. It cannot fix them. If the doctor finds a blocked blood vessel, you cannot get it treated during this scan. You would need a different procedure to fix it.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Tell your doctor if there's a possibility you are pregnant and discuss any recent illnesses, medical conditions, medications you're taking, and allergies. You will be instructed not to eat or drink anything several hours beforehand and to avoid caffeinated products, Viagra or similar medication. If you have a known allergy to contrast material, your doctor may prescribe medications to reduce the risk of an allergic reaction. These medications must be taken at multiple intervals beginning 13 hours prior to your exam. Leave jewelry at home and wear loose, comfortable clothing. You may be asked to wear a gown. If you are breastfeeding, talk to your doctor about how to proceed.

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A CT scan generates images that can be reformatted in multiple planes. It can even generate three-dimensional images. Your doctor can review these images on a computer monitor, print them on film or via a 3D printer, or transfer them to a CD or DVD.

CT images of internal organs, bones, soft tissue, and blood vessels provide greater detail than traditional x-rays. This is especially true for soft tissues and blood vessels.

What are some common uses of the procedure?

Many physicians advocate the careful use of CCTA for patients who have:

- suspected abnormal anatomy of the coronary arteries.

- low or intermediate risk for coronary artery disease, including patients who have chest pain and normal, non-diagnostic or unclear lab and ECG results.

- low to intermediate risk atypical chest pain in the emergency department.

- non-acute chest pain.

- new or worsening symptoms with a previous normal stress test result.

- unclear or inconclusive stress test results.

- new onset heart failure with reduced heart function and low or medium risk for coronary artery disease.

- intermediate risk of coronary artery disease before non-coronary cardiac surgery.

- coronary artery bypass grafts.

For patients meeting the above indications, CCTA can provide important information about the presence and extent of plaque in the coronary arteries. Apart from identifying coronary artery narrowing as the cause of chest discomfort, it can also detect other possible causes of symptoms, such as a collapsed lung, blood clot in the vessels leading to the lungs, or aortic abnormalities. Your primary care physician or cardiac specialist, in consultation with a radiologist who would perform the test, will determine whether CCTA is appropriate for you.

How should I prepare?

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Metal objects, including jewelry, eyeglasses, dentures, and hairpins, may affect the CT images. Leave them at home or remove them prior to your exam. Some CT exams will require you to remove hearing aids and removable dental work. Women will need to remove bras containing metal underwire. You may need to remove any piercings, if possible.

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- diet pills and caffeinated drinks such as coffee, tea, energy drinks or sodas. These may increase heart rate and limit the ability of the exam to evaluate for plaque in the coronary arteries.

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You should also inform your physician of any medications your child is taking and if he/she has any allergies, especially to intravenous (IV) or oral contrast materials. The allergy information should also be discussed with the CT technologist or nurse at the time of the CT examination. If your child has a known contrast material allergy, you should inform the doctor and technologist prior to the exam.

Also inform your doctor of any recent illnesses or other medical conditions your child may have, and if there is a history of heart disease, asthma, diabetes, kidney disease or thyroid problems. Any of these conditions may influence the decision on whether contrast material will be given to your child for the CT examination.

Talk to your doctor if you have questions about the instructions you've been given.

Women should always inform their physician and the CT technologist if there is any possibility that they may be pregnant. See the CT Safety During Pregnancy page for more information.

If you are breastfeeding at the time of the exam, ask your doctor how to proceed. It may help to pump breast milk ahead of time. Keep it on hand for use until all contrast material has cleared from your body (about 24 hours after the test). However, the most recent American College of Radiology (ACR) Manual on Contrast Media reports that studies show the amount of contrast absorbed by the infant during breastfeeding is extremely low. For further information please consult the ACR Manual on Contrast Media and its references.

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The technologist may ask you to hold your breath during the scanning. Any motion, including breathing and body movements, can lead to artifacts on the images. This loss of image quality can resemble the blurring seen on a photograph taken of a moving object.

Inform your doctor if you have problems holding your breath for 5 to 15 seconds. Breathing during the scan creates blurring on the images and can result in an inconclusive exam.

When the exam is complete, the technologist will ask you to wait until they verify that the images are of high enough quality for accurate interpretation by the radiologist.

The doctor or nurse will remove your IV line before you go home.

Including all preparations, the CCTA scan usually takes about 15 minutes if the heart rate is slow and steady. It may take longer if the baseline heart rate is fast and beta-blocker is given to slow it down. If the beta-blocker is given by mouth it generally will require at least one hour to take effect. If the medication is injected into a vein (intravenously), it may still require multiple doses and up to 20 minutes to reach the slower heart rate.

What will I experience during and after the procedure?

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Though the scan is painless, you may have some discomfort from remaining still for several minutes or from placement of an IV. If you have a hard time staying still, are very nervous, anxious, or in pain, you may find a CT exam stressful. The technologist or nurse, under the direction of a doctor, may offer you some medication to help you tolerate the CT exam.

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The medication given to slow the heart rate has been known to cause some patients to feel dizzy when they stand suddenly due to a lowering of blood pressure. Therefore, you will often be asked to sit up slowly on the table prior to standing. The dizziness is slight and only happens rarely. You may also have your blood pressure taken before the exam, during and following the examination if medications are given. The nitroglycerin medication may also give you a headache; this is not dangerous and will wear off quickly.

When you enter the CT scanner, you may see special light lines projected onto your body. These lines help ensure that you are in the correct position on the exam table. With modern CT scanners, you may hear slight buzzing, clicking and whirring sounds. These occur as the CT scanner's internal parts, not usually visible to you, revolve around you during the imaging process.

You will be alone in the exam room during the CT scan, unless there are special circumstances. For example, sometimes a parent wearing a lead shield may stay in the room with their child. However, the technologist will always be able to see, hear and speak with you through a built-in intercom system.

After a CT exam, the technologist will remove your intravenous line. They will cover the tiny hole made by the needle with a small dressing. You can return to your normal activities immediately.

Who interprets the results and how do I get them?

A radiologist, a doctor specially trained to supervise and interpret radiology exams, will analyze the images. The radiologist will send an official report to the doctor who ordered the exam.

If you are actively having chest pain, your results will be given to the emergency room doctor by the radiologist, and a preliminary result will be reported right away.

You may need a follow-up exam. If so, your doctor will explain why. Sometimes a follow-up exam further evaluates a potential issue with more views or a special imaging technique. It may also see if there has been any change in an issue over time. Follow-up exams are often the best way to see if treatment is working or if a problem needs attention.

What are the benefits vs. risks?

Benefits

- CCTA is not invasive. An alternative test, cardiac catheterization with a coronary angiogram, is invasive, has more complications related to the placement of a long catheter into the groin or wrist arteries extending all the way to the heart, and the movement of the catheter in the blood vessels. Invasive catheterization requires more time for the patient to recover.

- A major advantage of CT is that it is able to view bone, soft tissue and blood vessels all at the same time. It is therefore suited to identify other reasons for your discomfort such as an injury to the aorta or a blood clot in the lungs.

- Unlike conventional x-rays, CT scanning provides very detailed images of many types of tissue.

- CT examinations are fast and simple.

- CT has been shown to be cost-effective for a wide range of medical problems.

- CT is less sensitive to patient movement than MRI.

- Unlike MRI, an implanted medical device of any kind will not prevent you from having a CT scan.

- No radiation remains in a patient's body after a CT exam.

- The x-rays used for CT scanning should have no immediate side effects.

Risks

- In some people with abnormal kidney function, the contrast material used in CT scanning may worsen kidney function.

- If contrast material leaks out from the vessel being injected and spreads under the skin where the IV is placed, skin damage or damage to blood vessels and nerves, though unlikely, can result. If you feel any pain in your arm at the location of the IV during contrast material injection, you should immediately inform the technologist.

- There is always a slight chance of cancer from excessive exposure to radiation. However, the benefit of an accurate diagnosis far outweighs the risk involved with CT scanning.

- The radiation dose for this procedure varies. See the Radiation Dose page for more information.

- Women should always tell their doctor and x-ray or CT technologist if there is any chance they are pregnant. See the Radiation Safety page for more information about pregnancy and x-rays.

- Doctors do not generally recommend CT scanning for pregnant women unless medically necessary because of potential risk to the unborn baby.

- IV contrast manufacturers indicate mothers should not breastfeed their babies for 24-48 hours after contrast material is given. However, the most recent American College of Radiology (ACR) Manual on Contrast Media reports that studies show the amount of contrast absorbed by the infant during breastfeeding is extremely low. For further information please consult the ACR Manual on Contrast Media and its references.

- The risk of serious allergic reaction to contrast materials that contain iodine is rare, and radiology departments are well-equipped to deal with them.

What are the limitations of coronary CTA?

A person who is very large may not fit into the opening of a conventional CT scanner. Or, they may be over the weight limit-usually 450 pounds-for the moving table.

Patients who are extremely overweight or who have abnormal heart rhythms may not be good candidates for this test because image quality may be compromised.

Unlike CCTA, which is only a diagnostic test, invasive coronary angiography can be used for both diagnosis and treatment in a single session. If a narrowing or blockage is found during a CCTA, it cannot be treated at the same time. Patients with a high risk of coronary artery disease and typical symptoms might undergo coronary angiography instead of CCTA because they are more likely to need treatment.

CCTA can be difficult to read if there are many areas of old, calcified (hardened) plaque, which can be the case in older patients.

AI rewrite

Last reviewed on March 11, 2024

Coronary CTA

Coronary CTA is also called coronary CT angiography or CCTA. It is a heart imaging test. It uses a CT scan and a dye that contains iodine. The dye is given through an IV. An IV is a small tube placed in a vein.

This test looks at the arteries that bring blood to your heart. It can show if these arteries are narrowed. The CT scan makes pictures of the inside of your body. These pictures can be changed into 3D pictures. Your doctor can view them on a screen, print them, or save them to electronic media.

Tell your doctor if you may be pregnant. Tell your doctor about recent illnesses, health problems, medicines you take, and allergies.

You may be told not to eat or drink for several hours before the test. You may also be told to avoid caffeine, Viagra, or similar medicines.

If you are allergic to contrast dye, your doctor may give you medicine to lower the chance of an allergic reaction. You must take this medicine at several times, starting 13 hours before your test.

Leave jewelry at home. Wear loose, comfortable clothes. You may be asked to wear a gown. If you are breastfeeding, talk with your doctor about what to do.

- What is coronary CTA?

- What are common uses of this test?

- How should I prepare?

- What does the equipment look like?

- How does the test work?

- How is the test done?

- What will I feel during and after the test?

- Who reads the results, and how do I get them?

- What are the benefits and risks?

- What are the limits of coronary CTA?

What is coronary CTA?

Coronary computed tomography angiography, or CCTA, is a test that takes pictures of the heart. It helps show if plaque has narrowed the coronary arteries. These are the blood vessels that bring blood to the heart.

Plaque is made of things like fat, cholesterol, and calcium. It builds up on the inside walls of the arteries over time. Plaque can slow blood flow. In some cases, it can block blood flow completely.

For a CCTA scan, you get contrast dye through an IV. This dye contains iodine. It helps make the best pictures of the heart's blood vessels.

Computed tomography is also called CT or CAT scan. It is a medical imaging test. Like regular x-rays, it makes pictures of the inside of the body.

A CT scan makes images that can be viewed in many ways. It can also make 3D images. Your doctor can look at the images on a computer screen. The images can be printed on film, printed with a 3D printer, or saved to a CD or DVD.

CT pictures show organs, bones, soft tissue, and blood vessels in more detail than regular x-rays. This is especially true for soft tissue and blood vessels.

What are common uses of this test?

Doctors may use CCTA carefully for people who have:

- possible unusual shape or position of the coronary arteries

- low or medium risk for coronary artery disease, including people with chest pain and normal, unclear, or non-diagnostic lab and ECG results

- low to medium risk chest pain that is not typical, in the emergency department

- chest pain that is not sudden or severe

- new or worse symptoms after a normal stress test

- unclear stress test results

- new heart failure with reduced heart function and low or medium risk for coronary artery disease

- medium risk of coronary artery disease before heart surgery that is not on the coronary arteries

- coronary artery bypass grafts

For people who meet these reasons, CCTA can give important information. It can show if plaque is present and how much plaque is in the coronary arteries.

CCTA can show if narrowed coronary arteries are causing chest discomfort. It can also find other possible causes of symptoms. These may include a collapsed lung, a blood clot in blood vessels going to the lungs, or problems with the aorta.

Your primary care doctor or heart doctor will talk with the radiologist who performs the test. They will decide if CCTA is right for you.

How should I prepare?

Wear loose, comfortable clothes to your test. You may need to change into a gown.

Metal objects can affect CT pictures. These include jewelry, eyeglasses, dentures, and hairpins. Leave them at home or take them off before the test. For some CT tests, you may need to remove hearing aids and removable dental work. Women may need to remove bras with metal underwire. You may need to remove piercings, if possible.

Your doctor may tell you not to eat or drink for a few hours before the test if contrast dye will be used.

Tell your doctor about all medicines you take. Tell your doctor if you have allergies. If you are allergic to contrast dye, your doctor may give you medicine, often a steroid, to lower the chance of an allergic reaction. To avoid delays, contact your doctor well before your test date.

Also tell your doctor about recent illnesses or other health problems. Tell your doctor if you have a history of heart disease, asthma, diabetes, kidney disease, or thyroid problems. These problems may increase the risk of a bad reaction.

On the day before and the day of your test, you may be asked to avoid:

- diet pills and drinks with caffeine, such as coffee, tea, energy drinks, or soda. These can raise your heart rate. A fast heart rate can make it harder to see plaque in the coronary arteries.

- Viagra or similar medicines. These do not work safely with medicines you may get during the test.

The night before the test, you may be asked to take a beta blocker medicine. This medicine lowers your heart rate. A slower heart rate can help make better pictures.

Your child may be asked not to eat or drink for several hours before the test. This is especially true if a sedative or anesthesia will be used. In general, children who have been sick recently will not be given sedation or anesthesia. If this happens, or if you think your child may be getting sick, talk with your doctor about rescheduling the CT test.

Tell your doctor about any medicines your child takes. Tell your doctor if your child has allergies, especially to IV or oral contrast dye. Also tell the CT technologist or nurse about these allergies at the time of the test. If your child has a known contrast dye allergy, tell the doctor and technologist before the test.

Also tell your doctor about any recent illnesses or other health problems your child has. Tell the doctor if your child has a history of heart disease, asthma, diabetes, kidney disease, or thyroid problems. These problems may affect the decision to give contrast dye to your child.

Talk to your doctor if you have questions about the instructions you were given.

Women should always tell their doctor and the CT technologist if they may be pregnant.

If you are breastfeeding at the time of the test, ask your doctor what to do. It may help to pump breast milk before the test. Keep it for use until all contrast dye has left your body, about 24 hours after the test. However, the most recent American College of Radiology Manual on Contrast Media says studies show that very little contrast dye is absorbed by the baby during breastfeeding.

What does the equipment look like?

The CT scanner is usually a large, donut-shaped machine. It has a short tunnel in the center. You lie on a narrow table. The table slides in and out of the tunnel.

Inside the scanner, an x-ray tube and x-ray detectors move around you. They are across from each other in a ring called a gantry.

The computer that works with the pictures is in another room. This is called the control room. The technologist runs the scanner from there and watches your test. The technologist can see you. They can also hear and talk to you through a speaker and microphone.

CCTA is much like a regular CT scan. The main differences are that the CT scanner is fast and a heart monitor is used to check your heart rate.

How does the test work?

During the test, x-rays pass through your body. Special detectors in the scanner pick them up. In general, more detectors, especially 64 or more, can make clearer pictures. For this reason, CCTA is often called multi-detector or multi-slice CT scanning.

The information from the CCTA test is used to look at:

- the shape and position of the coronary arteries

- plaque

- narrowing of the blood vessel

- heart function in some cases

The radiologist uses a computer to make 3D pictures and pictures from different views. This helps the doctor fully check the heart and coronary arteries.

Contrast dye is used to show the blood vessels clearly. It makes the blood vessels look bright white on the pictures.

How is the test done?

A nurse will give you a gown to wear during the test.

The technologist will clean three small areas on your chest. They will place electrodes on these spots. Electrodes are small, sticky discs. Men may need a small area of chest hair shaved so the electrodes will stick. The electrodes connect to an ECG monitor. This shows your heart’s electrical activity during the test.

A nurse or technologist will place an IV line in a vein in your arm. Contrast dye will be given through the IV during the test.

While you are in the CT scanner, you may get a beta blocker medicine through the IV or by mouth. This helps slow your heart rate and can improve picture quality. You may also get nitroglycerin. Nitroglycerin helps widen the coronary arteries so they can be seen better. It may be given as a tablet, a spray under your tongue, or a patch on your skin.

You will lie on the scan table. You may be asked to raise your arms over your head for the test. This helps make better pictures.

Next, the table will move quickly through the scanner. This finds the right starting place for the scan. Then the table will move slowly through the machine for the CT scan. The machine may scan you more than once, depending on the type of CT scan.

The technologist may ask you to hold your breath during the scan. Any movement, including breathing or body movement, can blur the pictures. This can look like a blurry photo of a moving object.

Tell your doctor if you have trouble holding your breath for 5 to 15 seconds. Breathing during the scan can blur the pictures. This can make the test unclear.

When the test is done, the technologist will ask you to wait. They need to check that the pictures are clear enough for the radiologist to read.

The doctor or nurse will remove your IV before you go home.

With all preparation included, the CCTA scan usually takes about 15 minutes if your heart rate is slow and steady. It may take longer if your heart rate is fast and a beta blocker is given to slow it down. If the beta blocker is given by mouth, it usually takes at least 1 hour to work. If it is given through an IV, it may still take several doses and up to 20 minutes to slow your heart rate.

What will I feel during and after the test?

Most CT tests are fast, easy, and painless. You may feel a needle stick when the IV is placed.

The scan itself does not hurt. You may feel some discomfort from staying still for several minutes or from the IV. If it is hard for you to stay still, or if you are very nervous, anxious, or in pain, the CT test may feel stressful. The technologist or nurse may offer medicine to help you get through the test, under a doctor’s direction.

If the test uses iodine contrast dye, your doctor will check you for long-term or sudden kidney disease. The contrast dye may be given through a vein. You will feel a pin prick when the nurse places the needle.

When the dye is injected, you may feel warm or flushed. You may have a metal taste in your mouth. This will go away. You may feel like you need to urinate. These are side effects of the dye injection. They go away quickly.

The medicine used to slow your heart rate can make some people feel dizzy when they stand up suddenly. This happens because it can lower blood pressure. You will often be asked to sit up slowly on the table before standing. The dizziness is mild and happens only rarely. Your blood pressure may be checked before, during, and after the test if you get these medicines. Nitroglycerin may give you a headache. This is not dangerous and will go away quickly.

When you enter the CT scanner, you may see special light lines on your body. These lines help make sure you are in the right position on the table. With modern CT scanners, you may hear soft buzzing, clicking, and whirring sounds. These sounds happen as parts inside the scanner move around you.

You will be alone in the exam room during the CT scan, unless there is a special reason. For example, sometimes a parent wearing a lead shield may stay in the room with a child. The technologist can always see, hear, and talk with you through an intercom.

After the CT test, the technologist will remove your IV. They will cover the small needle hole with a small bandage. You can return to your normal activities right away.

Who reads the results, and how do I get them?

A radiologist will read your images. A radiologist is a doctor trained to supervise and read imaging tests. The radiologist will send an official report to the doctor who ordered the test.

If you are having chest pain at the time, the radiologist will give your results to the emergency room doctor. A first result will be reported right away.

You may need a follow-up test. If so, your doctor will explain why. A follow-up test may look more closely at a possible problem with more pictures or a special imaging method. It may also check if a problem has changed over time. Follow-up tests are often the best way to see if treatment is working or if a problem needs care.

What are the benefits and risks?

Benefits

- CCTA is not invasive. Another test, called cardiac catheterization with a coronary angiogram, is invasive. It has more complications linked to placing a long tube into an artery in the groin or wrist and moving it to the heart. It also takes more time to recover from.

- CT can show bone, soft tissue, and blood vessels at the same time. This can help find other causes of discomfort, such as an injury to the aorta or a blood clot in the lungs.

- CT gives very detailed pictures of many kinds of tissue. It gives more detail than regular x-rays.

- CT tests are fast and simple.

- CT has been shown to be cost-effective for many medical problems.

- CT is less affected by patient movement than MRI.

- Unlike MRI, having any kind of implanted medical device will not stop you from having a CT scan.

- No radiation stays in your body after a CT test.

- The x-rays used for CT scanning should have no immediate side effects.

Risks

- In some people with abnormal kidney function, the contrast dye used in CT may make kidney function worse.

- If contrast dye leaks out of the blood vessel and spreads under the skin where the IV is placed, it can cause skin damage or damage to blood vessels and nerves. This is unlikely. If you feel any pain in your arm where the IV is during the dye injection, tell the technologist right away.

- There is always a small chance of cancer from too much radiation exposure. But the benefit of a correct diagnosis is much greater than the risk from CT scanning.

- The radiation dose for this test can vary.

- Women should always tell their doctor and x-ray or CT technologist if there is any chance they are pregnant.

- Doctors usually do not recommend CT scans for pregnant women unless the scan is medically needed. This is because of possible risk to the unborn baby.

- Makers of IV contrast dye say mothers should not breastfeed their babies for 24 to 48 hours after getting contrast dye. However, the most recent American College of Radiology Manual on Contrast Media says studies show that very little contrast dye is absorbed by the baby during breastfeeding.

- A serious allergic reaction to iodine contrast dye is rare. Radiology departments are well prepared to treat it.

What are the limits of coronary CTA?

A very large person may not fit into a regular CT scanner. Or they may be over the weight limit for the moving table. The weight limit is usually 450 pounds.

People who are extremely overweight or who have abnormal heart rhythms may not be good candidates for this test. The pictures may not be clear enough.

CCTA is only a test to diagnose a problem. It cannot treat a narrowing or blockage at the same time. Invasive coronary angiography can be used for both diagnosis and treatment during one visit. If a narrowing or blockage is found during CCTA, it cannot be treated during that same test.

People with a high risk of coronary artery disease and typical symptoms may have coronary angiography instead of CCTA. This is because they may be more likely to need treatment.

CCTA can be hard to read if there are many areas of old, calcified plaque. Calcified plaque is hardened plaque. This can happen in older patients.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

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Coronary CTA

Coronary computed tomography angiography (also called coronary CT angiography or CCTA) uses an injection of iodine-containing contrast material and CT scanning to examine the arteries that supply blood to the heart and determine whether they have been narrowed. The images generated during a CT scan can be reformatted to create three-dimensional (3D) images that may be viewed on a monitor, printed on film or by a 3D printer, or transferred to electronic media.

Tell your doctor if there's a possibility you are pregnant and discuss any recent illnesses, medical conditions, medications you're taking, and allergies. You will be instructed not to eat or drink anything several hours beforehand and to avoid caffeinated products, Viagra or similar medication. If you have a known allergy to contrast material, your doctor may prescribe medications to reduce the risk of an allergic reaction. These medications must be taken at multiple intervals beginning 13 hours prior to your exam. Leave jewelry at home and wear loose, comfortable clothing. You may be asked to wear a gown. If you are breastfeeding, talk to your doctor about how to proceed.

- What is coronary CTA?

- What are some common uses of the procedure?

- How should I prepare?

- What does the equipment look like?

- How does the procedure work?

- How is the procedure performed?

- What will I experience during and after the procedure?

- Who interprets the results and how do I get them?

- What are the benefits vs. risks?

- What are the limitations of coronary CTA?

What is coronary CTA?

Coronary computed tomography angiography (CCTA) is a heart imaging test that helps determine if plaque buildup has narrowed the coronary arteries, the blood vessels that supply the heart. Plaque is made of various substances such as fat, cholesterol and calcium that deposit along the inner lining of the arteries. Plaque, which builds up over time, can reduce or in some cases completely block blood flow. Patients undergoing a CCTA scan receive an iodine-containing contrast material as an intravenous (IV) injection to ensure the best possible images of the heart blood vessels.

Computed tomography, more commonly known as a CT or CAT scan, is a diagnostic medical imaging test. Like traditional x-rays, it produces multiple images or pictures of the inside of the body.

A CT scan generates images that can be reformatted in multiple planes. It can even generate three-dimensional images. Your doctor can review these images on a computer monitor, print them on film or via a 3D printer, or transfer them to a CD or DVD.

CT images of internal organs, bones, soft tissue, and blood vessels provide greater detail than traditional x-rays. This is especially true for soft tissues and blood vessels.

What are some common uses of the procedure?

Many physicians advocate the careful use of CCTA for patients who have:

- suspected abnormal anatomy of the coronary arteries.

- low or intermediate risk for coronary artery disease, including patients who have chest pain and normal, non-diagnostic or unclear lab and ECG results.

- low to intermediate risk atypical chest pain in the emergency department.

- non-acute chest pain.

- new or worsening symptoms with a previous normal stress test result.

- unclear or inconclusive stress test results.

- new onset heart failure with reduced heart function and low or medium risk for coronary artery disease.

- intermediate risk of coronary artery disease before non-coronary cardiac surgery.

- coronary artery bypass grafts.

For patients meeting the above indications, CCTA can provide important information about the presence and extent of plaque in the coronary arteries. Apart from identifying coronary artery narrowing as the cause of chest discomfort, it can also detect other possible causes of symptoms, such as a collapsed lung, blood clot in the vessels leading to the lungs, or aortic abnormalities. Your primary care physician or cardiac specialist, in consultation with a radiologist who would perform the test, will determine whether CCTA is appropriate for you.

How should I prepare?

Wear comfortable, loose-fitting clothing to your exam. You may need to change into a gown for the procedure.

Metal objects, including jewelry, eyeglasses, dentures, and hairpins, may affect the CT images. Leave them at home or remove them prior to your exam. Some CT exams will require you to remove hearing aids and removable dental work. Women will need to remove bras containing metal underwire. You may need to remove any piercings, if possible.

Your doctor may instruct you to not eat or drink anything for a few hours before your exam if it will use contrast material. Tell your doctor about all medications you are taking and if you have any allergies. If you have a known allergy to contrast material, your doctor may prescribe medications (usually a steroid) to reduce the risk of an allergic reaction. To avoid unnecessary delays, contact your doctor well before the date of your exam.

Also tell your doctor about any recent illnesses or other medical conditions and whether you have a history of heart disease, asthma, diabetes, kidney disease, or thyroid problems. Any of these conditions may increase the risk of an adverse effect.

On the day before and day of your exam, you may be asked to avoid:

- diet pills and caffeinated drinks such as coffee, tea, energy drinks or sodas. These may increase heart rate and limit the ability of the exam to evaluate for plaque in the coronary arteries.

- Viagra or any similar medication. They are not compatible with the medications you will receive during the procedure.

On the night before the procedure, you may be asked to take a beta blocker medication to lower your heart rate to optimize the quality of the exam.

Your child may be asked not to eat or drink anything for several hours beforehand, especially if a sedative or anesthesia will be used in the exam. In general, children who have recently been ill will not be sedated or anesthetized. If this is the case, or if you suspect that your child may be getting sick, you should talk with your physician about rescheduling the CT exam.

You should also inform your physician of any medications your child is taking and if he/she has any allergies, especially to intravenous (IV) or oral contrast materials. The allergy information should also be discussed with the CT technologist or nurse at the time of the CT examination. If your child has a known contrast material allergy, you should inform the doctor and technologist prior to the exam.

Also inform your doctor of any recent illnesses or other medical conditions your child may have, and if there is a history of heart disease, asthma, diabetes, kidney disease or thyroid problems. Any of these conditions may influence the decision on whether contrast material will be given to your child for the CT examination.

Talk to your doctor if you have questions about the instructions you've been given.

Women should always inform their physician and the CT technologist if there is any possibility that they may be pregnant. See the CT Safety During Pregnancy page for more information.

If you are breastfeeding at the time of the exam, ask your doctor how to proceed. It may help to pump breast milk ahead of time. Keep it on hand for use until all contrast material has cleared from your body (about 24 hours after the test). However, the most recent American College of Radiology (ACR) Manual on Contrast Media reports that studies show the amount of contrast absorbed by the infant during breastfeeding is extremely low. For further information please consult the ACR Manual on Contrast Media and its references.

What does the equipment look like?

The CT scanner is typically a large, donut-shaped machine with a short tunnel in the center. You will lie on a narrow table that slides in and out of this short tunnel. Rotating around you, the x-ray tube and electronic x-ray detectors are located opposite each other in a ring, called a gantry. The computer workstation that processes the imaging information is in a separate control room. This is where the technologist operates the scanner and monitors your exam in direct visual contact. The technologist will be able to hear and talk to you using a speaker and microphone.

CCTA is very much like a normal CT scan. The only difference is the speed of the CT scanner and the use of a heart monitor to determine your heart rate.

How does the procedure work?

During the examination, x-rays pass through the body and are picked up by special detectors in the scanner. Typically, higher numbers (especially 64 or more) of these detectors result in clearer final images. For that reason, CCTA often is referred to as multi-detector or multi-slice CT scanning. The information collected during the CCTA examination is used to identify the coronary artery anatomy, plaque, narrowing of the vessel, and, in certain cases, heart function. The radiologist will use the computer to create three-dimensional images and images in various planes to completely evaluate the heart and coronary arteries.

Procedures use contrast material to clearly define the blood vessels being examined by making them appear bright white.

How is the procedure performed?

The nurse will give you a gown to wear during the procedure.

The technologist will clean three small areas of your chest and place electrodes (small, sticky discs) on these areas. Men may require a small area of hair to be shaved on their chest to help the electrodes stick. The electrodes are attached to an electrocardiograph (ECG) monitor, which shows your heart's electrical activity during the test.

A nurse or technologist will insert an intravenous (IV) line into a vein in your arm to administer contrast material during your procedure. While in the CT scanner, you may be given a beta blocker medication through the same IV line or by mouth to help slow your heart rate in order to improve image quality. Nitroglycerin, to dilate and improve visualization of the coronary arteries, may also be administered as a tablet, a spray underneath your tongue or a patch on your skin.

While lying on the scanning table, you may be asked to raise your arms over your head for the duration of the exam. This will help improve image quality.

Next, the table will move quickly through the scanner to determine the correct starting position for the scans. Then, the table will move slowly through the machine for the actual CT scan. Depending on the type of CT scan, the machine may make several passes.

The technologist may ask you to hold your breath during the scanning. Any motion, including breathing and body movements, can lead to artifacts on the images. This loss of image quality can resemble the blurring seen on a photograph taken of a moving object.

Inform your doctor if you have problems holding your breath for 5 to 15 seconds. Breathing during the scan creates blurring on the images and can result in an inconclusive exam.

When the exam is complete, the technologist will ask you to wait until they verify that the images are of high enough quality for accurate interpretation by the radiologist.

The doctor or nurse will remove your IV line before you go home.

Including all preparations, the CCTA scan usually takes about 15 minutes if the heart rate is slow and steady. It may take longer if the baseline heart rate is fast and beta-blocker is given to slow it down. If the beta-blocker is given by mouth it generally will require at least one hour to take effect. If the medication is injected into a vein (intravenously), it may still require multiple doses and up to 20 minutes to reach the slower heart rate.

What will I experience during and after the procedure?

Other than the needle stick when the IV line is placed, most CT exams are fast, easy and painless.

Though the scan is painless, you may have some discomfort from remaining still for several minutes or from placement of an IV. If you have a hard time staying still, are very nervous, anxious, or in pain, you may find a CT exam stressful. The technologist or nurse, under the direction of a doctor, may offer you some medication to help you tolerate the CT exam.

If the exam uses iodinated contrast material, your doctor will screen you for chronic or acute kidney disease. The doctor may administer contrast material intravenously (by vein), so you will feel a pin prick when the nurse inserts the needle into your vein. You may feel warm or flushed as the contrast is injected. You also may have a metallic taste in your mouth. This will pass. You may feel a need to urinate. However, these are only side effects of the contrast injection, and they subside quickly.

The medication given to slow the heart rate has been known to cause some patients to feel dizzy when they stand suddenly due to a lowering of blood pressure. Therefore, you will often be asked to sit up slowly on the table prior to standing. The dizziness is slight and only happens rarely. You may also have your blood pressure taken before the exam, during and following the examination if medications are given. The nitroglycerin medication may also give you a headache; this is not dangerous and will wear off quickly.

When you enter the CT scanner, you may see special light lines projected onto your body. These lines help ensure that you are in the correct position on the exam table. With modern CT scanners, you may hear slight buzzing, clicking and whirring sounds. These occur as the CT scanner's internal parts, not usually visible to you, revolve around you during the imaging process.

You will be alone in the exam room during the CT scan, unless there are special circumstances. For example, sometimes a parent wearing a lead shield may stay in the room with their child. However, the technologist will always be able to see, hear and speak with you through a built-in intercom system.

After a CT exam, the technologist will remove your intravenous line. They will cover the tiny hole made by the needle with a small dressing. You can return to your normal activities immediately.

Who interprets the results and how do I get them?

A radiologist, a doctor specially trained to supervise and interpret radiology exams, will analyze the images. The radiologist will send an official report to the doctor who ordered the exam.

If you are actively having chest pain, your results will be given to the emergency room doctor by the radiologist, and a preliminary result will be reported right away.

You may need a follow-up exam. If so, your doctor will explain why. Sometimes a follow-up exam further evaluates a potential issue with more views or a special imaging technique. It may also see if there has been any change in an issue over time. Follow-up exams are often the best way to see if treatment is working or if a problem needs attention.

What are the benefits vs. risks?

Benefits

- CCTA is not invasive. An alternative test, cardiac catheterization with a coronary angiogram, is invasive, has more complications related to the placement of a long catheter into the groin or wrist arteries extending all the way to the heart, and the movement of the catheter in the blood vessels. Invasive catheterization requires more time for the patient to recover.

- A major advantage of CT is that it is able to view bone, soft tissue and blood vessels all at the same time. It is therefore suited to identify other reasons for your discomfort such as an injury to the aorta or a blood clot in the lungs.

- Unlike conventional x-rays, CT scanning provides very detailed images of many types of tissue.

- CT examinations are fast and simple.

- CT has been shown to be cost-effective for a wide range of medical problems.

- CT is less sensitive to patient movement than MRI.

- Unlike MRI, an implanted medical device of any kind will not prevent you from having a CT scan.

- No radiation remains in a patient's body after a CT exam.

- The x-rays used for CT scanning should have no immediate side effects.

Risks

- In some people with abnormal kidney function, the contrast material used in CT scanning may worsen kidney function.

- If contrast material leaks out from the vessel being injected and spreads under the skin where the IV is placed, skin damage or damage to blood vessels and nerves, though unlikely, can result. If you feel any pain in your arm at the location of the IV during contrast material injection, you should immediately inform the technologist.

- There is always a slight chance of cancer from excessive exposure to radiation. However, the benefit of an accurate diagnosis far outweighs the risk involved with CT scanning.

- The radiation dose for this procedure varies. See the Radiation Dose page for more information.

- Women should always tell their doctor and x-ray or CT technologist if there is any chance they are pregnant. See the Radiation Safety page for more information about pregnancy and x-rays.

- Doctors do not generally recommend CT scanning for pregnant women unless medically necessary because of potential risk to the unborn baby.

- IV contrast manufacturers indicate mothers should not breastfeed their babies for 24-48 hours after contrast material is given. However, the most recent American College of Radiology (ACR) Manual on Contrast Media reports that studies show the amount of contrast absorbed by the infant during breastfeeding is extremely low. For further information please consult the ACR Manual on Contrast Media and its references.

- The risk of serious allergic reaction to contrast materials that contain iodine is rare, and radiology departments are well-equipped to deal with them.

What are the limitations of coronary CTA?

A person who is very large may not fit into the opening of a conventional CT scanner. Or, they may be over the weight limit-usually 450 pounds-for the moving table.

Patients who are extremely overweight or who have abnormal heart rhythms may not be good candidates for this test because image quality may be compromised.

Unlike CCTA, which is only a diagnostic test, invasive coronary angiography can be used for both diagnosis and treatment in a single session. If a narrowing or blockage is found during a CCTA, it cannot be treated at the same time. Patients with a high risk of coronary artery disease and typical symptoms might undergo coronary angiography instead of CCTA because they are more likely to need treatment.

CCTA can be difficult to read if there are many areas of old, calcified (hardened) plaque, which can be the case in older patients.

AI rewrite

Last reviewed on March 11, 2024

Coronary CTA

A coronary CT angiography is a heart test. It is also called coronary CT angiography or CCTA. The test uses a special dye and a CT scanner. The dye has iodine in it. A nurse puts the dye into your vein. The test looks at the arteries that bring blood to your heart. It shows if these arteries are too narrow. The CT scan makes pictures. These pictures can be turned into 3D images. Your doctor can see them on a screen. They can also be printed or saved on a disc.

Tell your doctor if you might be pregnant. Talk about any recent sickness, health problems, medicines you take, and allergies. You will be told not to eat or drink for a few hours before the test. You should also stay away from caffeine, Viagra, or similar drugs. If you are allergic to the dye, your doctor may give you medicine to help. You must take this medicine at set times. It starts 13 hours before your test. Leave your jewelry at home. Wear loose, comfy clothes. You may need to wear a gown. If you are breastfeeding, ask your doctor what to do.

- What is coronary CTA?

- What is the test used for?

- How should I get ready?

- What does the machine look like?

- How does the test work?

- How is the test done?

- What will I feel during and after the test?

- Who reads the results and how do I get them?

- What are the benefits and risks?

- What are the limits of coronary CTA?

What is coronary CTA?

Coronary CT angiography (CCTA) is a heart test. It helps show if plaque has made your heart arteries too narrow. These arteries bring blood to your heart. Plaque is made of fat, cholesterol, and calcium. It builds up on the inside walls of arteries. This takes time. Plaque can slow blood flow. Sometimes it blocks blood flow all the way. During this test, a nurse puts dye into your vein. The dye has iodine in it. It helps make clear pictures of your heart's blood vessels.

A CT scan is a kind of medical test. People also call it a CAT scan. Like normal x-rays, it makes many pictures of the inside of your body.

A CT scan makes pictures from many angles. It can even make 3D pictures. Your doctor can look at these pictures on a screen. They can also be printed or saved on a CD or DVD.

CT pictures show your organs, bones, soft parts, and blood vessels. They give more detail than normal x-rays. This is most true for soft parts and blood vessels.

What is the test used for?

Many doctors say CCTA can help patients who:

- may have heart arteries that are shaped in an odd way.
- have a low or medium chance of heart artery disease. This includes people with chest pain but normal or unclear test results.
- have low to medium chance of unusual chest pain in the emergency room.
- have chest pain that is not sudden.
- have new or worse symptoms but had a normal stress test before.
- had unclear stress test results.
- have new heart failure with weak heart function and a low or medium chance of heart artery disease.
- have a medium chance of heart artery disease before heart surgery that is not on the arteries.
- have had heart bypass surgery.

For these patients, CCTA can show if there is plaque in the heart arteries. It shows how much plaque there is. It can also find other reasons for your chest pain. These may be a collapsed lung, a blood clot in the lung vessels, or problems with the aorta. Your doctor and a radiologist will decide if this test is right for you.

How should I get ready?

Wear comfy, loose clothes to your test. You may need to change into a gown.

Metal can change the CT pictures. This includes jewelry, glasses, dentures, and hairpins. Leave them at home or take them off before your test. Some tests need you to take off hearing aids and dental work that comes out. Women need to take off bras with metal wires. You may need to take out piercings if you can.

Your doctor may tell you not to eat or drink for a few hours before your test if it uses dye. Tell your doctor about all the medicines you take. Tell them about any allergies. If you are allergic to the dye, your doctor may give you medicine to help. This is often a steroid. Call your doctor well before your test day so there are no delays.

Also tell your doctor about any recent sickness or health problems. Say if you have a history of heart disease, asthma, diabetes, kidney disease, or thyroid problems. Any of these can raise your risk of a bad effect.

On the day before and the day of your test, you may be asked to stay away from:

- diet pills and drinks with caffeine, like coffee, tea, energy drinks, or soda. These can speed up your heart. That makes it harder to check for plaque.
- Viagra or any similar drug. They do not mix well with the medicines you will get during the test.

The night before, you may be asked to take a beta blocker medicine. This slows your heart rate. It makes the test better.

Your child may be asked not to eat or drink for a few hours before. This is true if they will get medicine to make them sleepy. In most cases, a child who has been sick will not get this medicine. If your child is or may be sick, talk to your doctor about picking a new test day.

Tell your doctor what medicines your child takes. Say if your child has any allergies. This is most important for allergies to dye, by vein or by mouth. Tell the CT staff about these allergies at the test too. If your child is allergic to dye, tell the doctor and staff before the test.

Also tell your doctor about any recent sickness or health problems your child has. Say if your child has a history of heart disease, asthma, diabetes, kidney disease, or thyroid problems. Any of these can change whether your child gets dye for the test.

Talk to your doctor if you have questions about your instructions.

Women should always tell their doctor and the CT staff if they might be pregnant. See the CT Safety During Pregnancy page for more.

If you are breastfeeding, ask your doctor what to do. It may help to pump milk before the test. Keep it ready until all the dye is out of your body. This takes about 24 hours. But the newest American College of Radiology (ACR) guide says studies show the baby gets very little dye from breast milk. For more, see the ACR Manual on Contrast Media.

What does the machine look like?

The CT scanner is a big machine. It is shaped like a donut with a short tunnel in the middle. You lie on a narrow table. The table slides in and out of the tunnel. The x-ray parts spin around you inside a ring. This ring is called a gantry. The computer is in a separate room. This is where the staff runs the scanner and watches you. The staff can see you, hear you, and talk to you with a speaker and a microphone.

CCTA is a lot like a normal CT scan. The only difference is the speed of the scanner. It also uses a heart monitor to check your heart rate.

How does the test work?

During the test, x-rays go through your body. Special parts in the scanner pick them up. More of these parts (64 or more) make clearer pictures. That is why CCTA is also called multi-slice CT scanning. The test shows the heart arteries, plaque, and any narrow spots. Sometimes it shows how the heart works. The radiologist uses the computer to make 3D pictures and pictures from many angles. This helps them check your heart and arteries.

The test uses dye to show the blood vessels clearly. The dye makes them look bright white.

How is the test done?

The nurse will give you a gown to wear.

The staff will clean three small spots on your chest. They will place small sticky discs called electrodes on these spots. Men may need a small bit of chest hair shaved. This helps the discs stick. The discs connect to a heart monitor (ECG). It shows your heart's activity during the test.

A nurse or staff member will put an IV line into a vein in your arm. This is to give you dye during the test. While in the scanner, you may get a beta blocker medicine. It can go through the IV or by mouth. It slows your heart rate to make better pictures. You may also get nitroglycerin. It opens the heart arteries so they show up better. It can be a pill, a spray under your tongue, or a patch on your skin.

While you lie on the table, you may be asked to raise your arms over your head. You will hold them there for the test. This makes the pictures better.

Next, the table moves quickly through the scanner. This finds the right starting spot. Then the table moves slowly through for the real scan. The machine may make a few passes.

The staff may ask you to hold your breath during the scan. Any motion can blur the pictures. This includes breathing and moving your body. The blur looks like a photo of something moving.

Tell your doctor if you have trouble holding your breath for 5 to 15 seconds. Breathing during the scan blurs the pictures. This can make the test unclear.

When the test is done, the staff will check that the pictures are good enough. The radiologist needs clear pictures to read them well.

The doctor or nurse will take out your IV before you go home.

With all the prep, the CCTA scan usually takes about 15 minutes. This is if your heart rate is slow and steady. It may take longer if your heart rate is fast and you need a beta blocker. If you take the beta blocker by mouth, it takes at least one hour to work. If it goes in your vein, it may need more than one dose. It can take up to 20 minutes to slow your heart.

What will I feel during and after the test?

Most CT tests are fast, easy, and painless. The only sting is when the IV goes in.

The scan does not hurt. But you may feel uncomfortable from lying still. The IV may also bother you. Some people find the test hard if they are nervous, anxious, or in pain. The staff may give you medicine to help you get through it. A doctor will direct this.

If the test uses iodine dye, your doctor will check you for kidney disease. When the dye goes in your vein, you will feel a pin prick. You may feel warm or flushed. You may get a metal taste in your mouth. This will pass. You may feel like you need to pee. These are just side effects of the dye. They go away fast.

The medicine that slows your heart can make some people dizzy when they stand up fast. This is because it lowers blood pressure. So you may be asked to sit up slowly before you stand. The dizziness is mild and rare. Your blood pressure may be checked before, during, and after the test if you get medicine. Nitroglycerin may give you a headache. This is not dangerous. It will go away fast.

When you go into the scanner, you may see light lines on your body. These lines help put you in the right spot. With newer scanners, you may hear soft buzzing, clicking, and whirring. These come from the parts that spin around you.

You will be alone in the room during the scan. There may be special cases. For example, a parent may stay with a child while wearing a lead shield. The staff can always see, hear, and talk to you through an intercom.

After the test, the staff will take out your IV. They will cover the tiny needle hole with a small bandage. You can go back to your normal day right away.

Who reads the results and how do I get them?

A radiologist will look at the pictures. This is a doctor with special training to read these tests. The radiologist will send a report to the doctor who ordered the test.

If you are having chest pain right now, the radiologist will give your results to the emergency room doctor. A first result will be reported right away.

You may need a follow-up test. If so, your doctor will tell you why. Sometimes a follow-up test checks a possible problem with more views or a special way of imaging. It may also see if something has changed over time. Follow-up tests are often the best way to see if treatment is working. They can also show if a problem needs care.

What are the benefits and risks?

Benefits

- CCTA does not go inside your body. Another test, cardiac catheterization, does go inside. It puts a long tube into an artery in your groin or wrist. The tube goes all the way to your heart. This test has more problems and needs more time to heal.

- CT can show bone, soft parts, and blood vessels all at once. So it can find other reasons for your pain. These can be a hurt aorta or a blood clot in the lungs.

- CT gives very detailed pictures of many kinds of tissue. Normal x-rays cannot do this.

- CT tests are fast and simple.

- CT is a good value for many health problems.

- CT handles patient movement better than MRI.

- An implanted medical device will not stop you from having a CT scan. MRI is different.

- No radiation stays in your body after a CT test.

- The x-rays used for CT should have no quick side effects.

Risks

- In some people with kidney problems, the dye may make kidney function worse.

- The dye can leak out of the vein under your skin. This is unlikely. But it can hurt your skin, blood vessels, or nerves. Tell the staff right away if your arm hurts where the IV is during the dye injection.

- There is always a small chance of cancer from too much radiation. But a correct diagnosis is worth far more than this small risk.

- The amount of radiation varies. See the Radiation Dose page for more.

- Women should always tell their doctor and the CT staff if they might be pregnant. See the Radiation Safety page for more about pregnancy and x-rays.

- Doctors do not usually do CT scans on pregnant women unless they really need to. This is because of risk to the baby.

- Dye makers say mothers should not breastfeed for 24 to 48 hours after the dye is given. But the newest ACR guide says studies show the baby gets very little dye from breast milk. For more, see the ACR Manual on Contrast Media.

- A bad allergic reaction to iodine dye is rare. Radiology teams are well prepared to handle it.

What are the limits of coronary CTA?

A very large person may not fit in the opening of a normal CT scanner. They may also be over the table's weight limit. This limit is usually 450 pounds.

People who are very overweight may not be good for this test. The same is true for people with odd heart rhythms. This is because the pictures may not be clear.

CCTA only finds problems. It cannot treat them. Invasive coronary angiography can find and treat problems in one visit. If a narrow spot or block is found during a CCTA, it cannot be fixed at the same time. People with a high chance of heart artery disease and clear symptoms may get coronary angiography instead. This is because they are more likely to need treatment. CCTA can be hard to read if there is a lot of old, hardened plaque. This can happen in older patients.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

A CT angiogram is a test that checks for blockages and other problems in your arteries. It's a CT scan with the added benefit of contrast dye. A healthcare provider injects the dye into a vein in your arm. The dye flows through your blood vessels and makes them look bright on the CT images. This means it's easier to see blood flow and spot issues.

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A computed tomography (CT) angiogram is a test that makes detailed pictures of your blood vessels and nearby tissues. The most common reason to have a CT angiogram is to see if you have narrowed or blocked coronary arteries (coronary artery disease). When done for these reasons, the test is sometimes called a coronary CT angiogram, or CCTA.

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But a CT angiogram is useful for many other purposes, too. For example, it can help diagnose:

Your provider may also use a CT angiogram to plan a procedure or surgery. For example, they might look at your coronary arteries before a procedure such as a heart valve replacement, heart bypass surgery or stent placement.

Technically, a CT angiogram is the picture the test produces. And CT angiography is the name of the imaging technique. But people often use both terms to refer to the test you go in to get. You might also see this test called CTA.

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All CT scans produce detailed, 3D images of the inside of your body. This is because a scanner takes hundreds of 2D (flat) pictures. Then, a computer puts all these pictures together to form high-quality 3D images. But contrast dye is what makes the difference between a CT scan and CT angiography.

The combination of CT scanning and contrast dye produces "lit up" images of your blood vessels. Your healthcare provider can rotate each 3D image. It's a bit like holding a model of your blood vessel in their hand. They can see each part of the vessel from different angles. This lets them get a close-up view to pinpoint any problems and plan treatment.

CT angiography is less invasive and has fewer risks than traditional angiography. For a traditional angiogram, a catheter moves inside your blood vessels until it reaches the part providers want to see. You don't need this catheter for a CT angiogram. Based upon your age and other medical conditions, you may not be an ideal candidate for a CT angiogram and may still require a more invasive traditional angiography procedure.

The healthcare provider who requested your CT angiogram will tell you how to prepare. In general, you should tell your provider:

Your healthcare provider will give you specific instructions to follow. These will tell you:

When you arrive, you'll put on a hospital gown. You'll also need to remove any jewelry or metal objects that you're wearing. These could interfere with the test.

CT angiography is typically quick and painless. However, it does require an IV catheter to be inserted into your arm. You're awake and don't need anesthesia. You'll lie on an exam table that moves through a large ring that looks a bit like a doughnut. You're not as enclosed as you would be for an MRI.

During the test, you'll need to stay very still so the scanner can take good pictures. A specially trained healthcare provider called a radiologic technologist will perform your test. They'll explain each step and tell you exactly what you need to do (like when to hold your breath). Here are the general steps:

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You can expect to be in the testing room for about 20 to 60 minutes, including prep time. You'll spend about one to two minutes in the scanner. This should go by fast. But you might need to go through the scanner more than once so your provider can get all the pictures they need.

Depending on your needs and concerns, CT angiography may have some disadvantages. These include:

When your test is done, you can return to your usual routine. You should drink lots of water and other liquids. This helps flush the contrast dye out of your system.

CT angiography produces detailed images of your blood vessels and nearby tissues (including capillary beds). These images can show a wide range of issues, like plaque buildup, aneurysms and more. A radiologist will review and interpret these images. They'll share the findings with the healthcare provider who requested your test. Your provider will schedule a time to go over the results with you.

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You may need follow-up visits with your provider. You may also need further testing. Your provider will explain next steps. Don't hesitate to ask questions about the results and what they mean for you.

Call your provider if:

No matter how many times someone says a test is quick or painless, you still might be nervous. That's OK. It's normal to get a bit worked up before any sort of exam. But remind yourself that a CT angiogram is there to help you. It lets your provider see what's going on inside your blood vessels. The results can help you and your provider plan any needed treatments to support your health.

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AI rewrite

A CT angiogram is a test that checks for blockages and other problems in your arteries. It is a CT scan that uses a special dye called contrast dye. A healthcare provider puts the dye into a vein in your arm. The dye moves through your blood vessels. It makes them look bright on the CT pictures. This helps your provider see your blood flow and find problems.

A CT angiogram makes clear pictures of your blood vessels and the tissue near them. The most common reason to have this test is to check for narrow or blocked heart arteries. This is called coronary artery disease. When the test checks your heart arteries, it is sometimes called a coronary CT angiogram, or CCTA.

A CT angiogram is helpful for many other reasons, too. For example, it can help your provider find health problems.

Your provider may also use a CT angiogram to plan a procedure or surgery. For example, they might look at your heart arteries before a heart valve replacement, heart bypass surgery, or stent placement.

The word "CT angiogram" means the picture the test makes. "CT angiography" means the way the picture is taken. But people often use both words to mean the test you go in to get. You might also see this test called CTA.

All CT scans make clear, 3D pictures of the inside of your body. The scanner takes hundreds of flat (2D) pictures. Then a computer puts them together to make high-quality 3D pictures. But the contrast dye is what makes a CT angiogram different from a plain CT scan.

CT scanning and contrast dye work together to make your blood vessels "light up." Your provider can turn each 3D picture around. It is a bit like holding a model of your blood vessel in their hand. They can look at each part from many sides. This gives them a close-up view to find problems and plan your care.

A CT angiogram is less invasive and has fewer risks than a traditional angiogram. For a traditional angiogram, a thin tube called a catheter moves inside your blood vessels until it reaches the part the provider wants to see. You do not need this tube for a CT angiogram. But your age and other health problems may make a CT angiogram a poor choice for you. If so, you may still need the more invasive traditional angiogram.

The provider who ordered your CT angiogram will tell you how to get ready. In general, you should tell your provider some things first.

Your provider will give you clear steps to follow. These will tell you what you need to do.

When you arrive, you will put on a hospital gown. You will need to take off any jewelry or metal objects you are wearing. These could get in the way of the test.

A CT angiogram is usually quick and does not hurt. But you will need an IV in your arm. You will be awake and will not need medicine to make you sleep. You will lie on a table that moves through a large ring. The ring looks a bit like a doughnut. You are not closed in as much as you would be for an MRI.

During the test, you will need to stay very still so the scanner can take good pictures. A trained provider called a radiologic technologist will do your test. They will explain each step. They will tell you just what to do, like when to hold your breath. There are some general steps to follow.

You will likely be in the testing room for about 20 to 60 minutes. This includes the time to get ready. You will be in the scanner for about one to two minutes. This should go by fast. But you might need to go through the scanner more than once. This lets your provider get all the pictures they need.

A CT angiogram may have some downsides. This depends on your needs and concerns.

When your test is done, you can go back to your normal day. You should drink lots of water and other liquids. This helps wash the dye out of your body.

A CT angiogram makes clear pictures of your blood vessels and the tissue near them. This includes tiny blood vessels called capillary beds. These pictures can show many problems, like plaque buildup, aneurysms, and more. A doctor called a radiologist will look at the pictures. They will share what they find with the provider who ordered your test. Your provider will set up a time to go over the results with you.

You may need follow-up visits with your provider. You may also need more tests. Your provider will explain the next steps. Feel free to ask questions about your results and what they mean for you.

Call your provider if you have any concerns.

A test may be quick and painless, but you might still feel nervous. That is OK. It is normal to feel worried before any exam. But remember that a CT angiogram is there to help you. It lets your provider see what is going on inside your blood vessels. The results can help you and your provider plan any care you need to support your health.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Original page

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AI rewrite

A CT angiogram is a test that checks your blood vessels for blockages and other problems. It is a CT scan that uses a special dye. A healthcare worker puts the dye into a vein in your arm. The dye flows through your blood. It makes your blood vessels look bright on the pictures. This makes it easy to see your blood flow and find problems.

Doctors often use this test to see if the blood vessels in your heart are narrow or blocked. When used for the heart, it is called a CCTA. The test can also find other issues, like plaque buildup or weak spots called aneurysms. Your doctor might use it to plan a surgery, like a heart valve replacement, bypass, or stent. You might also hear this test called a CTA.

A CT scanner takes hundreds of flat pictures of your body. A computer puts them together to make clear 3D pictures. The special dye makes your blood vessels light up. Your doctor can turn the 3D picture around to look at it from all sides. This helps them get a close look to find problems and plan treatment.

This test is easier on your body than a regular angiogram. A regular angiogram uses a long tube that goes deep inside your blood vessels. A CT angiogram does not need this long tube. But, depending on your age and health, you might still need a regular angiogram.

The doctor who ordered your test will tell you how to get ready. You will need to share some health details with your doctor. They will also give you specific rules to follow before the test.

When you arrive, you will put on a hospital gown. You must take off all jewelry and metal objects. Metal can mess up the pictures.

The test is fast and does not hurt. You will be awake and will not need medicine to make you sleep. A small IV tube will be put into your arm. You will lie on a table that moves through a large scanner. The scanner looks like a doughnut. It is open, so you will not feel trapped.

You must stay very still so the pictures are clear. A specially trained worker will run the machine. They will explain each step and tell you what to do, like when to hold your breath.

You will be in the room for 20 to 60 minutes. You will only be in the scanner for one or two minutes. This part goes fast. You might go through the scanner a few times so the worker can get all the pictures they need.

Depending on your health, this test may have some downsides. Your doctor will talk to you about them.

After the test, you can go back to your normal day. You should drink lots of water and other liquids. This helps wash the dye out of your body.

A special doctor called a radiologist will look at your pictures. They will share the results with your doctor. Your doctor will set up a time to talk with you about what the pictures show.

You may need to see your doctor again or get more tests. Your doctor will explain what to do next. Always ask questions if you do not understand your results. Call your doctor if you have any problems.

It is normal to feel nervous before a test. Just remember that a CT angiogram is there to help you. It lets your doctor see inside your blood vessels. The results will help you and your doctor plan the best treatment for your health.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

A CT angiogram is a test that checks for blockages and other problems in your arteries. It's a CT scan with the added benefit of contrast dye. A healthcare provider injects the dye into a vein in your arm. The dye flows through your blood vessels and makes them look bright on the CT images. This means it's easier to see blood flow and spot issues.

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A computed tomography (CT) angiogram is a test that makes detailed pictures of your blood vessels and nearby tissues. The most common reason to have a CT angiogram is to see if you have narrowed or blocked coronary arteries (coronary artery disease). When done for these reasons, the test is sometimes called a coronary CT angiogram, or CCTA.

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But a CT angiogram is useful for many other purposes, too. For example, it can help diagnose:

Your provider may also use a CT angiogram to plan a procedure or surgery. For example, they might look at your coronary arteries before a procedure such as a heart valve replacement, heart bypass surgery or stent placement.

Technically, a CT angiogram is the picture the test produces. And CT angiography is the name of the imaging technique. But people often use both terms to refer to the test you go in to get. You might also see this test called CTA.

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All CT scans produce detailed, 3D images of the inside of your body. This is because a scanner takes hundreds of 2D (flat) pictures. Then, a computer puts all these pictures together to form high-quality 3D images. But contrast dye is what makes the difference between a CT scan and CT angiography.

The combination of CT scanning and contrast dye produces "lit up" images of your blood vessels. Your healthcare provider can rotate each 3D image. It's a bit like holding a model of your blood vessel in their hand. They can see each part of the vessel from different angles. This lets them get a close-up view to pinpoint any problems and plan treatment.

CT angiography is less invasive and has fewer risks than traditional angiography. For a traditional angiogram, a catheter moves inside your blood vessels until it reaches the part providers want to see. You don't need this catheter for a CT angiogram. Based upon your age and other medical conditions, you may not be an ideal candidate for a CT angiogram and may still require a more invasive traditional angiography procedure.

The healthcare provider who requested your CT angiogram will tell you how to prepare. In general, you should tell your provider:

Your healthcare provider will give you specific instructions to follow. These will tell you:

When you arrive, you'll put on a hospital gown. You'll also need to remove any jewelry or metal objects that you're wearing. These could interfere with the test.

CT angiography is typically quick and painless. However, it does require an IV catheter to be inserted into your arm. You're awake and don't need anesthesia. You'll lie on an exam table that moves through a large ring that looks a bit like a doughnut. You're not as enclosed as you would be for an MRI.

During the test, you'll need to stay very still so the scanner can take good pictures. A specially trained healthcare provider called a radiologic technologist will perform your test. They'll explain each step and tell you exactly what you need to do (like when to hold your breath). Here are the general steps:

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You can expect to be in the testing room for about 20 to 60 minutes, including prep time. You'll spend about one to two minutes in the scanner. This should go by fast. But you might need to go through the scanner more than once so your provider can get all the pictures they need.

Depending on your needs and concerns, CT angiography may have some disadvantages. These include:

When your test is done, you can return to your usual routine. You should drink lots of water and other liquids. This helps flush the contrast dye out of your system.

CT angiography produces detailed images of your blood vessels and nearby tissues (including capillary beds). These images can show a wide range of issues, like plaque buildup, aneurysms and more. A radiologist will review and interpret these images. They'll share the findings with the healthcare provider who requested your test. Your provider will schedule a time to go over the results with you.

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You may need follow-up visits with your provider. You may also need further testing. Your provider will explain next steps. Don't hesitate to ask questions about the results and what they mean for you.

Call your provider if:

No matter how many times someone says a test is quick or painless, you still might be nervous. That's OK. It's normal to get a bit worked up before any sort of exam. But remind yourself that a CT angiogram is there to help you. It lets your provider see what's going on inside your blood vessels. The results can help you and your provider plan any needed treatments to support your health.

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Cleveland Clinic's health articles are based on evidence-backed information and review by medical professionals to ensure accuracy, reliability and up-to-date clinical standards.

When your heart needs some help, the cardiology experts at Cleveland Clinic are here for you. We diagnose and treat the full spectrum of cardiovascular diseases.

AI rewrite

A CT angiogram is a test that checks your arteries. It looks for blockages and other problems. It is a CT scan that uses contrast dye.

A healthcare provider puts the dye into a vein in your arm. The dye moves through your blood vessels. It makes them show up bright on the CT pictures. This helps your provider see blood flow and find problems.

A CT angiogram makes detailed pictures of your blood vessels and nearby tissues. A common reason for this test is to check for narrow or blocked heart arteries. This is called coronary artery disease. When the test checks the heart arteries, it may be called a coronary CT angiogram or CCTA.

A CT angiogram can also be used for other reasons. Your provider may use it to help diagnose problems in your blood vessels. Your provider may also use it to help plan a procedure or surgery. For example, they may look at your heart arteries before a heart valve replacement, heart bypass surgery or stent placement.

The picture made by the test is called a CT angiogram. The way the pictures are made is called CT angiography. Many people use both names to mean the same test. You may also see it called CTA.

All CT scans make detailed 3D pictures of the inside of your body. The scanner takes many flat pictures. A computer puts them together to make 3D pictures. The contrast dye is what makes CT angiography different from a regular CT scan.

The CT scan and dye work together to make your blood vessels look bright. Your provider can turn the 3D picture to see it from many angles. This helps them look closely at the blood vessel. It can help them find problems and plan treatment.

CT angiography is less invasive and has fewer risks than traditional angiography. In traditional angiography, a catheter moves inside your blood vessels to the area your provider wants to see. You do not need this catheter for a CT angiogram. But based on your age and other health problems, a CT angiogram may not be right for you. You may still need traditional angiography.

The provider who ordered your CT angiogram will tell you how to get ready. In general, you should tell your provider about your health history, allergies, medicines and any concerns you have. Your provider will give you specific instructions. These will tell you what to do before the test.

When you arrive, you will put on a hospital gown. You will need to take off jewelry or metal items. These can affect the test.

CT angiography is usually quick and painless. But it does require an IV catheter in your arm. You will be awake. You do not need anesthesia.

You will lie on an exam table. The table moves through a large ring that looks a bit like a doughnut. You are not as closed in as you would be for an MRI.

During the test, you must stay very still. This helps the scanner take clear pictures. A radiologic technologist will do your test. This is a specially trained healthcare provider. They will explain each step. They will tell you what to do, such as when to hold your breath.

You can expect to be in the testing room for about 20 to 60 minutes. This includes prep time. You will spend about one to two minutes in the scanner. You may need to go through the scanner more than once. This helps your provider get all the pictures they need.

CT angiography may have some disadvantages. These depend on your needs and concerns. Talk with your provider about what this means for you.

When the test is done, you can go back to your usual routine. Drink lots of water and other liquids. This helps flush the contrast dye out of your body.

CT angiography makes detailed pictures of your blood vessels and nearby tissues, including capillary beds. These pictures can show many problems, such as plaque buildup, aneurysms and more.

A radiologist will review the pictures. They will explain what the pictures show. They will share the results with the provider who ordered your test. Your provider will set a time to go over the results with you.

You may need follow-up visits with your provider. You may also need more tests. Your provider will explain the next steps. Ask questions about your results and what they mean for you.

Call your provider if they told you to watch for symptoms or if you have concerns after the test.

It is normal to feel nervous before a test. A CT angiogram is done to help you. It lets your provider see what is happening inside your blood vessels. The results can help you and your provider plan any treatment you may need.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Cardiac Computed Tomography Angiography (CCTA)

Quick Facts

- A CCTA scan uses X-rays to take images of your heart and blood vessels.
- The test helps look for narrowed arteries and blood vessels.
- The test might be used when other heart tests don't provide enough details for a diagnosis.

What is a cardiac CT scan?

A cardiac CT angiography (CCTA) scan uses X-rays to take images of your heart and blood vessels. A computer combines the images to create a 3D image of your heart.

A CCTA scan is used to see narrowed arteries and blood vessels that supply blood to the heart or other parts of the body.

A multidetector CT (MDCT) is faster than a typical CCTA. It can produce better images with less radiation exposure.

Why do people need a CCTA scan?

You may need a CCTA scan when other tests don't provide enough information about your heart.

The test can help your health care team gather more information on:

- Your heart's structure
- How well your heart pumps blood
- Heart muscle scarring caused by a heart attack
- Fluid in the pericardial sac that covers the surface of the heart
- The amount of plaque buildup and narrowing of your coronary arteries
- Anything abnormal in the large blood vessels leaving the heart
- Your risk for a heart attack

Can a CCTA scan show if I have heart disease?

When contrast dye is injected during the CCTA scan, it can show blockages in your heart arteries. This helps see if chest discomfort (angina) is caused by a lack of blood flow to the heart due to blocked arteries. If the arteries are normal, your health care team can look into other causes of the chest pain.

With contrast dye, the CCTA scan can also check conditions including:

- If coronary artery bypass grafts are open after bypass surgery
- For congenital heart defects
- How your ventricles are working

Can I have a CCTA scan instead of a coronary angiogram?

A CCTA scan does not replace a coronary angiogram or cardiac catheterization. Coronary angiography is considered the best method for showing blockages in the coronary arteries. It also gives information about how your heart is working. In some cases, a CCTA may be done instead of a coronary angiogram. This is based on patient factors and the abilities of the hospital or outpatient center.

What are the risks with having a CCTA scan?

A CCTA scan exposes you to radiation. Talk with your health care team about the safety and risks for any test you're going to have.

Tell your health care team if you're pregnant. If the test is not urgent, ask them about delaying it until after your pregnancy. Also, if you have kidney problems, your health care team may avoid using contrast dye. They should check your kidney function before the test.

Some people have allergic reactions to the contrast dye that's used in the test. Before the test, tell your health care professional if you're allergic to dyes. Depending on the reaction you had before, you may be given medications before the test to reduce the

chance of an allergic reaction. In the case of a severe allergic reaction, contrast dye will not be used. Instead, a test that doesn't require contrast may be used.

What happens after the CCTA scan?

- Your health care professional will let you know when to resume normal activities.

- After your health care team gets your test results, they will discuss them and next steps with you.

AI rewrite

Cardiac Computed Tomography Angiography (CCTA)

Quick Facts

- A CCTA scan uses X-rays to take pictures of your heart and blood vessels.

- The test helps look for narrow arteries and blood vessels.

- The test may be used when other heart tests do not give enough details.

What is a cardiac CT scan?

A cardiac CT angiography (CCTA) scan uses X-rays to take pictures of your heart and blood vessels. A computer puts the pictures together to make a 3D picture of your heart.

A CCTA scan is used to look for narrow arteries and blood vessels. These blood vessels bring blood to the heart or other parts of the body.

A multidetector CT (MDCT) is faster than a typical CCTA. It can make better pictures with less radiation.

Why do people need a CCTA scan?

You may need a CCTA scan when other tests do not give enough information about your heart.

The test can help your health care team learn more about:

- The shape and parts of your heart

- How well your heart pumps blood

- Scars in the heart muscle from a heart attack

- Fluid in the sac that covers the heart

- How much plaque has built up in your heart arteries and how narrow they are

- Anything unusual in the large blood vessels leaving the heart

- Your risk for a heart attack

Can a CCTA scan show if I have heart disease?

During the CCTA scan, contrast dye may be injected. This dye can show blockages in your heart arteries. This can help show if chest pain or discomfort, called angina, is caused by low blood flow to the heart from blocked arteries. If the arteries are normal, your health care team can look for other causes of chest pain.

With contrast dye, the CCTA scan can also check:

- If bypass grafts are open after bypass surgery

- Heart defects you were born with

- How your ventricles are working

Can I have a CCTA scan instead of a coronary angiogram?

A CCTA scan does not replace a coronary angiogram or cardiac catheterization. A coronary angiogram is considered the best way to show blockages in the heart arteries. It also gives information about how your heart is working.

In some cases, a CCTA may be done instead of a coronary angiogram. This depends on the patient and what the hospital or outpatient center can do.

What are the risks of having a CCTA scan?

A CCTA scan exposes you to radiation. Talk with your health care team about the safety and risks of any test you will have.

Tell your health care team if you are pregnant. If the test is not urgent, ask if you can wait until after your pregnancy.

Also tell your health care team if you have kidney problems. They may avoid using contrast dye. They should check how well your kidneys work before the test.

Some people have allergic reactions to the contrast dye used in the test. Before the test, tell your health care professional if you are allergic to dyes. Depending on your past reaction, you may be given medicine before the test. This can lower the chance of an allergic reaction.

If you had a severe allergic reaction, contrast dye will not be used. Instead, you may have a test that does not need contrast dye.

What happens after the CCTA scan?

- Your health care professional will tell you when you can go back to normal activities.

- After your health care team gets your test results, they will talk with you about the results and next steps.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Cardiac Computed Tomography Angiography (CCTA)

Quick Facts

- A CCTA scan uses X-rays to take images of your heart and blood vessels.
- The test helps look for narrowed arteries and blood vessels.
- The test might be used when other heart tests don't provide enough details for a diagnosis.

What is a cardiac CT scan?

A cardiac CT angiography (CCTA) scan uses X-rays to take images of your heart and blood vessels. A computer combines the images to create a 3D image of your heart.

A CCTA scan is used to see narrowed arteries and blood vessels that supply blood to the heart or other parts of the body.

A multidetector CT (MDCT) is faster than a typical CCTA. It can produce better images with less radiation exposure.

Why do people need a CCTA scan?

You may need a CCTA scan when other tests don't provide enough information about your heart.

The test can help your health care team gather more information on:

- Your heart's structure
- How well your heart pumps blood
- Heart muscle scarring caused by a heart attack
- Fluid in the pericardial sac that covers the surface of the heart
- The amount of plaque buildup and narrowing of your coronary arteries
- Anything abnormal in the large blood vessels leaving the heart
- Your risk for a heart attack

Can a CCTA scan show if I have heart disease?

When contrast dye is injected during the CCTA scan, it can show blockages in your heart arteries. This helps see if chest discomfort (angina) is caused by a lack of blood flow to the heart due to blocked arteries. If the arteries are normal, your health care team can look into other causes of the chest pain.

With contrast dye, the CCTA scan can also check conditions including:

- If coronary artery bypass grafts are open after bypass surgery
- For congenital heart defects
- How your ventricles are working

Can I have a CCTA scan instead of a coronary angiogram?

A CCTA scan does not replace a coronary angiogram or cardiac catheterization. Coronary angiography is considered the best method for showing blockages in the coronary arteries. It also gives information about how your heart is working. In some cases, a CCTA may be done instead of a coronary angiogram. This is based on patient factors and the abilities of the hospital or outpatient center.

What are the risks with having a CCTA scan?

A CCTA scan exposes you to radiation. Talk with your health care team about the safety and risks for any test you're going to have.

Tell your health care team if you're pregnant. If the test is not urgent, ask them about delaying it until after your pregnancy. Also, if you have kidney problems, your health care team may avoid using contrast dye. They should check your kidney function before the test.

Some people have allergic reactions to the contrast dye that's used in the test. Before the test, tell your health care professional if you're allergic to dyes. Depending on the reaction you had before, you may be given medications before the test to reduce the

chance of an allergic reaction. In the case of a severe allergic reaction, contrast dye will not be used. Instead, a test that doesn't require contrast may be used.

What happens after the CCTA scan?

- Your health care professional will let you know when to resume normal activities.

- After your health care team gets your test results, they will discuss them and next steps with you.

AI rewrite

Quick Facts

- A CCTA scan uses X-rays to take pictures of your heart and blood vessels.

- The test looks for narrow arteries and blood vessels.

- You might get this test if other heart tests do not give enough details.

What is a cardiac CT scan?

A CCTA scan uses X-rays to take pictures of your heart and blood vessels. A computer puts the pictures together to make a 3D image of your heart. The scan looks for narrow blood vessels that send blood to your heart and body. A type of scan called MDCT is faster than a normal CCTA. It takes better pictures and uses less radiation.

Why do people need a CCTA scan?

You may need this scan if your doctor needs more facts about your heart. The test helps your care team see:

- How your heart is built

- How well your heart pumps blood

- Scars on your heart muscle from a heart attack

- Fluid in the sac around your heart

- Plaque buildup and narrow areas in your heart arteries

- Problems in the large blood vessels that leave your heart

- Your risk for a heart attack

Can a CCTA scan show if I have heart disease?

During the scan, you might get a shot of contrast dye. This dye helps show blockages in your heart arteries. It helps doctors see if chest pain is caused by blocked blood flow. If your arteries are clear, your doctor will look for other causes of your chest pain.

With the dye, the scan can also check:

- If blood vessels from a past bypass surgery are still open

- For heart defects you were born with

- How well the lower chambers of your heart are working

Can I have a CCTA scan instead of a coronary angiogram?

This scan does not replace a regular coronary angiogram. An angiogram is still the best way to see blocked arteries and check how your heart works. But sometimes, a CCTA scan is used instead. This depends on your health and what the clinic can do.

What are the risks of a CCTA scan?

A CCTA scan uses radiation. Talk with your doctor about the safety and risks of this test.

Tell your doctor if you are pregnant. If the test is not a rush, you might wait until after you have your baby.

Tell your doctor if you have kidney problems. They will check how well your kidneys work before the test. They might not use the contrast dye if your kidneys are weak.

Some people are allergic to the contrast dye. Tell your doctor if you have ever had a bad reaction to dye. You might get medicine before the test to stop an allergic reaction. If you have a severe allergy, the doctor will not use the dye. They might use a different test instead.

What happens after the CCTA scan?

- Your doctor will tell you when you can go back to your normal routine.
- When your test results are ready, your care team will talk with you about them and your next steps.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Cardiac Computed Tomography Angiography (CCTA)

Quick Facts

- A CCTA scan uses X-rays to take images of your heart and blood vessels.
- The test helps look for narrowed arteries and blood vessels.
- The test might be used when other heart tests don't provide enough details for a diagnosis.

What is a cardiac CT scan?

A cardiac CT angiography (CCTA) scan uses X-rays to take images of your heart and blood vessels. A computer combines the images to create a 3D image of your heart.

A CCTA scan is used to see narrowed arteries and blood vessels that supply blood to the heart or other parts of the body.

A multidetector CT (MDCT) is faster than a typical CCTA. It can produce better images with less radiation exposure.

Why do people need a CCTA scan?

You may need a CCTA scan when other tests don't provide enough information about your heart.

The test can help your health care team gather more information on:

- Your heart's structure
- How well your heart pumps blood
- Heart muscle scarring caused by a heart attack
- Fluid in the pericardial sac that covers the surface of the heart
- The amount of plaque buildup and narrowing of your coronary arteries
- Anything abnormal in the large blood vessels leaving the heart
- Your risk for a heart attack

Can a CCTA scan show if I have heart disease?

When contrast dye is injected during the CCTA scan, it can show blockages in your heart arteries. This helps see if chest discomfort (angina) is caused by a lack of blood flow to the heart due to blocked arteries. If the arteries are normal, your health care team can look into other causes of the chest pain.

With contrast dye, the CCTA scan can also check conditions including:

- If coronary artery bypass grafts are open after bypass surgery
- For congenital heart defects
- How your ventricles are working

Can I have a CCTA scan instead of a coronary angiogram?

A CCTA scan does not replace a coronary angiogram or cardiac catheterization. Coronary angiography is considered the best method for showing blockages in the coronary arteries. It also gives information about how your heart is working. In some cases, a CCTA may be done instead of a coronary angiogram. This is based on patient factors and the abilities of the hospital or outpatient center.

What are the risks with having a CCTA scan?

A CCTA scan exposes you to radiation. Talk with your health care team about the safety and risks for any test you're going to have.

Tell your health care team if you're pregnant. If the test is not urgent, ask them about delaying it until after your pregnancy. Also, if you have kidney problems, your health care team may avoid using contrast dye. They should check your kidney function before the test.

Some people have allergic reactions to the contrast dye that's used in the test. Before the test, tell your health care professional if you're allergic to dyes. Depending on the reaction you had before, you may be given medications before the test to reduce the

chance of an allergic reaction. In the case of a severe allergic reaction, contrast dye will not be used. Instead, a test that doesn't require contrast may be used.

What happens after the CCTA scan?

- Your health care professional will let you know when to resume normal activities.

- After your health care team gets your test results, they will discuss them and next steps with you.

AI rewrite

Heart CT Scan (CCTA)

Quick Facts

- A CCTA scan uses X-rays to take pictures of your heart and blood vessels.

- The test helps look for narrow arteries and blood vessels.

- You may get this test when other heart tests do not give enough details.

What is a heart CT scan?

A heart CT scan (CCTA) uses X-rays to take pictures of your heart and blood vessels. A computer puts the pictures together. This makes a 3D picture of your heart.

A CCTA scan helps doctors see narrow arteries and blood vessels. These vessels carry blood to your heart and other parts of your body.

One kind of scan is called an MDCT. It is faster than a normal CCTA. It can make better pictures with less radiation.

Why do people need a CCTA scan?

You may need a CCTA scan when other tests do not tell enough about your heart.

The test can help your care team learn more about:

- How your heart is built

- How well your heart pumps blood

- Scars on your heart muscle from a heart attack

- Fluid in the sac that covers your heart

- How much plaque has built up and how narrow your heart arteries are

- Any problems in the large blood vessels that leave your heart

- Your chance of having a heart attack

Can a CCTA scan show if I have heart disease?

During the scan, a dye may be put into your body. This dye can show blockages in your heart arteries. It helps doctors find out if chest pain (angina) comes from blocked arteries that slow blood flow to the heart. If the arteries look normal, your care team can check for other causes of the chest pain.

With dye, the CCTA scan can also check:

- If bypass grafts are still open after bypass surgery

- For heart problems you were born with

- How well your heart's lower chambers (ventricles) work

Can I have a CCTA scan instead of a coronary angiogram?

A CCTA scan does not take the place of a coronary angiogram or cardiac catheterization. A coronary angiogram is the best way to show blockages in the heart arteries. It also tells how well your heart works. Sometimes a CCTA may be done instead. This depends on your health and on what the hospital or clinic can do.

What are the risks of a CCTA scan?

A CCTA scan exposes you to radiation. Talk with your care team about the safety and risks of any test you will have.

Tell your care team if you are pregnant. If the test is not urgent, ask if you can wait until after your pregnancy. Also tell them if you have kidney problems. If you do, they may not use the dye. They should check how well your kidneys work before the test.

Some people have allergic reactions to the dye used in the test. Before the test, tell your care team if you are allergic to dyes. Based on your past reaction, you may get medicine before the test. This medicine helps lower the chance of an allergic reaction. If you had a severe reaction before, the dye will not be used. Instead, you may get a test that does not need dye.

What happens after the CCTA scan?

- Your care team will tell you when you can go back to your normal activities.

- After your care team gets your test results, they will talk with you about them and the next steps.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

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Original page

A CT scan uses X-rays of the heart to produce detailed images of the heart and the heart arteries, which the doctor can then see on a computer screen. The images are picked up using detectors.

The greater the number of detectors - the UK's most commonly used version has 64 - the clearer the image and the more useful it is in helping the doctor to make a diagnosis.

There are two ways in which a CT scan can be used for the heart - one is a CT coronary angiogram and the other is a CT calcium score.

CT coronary angiogram

What is a CT coronary angiogram?

A CT coronary angiogram is used to directly visualise the heart arteries and the fatty deposits (plaques) that can develop within them. This fatty plaques can mean your heart is not getting the blood supply it needs, which can cause chest pain (angina) and heart attacks.

Similar to a conventional coronary angiogram, a CT coronary angiogram involves injecting an iodine-based dye into your bloodstream to highlight your blood vessels.

However, unlike the traditional angiogram, which involves inserting a catheter (a thin flexible tube) via the wrist or groin into the heart, a CT coronary angiogram, is non-invasive. This means that there are fewer potential complications such as bleeding or bruising.

You may also be given some medication to slow down your heartbeat, making it easier to take images.

Why would you have a CT coronary angiogram?

The CT coronary angiogram is the first line test recommended by UK guidelines for assessing patients presenting chest pain to outpatient cardiology or chest pain clinics. It will tell your cardiologist whether you have fatty plaques in your coronary arteries, the more plaques you have the more likely a heart attack is in the future.

It will also tell them whether these plaques are likely to be obstructing blood flow to the heart muscle and causing your chest pain.

A CT coronary angiogram can also be useful if you have heart failure but you doctor does not know why, if your doctor suspects you may have an abnormality in the structure of your heart, or if you are being considered for transcatheter aortic valve implantation (TAVI).

CT calcium score

What is CT calcium scoring?

This is a simpler CT scan of the heart. It does not provide detailed pictures of the heart or heart arteries but instead measures the amount of calcified, or hardened, plaques in the heart arteries. This is usually explained as a calcium score which provides an assessment of the number of fatty plaques in the heart arteries. Again, the more fatty plaques you have, the higher the risk of heart attacks.

Unlike a CT angiogram, a calcium score does not involve a dye.

Why would you have a CT calcium score?

CT calcium scoring is sometimes used in patients without any symptoms related to their heart, to better understand their risk of having a heart attack and whether they need to take preventative medicines such as aspirin and statins to lower their risk of having heart problems.

Usually, these decisions are made by assessing patients' risk factors. Sometimes this can be tricky, and a CT calcium score can help.

Patients without any symptoms of chest pain and a zero calcium have a very low risk of having a future heart attack. They may therefore not need to take preventative tablets such as aspirin or statins.

Patients with no symptoms but a high calcium score are at higher risk of heart attacks in the future and so may benefit from these preventative medications. At the moment this test is mainly used in private practice as a screening test.

Sometimes CT calcium scoring can be used in other conditions such as heart valve disease (in particular aortic stenosis, which is narrowing of your aortic valve) to assess how severe the disease is.

Who should not have a CT scan?

Unlike an MRI scan, CT scans expose you to radiation, so they are not suitable for pregnant women.

The CT angiogram may not be suitable for patients with advanced kidney problems because the dye can worsen kidney function.

It's also not suitable for anyone who's allergic to the dye or for anyone with severe asthma, because the dye can cause narrowing of airways.

What are the risks of CT scans?

The radiation levels associated with both CT coronary angiography and CT calcium scoring are low, similar to the annual background radiation that people are exposed to over the course of a year. However, because of this radiation, they should only be carried out only when really necessary.

To find out more, or to support British Heart Foundation's work, please visit www.bhf.org.uk. You can speak to one of our cardiac nurses by calling our helpline on 0808 802 1234 (freephone), Monday to Friday, 9am to 5pm. For general customer service enquiries, please call 0300 330 3322, Monday to Friday, 9am to 5pm.

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AI rewrite

A CT scan uses X-rays to take detailed pictures of your heart and its blood vessels. A doctor looks at these pictures on a computer. The machine uses parts called detectors to take the pictures. More detectors make a clearer picture. The most common machine in the UK uses 64 detectors.

There are two types of heart CT scans. One is a CT coronary angiogram. The other is a CT calcium score.

CT coronary angiogram

What is a CT coronary angiogram?

This test lets doctors see your heart's blood vessels. It shows if you have fatty build-ups, called plaques, inside them. These build-ups can block blood flow. This can cause chest pain, called angina, or a heart attack.

For this test, a special dye with iodine is put into your blood. This dye makes your blood vessels show up clearly on the scan.

An older type of test uses a long, thin tube put into your wrist or groin. This CT scan does not use that tube. Because it does not use a tube, you have less chance of bleeding or bruising.

You might also get medicine to slow down your heartbeat. A slower heartbeat makes the pictures clearer.

Why would you have a CT coronary angiogram?

In the UK, this is the first test doctors use if you go to a clinic for chest pain. It tells your doctor if you have fatty build-ups in your blood vessels. Having more build-ups means you have a higher chance of a heart attack.

The test also shows if these build-ups are blocking blood flow and causing your chest pain.

Your doctor might also use this test if you have heart failure and they do not know why. It is also used if your doctor thinks your heart is shaped differently. Finally, it is used if you might need a heart valve procedure called TAVI.

CT calcium score

What is CT calcium scoring?

This is a simpler heart scan. It does not give detailed pictures of the heart. Instead, it measures the amount of hard, fatty build-ups in your blood vessels. This gives you a number called a calcium score. A higher score means you have more build-ups and a higher chance of a heart attack. This test does not use any dye.

Why would you have a CT calcium score?

Doctors use this test for people who do not have heart symptoms. It helps them see your chance of having a heart attack. It also helps them decide if you need medicines, like aspirin or statins, to protect your heart.

If you have no symptoms and a score of zero, your chance of a heart attack is very low. You probably do not need these medicines.

If you have no symptoms but a high score, your chance of a heart attack is higher. These medicines could help you. Right now, this test is mostly used in private clinics.

Sometimes, this test is used to check other heart problems. For example, it can show how bad a narrowed heart valve is. This narrowing is called aortic stenosis.

Who should not have a CT scan?

CT scans use radiation. Because of this, pregnant women should not have them.

You should not have the scan with dye if you have bad kidney problems. The dye can make your kidneys worse.

You also should not have the dye if you are allergic to it or if you have severe asthma. The dye can make your airways narrow and make it hard to breathe.

What are the risks of CT scans?

Both types of CT scans use a small amount of radiation. It is about the same amount of radiation you get from the natural world in one year. Because there is some radiation, you should only have these scans when you really need them.

To learn more or to help the British Heart Foundation, please visit www.bhf.org.uk. You can talk to a heart nurse by calling 0808 802 1234 for free. The line is open Monday to Friday, from 9 am to 5 pm. For other questions, please call 0300 330 3322, Monday to Friday, from 9 am to 5 pm.

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Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

A CT scan uses X-rays of the heart to produce detailed images of the heart and the heart arteries, which the doctor can then see on a computer screen. The images are picked up using detectors.

The greater the number of detectors - the UK's most commonly used version has 64 - the clearer the image and the more useful it is in helping the doctor to make a diagnosis.

There are two ways in which a CT scan can be used for the heart - one is a CT coronary angiogram and the other is a CT calcium score.

CT coronary angiogram

What is a CT coronary angiogram?

A CT coronary angiogram is used to directly visualise the heart arteries and the fatty deposits (plaques) that can develop within them. This fatty plaques can mean your heart is not getting the blood supply it needs, which can cause chest pain (angina) and heart attacks.

Similar to a conventional coronary angiogram, a CT coronary angiogram involves injecting an iodine-based dye into your bloodstream to highlight your blood vessels.

However, unlike the traditional angiogram, which involves inserting a catheter (a thin flexible tube) via the wrist or groin into the heart, a CT coronary angiogram, is non-invasive. This means that there are fewer potential complications such as bleeding or bruising.

You may also be given some medication to slow down your heartbeat, making it easier to take images.

Why would you have a CT coronary angiogram?

The CT coronary angiogram is the first line test recommended by UK guidelines for assessing patients presenting chest pain to outpatient cardiology or chest pain clinics. It will tell your cardiologist whether you have fatty plaques in your coronary arteries, the more plaques you have the more likely a heart attack is in the future.

It will also tell them whether these plaques are likely to be obstructing blood flow to the heart muscle and causing your chest pain.

A CT coronary angiogram can also be useful if you have heart failure but you doctor does not know why, if your doctor suspects you may have an abnormality in the structure of your heart, or if you are being considered for transcatheter aortic valve implantation (TAVI).

CT calcium score

What is CT calcium scoring?

This is a simpler CT scan of the heart. It does not provide detailed pictures of the heart or heart arteries but instead measures the amount of calcified, or hardened, plaques in the heart arteries. This is usually explained as a calcium score which provides an assessment of the number of fatty plaques in the heart arteries. Again, the more fatty plaques you have, the higher the risk of heart attacks.

Unlike a CT angiogram, a calcium score does not involve a dye.

Why would you have a CT calcium score?

CT calcium scoring is sometimes used in patients without any symptoms related to their heart, to better understand their risk of having a heart attack and whether they need to take preventative medicines such as aspirin and statins to lower their risk of having heart problems.

Usually, these decisions are made by assessing patients' risk factors. Sometimes this can be tricky, and a CT calcium score can help.

Patients without any symptoms of chest pain and a zero calcium have a very low risk of having a future heart attack. They may therefore not need to take preventative tablets such as aspirin or statins.

Patients with no symptoms but a high calcium score are at higher risk of heart attacks in the future and so may benefit from these preventative medications. At the moment this test is mainly used in private practice as a screening test.

Sometimes CT calcium scoring can be used in other conditions such as heart valve disease (in particular aortic stenosis, which is narrowing of your aortic valve) to assess how severe the disease is.

Who should not have a CT scan?

Unlike an MRI scan, CT scans expose you to radiation, so they are not suitable for pregnant women.

The CT angiogram may not be suitable for patients with advanced kidney problems because the dye can worsen kidney function.

It's also not suitable for anyone who's allergic to the dye or for anyone with severe asthma, because the dye can cause narrowing of airways.

What are the risks of CT scans?

The radiation levels associated with both CT coronary angiography and CT calcium scoring are low, similar to the annual background radiation that people are exposed to over the course of a year. However, because of this radiation, they should only be carried out only when really necessary.

To find out more, or to support British Heart Foundation's work, please visit www.bhf.org.uk. You can speak to one of our cardiac nurses by calling our helpline on 0808 802 1234 (freephone), Monday to Friday, 9am to 5pm. For general customer service enquiries, please call 0300 330 3322, Monday to Friday, 9am to 5pm.

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AI rewrite

A CT scan uses X-rays to take detailed pictures of the heart and the heart arteries. The doctor can see these pictures on a computer screen. The pictures are taken using detectors.

More detectors can make the picture clearer. In the UK, the most common type has 64 detectors. Clearer pictures can help the doctor make a diagnosis.

There are two main CT scans used for the heart:

CT coronary angiogram

CT calcium score

CT coronary angiogram

What is a CT coronary angiogram?

A CT coronary angiogram is used to look closely at the heart arteries. It can show fatty deposits, called plaques, inside the arteries. These plaques can reduce blood flow to the heart. This can cause chest pain, called angina, and heart attacks.

For this test, an iodine-based dye is injected into your blood. The dye helps show your blood vessels.

This is similar to a standard coronary angiogram. But a standard angiogram uses a catheter. A catheter is a thin, flexible tube. It is put into the body through the wrist or groin and guided to the heart.

A CT coronary angiogram does not use a catheter. This means it is non-invasive. It may have fewer problems, such as bleeding or bruising.

You may also be given medicine to slow your heartbeat. This helps the scan take clearer pictures.

Why would you have a CT coronary angiogram?

A CT coronary angiogram is the first test recommended by UK guidelines for people who have chest pain and are seen in outpatient heart clinics or chest pain clinics.

It can show your heart doctor if you have fatty plaques in your coronary arteries. The more plaques you have, the more likely you may be to have a heart attack in the future.

It can also show if these plaques may be blocking blood flow to the heart muscle and causing your chest pain.

A CT coronary angiogram may also be useful if:

you have heart failure and your doctor does not know why

your doctor thinks you may have a problem with the structure of your heart

you are being considered for transcatheter aortic valve implantation, called TAVI

CT calcium score

What is CT calcium scoring?

This is a simpler CT scan of the heart. It does not give detailed pictures of the heart or heart arteries.

Instead, it measures the amount of hardened, or calcified, plaques in the heart arteries. This is called a calcium score. It helps show how many fatty plaques are in the heart arteries.

The more fatty plaques you have, the higher your risk of a heart attack.

A CT calcium score does not use dye.

Why would you have a CT calcium score?

CT calcium scoring is sometimes used for people who do not have heart symptoms. It can help doctors better understand their risk of having a heart attack.

It can also help doctors decide if they need preventive medicines. These may include aspirin and statins. These medicines can lower the risk of heart problems.

Usually, doctors make these decisions by looking at a person's risk factors. Sometimes this is not clear. A CT calcium score can help.

People with no chest pain and a calcium score of zero have a very low risk of a future heart attack. They may not need preventive tablets such as aspirin or statins.

People with no symptoms but a high calcium score have a higher risk of heart attacks in the future. They may benefit from these preventive medicines.

At the moment, this test is mainly used in private practice as a screening test.

Sometimes CT calcium scoring can be used for other conditions, such as heart valve disease. This includes aortic stenosis, which is narrowing of the aortic valve. The test can help show how severe the disease is.

Who should not have a CT scan?

CT scans expose you to radiation. MRI scans do not. Because of this, CT scans are not suitable for pregnant women.

A CT angiogram may not be suitable for people with advanced kidney problems. This is because the dye can make kidney function worse.

It is also not suitable for anyone who is allergic to the dye.

It is not suitable for anyone with severe asthma, because the dye can cause the airways to narrow.

What are the risks of CT scans?

The amount of radiation from both CT coronary angiography and CT calcium scoring is low. It is similar to the background radiation people are exposed to over one year.

But because CT scans use radiation, they should only be done when they are really needed.

To find out more, or to support British Heart Foundation's work, visit www.bhf.org.uk.

You can speak to one of our cardiac nurses by calling our helpline on 0808 802 1234. This is a freephone number. The helpline is open Monday to Friday, 9am to 5pm.

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Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

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Original page

A CT scan uses X-rays of the heart to produce detailed images of the heart and the heart arteries, which the doctor can then see on a computer screen. The images are picked up using detectors.

The greater the number of detectors - the UK's most commonly used version has 64 - the clearer the image and the more useful it is in helping the doctor to make a diagnosis.

There are two ways in which a CT scan can be used for the heart - one is a CT coronary angiogram and the other is a CT calcium score.

CT coronary angiogram

What is a CT coronary angiogram?

A CT coronary angiogram is used to directly visualise the heart arteries and the fatty deposits (plaques) that can develop within them. This fatty plaques can mean your heart is not getting the blood supply it needs, which can cause chest pain (angina) and heart attacks.

Similar to a conventional coronary angiogram, a CT coronary angiogram involves injecting an iodine-based dye into your bloodstream to highlight your blood vessels.

However, unlike the traditional angiogram, which involves inserting a catheter (a thin flexible tube) via the wrist or groin into the heart, a CT coronary angiogram, is non-invasive. This means that there are fewer potential complications such as bleeding or bruising.

You may also be given some medication to slow down your heartbeat, making it easier to take images.

Why would you have a CT coronary angiogram?

The CT coronary angiogram is the first line test recommended by UK guidelines for assessing patients presenting chest pain to outpatient cardiology or chest pain clinics. It will tell your cardiologist whether you have fatty plaques in your coronary arteries, the more plaques you have the more likely a heart attack is in the future.

It will also tell them whether these plaques are likely to be obstructing blood flow to the heart muscle and causing your chest pain.

A CT coronary angiogram can also be useful if you have heart failure but you doctor does not know why, if your doctor suspects you may have an abnormality in the structure of your heart, or if you are being considered for transcatheter aortic valve implantation (TAVI).

CT calcium score

What is CT calcium scoring?

This is a simpler CT scan of the heart. It does not provide detailed pictures of the heart or heart arteries but instead measures the amount of calcified, or hardened, plaques in the heart arteries. This is usually explained as a calcium score which provides an assessment of the number of fatty plaques in the heart arteries. Again, the more fatty plaques you have, the higher the risk of heart attacks.

Unlike a CT angiogram, a calcium score does not involve a dye.

Why would you have a CT calcium score?

CT calcium scoring is sometimes used in patients without any symptoms related to their heart, to better understand their risk of having a heart attack and whether they need to take preventative medicines such as aspirin and statins to lower their risk of having heart problems.

Usually, these decisions are made by assessing patients' risk factors. Sometimes this can be tricky, and a CT calcium score can help.

Patients without any symptoms of chest pain and a zero calcium have a very low risk of having a future heart attack. They may therefore not need to take preventative tablets such as aspirin or statins.

Patients with no symptoms but a high calcium score are at higher risk of heart attacks in the future and so may benefit from these preventative medications. At the moment this test is mainly used in private practice as a screening test.

Sometimes CT calcium scoring can be used in other conditions such as heart valve disease (in particular aortic stenosis, which is narrowing of your aortic valve) to assess how severe the disease is.

Who should not have a CT scan?

Unlike an MRI scan, CT scans expose you to radiation, so they are not suitable for pregnant women.

The CT angiogram may not be suitable for patients with advanced kidney problems because the dye can worsen kidney function.

It's also not suitable for anyone who's allergic to the dye or for anyone with severe asthma, because the dye can cause narrowing of airways.

What are the risks of CT scans?

The radiation levels associated with both CT coronary angiography and CT calcium scoring are low, similar to the annual background radiation that people are exposed to over the course of a year. However, because of this radiation, they should only be carried out only when really necessary.

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AI rewrite

A CT scan uses X-rays to take clear pictures of your heart and the arteries around it. The doctor can see these pictures on a computer screen. Small parts called detectors pick up the images.

More detectors make clearer pictures. The kind used most in the UK has 64 detectors. Clearer pictures help the doctor find out what is wrong.

There are two kinds of heart CT scans. One is a CT coronary angiogram. The other is a CT calcium score.

CT coronary angiogram

What is a CT coronary angiogram?

This scan lets the doctor see your heart arteries. It also shows fatty lumps, called plaques, that can build up inside them. These plaques can stop your heart from getting enough blood. This can cause chest pain (called angina) and heart attacks.

For this scan, a dye that has iodine in it is put into your blood. The dye makes your blood vessels easier to see.

A normal angiogram uses a thin, bendy tube. The tube goes in through your wrist or groin and up to your heart. A CT coronary angiogram does not need this tube. So there is less chance of problems like bleeding or bruising.

You may also get some medicine to slow your heartbeat. This makes it easier to take good pictures.

Why would you have a CT coronary angiogram?

UK guidelines say this is the first test to use when someone has chest pain and comes to a heart clinic. It tells your heart doctor if you have fatty plaques in your heart arteries. The more plaques you have, the more likely you are to have a heart attack later.

It also tells the doctor if these plaques are blocking blood flow to your heart muscle and causing your chest pain.

This scan can also help if you have heart failure and the doctor does not know why. It can help if the doctor thinks your heart is shaped in an unusual way. It can also help if you may need a heart valve put in through a tube. This is called TAVI.

CT calcium score

What is CT calcium scoring?

This is a simpler heart CT scan. It does not give detailed pictures of your heart or arteries. Instead, it measures how much hard plaque is in your heart arteries. This is shown as a calcium score. The score tells you how many fatty plaques are in your heart arteries. The more plaques you have, the higher your risk of a heart attack.

Unlike the CT angiogram, this test does not use a dye.

Why would you have a CT calcium score?

This test is sometimes used for people who have no heart symptoms. It helps the doctor learn their risk of a heart attack. It also helps the doctor decide if they need medicines to lower their risk. These medicines include aspirin and statins.

Usually, the doctor checks a person's risk factors to decide this. Sometimes this is hard to do. A CT calcium score can help.

People with no chest pain and a score of zero have a very low risk of a heart attack later. They may not need to take aspirin or statins.

People with no symptoms but a high score have a higher risk of a heart attack later. They may do better if they take these medicines. Right now, this test is mostly used in private clinics as a screening test.

Sometimes this test is used for other problems, like heart valve disease. One example is aortic stenosis, which means your aortic valve is too narrow. The test shows how bad the problem is.

Who should not have a CT scan?

An MRI scan does not use radiation, but a CT scan does. So CT scans are not safe for pregnant women.

The CT angiogram may not be safe for people with bad kidney problems. The dye can make kidney problems worse.

It is also not safe for anyone who is allergic to the dye. It is not safe for anyone with severe asthma. The dye can make the airways narrow.

What are the risks of CT scans?

Both kinds of scans use only a small amount of radiation. It is about the same as the radiation you get from your surroundings in one year. But because of this radiation, these scans should only be done when they are really needed.

To find out more, or to support British Heart Foundation's work, please visit www.bhf.org.uk. You can speak to one of our cardiac nurses by calling our helpline on 0808 802 1234 (freephone), Monday to Friday, 9am to 5pm. For general customer service enquiries, please call 0300 330 3322, Monday to Friday, 9am to 5pm.

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Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

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Original page

Cardiac computed tomography (CT) is a non-invasive test that can be used to evaluate various parts of the heart.

A CT scan, also known as a CAT scan, is a noninvasive test where an X-ray machine rotates around a patient's body while they lie on a table. Unlike traditional X-rays, a CT scanner makes highly detailed images of the internal organs and, in many cases, is able to distinguish healthy tissue from diseased tissue.

Because the heart is moving, creating a cardiac CT scan is more difficult than for other body parts, like the brain. However, new CT technology at Brigham and Women's Hospital (BWH) is able to accurately image the heart in most patients.

BWH offers one of the few multidisciplinary cardiovascular imaging programs in the country that includes cardiologists, radiologists, and other imaging experts. The BWH Cardiovascular Imaging Program combines the treatment, education, and research expertise of multiple disciplines - cardiology, radiology, nuclear medicine, molecular biology, medical physics, and chemistry - and incorporates the use of all available imaging types, including echocardiography, cardiac CT, cardiac MRI, nuclear cardiology, PET/CT, CT/MRI, and ultrasound.

Our patients with heart and vascular conditions have access to top specialists throughout the fields of cardiovascular medicine, cardiac surgery, cardiac imaging, vascular surgery, and cardiac anesthesia. These physicians practice at the BWH Heart & Vascular Center, consistently ranked as one of the top Heart and Vascular hospitals in U.S. News and World Report's annual "America's Best Hospitals" survey.

Cardiac CT can be used to evaluate the cause of chest discomfort and/or shortness of breath in patients who do not have a prior history of coronary artery disease, such as prior heart attacks or coronary stents. Cardiac CT also can be used to evaluate for the presence or absence of coronary artery disease in patients who had stress tests with uncertain findings.

Other uses of cardiac CT:

The two types of cardiac CT most commonly performed are:

Coronary CT angiography

This is the most common type of cardiac CT. Coronary CT angiography is performed to visualize the blood vessels (coronary arteries) that supply the heart muscle. The presence of fatty deposits (plaque) within these arteries indicates the presence of coronary artery disease. In some individuals, plaque buildup can limit blood flow to the heart muscle and result in symptoms of chest discomfort or shortness of breath, typically during physical activity. However, these symptoms often can be caused by other reasons and are not related to heart disease.

Coronary artery calcium scan

A coronary artery calcium scan is performed to detect the presence (and if present, quantify the amount) of coronary artery calcium in patients who do not have any known coronary artery disease. The presence of calcium indicates the presence of coronary plaque.

The physician can use the information from this test, together with other risk factor information, such as blood pressure and cholesterol levels, to determine a patient's future risk of having a heart attack or stroke. Patients with coronary plaque have a higher risk of future events and are more likely to benefit from treatment with cholesterol-lowering medications or aspirin than those who have no evidence of plaque.

The Heart & Vascular Center is located in the Shapiro Cardiovascular Center, across the street from BWH's main 75 Francis Street entrance. The Heart & Vascular Center brings together the full range of cardiovascular services in one location, fostering seamless and coordinated care for all patients.

Prior to your imaging procedure, you will check in at the Cardiovascular Imaging Center in the Shapiro Cardiovascular Center. During your procedure, family and friends can wait in the Shapiro Family Center.

In addition to our cardiovascular imaging experts, patients also benefit from the teamwork of cardiologists, interventional cardiologists, cardiac surgeons, cardiac electrophysiologists, and anesthesiologists, all experts in cardiovascular disorders. These specialists work alongside nurses, physician assistants, physical therapists, dietitians, and social workers to achieve better outcomes for patients.

Learn more about cardiac catheterization in our health library.

Visit the Kessler Health Education Library in the Bretholtz Center for Patients and Families.

Access a complete directory of patient and family services.

Learn about the Watkins Clinic in the Shapiro Cardiovascular Center.

For over a century, a leader in patient care, medical education and research, with expertise in virtually every specialty of medicine and surgery.

About BWH

AI rewrite

Cardiac computed tomography (CT) is a test that looks at different parts of the heart. It does not involve surgery.

A CT scan is also called a CAT scan. During the test, you lie on a table. An X-ray machine moves around your body. A CT scan makes very detailed pictures of organs inside the body. In many cases, it can show the difference between healthy tissue and diseased tissue.

The heart is always moving. This makes a heart CT scan harder than CT scans of other body parts, such as the brain. New CT technology at Brigham and Women’s Hospital (BWH) can take accurate pictures of the heart in most patients.

BWH has a cardiovascular imaging program. The team includes heart doctors, radiologists, and other imaging experts. The program brings together care, education, and research from many areas. These include cardiology, radiology, nuclear medicine, molecular biology, medical physics, and chemistry. The program uses many types of imaging. These include echocardiography, cardiac CT, cardiac MRI, nuclear cardiology, PET/CT, CT/MRI, and ultrasound.

Patients with heart and blood vessel conditions can see specialists in many areas. These include cardiovascular medicine, heart surgery, heart imaging, vascular surgery, and heart anesthesia. These doctors work at the BWH Heart & Vascular Center. This center is often ranked as one of the top heart and vascular hospitals in the U.S. News and World Report “America’s Best Hospitals” yearly survey.

Cardiac CT can help find the cause of chest discomfort or shortness of breath. It may be used in patients who have not had coronary artery disease before. This includes patients who have not had a heart attack or coronary stents. Cardiac CT can also help check for coronary artery disease in patients whose stress test results were not clear.

The two types of cardiac CT done most often are coronary CT angiography and coronary artery calcium scan.

Coronary CT angiography is the most common type of cardiac CT. It is used to see the blood vessels that bring blood to the heart muscle. These blood vessels are called coronary arteries. Fatty buildup, called plaque, in these arteries means coronary artery disease is present. In some people, plaque can limit blood flow to the heart muscle. This can cause chest discomfort or shortness of breath, often during physical activity. These symptoms can also have other causes and may not be related to heart disease.

A coronary artery calcium scan checks for calcium in the coronary arteries. It is used for patients who do not have known coronary artery disease. If calcium is found, the test can measure how much is there. Calcium means there is plaque in the coronary arteries.

The doctor can use the test results along with other risk factors. These include blood pressure and cholesterol levels. This information helps estimate a patient’s future risk of heart attack or stroke. Patients with coronary plaque have a higher risk of future events. They are more likely to benefit from treatment with cholesterol-lowering medicines or aspirin than patients with no signs of plaque.

The Heart & Vascular Center is in the Shapiro Cardiovascular Center. It is across the street from BWH’s main entrance at 75 Francis Street. The Heart & Vascular Center brings many heart and blood vessel services together in one place. This helps make care easier and better coordinated for patients.

Before your imaging test, you will check in at the Cardiovascular Imaging Center in the Shapiro Cardiovascular Center. During your test, family and friends can wait in the Shapiro Family Center.

Patients also benefit from teamwork among many experts. These include heart doctors, interventional heart doctors, heart surgeons, heart rhythm doctors, and anesthesiologists. They work with nurses, physician assistants, physical therapists, dietitians, and social workers to help patients have better results.

Learn more about cardiac catheterization in our health library.

Visit the Kessler Health Education Library in the Bretholtz Center for Patients and Families.

Access a complete directory of patient and family services.

Learn about the Watkins Clinic in the Shapiro Cardiovascular Center.

For over 100 years, BWH has been a leader in patient care, medical education, and research. BWH has expertise in almost every area of medicine and surgery.

About BWH.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 29 of 77 · Item ID B-3UEA3GE8N6

Original page

Cardiac computed tomography (CT) is a non-invasive test that can be used to evaluate various parts of the heart.

A CT scan, also known as a CAT scan, is a noninvasive test where an X-ray machine rotates around a patient's body while they lie on a table. Unlike traditional X-rays, a CT scanner makes highly detailed images of the internal organs and, in many cases, is able to distinguish healthy tissue from diseased tissue.

Because the heart is moving, creating a cardiac CT scan is more difficult than for other body parts, like the brain. However, new CT technology at Brigham and Women's Hospital (BWH) is able to accurately image the heart in most patients.

BWH offers one of the few multidisciplinary cardiovascular imaging programs in the country that includes cardiologists, radiologists, and other imaging experts. The BWH Cardiovascular Imaging Program combines the treatment, education, and research expertise of multiple disciplines - cardiology, radiology, nuclear medicine, molecular biology, medical physics, and chemistry - and incorporates the use of all available imaging types, including echocardiography, cardiac CT, cardiac MRI, nuclear cardiology, PET/CT, CT/MRI, and ultrasound.

Our patients with heart and vascular conditions have access to top specialists throughout the fields of cardiovascular medicine, cardiac surgery, cardiac imaging, vascular surgery, and cardiac anesthesia. These physicians practice at the BWH Heart & Vascular Center, consistently ranked as one of the top Heart and Vascular hospitals in U.S. News and World Report's annual "America's Best Hospitals" survey.

Cardiac CT can be used to evaluate the cause of chest discomfort and/or shortness of breath in patients who do not have a prior history of coronary artery disease, such as prior heart attacks or coronary stents. Cardiac CT also can be used to evaluate for the presence or absence of coronary artery disease in patients who had stress tests with uncertain findings.

Other uses of cardiac CT:

The two types of cardiac CT most commonly performed are:

Coronary CT angiography

This is the most common type of cardiac CT. Coronary CT angiography is performed to visualize the blood vessels (coronary arteries) that supply the heart muscle. The presence of fatty deposits (plaque) within these arteries indicates the presence of coronary artery disease. In some individuals, plaque buildup can limit blood flow to the heart muscle and result in symptoms of chest discomfort or shortness of breath, typically during physical activity. However, these symptoms often can be caused by other reasons and are not related to heart disease.

Coronary artery calcium scan

A coronary artery calcium scan is performed to detect the presence (and if present, quantify the amount) of coronary artery calcium in patients who do not have any known coronary artery disease. The presence of calcium indicates the presence of coronary plaque.

The physician can use the information from this test, together with other risk factor information, such as blood pressure and cholesterol levels, to determine a patient's future risk of having a heart attack or stroke. Patients with coronary plaque have a higher risk of future events and are more likely to benefit from treatment with cholesterol-lowering medications or aspirin than those who have no evidence of plaque.

The Heart & Vascular Center is located in the Shapiro Cardiovascular Center, across the street from BWH's main 75 Francis Street entrance. The Heart & Vascular Center brings together the full range of cardiovascular services in one location, fostering seamless and coordinated care for all patients.

Prior to your imaging procedure, you will check in at the Cardiovascular Imaging Center in the Shapiro Cardiovascular Center. During your procedure, family and friends can wait in the Shapiro Family Center.

In addition to our cardiovascular imaging experts, patients also benefit from the teamwork of cardiologists, interventional cardiologists, cardiac surgeons, cardiac electrophysiologists, and anesthesiologists, all experts in cardiovascular disorders. These specialists work alongside nurses, physician assistants, physical therapists, dietitians, and social workers to achieve better outcomes for patients.

Learn more about cardiac catheterization in our health library.

Visit the Kessler Health Education Library in the Bretholtz Center for Patients and Families.

Access a complete directory of patient and family services.

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For over a century, a leader in patient care, medical education and research, with expertise in virtually every specialty of medicine and surgery.

About BWH

AI rewrite

Cardiac computed tomography (CT) is a test that looks at the parts of your heart. It does not go inside your body.

A CT scan is also called a CAT scan. During the test, an X-ray machine moves around your body while you lie on a table. A CT scan is not like a normal X-ray. It makes very clear pictures of the organs inside you. Many times, it can tell healthy tissue from sick tissue.

The heart is hard to scan because it moves. This makes a heart CT harder than a scan of other body parts, like the brain. But Brigham and Women's Hospital (BWH) has new CT tools. These tools can make clear pictures of the heart in most people.

BWH has one of the few heart imaging programs in the country that uses a team of many experts. The team has heart doctors, X-ray doctors, and other imaging experts. The BWH Cardiovascular Imaging Program brings together many fields. These include heart care, X-ray, nuclear medicine, molecular biology, medical physics, and chemistry. The team uses all kinds of imaging. These include echocardiography, cardiac CT, cardiac MRI, nuclear cardiology, PET/CT, CT/MRI, and ultrasound.

Our patients with heart and blood vessel problems can see top experts. These experts work in heart care, heart surgery, heart imaging, blood vessel surgery, and heart anesthesia. They work at the BWH Heart & Vascular Center. This center is often named one of the best heart hospitals in the U.S. News and World Report "America's Best Hospitals" list.

Cardiac CT can help find the cause of chest pain or trouble breathing. It is used for people who never had heart artery disease before. This means they never had a heart attack or a heart stent. Cardiac CT can also check for heart artery disease in people whose stress test results were not clear.

Other ways cardiac CT is used:

The two most common kinds of cardiac CT are:

Coronary CT angiography

This is the most common kind of cardiac CT. It is done to look at the blood vessels that feed the heart muscle. These are called coronary arteries. Fatty buildup (called plaque) in these arteries shows heart artery disease. In some people, plaque can block blood flow to the heart muscle. This can cause chest pain or trouble breathing, often during exercise. But these signs can come from other causes too. They may not be from heart disease.

Coronary artery calcium scan

This scan checks for calcium in the heart arteries. It is done for people who do not have known heart artery disease. If calcium is there, the scan also measures how much. Calcium is a sign that plaque is there.

Your doctor can use this test with other facts about your health. These facts include your blood pressure and cholesterol levels. Together, they help show your future risk of a heart attack or stroke. People with plaque have a higher risk later. They are more likely to be helped by cholesterol-lowering medicine or aspirin than people with no plaque.

The Heart & Vascular Center is in the Shapiro Cardiovascular Center. It is across the street from the main BWH entrance at 75 Francis Street. The Heart & Vascular Center puts all heart and blood vessel care in one place. This helps patients get smooth, well-planned care.

Before your imaging test, you will check in at the Cardiovascular Imaging Center in the Shapiro Cardiovascular Center. While you have your test, your family and friends can wait in the Shapiro Family Center.

You will get help from our heart imaging experts. You will also get help from a team of other experts. This team has heart doctors, heart procedure doctors, heart surgeons, heart rhythm doctors, and anesthesia doctors. They all know a lot about heart problems. They work with nurses, physician assistants, physical therapists, dietitians, and social workers. Together, they help patients get better.

Learn more about cardiac catheterization in our health library.

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About BWH

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 30 of 77 · Item ID B-T7A9TGIQHG

Original page

Cardiac computed tomography (CT) is a non-invasive test that can be used to evaluate various parts of the heart.

A CT scan, also known as a CAT scan, is a noninvasive test where an X-ray machine rotates around a patient's body while they lie on a table. Unlike traditional X-rays, a CT scanner makes highly detailed images of the internal organs and, in many cases, is able to distinguish healthy tissue from diseased tissue.

Because the heart is moving, creating a cardiac CT scan is more difficult than for other body parts, like the brain. However, new CT technology at Brigham and Women's Hospital (BWH) is able to accurately image the heart in most patients.

BWH offers one of the few multidisciplinary cardiovascular imaging programs in the country that includes cardiologists, radiologists, and other imaging experts. The BWH Cardiovascular Imaging Program combines the treatment, education, and research expertise of multiple disciplines - cardiology, radiology, nuclear medicine, molecular biology, medical physics, and chemistry - and incorporates the use of all available imaging types, including echocardiography, cardiac CT, cardiac MRI, nuclear cardiology, PET/CT, CT/MRI, and ultrasound.

Our patients with heart and vascular conditions have access to top specialists throughout the fields of cardiovascular medicine, cardiac surgery, cardiac imaging, vascular surgery, and cardiac anesthesia. These physicians practice at the BWH Heart & Vascular Center, consistently ranked as one of the top Heart and Vascular hospitals in U.S. News and World Report's annual "America's Best Hospitals" survey.

Cardiac CT can be used to evaluate the cause of chest discomfort and/or shortness of breath in patients who do not have a prior history of coronary artery disease, such as prior heart attacks or coronary stents. Cardiac CT also can be used to evaluate for the presence or absence of coronary artery disease in patients who had stress tests with uncertain findings.

Other uses of cardiac CT:

The two types of cardiac CT most commonly performed are:

Coronary CT angiography

This is the most common type of cardiac CT. Coronary CT angiography is performed to visualize the blood vessels (coronary arteries) that supply the heart muscle. The presence of fatty deposits (plaque) within these arteries indicates the presence of coronary artery disease. In some individuals, plaque buildup can limit blood flow to the heart muscle and result in symptoms of chest discomfort or shortness of breath, typically during physical activity. However, these symptoms often can be caused by other reasons and are not related to heart disease.

Coronary artery calcium scan

A coronary artery calcium scan is performed to detect the presence (and if present, quantify the amount) of coronary artery calcium in patients who do not have any known coronary artery disease. The presence of calcium indicates the presence of coronary plaque.

The physician can use the information from this test, together with other risk factor information, such as blood pressure and cholesterol levels, to determine a patient's future risk of having a heart attack or stroke. Patients with coronary plaque have a higher risk of future events and are more likely to benefit from treatment with cholesterol-lowering medications or aspirin than those who have no evidence of plaque.

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Prior to your imaging procedure, you will check in at the Cardiovascular Imaging Center in the Shapiro Cardiovascular Center. During your procedure, family and friends can wait in the Shapiro Family Center.

In addition to our cardiovascular imaging experts, patients also benefit from the teamwork of cardiologists, interventional cardiologists, cardiac surgeons, cardiac electrophysiologists, and anesthesiologists, all experts in cardiovascular disorders. These specialists work alongside nurses, physician assistants, physical therapists, dietitians, and social workers to achieve better outcomes for patients.

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For over a century, a leader in patient care, medical education and research, with expertise in virtually every specialty of medicine and surgery.

About BWH

AI rewrite

A cardiac CT scan is a safe test that looks at your heart. It does not use cuts or needles. It is also called a CAT scan. You will lie on a table while an X-ray machine moves around you. It takes very clear pictures of your inside organs. It is better than a regular X-ray at showing the difference between healthy parts and sick parts.

The heart is always moving. This makes it harder to take pictures of the heart than it is for the brain. But Brigham and Women's Hospital (BWH) has new machines that can take clear pictures of the heart for most people.

BWH has a special team for taking pictures of the heart. This team includes heart doctors, X-ray doctors, and other experts. They work together to treat patients and learn new things. They use many types of tests to look at the heart. These include echocardiograms, cardiac CTs, cardiac MRIs, nuclear tests, PET scans, and ultrasounds.

Patients with heart and blood vessel problems get care from top experts. These include heart doctors, surgeons, and anesthesia doctors. They work at the BWH Heart & Vascular Center. This center is ranked as one of the best heart hospitals in the country.

A cardiac CT scan can help find out why you have chest pain or trouble breathing. It is used for people who do not have a history of heart disease, like past heart attacks or heart stents. It is also used if you had a stress test and the results were not clear. This helps doctors see if you have heart disease.

There are other uses for a cardiac CT. The two most common types of cardiac CT scans are the coronary CT angiography and the coronary artery calcium scan.

The coronary CT angiography is the most common type. It looks at the blood vessels, called coronary arteries, that bring blood to your heart muscle. It checks for fatty buildup, called plaque, inside these blood vessels. Having plaque means you have heart disease. Plaque can block blood flow to your heart. This can cause chest pain or trouble breathing when you are active. But sometimes, these feelings are caused by other things and are not from heart disease.

The coronary artery calcium scan looks for calcium in your heart's blood vessels. It measures how much calcium is there. It is for people who do not already know they have heart disease. Finding calcium means you have plaque. Your doctor will use this test, along with your blood pressure and cholesterol numbers, to check your health risks. It helps them see if you might have a heart attack or stroke in the future. People with plaque have a higher risk. They are more likely to need aspirin or medicine to lower their cholesterol than people without plaque.

The Heart & Vascular Center is in the Shapiro Cardiovascular Center. It is across the street from the main BWH entrance at 75 Francis Street. All heart services are in this one place to make your care easy.

Before your test, you will check in at the Cardiovascular Imaging Center. This is inside the Shapiro Cardiovascular Center. While you have your test, your family and friends can wait in the Shapiro Family Center.

Our picture experts work with a large team to give you the best care. This team includes heart doctors, heart surgeons, doctors who treat heart rhythms, and anesthesia doctors. They also work with nurses, physician assistants, physical therapists, dietitians, and social workers to help you get better.

You can learn more about a test called cardiac catheterization in our health library. You can also visit the Kessler Health Education Library in the Bretholtz Center for Patients and Families. We offer a full list of services for patients and families. You can also learn about the Watkins Clinic in the Shapiro Cardiovascular Center. Brigham and Women's Hospital has been a leader in patient care, teaching, and research for over 100 years. We have experts in almost every type of medicine and surgery.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 31 of 77 · Item ID B-RSNVNQ65QD

Original page

CT angiography - chest

Definition

CT angiography combines a CT scan with the injection of dye. This technique is able to create pictures of the blood vessels in the chest and upper abdomen. CT stands for computed tomography.

Alternative Names

Computed tomography angiography - thorax; CTA - lungs; Pulmonary embolism - CTA chest; Thoracic aortic aneurysm - CTA chest; Venous thromboembolism - CTA lung; Blood clot - CTA lung; Embolus - CTA lung; CT pulmonary angiogram

How the Test is Performed

You will be asked to lie on a narrow table that slides into the center of the CT scanner.

While inside the scanner, the machine's x-ray beam rotates around you.

A computer creates multiple separate images of the body area, called slices. These images can be stored, viewed on a monitor, or printed on film. Three-dimensional models of the chest area can be created by stacking the slices together.

You must be still during the exam, because movement causes blurred images. You may be told to hold your breath for short periods of time.

Complete scans usually take only a few minutes. The newest scanners can image your entire body, head to toe, in less than 30 seconds.

How to Prepare for the Test

Certain exams require a special dye, called contrast, to be delivered into the body before the test starts. Contrast helps certain areas show up better on x-rays.

- Contrast can be given through a vein (IV) in your hand or forearm. If contrast is used, you may also be asked not to eat or drink anything for 4 to 6 hours before the test.

- Let your health care provider know if you have ever had a reaction to contrast. You may need to take medications before the test in order to safely receive it.

- Before receiving the contrast, tell your provider if you take the diabetes medication metformin (Glucophage). You may need to take extra precautions.

The contrast can worsen kidney function problems in people with poorly functioning kidneys. Talk to your provider if you have a history of kidney problems.

Too much weight can damage the scanner. If you weigh more than 300 pounds (135 kilograms), talk to your provider about the weight limit before the test.

You will be asked to remove jewelry and wear a hospital gown during the study.

How the Test will Feel

The x-rays produced by the CT scan are painless. Some people may have discomfort from lying on the hard table.

If you have contrast through a vein, you may have a:

- Slight burning feeling

- Metallic taste in your mouth

- Warm flushing of your body

This is normal and usually goes away within a few seconds.

Why the Test is Performed

A chest CT angiogram may be done:

- For symptoms that suggest blood clots in the lungs, such as chest pain, rapid breathing, or shortness of breath

- After a chest injury or trauma

- Before surgery in the lung or chest

- To look for a possible site to insert a catheter for hemodialysis

- For swelling of the face or upper arms that cannot be explained

- To look for a suspected birth defect of the aorta or other blood vessels in the chest

- To look for a widening of an artery (aneurysm)

- To look for a tear in an artery (dissection)

- To assess certain blood vessels around the heart

Normal Results

Results are considered normal if no problems are seen.

What Abnormal Results Mean

A chest CT may show many disorders of the heart, lungs, or chest area, including:

- Suspected

blockage of the superior vena cava : This large vein moves blood from the upper half of the body to the heart. Blood clot(s) in the lungs .- Abnormalities of the blood vessels in the lungs or chest, such as

aortic arch syndrome . Aortic aneurysm (in the chest area) , also called thoracic aortic aneurysm.- Narrowing of part of the

major artery leading out of the heart (aorta). - Tear in the wall of an artery (dissection).

- Inflammation of the blood vessel walls (vasculitis).

- Problems with pulmonary vessels, including narrowing of pulmonary veins of abnormal connections between pulmonary arteries and veins (arteriovenous malformations).

Risks

Risks of CT scans include:

- Being exposed to radiation

- Allergic reaction to contrast dye

- Damage to kidneys from contrast dye

CT scans use more radiation than regular x-rays. Having many x-rays or CT scans over time may increase your risk for cancer. However, the risk from any one scan is small. You and your provider should weigh this risk against the benefits of getting a correct diagnosis for a medical problem. Most modern scanners use techniques to use less radiation.

Some people have allergies to contrast dye. Let your provider know if you have ever had an allergic reaction to injected contrast dye.

- The most common type of contrast given into a vein contains iodine. If you have an iodine allergy, you may have

nausea or vomiting ,sneezing ,itching , orhives if you get this type of contrast. - If you absolutely must be given such contrast, your provider may give you antihistamines (such as Benadryl) and/or steroids before the test.

- The kidneys help remove iodine out of the body. Those with kidney disease or diabetes may need to receive extra fluids after the test to help flush the iodine out of the body.

Rarely, the dye may cause a life-threatening allergic response called

Your provider may ask you to avoid the use of metformin for two days after the CT-angiogram.

Carmichael SP, Mowery NT, Martin RS, Meredith JW. Management of acute trauma. In: Townsend CM Jr, Beauchamp RD, Evers BM, Mattox KL, eds. Sabiston Textbook of Surgery. 21st ed. Philadelphia, PA: Elsevier; 2022:chap 17.

Gilman M. Congenital and developmental diseases of the lungs and airways. In: Digumarthy SR, Abbara S, Chung JH, eds. Problem Solving in Chest Imaging. Philadelphia, PA: Elsevier; 2020:chap 15.

Reekers JA. Angiography: principles, techniques and complications. In: Adam A, Dixon AK, Gillard JH, Schaefer-Prokop CM, eds. Grainger & Allison's Diagnostic Radiology: A Textbook of Medical Imaging. 7th ed. Philadelphia, PA: Elsevier; 2021:chap 78.

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AI rewrite

CT Angiography - Chest

What It Is

A CT angiography uses a CT scan along with a shot of dye. This test makes pictures of the blood vessels in your chest and upper belly. CT stands for computed tomography.

Other Names

Computed tomography angiography - thorax; CTA - lungs; Pulmonary embolism - CTA chest; Thoracic aortic aneurysm - CTA chest; Venous thromboembolism - CTA lung; Blood clot - CTA lung; Embolus - CTA lung; CT pulmonary angiogram

How the Test Is Done

You will lie on a narrow table. The table slides into the middle of the CT scanner.

While you are inside, the machine's x-ray beam spins around you.

A computer makes many pictures of your body. These pictures are called slices. The slices can be saved, shown on a screen, or printed. The slices can also be stacked to make 3-D pictures of your chest.

You must hold still during the test. If you move, the pictures will be blurry. You may be asked to hold your breath for a short time.

Most scans take only a few minutes. The newest machines can scan your whole body in less than 30 seconds.

How to Get Ready

Some tests need a special dye called contrast. The dye goes into your body before the test. It helps some areas show up better on x-rays.

- The dye can go in through a vein (IV) in your hand or arm. If you get the dye, you may be told not to eat or drink for 4 to 6 hours before the test.

- Tell your provider if you have ever had a bad reaction to the dye. You may need to take medicine before the test so you can get the dye safely.

- Tell your provider if you take the diabetes medicine metformin (Glucophage). You may need to take extra care.

The dye can make kidney problems worse in people whose kidneys do not work well. Talk to your provider if you have had kidney problems.

Too much weight can break the scanner. If you weigh more than 300 pounds (135 kilograms), ask your provider about the weight limit before the test.

You will be asked to take off jewelry. You will wear a hospital gown.

How the Test Will Feel

The x-rays do not hurt. Some people feel uncomfortable lying on the hard table.

If you get the dye through a vein, you may feel:

- A slight burning

- A metal taste in your mouth

- A warm feeling in your body

This is normal. It usually goes away in a few seconds.

Why the Test Is Done

A chest CT angiogram may be done:

- For signs of blood clots in the lungs, like chest pain, fast breathing, or trouble breathing
- After a chest injury
- Before surgery in the lung or chest
- To find a good spot to put a tube (catheter) for kidney dialysis
- For swelling of the face or upper arms with no clear cause
- To look for a birth defect of the aorta or other blood vessels in the chest
- To look for a widened artery (aneurysm)
- To look for a tear in an artery (dissection)
- To check certain blood vessels around the heart

Normal Results

Results are normal if no problems are seen.

What Abnormal Results Mean

A chest CT may show many problems of the heart, lungs, or chest, including:

- A possible blockage of the superior vena cava. This large vein moves blood from the upper body to the heart.
- Blood clots in the lungs.
- Problems with the blood vessels in the lungs or chest, such as aortic arch syndrome.
- A widened artery in the chest (thoracic aortic aneurysm).
- Narrowing of part of the main artery that leads out of the heart (aorta).
- A tear in the wall of an artery (dissection).
- Swelling of the blood vessel walls (vasculitis).
- Problems with the lung's blood vessels. This can include narrowed lung veins or odd links between lung arteries and veins (arteriovenous malformations).

Risks

Risks of CT scans include:

- Being exposed to radiation
- An allergic reaction to the dye
- Harm to the kidneys from the dye

CT scans use more radiation than regular x-rays. Having many x-rays or CT scans over time may raise your risk for cancer. But the risk from any one scan is small. You and your provider should weigh this risk against the good of getting the right diagnosis. Most new scanners use ways to lower the radiation.

Some people are allergic to the dye. Tell your provider if you have ever had an allergic reaction to dye that was put in a vein.

- The most common dye given into a vein has iodine in it. If you are allergic to iodine, the dye may make you feel sick to your stomach, vomit, sneeze, itch, or get hives.
- If you must have this dye, your provider may give you medicine first, such as antihistamines (like Benadryl) and/or steroids.
- The kidneys help clear iodine from the body. People with kidney disease or diabetes may need extra fluids after the test to help flush out the iodine.

Rarely, the dye can cause a life-threatening allergic reaction.

Your provider may ask you not to take metformin for two days after the CT angiogram.

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Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

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Original page

CT angiography - chest

Definition

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Alternative Names

Computed tomography angiography - thorax; CTA - lungs; Pulmonary embolism - CTA chest; Thoracic aortic aneurysm - CTA chest; Venous thromboembolism - CTA lung; Blood clot - CTA lung; Embolus - CTA lung; CT pulmonary angiogram

How the Test is Performed

You will be asked to lie on a narrow table that slides into the center of the CT scanner.

While inside the scanner, the machine's x-ray beam rotates around you.

A computer creates multiple separate images of the body area, called slices. These images can be stored, viewed on a monitor, or printed on film. Three-dimensional models of the chest area can be created by stacking the slices together.

You must be still during the exam, because movement causes blurred images. You may be told to hold your breath for short periods of time.

Complete scans usually take only a few minutes. The newest scanners can image your entire body, head to toe, in less than 30 seconds.

How to Prepare for the Test

Certain exams require a special dye, called contrast, to be delivered into the body before the test starts. Contrast helps certain areas show up better on x-rays.

- Contrast can be given through a vein (IV) in your hand or forearm. If contrast is used, you may also be asked not to eat or drink anything for 4 to 6 hours before the test.

- Let your health care provider know if you have ever had a reaction to contrast. You may need to take medications before the test in order to safely receive it.

- Before receiving the contrast, tell your provider if you take the diabetes medication metformin (Glucophage). You may need to take extra precautions.

The contrast can worsen kidney function problems in people with poorly functioning kidneys. Talk to your provider if you have a history of kidney problems.

Too much weight can damage the scanner. If you weigh more than 300 pounds (135 kilograms), talk to your provider about the weight limit before the test.

You will be asked to remove jewelry and wear a hospital gown during the study.

How the Test will Feel

The x-rays produced by the CT scan are painless. Some people may have discomfort from lying on the hard table.

If you have contrast through a vein, you may have a:

- Slight burning feeling

- Metallic taste in your mouth

- Warm flushing of your body

This is normal and usually goes away within a few seconds.

Why the Test is Performed

A chest CT angiogram may be done:

- For symptoms that suggest blood clots in the lungs, such as chest pain, rapid breathing, or shortness of breath

- After a chest injury or trauma

- Before surgery in the lung or chest

- To look for a possible site to insert a catheter for hemodialysis

- For swelling of the face or upper arms that cannot be explained

- To look for a suspected birth defect of the aorta or other blood vessels in the chest

- To look for a widening of an artery (aneurysm)

- To look for a tear in an artery (dissection)

- To assess certain blood vessels around the heart

Normal Results

Results are considered normal if no problems are seen.

What Abnormal Results Mean

A chest CT may show many disorders of the heart, lungs, or chest area, including:

- Suspected

blockage of the superior vena cava : This large vein moves blood from the upper half of the body to the heart. Blood clot(s) in the lungs .- Abnormalities of the blood vessels in the lungs or chest, such as

aortic arch syndrome . Aortic aneurysm (in the chest area) , also called thoracic aortic aneurysm.- Narrowing of part of the

major artery leading out of the heart (aorta). - Tear in the wall of an artery (dissection).

- Inflammation of the blood vessel walls (vasculitis).

- Problems with pulmonary vessels, including narrowing of pulmonary veins of abnormal connections between pulmonary arteries and veins (arteriovenous malformations).

Risks

Risks of CT scans include:

- Being exposed to radiation

- Allergic reaction to contrast dye

- Damage to kidneys from contrast dye

CT scans use more radiation than regular x-rays. Having many x-rays or CT scans over time may increase your risk for cancer. However, the risk from any one scan is small. You and your provider should weigh this risk against the benefits of getting a correct diagnosis for a medical problem. Most modern scanners use techniques to use less radiation.

Some people have allergies to contrast dye. Let your provider know if you have ever had an allergic reaction to injected contrast dye.

- The most common type of contrast given into a vein contains iodine. If you have an iodine allergy, you may have

nausea or vomiting ,sneezing ,itching , orhives if you get this type of contrast. - If you absolutely must be given such contrast, your provider may give you antihistamines (such as Benadryl) and/or steroids before the test.

- The kidneys help remove iodine out of the body. Those with kidney disease or diabetes may need to receive extra fluids after the test to help flush the iodine out of the body.

Rarely, the dye may cause a life-threatening allergic response called

Your provider may ask you to avoid the use of metformin for two days after the CT-angiogram.

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AI rewrite

Chest CT Angiogram

What is it?

A chest CT angiogram is a test that takes pictures of the blood vessels in your chest and upper belly. It uses a CT scanner (a special x-ray machine) and a special dye.

Other Names for this Test

CTA of the chest, CTA of the lungs, or CT pulmonary angiogram.

How the Test is Done

You will lie on a narrow table. The table slides into the center of the CT scanner. The x-ray machine moves around you.

A computer takes many pictures of your body. These pictures are called slices. The computer puts the slices together to make a 3D model of your chest.

You must hold very still. Moving makes the pictures blurry. You may need to hold your breath for a short time.

The scan only takes a few minutes. Some new machines take less than 30 seconds.

How to Get Ready

You may need a special dye called contrast. This dye helps your blood vessels show up better on the x-rays.

* The dye is put into a vein in your hand or arm through an IV. If you get this dye, you may need to stop eating and drinking for 4 to 6 hours before the test.

* Tell your doctor if you have ever had a bad reaction to contrast dye. You might need medicine before the test to keep you safe.

* Tell your doctor if you take a diabetes medicine called metformin (Glucophage). You may need to take special steps.

The dye can harm your kidneys if they do not work well. Tell your doctor if you have kidney problems.

The scanner has a weight limit. If you weigh more than 300 pounds (135 kilograms), tell your doctor before the test.

You will need to take off your jewelry and wear a hospital gown.

How the Test Feels

The CT scan does not hurt. The table may feel hard and a little uncomfortable.

If you get the dye through an IV, you might feel a:

* Slight burn

* Metal taste in your mouth

* Warm or flushed feeling in your body

This is normal and goes away in a few seconds.

Why the Test is Done

Your doctor may order this test to:

* Check for blood clots in the lungs if you have chest pain or trouble breathing

* Look at your chest after an injury

* Plan for chest or lung surgery

* Find a spot to place a tube for kidney dialysis

* Find out why your face or arms are swollen

* Look for blood vessel problems you were born with

* Check for a bulging blood vessel (aneurysm)

* Check for a torn blood vessel (dissection)

* Look at the blood vessels around your heart

Normal Results

A normal result means no problems were found.

What Abnormal Results Mean

The test may show problems in your heart, lungs, or chest. These problems can include:

* A block in the large vein that brings blood from your upper body to your heart

* Blood clots in your lungs

* Problems with the blood vessels in your chest or lungs

* A bulging blood vessel in your chest (aneurysm)

* A narrow spot in the main artery that leaves your heart (aorta)

* A tear in a blood vessel wall

* Swelling in the blood vessel walls

* Narrow lung veins or abnormal connections between your blood vessels

Risks of the Test

Risks include:

* Radiation exposure

* An allergic reaction to the dye

* Kidney damage from the dye

CT scans use more radiation than regular x-rays. Having many scans over time can slightly raise your risk for cancer. But the risk from one scan is very small. The benefit of finding out what is wrong is usually greater than the risk. New machines use less radiation.

Some people are allergic to the dye. Tell your doctor if you have ever had a bad reaction to it.

* The most common dye has iodine in it. If you are allergic to iodine, the dye might make you feel sick to your stomach or throw up. It might also cause sneezing, itching, or hives.

* If you must have this dye, your doctor can give you allergy medicine first.

Your kidneys clean the iodine out of your body. If you have kidney disease or diabetes, you may need extra fluids after the test. This helps wash the dye out.

In very rare cases, the dye can cause a severe allergic reaction that can be life-threatening.

Your doctor may ask you to stop taking your metformin medicine for two days after the test.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

New to MyHealth?

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CT Coronary Angiogram

Our Approach

A computed tomography (CT) coronary angiogram is an innovative approach to investigating chest pain and your coronary arteries. Instead of immediately performing a conventional invasive angiogram to look for blockages or narrowing, we can create a 3-D image of your arteries using X-rays. If they are clear, you may not need invasive testing.

We use the latest CT scanners when looking for coronary artery disease, offering accurate results and low radiation exposure. If you still need cardiac catheterization after the CT scan, our team offers a range of procedures for diagnosis and treatment.

What Is a CT Coronary Angiogram?

To request an appointment with an interventional cardiologist, call 650-725-2621.

About Cardiac CT

Cardiac CT, or computed tomography, uses X-ray technology to create high-resolution, 3-D images of your heart. A CT coronary angiogram is one type of cardiac CT, focusing on the arteries that supply your heart muscle with blood.

Often, an angiogram with cardiac catheterization is the best choice for determining if these arteries are restricted or blocked. For some people, a coronary artery scan can provide a noninvasive alternative. The scan may provide an appropriate option if you face a lower risk for obstructive coronary artery disease, particularly if you:

- Received inconclusive results after undergoing a stress test or other initial evaluation, such as an electrocardiogram (ECG)

- Have unusual symptoms, such as chest pain that does not become worse with exertion

AI rewrite

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MyHealth for Mobile

CT Coronary Angiogram

Our Approach

A CT coronary angiogram is a new way to look at your chest pain and your heart arteries. CT stands for computed tomography. The old way to check for blockages used an invasive test. That means a tool goes inside your body. Instead, we can use X-rays to make a 3-D picture of your arteries. If your arteries are clear, you may not need the invasive test.

We use the newest CT scanners to look for heart artery disease. These scanners give clear results. They also use only a small amount of radiation. You may still need a test called cardiac catheterization after the CT scan. If so, our team can do many kinds of tests and treatments.

What Is a CT Coronary Angiogram?

To make an appointment with a heart doctor, call 650-725-2621.

About Cardiac CT

Cardiac CT uses X-rays to make clear, 3-D pictures of your heart. CT means computed tomography. A CT coronary angiogram is one kind of cardiac CT. It looks at the arteries that carry blood to your heart muscle.

Often, the best way to check these arteries is an angiogram with cardiac catheterization. This test checks if the arteries are narrow or blocked. But for some people, a heart artery scan is a good choice. This scan does not go inside your body. The scan may be right for you if you have a lower risk for blocked heart arteries. This is true if you:

- Had unclear results from a stress test or another first test, like an electrocardiogram (ECG)

- Have unusual symptoms, like chest pain that does not get worse when you are active

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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CT Coronary Angiogram

Our Approach

A computed tomography (CT) coronary angiogram is an innovative approach to investigating chest pain and your coronary arteries. Instead of immediately performing a conventional invasive angiogram to look for blockages or narrowing, we can create a 3-D image of your arteries using X-rays. If they are clear, you may not need invasive testing.

We use the latest CT scanners when looking for coronary artery disease, offering accurate results and low radiation exposure. If you still need cardiac catheterization after the CT scan, our team offers a range of procedures for diagnosis and treatment.

What Is a CT Coronary Angiogram?

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About Cardiac CT

Cardiac CT, or computed tomography, uses X-ray technology to create high-resolution, 3-D images of your heart. A CT coronary angiogram is one type of cardiac CT, focusing on the arteries that supply your heart muscle with blood.

Often, an angiogram with cardiac catheterization is the best choice for determining if these arteries are restricted or blocked. For some people, a coronary artery scan can provide a noninvasive alternative. The scan may provide an appropriate option if you face a lower risk for obstructive coronary artery disease, particularly if you:

- Received inconclusive results after undergoing a stress test or other initial evaluation, such as an electrocardiogram (ECG)

- Have unusual symptoms, such as chest pain that does not become worse with exertion

AI rewrite

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MyHealth for Mobile

CT Coronary Angiogram

Our Approach

A CT coronary angiogram is a newer way to check chest pain and the arteries of your heart. CT stands for computed tomography.

Instead of doing an invasive angiogram right away to look for blocked or narrow arteries, we can use X-rays to make a 3-D picture of your arteries. If your arteries are clear, you may not need invasive testing.

We use the latest CT scanners to look for coronary artery disease. These scanners give accurate results and use low radiation. If you still need cardiac catheterization after the CT scan, our team offers procedures to help find and treat heart problems.

What Is a CT Coronary Angiogram?

To ask for an appointment with an interventional cardiologist, call 650-725-2621.

About Cardiac CT

Cardiac CT uses X-rays to make clear, 3-D pictures of your heart. A CT coronary angiogram is one kind of cardiac CT. It looks at the arteries that bring blood to your heart muscle.

Often, an angiogram with cardiac catheterization is the best test to see if these arteries are narrowed or blocked. For some people, a coronary artery scan can be a choice that does not go inside the body.

The scan may be a good option if you have a lower risk for obstructive coronary artery disease. This may be true if you:

- Had unclear results after a stress test or another first test, such as an electrocardiogram (ECG)

- Have unusual symptoms, such as chest pain that does not get worse with activity

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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We use the latest CT scanners when looking for coronary artery disease, offering accurate results and low radiation exposure. If you still need cardiac catheterization after the CT scan, our team offers a range of procedures for diagnosis and treatment.

What Is a CT Coronary Angiogram?

To request an appointment with an interventional cardiologist, call 650-725-2621.

About Cardiac CT

Cardiac CT, or computed tomography, uses X-ray technology to create high-resolution, 3-D images of your heart. A CT coronary angiogram is one type of cardiac CT, focusing on the arteries that supply your heart muscle with blood.

Often, an angiogram with cardiac catheterization is the best choice for determining if these arteries are restricted or blocked. For some people, a coronary artery scan can provide a noninvasive alternative. The scan may provide an appropriate option if you face a lower risk for obstructive coronary artery disease, particularly if you:

- Received inconclusive results after undergoing a stress test or other initial evaluation, such as an electrocardiogram (ECG)

- Have unusual symptoms, such as chest pain that does not become worse with exertion

AI rewrite

Are you new to MyHealth? You can manage your care from anywhere. You can look at your health details on any device. You can message your clinic, see lab results, set up a visit, and pay your bill. You can sign in if you have an access code, or ask for one if you do not. You can also get more details about the MyHealth mobile app.

A computed tomography (CT) coronary angiogram is a new way to check chest pain and your coronary arteries. These are the blood vessels that feed your heart. Instead of doing an invasive test right away, we use X-rays to make a 3-D picture of your arteries. An invasive test means putting a small tube inside your body. If your X-ray pictures are clear, you may not need the invasive test.

We use the newest CT scanners to look for heart disease. They give exact results and use low amounts of radiation. If you still need the tube test, called cardiac catheterization, our team offers many ways to find and treat your problem.

To ask for an appointment with an interventional cardiologist (a special heart doctor), call 650-725-2621.

Cardiac CT uses X-rays to make clear, 3-D pictures of your heart. A CT coronary angiogram is one type of cardiac CT. It focuses on the arteries that bring blood to your heart muscle.

Often, an invasive angiogram with a tube is the best way to see if these arteries are narrow or blocked. But for some people, a CT scan is a good choice because it does not use tubes. This scan may be a good option if you have a lower risk for blocked arteries. It is especially helpful if:

- You had a stress test or an electrocardiogram (ECG) and the results were not clear.

- You have unusual signs, like chest pain that does not get worse when you are active.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Your health care team will take a medical history, ask about your symptoms and listen to your heart with a stethoscope.

Your care team also may order tests to determine whether you have coronary artery disease and to what extent.

The results of these tests can also help guide treatment decisions.

Tests may include:

- Routine blood tests to check the level of fats, cholesterol, sugar and proteins in your blood. These are risk factors for heart disease that can be modified with lifestyle changes and, if needed, medication.

- Electrocardiogram (ECG), which records your heart's electrical activity and shows how fast or evenly the heart is beating. It can also show whether there is enough blood supply to the heart or if it is already damaged.

- Echocardiogram to look at the structure and overall function of your heart.

- Stress testing, which involves exercising, usually on a treadmill or stationary bike (or taking medicine to simulate exercise if you are unable to be active), to non-invasively evaluate blocked arteries in the heart.

- Chest X-ray to look at the heart and lungs and to see if there are abnormalities that might explain your symptoms.

- Computed tomography (CT) angiography scan of the heart that shows pictures of the heart's arteries and whether there is a buildup of plaque, even in the early stages before the plaque hardens.

- Coronary angiogram is done in the cardiac catheterization laboratory. It involves threading a thin tube or catheter into an artery, usually in the wrist or leg, and up to the heart. Dye is injected into the artery to evaluate it for any blockage. This test usually is recommended when a non-invasive one is abnormal, or a patient's symptoms strongly suggest CAD, or after a heart attack.

What Do the Results Mean?

Your care team might tell you that you have "less than 70% blockage in an artery." That means that you have non-obstructive CAD. In other words, blood flow to the heart muscle is not reduced at rest but may be limited during intense exercise or exertion.

In this case, risk reduction through lifestyle and medication is the best treatment.

Or your care team might say you have "over 70% blockage in one artery." This means that you have a severe blockage in a coronary artery. Blood flow to a portion (or portions) of the heart muscle is greatly reduced and would likely explain any chest pain or shortness of breath

you may have been feeling. In this case, your care team might recommend stepped up medical treatment, and possibly a procedure or surgery.

AI rewrite

Your health care team will ask about your past health and your symptoms. They will also listen to your heart with a stethoscope.

Your team may order tests to see if you have heart disease and how bad it is. The results of these tests help your team know how to treat you.

Tests may include:

- Blood tests. These check the levels of fats, cholesterol, sugar, and proteins in your blood. High levels can lead to heart disease. You can lower your risk by changing your habits and taking medicine if needed.

- ECG (electrocardiogram). This records your heart's electrical signals. It shows how fast and steady your heart beats. It can show if your heart gets enough blood or if it is already damaged.

- Echocardiogram. This looks at the shape of your heart and how well it works.

- Stress test. This checks for blocked blood vessels in your heart. You will walk on a treadmill or ride a stationary bike. If you cannot exercise, you will get medicine to make your heart work hard like it is exercising.

- Chest X-ray. This takes a picture of your heart and lungs. It helps find out what is causing your symptoms.

- CT (computed tomography) scan of the heart. This takes pictures of your heart's blood vessels. It shows if there is a buildup of plaque. Plaque is a fatty deposit. The scan can find plaque early, even before it gets hard.

- Coronary angiogram. This is done in a special heart lab. A doctor puts a thin tube into a blood vessel in your wrist or leg. The tube is moved up to your heart. Dye is put into the tube to check for blockages. You might need this test if other tests are not normal. You might also need it if your symptoms point to heart disease or if you had a heart attack.

What do the results mean?

Your team might say you have "less than 70% blockage in an artery." This means you have non-obstructive heart disease. Blood flow to your heart is fine when you rest. But it might slow down when you exercise hard. For this, the best treatment is changing your habits and taking medicine.

Your team might say you have "over 70% blockage in one artery." This means you have a severe blockage. Blood flow to parts of your heart is very low. This low blood flow likely causes any chest pain or shortness of breath you feel. In this case, your team may give you more medicine. You might also need a procedure or surgery.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

Item 37 of 77 · Item ID B-3WTOOBMZVR

Original page

Your health care team will take a medical history, ask about your symptoms and listen to your heart with a stethoscope.

Your care team also may order tests to determine whether you have coronary artery disease and to what extent.

The results of these tests can also help guide treatment decisions.

Tests may include:

- Routine blood tests to check the level of fats, cholesterol, sugar and proteins in your blood. These are risk factors for heart disease that can be modified with lifestyle changes and, if needed, medication.

- Electrocardiogram (ECG), which records your heart's electrical activity and shows how fast or evenly the heart is beating. It can also show whether there is enough blood supply to the heart or if it is already damaged.

- Echocardiogram to look at the structure and overall function of your heart.

- Stress testing, which involves exercising, usually on a treadmill or stationary bike (or taking medicine to simulate exercise if you are unable to be active), to non-invasively evaluate blocked arteries in the heart.

- Chest X-ray to look at the heart and lungs and to see if there are abnormalities that might explain your symptoms.

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What Do the Results Mean?

Your care team might tell you that you have "less than 70% blockage in an artery." That means that you have non-obstructive CAD. In other words, blood flow to the heart muscle is not reduced at rest but may be limited during intense exercise or exertion.

In this case, risk reduction through lifestyle and medication is the best treatment.

Or your care team might say you have "over 70% blockage in one artery." This means that you have a severe blockage in a coronary artery. Blood flow to a portion (or portions) of the heart muscle is greatly reduced and would likely explain any chest pain or shortness of breath

you may have been feeling. In this case, your care team might recommend stepped up medical treatment, and possibly a procedure or surgery.

AI rewrite

Your health care team will ask about your health history. They will ask about your symptoms. They will listen to your heart with a stethoscope.

Your care team may also order tests. These tests can help find out if you have coronary artery disease. They can also show how serious it is. The test results can help your care team choose the best treatment.

Tests may include:

- Routine blood tests. These check the levels of fats, cholesterol, sugar, and proteins in your blood. These can be risk factors for heart disease. You may be able to change these risk factors with healthy habits and, if needed, medicine.

- Electrocardiogram, also called an ECG. This test records your heart’s electrical activity. It shows how fast and how evenly your heart is beating. It can also show if your heart is getting enough blood or if it is damaged.

- Echocardiogram. This test looks at the shape and function of your heart.

- Stress testing. This test usually means you exercise on a treadmill or stationary bike. If you cannot exercise, you may take medicine that acts like exercise on your heart. This test helps check for blocked arteries in the heart without surgery.

- Chest X-ray. This test looks at your heart and lungs. It can help find problems that may explain your symptoms.

- Computed tomography angiography, also called a CT angiography scan. This scan takes pictures of the heart's arteries. It can show if plaque has built up, even in early stages before the plaque hardens.

- Coronary angiogram. This test is done in a cardiac catheterization lab. A thin tube, called a catheter, is put into an artery, usually in the wrist or leg. The tube is moved up to the heart. Dye is put into the artery to check for blockages. This test is usually recommended when a non-surgery test is abnormal, when symptoms strongly suggest coronary artery disease, or after a heart attack.

Your care team might tell you that you have “less than 70% blockage in an artery.” This means you have non-obstructive coronary artery disease. Blood flow to the heart muscle is not reduced when you are resting. But blood flow may be limited during hard exercise or activity.

In this case, the best treatment is to lower your risk. This may be done with healthy lifestyle changes and medicine.

Your care team might say you have “over 70% blockage in one artery.” This means you have a severe blockage in a coronary artery. Blood flow to part of the heart muscle is greatly reduced. This would likely explain chest pain or shortness of breath you may have had.

In this case, your care team might recommend stronger medical treatment. They may also recommend a procedure or surgery.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 38 of 77 · Item ID B-0PRKGYC4OM

Original page

Your health care team will take a medical history, ask about your symptoms and listen to your heart with a stethoscope.

Your care team also may order tests to determine whether you have coronary artery disease and to what extent.

The results of these tests can also help guide treatment decisions.

Tests may include:

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- Electrocardiogram (ECG), which records your heart's electrical activity and shows how fast or evenly the heart is beating. It can also show whether there is enough blood supply to the heart or if it is already damaged.

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you may have been feeling. In this case, your care team might recommend stepped up medical treatment, and possibly a procedure or surgery.

AI rewrite

Your health care team will ask about your health history. They will ask about your symptoms. They will listen to your heart with a stethoscope.

Your care team may also order tests. These tests can show if you have coronary artery disease. They can also show how bad it is. The results help your team choose the best treatment.

You may have these tests:

- Routine blood tests. These check the levels of fats, cholesterol, sugar, and proteins in your blood. High levels are risk factors for heart disease. You can lower them by changing how you live. You can also lower them with medicine if you need it.

- Electrocardiogram (ECG). This test records your heart's electrical activity. It shows how fast and how evenly your heart beats. It can also show if your heart gets enough blood or if it is already hurt.

- Echocardiogram. This test looks at the shape of your heart and how well it works.

- Stress testing. For this test, you exercise. You usually walk on a treadmill or ride a bike that stays in place. If you cannot exercise, you can take medicine that acts like exercise. This test checks for blocked arteries in the heart. It does not go inside your body.

- Chest X-ray. This test looks at your heart and lungs. It can show problems that may explain your symptoms.

- Computed tomography (CT) angiography scan. This test takes pictures of the heart's arteries. It shows if there is plaque buildup. It can find plaque early, even before it gets hard.

- Coronary angiogram. This test is done in a special lab. The team puts a thin tube into an artery. The tube usually goes in your wrist or leg. Then it goes up to your heart. The team puts dye into the artery. The dye shows if there is a blockage. This test is often used when another test is not normal. It may also be used when your symptoms point strongly to CAD. It may be used after a heart attack.

What Do the Results Mean?

Your care team might say you have "less than 70% blockage in an artery." This means you have non-obstructive CAD. Blood flow to your heart muscle is fine when you rest. But it may be low when you exercise hard or work hard.

If this is the case, the best treatment is to lower your risk. You do this through how you live and with medicine.

Or your care team might say you have "over 70% blockage in one artery." This means you have a bad blockage in a heart artery. Blood flow to part of your heart muscle is much lower than normal. This may explain any chest pain or shortness of breath you have felt. If this is the case, your care team might want stronger treatment. They may suggest a procedure or surgery.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

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Original page

TAVR is a procedure to replace your heart's aortic valve. It's an alternative to open-heart surgery. TAVR treats a narrowed aortic valve (aortic stenosis), which can overwork your heart and increase your risk of serious complications. TAVR gives you a new valve so you can feel better and protect your heart.

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TAVR (transcatheter aortic valve replacement) is a minimally invasive procedure that replaces your aortic valve without open-heart surgery. An interventional cardiologist inserts a thin tube (catheter) into an artery, typically in your groin. They guide the catheter to your heart. The catheter delivers your new valve, placing it inside your old one.

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Receiving a new aortic valve can help you feel better and live longer. It can also protect your heart. That's because your heart pumps blood through your aortic valve and out to your entire body. If the valve isn't working right, your heart is forced to work harder to push blood through. This extra workload can damage your heart over time, leading to heart failure.

Transcatheter aortic valve implantation (TAVI) is another name for TAVR.

TAVR treats aortic stenosis. This is the narrowing of your aortic valve or the area around it. Narrowing usually happens because of:

You may need to fast (no food or drinks other than water) and stop taking certain medications before having TAVR. Your healthcare provider will tell you exactly what to do. Follow their guidance and ask if there's anything you don't understand.

Your provider may also run some tests, including blood and imaging tests, to help plan the procedure.

At the start of the procedure, you'll receive sedation and/or anesthesia. You'll also receive medicine to prevent blood clots.

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To perform TAVR, an interventional cardiologist will:

TAVR typically takes about an hour or two from start to finish.

Transcatheter aortic valve replacement has several advantages over surgery. These include:

Many people who need aortic valve replacement are "high-risk" because of severe stenosis or other health conditions. That means they have a greater risk of complications or death from open-heart surgery. TAVR is an excellent alternative for them.

Despite this, some people may still benefit from an aortic valve replacement surgery. Talk to your healthcare provider about what's right for you.

With advancements in valve and catheter technology, TAVR complications don't happen often. But when problems do arise, they may include:

TAVR has many benefits. But it's not right for everyone. The most common reasons your provider may not recommend TAVR include:

After you wake up from surgery, you'll need to stay in bed for at least several hours. This is to prevent bleeding from the spot where the catheter went in. Your care team will make sure the stitches are secure before you get up.

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A provider may prescribe medicine to keep you from getting an infection or blood clot after the procedure.

Most people who have transcatheter aortic valve replacement can leave the hospital within one day. Some may stay two days or more. It depends on how your procedure went and your overall health.

Your healthcare provider will explain how long your recovery is likely to take. In general, most people can return to most of their normal activities within days. For some, it may take a little longer.

You'll need to wait at least one week before moving heavy items or doing other physical activity. You can typically go back to work two weeks after TAVR and drive after one month. Complete recovery takes six to 10 weeks.

Your healthcare provider will likely refer you to a cardiac rehab program. This is a prescribed workout plan that involves a team of providers from several different fields. You'll start rehab within several days of your TAVR procedure to increase your heart's strength and endurance.

Your healthcare provider will tell you when you should see them for follow-up care. You can expect a checkup a month after your TAVR procedure and then once a year after that. Regular follow-up care is vital. It allows your provider to check your heart and see how you're feeling. They may also run tests to make sure your new valve is working well.

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After TAVR, contact your provider if you have:

Someone's going to put a new valve into your heart? That can feel scary. But keep in mind that you're not the first person who's had this procedure. Thousands of people have had TAVR. You can also give yourself some peace of mind by choosing a provider with a lot of experience performing this procedure.

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It can be scary and overwhelming when something is happening with your heart valves. Cleveland Clinic heart specialists are ready to get you the help you need.

AI rewrite

TAVR is a procedure to replace your heart’s aortic valve. It is another choice instead of open-heart surgery.

TAVR treats a narrowed aortic valve. This is called aortic stenosis. A narrow valve can make your heart work too hard. It can also raise your risk of serious problems. TAVR gives you a new valve. This can help you feel better and help protect your heart.

TAVR stands for transcatheter aortic valve replacement. It is a less invasive procedure. This means doctors do not need to do open-heart surgery.

A heart doctor called an interventional cardiologist does the procedure. They put a thin tube, called a catheter, into an artery. This is often done through your groin. The doctor guides the catheter to your heart. The catheter carries your new valve. The new valve is placed inside your old valve.

A new aortic valve can help you feel better and live longer. It can also protect your heart. Your heart pumps blood through your aortic valve to the rest of your body. If the valve does not work well, your heart has to work harder to push blood through. Over time, this extra work can hurt your heart. It can lead to heart failure.

TAVR is also called TAVI. TAVI stands for transcatheter aortic valve implantation.

TAVR treats aortic stenosis. This means your aortic valve, or the area around it, has become narrow.

Before TAVR, you may need to fast. This means no food or drinks except water. You may also need to stop taking some medicines. Your healthcare provider will tell you exactly what to do. Follow their instructions. Ask questions if you do not understand something.

Your provider may also do tests before the procedure. These may include blood tests and imaging tests. These tests help your care team plan the procedure.

At the start of the procedure, you will get sedation and/or anesthesia. This helps you relax or sleep. You will also get medicine to help prevent blood clots.

TAVR usually takes about one to two hours from start to finish.

TAVR has several advantages over open-heart surgery.

Many people who need a new aortic valve are “high-risk.” This may be because of severe narrowing of the valve or other health problems. This means they have a higher risk of problems or death from open-heart surgery. TAVR is a very good choice for many of these people.

Still, some people may do better with aortic valve replacement surgery. Talk with your healthcare provider about what is right for you.

With better valve and catheter technology, problems from TAVR do not happen often. But problems can still happen.

TAVR has many benefits. But it is not right for everyone. Your provider may not recommend TAVR for some reasons.

After you wake up from the procedure, you will need to stay in bed for at least several hours. This helps prevent bleeding from the place where the catheter went in. Your care team will make sure the stitches are secure before you get up.

Your provider may prescribe medicine after the procedure. This medicine may help prevent an infection or a blood clot.

Most people who have TAVR can leave the hospital within one day. Some people may stay two days or longer. This depends on how the procedure went and your overall health.

Your healthcare provider will tell you how long your recovery may take. Most people can return to most normal activities within a few days. For some people, it may take longer.

You will need to wait at least one week before lifting heavy items or doing other physical activity. You can usually go back to work two weeks after TAVR. You can usually drive after one month. Full recovery takes six to 10 weeks.

Your healthcare provider will likely send you to a cardiac rehab program. This is a workout plan ordered by your provider. A team of providers from different fields helps with the plan. You will start rehab within several days after TAVR. Rehab helps make your heart stronger and helps build your endurance.

Your healthcare provider will tell you when to come back for follow-up care. You can expect a checkup one month after TAVR. After that, you can expect a checkup once a year. Regular follow-up care is very important. It lets your provider check your heart and see how you are feeling. They may also do tests to make sure your new valve is working well.

After TAVR, contact your provider if you have any symptoms or concerns they told you to watch for.

Having a new valve placed in your heart can feel scary. But many people have had TAVR. You can feel more at ease by choosing a provider with a lot of experience doing this procedure.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 40 of 77 · Item ID B-CG1WU16HYM

Original page

TAVR is a procedure to replace your heart's aortic valve. It's an alternative to open-heart surgery. TAVR treats a narrowed aortic valve (aortic stenosis), which can overwork your heart and increase your risk of serious complications. TAVR gives you a new valve so you can feel better and protect your heart.

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To perform TAVR, an interventional cardiologist will:

TAVR typically takes about an hour or two from start to finish.

Transcatheter aortic valve replacement has several advantages over surgery. These include:

Many people who need aortic valve replacement are "high-risk" because of severe stenosis or other health conditions. That means they have a greater risk of complications or death from open-heart surgery. TAVR is an excellent alternative for them.

Despite this, some people may still benefit from an aortic valve replacement surgery. Talk to your healthcare provider about what's right for you.

With advancements in valve and catheter technology, TAVR complications don't happen often. But when problems do arise, they may include:

TAVR has many benefits. But it's not right for everyone. The most common reasons your provider may not recommend TAVR include:

After you wake up from surgery, you'll need to stay in bed for at least several hours. This is to prevent bleeding from the spot where the catheter went in. Your care team will make sure the stitches are secure before you get up.

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A provider may prescribe medicine to keep you from getting an infection or blood clot after the procedure.

Most people who have transcatheter aortic valve replacement can leave the hospital within one day. Some may stay two days or more. It depends on how your procedure went and your overall health.

Your healthcare provider will explain how long your recovery is likely to take. In general, most people can return to most of their normal activities within days. For some, it may take a little longer.

You'll need to wait at least one week before moving heavy items or doing other physical activity. You can typically go back to work two weeks after TAVR and drive after one month. Complete recovery takes six to 10 weeks.

Your healthcare provider will likely refer you to a cardiac rehab program. This is a prescribed workout plan that involves a team of providers from several different fields. You'll start rehab within several days of your TAVR procedure to increase your heart's strength and endurance.

Your healthcare provider will tell you when you should see them for follow-up care. You can expect a checkup a month after your TAVR procedure and then once a year after that. Regular follow-up care is vital. It allows your provider to check your heart and see how you're feeling. They may also run tests to make sure your new valve is working well.

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After TAVR, contact your provider if you have:

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It can be scary and overwhelming when something is happening with your heart valves. Cleveland Clinic heart specialists are ready to get you the help you need.

AI rewrite

TAVR is a way to replace your heart's aortic valve. It is another option besides open-heart surgery. TAVR treats a narrowed aortic valve. This problem is called aortic stenosis. A narrowed valve makes your heart work too hard. It can lead to serious health problems. TAVR gives you a new valve. This can help you feel better and protect your heart.

TAVR stands for transcatheter aortic valve replacement. It is a procedure that does not need open-heart surgery. A heart doctor puts a thin tube into a blood vessel. The tube is called a catheter. It usually goes in through your groin. The doctor moves the catheter up to your heart. The catheter carries your new valve. The doctor places the new valve inside your old one.

A new aortic valve can help you feel better and live longer. It can also protect your heart. Your heart pumps blood through your aortic valve. From there, blood goes out to your whole body. If the valve does not work right, your heart must work harder to push blood through. Over time, this extra work can hurt your heart. It can lead to heart failure.

TAVR has another name. It is called transcatheter aortic valve implantation, or TAVI.

TAVR treats aortic stenosis. This is when your aortic valve, or the area around it, gets too narrow. This narrowing usually happens because of:

Before TAVR, you may need to fast. This means no food or drinks except water. You may also need to stop taking some medicines. Your healthcare provider will tell you exactly what to do. Follow their advice. Ask if there is anything you do not understand.

Your provider may also run some tests. These may include blood tests and imaging tests. The tests help your provider plan the procedure.

When the procedure starts, you will get medicine to make you sleepy or numb. You will also get medicine to stop blood clots.

To do TAVR, a heart doctor will:

TAVR usually takes about one to two hours.

TAVR has some good points compared with surgery. These include:

Many people who need a new aortic valve are "high-risk." This may be because of severe narrowing or other health problems. High-risk means they have a greater chance of problems or death from open-heart surgery. TAVR is a good choice for these people.

Even so, some people may do better with open-heart surgery. Talk to your healthcare provider about what is best for you.

Valve and catheter tools have gotten better over time. So problems with TAVR do not happen often. But when problems do happen, they may include:

TAVR has many benefits. But it is not right for everyone. The most common reasons your provider may not suggest TAVR include:

After you wake up, you will need to stay in bed for at least a few hours. This helps stop bleeding where the catheter went in. Your care team will make sure your stitches are secure before you get up.

Your provider may give you medicine to stop an infection or blood clot after the procedure.

Most people who have TAVR can leave the hospital within one day. Some people stay two days or more. It depends on how your procedure went and your health.

Your provider will tell you how long your recovery may take. Most people can go back to most normal activities within a few days. For some people, it may take a little longer.

Wait at least one week before you lift heavy things or do other hard activity. Most people can go back to work two weeks after TAVR. You can usually drive after one month. Full recovery takes six to 10 weeks.

Your provider will likely send you to a cardiac rehab program. This is a workout plan made just for you. A team of providers from different fields will help you. You will start rehab within a few days of your TAVR. Rehab helps make your heart stronger.

Your provider will tell you when to come back for follow-up care. You can expect a checkup one month after TAVR. After that, you will have a checkup once a year. Regular follow-up care is very important. It lets your provider check your heart and see how you feel. They may also run tests to make sure your new valve is working well.

After TAVR, call your provider if you have:

Someone is going to put a new valve in your heart? That can feel scary. But remember, you are not the first person to have this procedure. Thousands of people have had TAVR. You can also feel calmer by choosing a provider who has done this procedure many times.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 41 of 77 · Item ID B-VUHT6T0UZS

Original page

TAVR is a procedure to replace your heart's aortic valve. It's an alternative to open-heart surgery. TAVR treats a narrowed aortic valve (aortic stenosis), which can overwork your heart and increase your risk of serious complications. TAVR gives you a new valve so you can feel better and protect your heart.

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Your provider may also run some tests, including blood and imaging tests, to help plan the procedure.

At the start of the procedure, you'll receive sedation and/or anesthesia. You'll also receive medicine to prevent blood clots.

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To perform TAVR, an interventional cardiologist will:

TAVR typically takes about an hour or two from start to finish.

Transcatheter aortic valve replacement has several advantages over surgery. These include:

Many people who need aortic valve replacement are "high-risk" because of severe stenosis or other health conditions. That means they have a greater risk of complications or death from open-heart surgery. TAVR is an excellent alternative for them.

Despite this, some people may still benefit from an aortic valve replacement surgery. Talk to your healthcare provider about what's right for you.

With advancements in valve and catheter technology, TAVR complications don't happen often. But when problems do arise, they may include:

TAVR has many benefits. But it's not right for everyone. The most common reasons your provider may not recommend TAVR include:

After you wake up from surgery, you'll need to stay in bed for at least several hours. This is to prevent bleeding from the spot where the catheter went in. Your care team will make sure the stitches are secure before you get up.

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A provider may prescribe medicine to keep you from getting an infection or blood clot after the procedure.

Most people who have transcatheter aortic valve replacement can leave the hospital within one day. Some may stay two days or more. It depends on how your procedure went and your overall health.

Your healthcare provider will explain how long your recovery is likely to take. In general, most people can return to most of their normal activities within days. For some, it may take a little longer.

You'll need to wait at least one week before moving heavy items or doing other physical activity. You can typically go back to work two weeks after TAVR and drive after one month. Complete recovery takes six to 10 weeks.

Your healthcare provider will likely refer you to a cardiac rehab program. This is a prescribed workout plan that involves a team of providers from several different fields. You'll start rehab within several days of your TAVR procedure to increase your heart's strength and endurance.

Your healthcare provider will tell you when you should see them for follow-up care. You can expect a checkup a month after your TAVR procedure and then once a year after that. Regular follow-up care is vital. It allows your provider to check your heart and see how you're feeling. They may also run tests to make sure your new valve is working well.

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It can be scary and overwhelming when something is happening with your heart valves. Cleveland Clinic heart specialists are ready to get you the help you need.

AI rewrite

TAVR is a procedure to replace your heart's aortic valve. It is used instead of open-heart surgery. TAVR treats a narrowed valve, which is called aortic stenosis. A narrowed valve makes your heart work too hard to pump blood to your body. This extra work can damage your heart over time and lead to heart failure. TAVR gives you a new valve. This helps you feel better, live longer, and protects your heart. Another name for TAVR is TAVI.

TAVR stands for transcatheter aortic valve replacement. A heart doctor puts a thin tube into a blood vessel, usually in your groin. The doctor guides the tube up to your heart. The tube carries your new valve. The doctor places the new valve right inside your old one.

Before TAVR, you may need to stop eating and drinking. You may only be allowed to drink water. You might also need to stop taking some medicines. Your doctor will tell you exactly what to do. Follow their rules and ask questions if you are confused. Your doctor may also do blood tests and take pictures of your heart to plan the procedure.

The procedure takes about one or two hours. You will get medicine to help you sleep or relax so you do not feel pain. You will also get medicine to stop blood clots.

TAVR has many benefits over open-heart surgery. Many people who need a new valve have other health problems. This means open-heart surgery is too dangerous for them. TAVR is a great choice for these people. Still, some people may do better with regular surgery. Talk to your doctor about what is best for you.

Because of new tools, problems with TAVR do not happen often. However, problems can still happen. TAVR is also not right for everyone. Your doctor will check if it is safe for you.

When you wake up, you must stay in bed for a few hours. This stops bleeding where the tube went in. Your care team will check your stitches before you get up. You may get medicine to stop infections and blood clots. Most people go home in one or two days. This depends on your health.

Your doctor will tell you how long it will take to heal. Most people can do normal things again in a few days. You must wait at least one week before you lift heavy things or exercise hard. You can usually go back to work in two weeks. You can drive after one month. It takes six to 10 weeks to fully heal.

Your doctor will likely send you to a heart rehab program. This is a safe workout plan with a care team. You will start this a few days after TAVR to make your heart strong.

You will see your doctor for a checkup one month after TAVR. After that, you will go once a year. These visits are very important. Your doctor will check your heart and test your new valve. Call your doctor right away if you have any problems or feel worse after you go home.

Getting a new heart valve can feel scary. Remember that thousands of people have had TAVR. You can feel safer by picking a doctor who has done this procedure many times.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 42 of 77 · Item ID B-N1U322N64K

Original page

Transcatheter Aortic Valve Replacement (TAVR)

Our Approach

Transcatheter aortic valve replacement (TAVR) is a newer, minimally invasive approach for treating aortic valve disease. The TAVR procedure has expanded the possibilities for replacing the aortic valve, and we continue to help shape its evolution.

For people who cannot undergo surgery to replace narrowed valves, TAVR first provided an effective option where none existed. Recently, it also presented a minimally invasive alternative for people who could endure an operation, but faced moderate risk. Now, we are investigating its possible use in an even wider range of people.

WHAT WE OFFER YOU FOR TAVR

- Deep experience, with our program as the first in Northern California and the second in the state performing TAVR. We complete more than 250 procedures a year, the region's highest volume.

- Nationally recognized expertise, with our doctors offering TAVR since its beginning. We have years of experience working with the procedure and contributing to its development.

- Team approach that includes surgeons with previous valve replacement experience and interventional cardiologists who have mastered minimally invasive treatment. They partner with top imaging specialists.

- Access to the latest valves by participating in clinical trials. TAVR constantly evolves. We receive valves still being studied, as well as those released in limited quantities upon FDA approval.

- Additional types of TAVR, with our team often able to do a second procedure when an earlier replacement valve wears out.

- Active research program, starting with TAVR's first clinical trials. A Stanford doctor co-authored the paper on TAVR and moderate surgical risk, and we now participate in a study of low risk.

What Is TAVR?

To schedule an appointment with the TAVR program, please call 650-725-2687

To schedule an appointment with the Valvular Heart Disease Center, please call 650-723-6459

Transcatheter Aortic Valve Replacement (TAVR)

TAVR uses minimally invasive tubes instead of surgery to treat aortic valve stenosis, with our doctors helping to refine the procedure and expand its use.

transcatheter aortic valve replacement

Heart valve

aortic valve

aortic stenosis

Aortic valve stenosis

Transcatheter aortic valve implantation

TAVI

percutaneous aortic valve replacement

AI rewrite

Transcatheter Aortic Valve Replacement (TAVR)

Our Approach

Transcatheter aortic valve replacement is also called TAVR. It is a newer way to treat aortic valve disease. It uses small tubes instead of open-heart surgery. This is called a minimally invasive procedure.

TAVR has made it possible for more people to get a new aortic valve. We continue to help improve this procedure.

At first, TAVR helped people who could not have surgery for a narrowed aortic valve. It gave them a treatment option when they did not have one before.

Later, TAVR became an option for people who could have surgery but had a moderate risk from surgery. Now, we are studying if TAVR can help even more people.

What We Offer You for TAVR

- Deep experience. Our program was the first in Northern California and the second in the state to perform TAVR. We do more than 250 TAVR procedures each year. This is the highest number in the region.

- Nationally known skill. Our doctors have offered TAVR since it began. They have many years of experience with the procedure. They have also helped develop it.

- A team approach. Your care team includes surgeons with experience in valve replacement. It also includes heart doctors who are experts in minimally invasive treatment. They work with top imaging specialists.

- Access to the latest valves. We take part in clinical trials. TAVR keeps changing and improving. We receive valves that are still being studied. We also receive some valves after FDA approval when they are available in limited amounts.

- More types of TAVR. Our team can often do a second procedure if an earlier replacement valve wears out.

- Active research. We have taken part in TAVR research since its first clinical trials. A Stanford doctor helped write the paper on TAVR for people with moderate surgical risk. We now take part in a study for people with low risk.

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Transcatheter Aortic Valve Replacement (TAVR)

TAVR uses small tubes instead of surgery to treat aortic valve stenosis. Aortic valve stenosis means the aortic valve is narrowed. Our doctors help improve the procedure and study how it can be used for more people.

Transcatheter aortic valve replacement

Heart valve

Aortic valve

Aortic stenosis

Aortic valve stenosis

Transcatheter aortic valve implantation

TAVI

Percutaneous aortic valve replacement

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 43 of 77 · Item ID B-FS6VFPTOMJ

Original page

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transcatheter aortic valve replacement

Heart valve

aortic valve

aortic stenosis

Aortic valve stenosis

Transcatheter aortic valve implantation

TAVI

percutaneous aortic valve replacement

AI rewrite

Transcatheter Aortic Valve Replacement (TAVR) is a newer way to treat a narrowed heart valve. This problem is called aortic valve stenosis. TAVR does not use open-heart surgery. Instead, doctors use small tubes to put a new aortic valve in place. TAVR is also known as TAVI (Transcatheter Aortic Valve Implantation) or percutaneous aortic valve replacement.

At first, TAVR was only for people who could not have regular surgery. Later, it became a choice for people who had a moderate risk for surgery. Now, doctors are studying if it can help even more people, including those with a low risk.

Our team has a lot of experience with this treatment. We were the first program in Northern California to do TAVR, and the second in the state. We do more than 250 of these procedures each year. This is the most in our area. Our doctors have been doing TAVR since it began and have helped make it better.

When you come to us, a whole team works together to care for you. This team includes heart surgeons, heart doctors who are experts in treatments using small tubes, and doctors who take detailed pictures of your heart.

Because we take part in research studies, we can offer the newest valves. You may be able to get new valves that are still being tested or were just approved by the FDA. If you already have a replacement valve and it wears out, our team can often do a second TAVR procedure to help you. Our doctors are very active in research and help lead studies to improve heart valve care.

To make an appointment with the TAVR program, please call 650-725-2687. To make an appointment with the Valvular Heart Disease Center, please call 650-723-6459.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Transcatheter Aortic Valve Replacement (TAVR)

Our Approach

Transcatheter aortic valve replacement (TAVR) is a newer, minimally invasive approach for treating aortic valve disease. The TAVR procedure has expanded the possibilities for replacing the aortic valve, and we continue to help shape its evolution.

For people who cannot undergo surgery to replace narrowed valves, TAVR first provided an effective option where none existed. Recently, it also presented a minimally invasive alternative for people who could endure an operation, but faced moderate risk. Now, we are investigating its possible use in an even wider range of people.

WHAT WE OFFER YOU FOR TAVR

- Deep experience, with our program as the first in Northern California and the second in the state performing TAVR. We complete more than 250 procedures a year, the region's highest volume.

- Nationally recognized expertise, with our doctors offering TAVR since its beginning. We have years of experience working with the procedure and contributing to its development.

- Team approach that includes surgeons with previous valve replacement experience and interventional cardiologists who have mastered minimally invasive treatment. They partner with top imaging specialists.

- Access to the latest valves by participating in clinical trials. TAVR constantly evolves. We receive valves still being studied, as well as those released in limited quantities upon FDA approval.

- Additional types of TAVR, with our team often able to do a second procedure when an earlier replacement valve wears out.

- Active research program, starting with TAVR's first clinical trials. A Stanford doctor co-authored the paper on TAVR and moderate surgical risk, and we now participate in a study of low risk.

What Is TAVR?

To schedule an appointment with the TAVR program, please call 650-725-2687

To schedule an appointment with the Valvular Heart Disease Center, please call 650-723-6459

Transcatheter Aortic Valve Replacement (TAVR)

TAVR uses minimally invasive tubes instead of surgery to treat aortic valve stenosis, with our doctors helping to refine the procedure and expand its use.

transcatheter aortic valve replacement

Heart valve

aortic valve

aortic stenosis

Aortic valve stenosis

Transcatheter aortic valve implantation

TAVI

percutaneous aortic valve replacement

AI rewrite

Transcatheter Aortic Valve Replacement (TAVR)

Our Approach

TAVR is a newer way to fix a sick aortic valve in your heart. It does not need open surgery. Instead, it uses small tubes. This makes it easier on your body. TAVR has given doctors new ways to replace the aortic valve. We keep helping make it better.

Some people are too sick to have surgery. For them, TAVR was the first good way to fix the valve. Before, they had no choice at all. Later, TAVR also became a good choice for people who could have surgery but faced some risk. Now, we are studying how to use TAVR for even more people.

WHAT WE OFFER YOU FOR TAVR

- Lots of experience. We were the first program in Northern California to do TAVR. We were the second in the whole state. We do more than 250 of these each year. That is the most in our area.

- Skilled, well-known doctors. Our doctors have done TAVR since it started. They have many years of practice. They have also helped make TAVR better.

- A team that works together. Our team has surgeons who have replaced valves before. It also has heart doctors who are experts in this small-tube method. They work with top imaging experts who take clear pictures of your heart.

- New valves through research studies. TAVR keeps changing. We can use valves that are still being studied. We can also use new valves that just got FDA approval but are not yet widely available.

- More kinds of TAVR. Sometimes a valve we put in wears out over time. Our team can often do a second TAVR to fix it.

- An active research program. We took part in the first TAVR studies. A Stanford doctor helped write the paper on TAVR for people with some surgery risk. Now we are part of a study for people with low risk.

What Is TAVR?

To make an appointment with the TAVR program, please call 650-725-2687.

To make an appointment with the Valvular Heart Disease Center, please call 650-723-6459.

Transcatheter Aortic Valve Replacement (TAVR)

TAVR uses small tubes instead of surgery. It treats a tight aortic valve. Our doctors help make this method better. They also help more people get it.

transcatheter aortic valve replacement

Heart valve

aortic valve

aortic stenosis

Aortic valve stenosis

Transcatheter aortic valve implantation

TAVI

percutaneous aortic valve replacement

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Transcatheter Aortic Valve Replacement (TAVR)

On this page

What is transcatheter aortic valve replacement (TAVR)?

Transcatheter aortic valve replacement (TAVR), sometimes called transcatheter aortic valve implantation (TAVI), is a minimally invasive procedure for treating aortic valve stenosis. As blood exits the heart, it passes through the aortic valve. In patients with aortic stenosis, the valve is stiff and narrow and doesn't open and close properly. This makes it harder for the heart to pump blood to the body.

As a result, patients feel short of breath and fatigued, and are at high risk for a heart attack.

The only treatment for aortic valve stenosis is to replace the faulty valve, which traditionally required open-heart surgery. For many patients, however, there is an equally effective, much less invasive alternative - TAVR.

How is the TAVR procedure done?

Instead of opening the chest to remove and replace the damaged valve, the doctor threads a catheter, a thin, flexible tube, through a blood vessel to reach the heart. The catheter is usually threaded through the groin or the chest. Then, the doctor inserts a new valve using the catheter, positioning it inside of the faulty valve to restore healthy blood flow. The TAVR procedure takes much less time than open-heart surgery, and patients recover more quickly, with less pain and scarring.

UCSF has a comprehensive valve program that offers catheter-based treatment options for all four valves in the heart.

Evaluation for TAVR

Initially, TAVR was intended for patients who were at too high a risk to undergo open-heart surgery. Now, it is approved for anyone with severe aortic stenosis. To determine whether it's a good option for you, our team of heart specialists will evaluate your heart health as well as your overall health. You may have all or some of the following tests as part of the evaluation:

- Blood tests. These tests help doctors assess your health. They will likely include a red blood cell count, which indicates how much oxygen is being delivered to your body's cells; electrolyte levels, which can be disturbed by an underlying illness; and tests to measure substances in the bloodstream that reflect how well your kidneys, liver and thyroid are working.

- Chest X-ray. This exam can show the size of your heart and reveal whether you have fluid buildup in your lungs.

- Electrocardiogram (EKG). This is a painless test to assess the heart's electrical activity. Electrodes are taped to your chest, arms and legs to record and measure the signals that cause your heart to beat.

- Transesophageal echocardiogram (TEE). During a TEE, an ultrasound probe is inserted into your esophagus to capture clear images of your heart's chambers and valves. These images are used to assess how blood flows through your heart.

- Cardiac MRI or CT scan. These exams provide detailed images of your blood vessels, the blood flow through your heart, and your heart's structure and motion.

- Right heart catheterization. This exam may be performed to collect more information about how well your heart is pumping. A catheter is inserted through a blood vessel and threaded into the heart, where it measures the blood pressure in your heart and in the main arteries of your lungs. The catheter can also measure cardiac output and blood oxygen levels.

If your doctor suspects that you have coronary artery disease (CAD) in addition to aortic stenosis, you may also have these tests:

- Coronary angiogram. This provides images of the heart's blood vessels to detect blockages. During this test, a catheter is inserted into a blood vessel (usually in the groin area) and threaded up to the heart. A contrast dye (a harmless substance that shows up well on X-rays) is injected through the catheter, and X-ray images track the blood flow to your heart muscle.

- Exercise stress test. This shows your heart's response to exertion. You'll either be given medication that makes your heart work extra hard, or asked to exercise (on a treadmill or stationary bike), while we track your vital signs and monitor your heart's response.

If your tests indicate that you would benefit from TAVR, our cardiac care team will develop a treatment plan for you. Each patient's team includes a cardiologist, cardiac surgeon, interventional cardiologist, nurse practitioner and imaging specialist.

Preparing for a TAVR

Our team will set you up for an appointment with the UCSF Prepare Program. The program staff are dedicated to preparing you for your scheduled procedure, including ensuring you complete all required testing. They will review your medications to be certain you have the correct instructions regarding their use and will discuss your pain control options. They will share all their communication and information with the TAVR team.

How you can prepare for your TAVR procedure:

- Talk to your doctor about your medications. Find out which ones you should or shouldn't take on the days leading up to your procedure, on the day itself, and on the following days.

- Get a dental checkup. Oral bacteria can infect the heart valve.

- Make a recovery plan. It's important to work out details, such as who will take you home, stay with you at home, and do things like help you prepare meals.

TAVR Procedure

We perform TAVR procedures in a state-of-the-art operating room at the UCSF Helen Diller Medical Center at Parnassus Heights. Before the procedure, you will have an echocardiogram, then we will bring you to the operating room.

Our team will have already determined the type of anesthesia they'll use for you, as the best choice depends on certain health factors. You may be given general anesthesia where you are fully asleep, or a sedative that leaves you awake but relaxed and not feeling pain.

Once you are under anesthesia, the doctor will make a small incision in either your groin or chest and insert a catheter into the major artery there. The doctor will use this to convey a new valve to your heart. The new valve is collapsed so it can pass easily through the catheter to your heart.

As part of the procedure, we will place a temporary pacemaker to ensure your heartbeat stays regular. In many patients, we also place a filtration device inside a blood vessel in your neck. This device lowers the risk of stroke by capturing any tiny bits of plaque that come loose during the procedure, preventing them from reaching the brain. The device is inserted through a small incision in your wrist.

Once the replacement valve is inside your damaged valve, the new valve is expanded to full size. It then pushes aside the "leaflets" of the old valve - the flaps that open and close - and takes over the job of regulating blood flow from the heart. The new valve is typically made of bovine (cow) or porcine (pig) tissue; both are compatible with human tissue.

After we place the valve, we remove the temporary pacemaker. If we detect a slow heart rate, however, we may wait until the following day to remove it. The filtration device in your neck is also removed. The whole operation takes about two to three hours - half as long as open-heart surgery.

Another echocardiogram is performed immediately after the procedure to assess how the new valve is functioning. Once that's completed, you'll be taken to the recovery area.

Recovery from TAVR

Immediately after the procedure, you'll recover in the post-anesthesia care unit. You'll then be transferred to either our transitional care or intensive care unit, depending on the level of care you need.

Patients are encouraged to be up and walking within 24 hours. Most patients spend one to two nights in the hospital, where they are closely monitored. During your hospital stay, your cardiac care team will conduct several follow-up tests, including a chest X-ray, EKG and echocardiogram, to ensure your heart is functioning properly.

Before you go home, your team will talk with you about caring for yourself. They'll cover topics such as diet, exercise, care of your incision sites, and any new medications.

After discharge, you'll need to have regular checkups with your primary care doctor, local cardiologist and the heart team, as necessary. These checkups are important to ensure all is well with your new valve and to detect any potential problems early.

Particularly during the first weeks of recovery at home, you'll need to be mindful of how you're feeling. If you experience unusual headaches, dizziness or other concerning symptoms, immediately call the doctor who performed your procedure or dial 911.

We advise patients not to drive for at least 72 hours after the procedure (to allow the incisions to heal) and to avoid strenuous physical activity for 10 days. TAVR recovery is much faster than recovery from open-heart surgery, and most patients can resume normal activities within two weeks. However, it may take one to two months for a full recovery.

You can talk to one of our social workers about the help and care you may need at home as you heal. Some patients may require a homecare nurse or a cardiac rehabilitation program after a TAVR.

This program includes a team of health care professionals who provide an individualized exercise plan, education and support to help you improve your heart health and quality of life. Ask your doctor whether cardiac rehab is right for you.

Awards & recognition

One of the nation's best for heart & vascular care

Rated high-performing hospital for aortic valve surgery

Rated high-performing hospital for transcatheter aortic valve replacement

Related services & conditions

Specialties

Conditions

- Aortic Valve Disease

UCSF Health medical specialists have reviewed this information. It is for educational purposes only and is not intended to replace the advice of your doctor or other health care provider. We encourage you to discuss any questions or concerns you may have with your provider.

AI rewrite

Transcatheter Aortic Valve Replacement (TAVR)

On this page

What is transcatheter aortic valve replacement (TAVR)?

TAVR is a way to fix a bad heart valve. It is sometimes called transcatheter aortic valve implantation (TAVI). It does not need a big surgery. TAVR treats a problem called aortic valve stenosis.

When blood leaves your heart, it goes through the aortic valve. If you have aortic stenosis, this valve is stiff and narrow. It does not open and close the right way. This makes it harder for your heart to pump blood to your body.

Because of this, you may feel short of breath and very tired. You also have a high risk of a heart attack.

The only way to treat aortic valve stenosis is to replace the bad valve. In the past, this needed open-heart surgery. But for many people, there is another way that works just as well. It is much easier on your body. This is TAVR.

How is the TAVR procedure done?

With TAVR, the doctor does not open your chest to take out and replace the bad valve. Instead, the doctor puts a thin, soft tube (called a catheter) into a blood vessel. The doctor pushes the tube through the blood vessel until it reaches your heart. The tube usually goes in through your groin or your chest.

Then the doctor uses the tube to put in a new valve. The doctor places the new valve inside the bad valve. This helps blood flow well again.

TAVR takes much less time than open-heart surgery. People heal faster. They have less pain and smaller scars.

UCSF has a full valve program. We offer tube-based treatments for all four valves in the heart.

Evaluation for TAVR

At first, TAVR was only for people who were too sick for open-heart surgery. Now it is approved for anyone with severe aortic stenosis.

To see if TAVR is right for you, our heart doctors will check your heart and your overall health. You may have some or all of these tests:

- Blood tests. These tests help doctors learn about your health. They may check your red blood cells, which show how much oxygen reaches your body. They may check your electrolyte levels, which can change when you are sick. They also check how well your kidneys, liver, and thyroid are working.

- Chest X-ray. This shows the size of your heart. It can also show if you have fluid in your lungs.

- Electrocardiogram (EKG). This test does not hurt. It checks the electrical activity of your heart. Small pads go on your chest, arms, and legs. They record the signals that make your heart beat.

- Transesophageal echocardiogram (TEE). For this test, a small probe goes into your esophagus (the tube that leads to your stomach). It takes clear pictures of your heart's chambers and valves. These pictures show how blood flows through your heart.

- Cardiac MRI or CT scan. These tests give clear pictures of your blood vessels, your blood flow, and how your heart looks and moves.

- Right heart catheterization. This test gives more facts about how well your heart pumps. The doctor puts a tube into a blood vessel and pushes it into your heart. The tube measures the blood pressure in your heart and in the main arteries of your lungs. It can also measure how much blood your heart pumps and the oxygen in your blood.

If your doctor thinks you also have coronary artery disease (CAD), you may have these tests too:

- Coronary angiogram. This shows pictures of the heart's blood vessels to find blockages. The doctor puts a tube into a blood vessel (usually in the groin). The doctor pushes it up to the heart. Then a special dye goes in through the tube. This dye is safe and shows up well on X-rays. X-rays then track the blood flow to your heart muscle.

- Exercise stress test. This shows how your heart works when it has to work hard. You may get medicine that makes your heart work harder. Or you may exercise on a treadmill or bike. While you do this, we check your vital signs and watch your heart.

If your tests show that TAVR can help you, our heart care team will make a plan for you. Your team will include a cardiologist, a heart surgeon, an interventional cardiologist, a nurse practitioner, and an imaging specialist.

Preparing for a TAVR

Our team will set up an appointment for you with the UCSF Prepare Program. The staff there will help you get ready for your procedure. They will make sure you finish all the tests you need. They will check your medicines so you know exactly how to take them. They will talk with you about ways to control pain. They will share all this information with the TAVR team.

How you can get ready for your TAVR:

- Talk to your doctor about your medicines. Ask which ones you should or should not take in the days before your procedure, on the day of, and in the days after.

- Get a dental checkup. Germs in your mouth can infect the heart valve.

- Make a recovery plan. Plan ahead. Decide who will take you home, who will stay with you, and who will help with things like meals.

TAVR Procedure

We do TAVR in a modern operating room at the UCSF Helen Diller Medical Center at Parnassus Heights. Before the procedure, you will have an echocardiogram. Then we will take you to the operating room.

Our team will have already chosen the type of anesthesia for you. The best choice depends on your health. You may get general anesthesia, which puts you fully to sleep. Or you may get medicine that keeps you awake but relaxed and free of pain.

Once the anesthesia works, the doctor makes a small cut in your groin or chest. The doctor puts a tube into the large artery there. The doctor uses this tube to move a new valve to your heart. The new valve is folded up small so it can pass easily through the tube.

During the procedure, we place a temporary pacemaker. This keeps your heartbeat steady. In many people, we also place a filter device inside a blood vessel in your neck. This device lowers the risk of stroke. It catches any tiny bits of plaque that come loose during the procedure. This stops them from reaching the brain. The device goes in through a small cut in your wrist.

Once the new valve is inside your bad valve, we open it to full size. The new valve pushes aside the flaps of the old valve. These flaps are called leaflets. They open and close. The new valve then takes over the job of controlling blood flow from the heart. The new valve is usually made from cow or pig tissue. Both work well with human tissue.

After we place the valve, we take out the temporary pacemaker. But if your heart rate is slow, we may wait until the next day to take it out. We also take out the filter device in your neck.

The whole procedure takes about two to three hours. That is half as long as open-heart surgery.

Right after the procedure, we do another echocardiogram. This checks how the new valve is working. After that, we take you to the recovery area.

Recovery from TAVR

Right after the procedure, you will rest in the post-anesthesia care unit. Then we will move you to our transitional care unit or our intensive care unit. This depends on how much care you need.

We want you to get up and walk within 24 hours. Most people stay in the hospital one to two nights. We watch you closely. During your stay, your heart care team will do several follow-up tests. These include a chest X-ray, an EKG, and an echocardiogram. These tests make sure your heart is working well.

Before you go home, your team will talk with you about how to take care of yourself. They will cover your diet, exercise, care of your cut sites, and any new medicines.

After you go home, you will need regular checkups. These will be with your primary care doctor, your local cardiologist, and the heart team, as needed. These checkups make sure your new valve is working well. They also help find any problems early.

Be careful about how you feel, especially in the first weeks at home. If you have unusual headaches, dizziness, or other worrying signs, call the doctor who did your procedure right away. Or call 911.

We tell people not to drive for at least 72 hours after the procedure. This gives your cuts time to heal. Do not do hard physical activity for 10 days. TAVR recovery is much faster than open-heart surgery. Most people can go back to normal activities within two weeks. But a full recovery may take one to two months.

You can talk to one of our social workers about the help you may need at home while you heal. Some people may need a homecare nurse or a cardiac rehab program after TAVR.

A cardiac rehab program has a team of health care workers. They give you an exercise plan made just for you. They also give you education and support. This helps you improve your heart health and your quality of life. Ask your doctor if cardiac rehab is right for you.

Awards & recognition

One of the nation's best for heart and vascular care

Rated high-performing hospital for aortic valve surgery

Rated high-performing hospital for transcatheter aortic valve replacement

Related services & conditions

Specialties

Conditions

- Aortic Valve Disease

UCSF Health medical specialists have reviewed this information. It is for learning only. It is not meant to replace the advice of your doctor or other health care provider. We hope you will talk about any questions or concerns with your provider.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

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Original page

Transcatheter Aortic Valve Replacement (TAVR)

On this page

What is transcatheter aortic valve replacement (TAVR)?

Transcatheter aortic valve replacement (TAVR), sometimes called transcatheter aortic valve implantation (TAVI), is a minimally invasive procedure for treating aortic valve stenosis. As blood exits the heart, it passes through the aortic valve. In patients with aortic stenosis, the valve is stiff and narrow and doesn't open and close properly. This makes it harder for the heart to pump blood to the body.

As a result, patients feel short of breath and fatigued, and are at high risk for a heart attack.

The only treatment for aortic valve stenosis is to replace the faulty valve, which traditionally required open-heart surgery. For many patients, however, there is an equally effective, much less invasive alternative - TAVR.

How is the TAVR procedure done?

Instead of opening the chest to remove and replace the damaged valve, the doctor threads a catheter, a thin, flexible tube, through a blood vessel to reach the heart. The catheter is usually threaded through the groin or the chest. Then, the doctor inserts a new valve using the catheter, positioning it inside of the faulty valve to restore healthy blood flow. The TAVR procedure takes much less time than open-heart surgery, and patients recover more quickly, with less pain and scarring.

UCSF has a comprehensive valve program that offers catheter-based treatment options for all four valves in the heart.

Evaluation for TAVR

Initially, TAVR was intended for patients who were at too high a risk to undergo open-heart surgery. Now, it is approved for anyone with severe aortic stenosis. To determine whether it's a good option for you, our team of heart specialists will evaluate your heart health as well as your overall health. You may have all or some of the following tests as part of the evaluation:

- Blood tests. These tests help doctors assess your health. They will likely include a red blood cell count, which indicates how much oxygen is being delivered to your body's cells; electrolyte levels, which can be disturbed by an underlying illness; and tests to measure substances in the bloodstream that reflect how well your kidneys, liver and thyroid are working.

- Chest X-ray. This exam can show the size of your heart and reveal whether you have fluid buildup in your lungs.

- Electrocardiogram (EKG). This is a painless test to assess the heart's electrical activity. Electrodes are taped to your chest, arms and legs to record and measure the signals that cause your heart to beat.

- Transesophageal echocardiogram (TEE). During a TEE, an ultrasound probe is inserted into your esophagus to capture clear images of your heart's chambers and valves. These images are used to assess how blood flows through your heart.

- Cardiac MRI or CT scan. These exams provide detailed images of your blood vessels, the blood flow through your heart, and your heart's structure and motion.

- Right heart catheterization. This exam may be performed to collect more information about how well your heart is pumping. A catheter is inserted through a blood vessel and threaded into the heart, where it measures the blood pressure in your heart and in the main arteries of your lungs. The catheter can also measure cardiac output and blood oxygen levels.

If your doctor suspects that you have coronary artery disease (CAD) in addition to aortic stenosis, you may also have these tests:

- Coronary angiogram. This provides images of the heart's blood vessels to detect blockages. During this test, a catheter is inserted into a blood vessel (usually in the groin area) and threaded up to the heart. A contrast dye (a harmless substance that shows up well on X-rays) is injected through the catheter, and X-ray images track the blood flow to your heart muscle.

- Exercise stress test. This shows your heart's response to exertion. You'll either be given medication that makes your heart work extra hard, or asked to exercise (on a treadmill or stationary bike), while we track your vital signs and monitor your heart's response.

If your tests indicate that you would benefit from TAVR, our cardiac care team will develop a treatment plan for you. Each patient's team includes a cardiologist, cardiac surgeon, interventional cardiologist, nurse practitioner and imaging specialist.

Preparing for a TAVR

Our team will set you up for an appointment with the UCSF Prepare Program. The program staff are dedicated to preparing you for your scheduled procedure, including ensuring you complete all required testing. They will review your medications to be certain you have the correct instructions regarding their use and will discuss your pain control options. They will share all their communication and information with the TAVR team.

How you can prepare for your TAVR procedure:

- Talk to your doctor about your medications. Find out which ones you should or shouldn't take on the days leading up to your procedure, on the day itself, and on the following days.

- Get a dental checkup. Oral bacteria can infect the heart valve.

- Make a recovery plan. It's important to work out details, such as who will take you home, stay with you at home, and do things like help you prepare meals.

TAVR Procedure

We perform TAVR procedures in a state-of-the-art operating room at the UCSF Helen Diller Medical Center at Parnassus Heights. Before the procedure, you will have an echocardiogram, then we will bring you to the operating room.

Our team will have already determined the type of anesthesia they'll use for you, as the best choice depends on certain health factors. You may be given general anesthesia where you are fully asleep, or a sedative that leaves you awake but relaxed and not feeling pain.

Once you are under anesthesia, the doctor will make a small incision in either your groin or chest and insert a catheter into the major artery there. The doctor will use this to convey a new valve to your heart. The new valve is collapsed so it can pass easily through the catheter to your heart.

As part of the procedure, we will place a temporary pacemaker to ensure your heartbeat stays regular. In many patients, we also place a filtration device inside a blood vessel in your neck. This device lowers the risk of stroke by capturing any tiny bits of plaque that come loose during the procedure, preventing them from reaching the brain. The device is inserted through a small incision in your wrist.

Once the replacement valve is inside your damaged valve, the new valve is expanded to full size. It then pushes aside the "leaflets" of the old valve - the flaps that open and close - and takes over the job of regulating blood flow from the heart. The new valve is typically made of bovine (cow) or porcine (pig) tissue; both are compatible with human tissue.

After we place the valve, we remove the temporary pacemaker. If we detect a slow heart rate, however, we may wait until the following day to remove it. The filtration device in your neck is also removed. The whole operation takes about two to three hours - half as long as open-heart surgery.

Another echocardiogram is performed immediately after the procedure to assess how the new valve is functioning. Once that's completed, you'll be taken to the recovery area.

Recovery from TAVR

Immediately after the procedure, you'll recover in the post-anesthesia care unit. You'll then be transferred to either our transitional care or intensive care unit, depending on the level of care you need.

Patients are encouraged to be up and walking within 24 hours. Most patients spend one to two nights in the hospital, where they are closely monitored. During your hospital stay, your cardiac care team will conduct several follow-up tests, including a chest X-ray, EKG and echocardiogram, to ensure your heart is functioning properly.

Before you go home, your team will talk with you about caring for yourself. They'll cover topics such as diet, exercise, care of your incision sites, and any new medications.

After discharge, you'll need to have regular checkups with your primary care doctor, local cardiologist and the heart team, as necessary. These checkups are important to ensure all is well with your new valve and to detect any potential problems early.

Particularly during the first weeks of recovery at home, you'll need to be mindful of how you're feeling. If you experience unusual headaches, dizziness or other concerning symptoms, immediately call the doctor who performed your procedure or dial 911.

We advise patients not to drive for at least 72 hours after the procedure (to allow the incisions to heal) and to avoid strenuous physical activity for 10 days. TAVR recovery is much faster than recovery from open-heart surgery, and most patients can resume normal activities within two weeks. However, it may take one to two months for a full recovery.

You can talk to one of our social workers about the help and care you may need at home as you heal. Some patients may require a homecare nurse or a cardiac rehabilitation program after a TAVR.

This program includes a team of health care professionals who provide an individualized exercise plan, education and support to help you improve your heart health and quality of life. Ask your doctor whether cardiac rehab is right for you.

Awards & recognition

One of the nation's best for heart & vascular care

Rated high-performing hospital for aortic valve surgery

Rated high-performing hospital for transcatheter aortic valve replacement

Related services & conditions

Specialties

Conditions

- Aortic Valve Disease

UCSF Health medical specialists have reviewed this information. It is for educational purposes only and is not intended to replace the advice of your doctor or other health care provider. We encourage you to discuss any questions or concerns you may have with your provider.

AI rewrite

Transcatheter aortic valve replacement is often called TAVR. It is a procedure to fix a stiff and narrow heart valve. This problem is called aortic stenosis. When your valve is stiff, it does not open and close the right way. Your heart has to work harder to pump blood to your body. This can make you feel tired and short of breath. It also raises your risk for a heart attack.

The only way to fix this is to replace the bad valve. In the past, this meant open-heart surgery. Now, doctors can use TAVR. It works just as well but is much easier on your body.

During TAVR, the doctor does not open your chest. Instead, they use a thin, flexible tube called a catheter. The doctor puts the tube into a blood vessel in your groin or chest. They guide the tube up to your heart. Then, they use the tube to place a new valve inside your old one. This helps your blood flow the right way again. TAVR takes less time than open-heart surgery. You will also heal faster and have less pain and scarring.

TAVR is approved for anyone with severe aortic stenosis. To see if TAVR is right for you, your heart doctors will check your health. You may need some of these tests:

* Blood tests: These check your red blood cells, oxygen levels, and how well your organs work.

* Chest X-ray: This shows the size of your heart. It also checks for fluid in your lungs.

* EKG: Stickers are placed on your skin to check your heart's electrical signals. It does not hurt.

* TEE: A doctor guides an ultrasound wand down your throat. This takes clear pictures of your heart and blood flow.

* MRI or CT scan: These take detailed pictures of your heart and blood vessels.

* Right heart catheterization: A thin tube is put into a blood vessel and moved to your heart. It measures blood pressure and oxygen levels.

If your doctor thinks you might have blocked blood vessels, you may need two more tests:

* Coronary angiogram: A tube is put into your groin and moved to your heart. The doctor injects a safe dye. X-rays track the dye to find blockages.

* Exercise stress test: You will walk on a treadmill, ride a bike, or take a special medicine. This shows how your heart works when it works hard.

If TAVR is right for you, a team of doctors and nurses will make a plan. Staff will help you get ready. Here is how you can prepare:

* Talk to your doctor about your medicines. Ask which ones you should take or stop taking before and after your procedure.

* Get a dental checkup. Bacteria in your mouth can travel to your heart and cause an infection.

* Make a plan for going home. Find someone to drive you home. You will also need someone to stay with you and help make meals.

Before the procedure, you will have an ultrasound of your heart. Then, you will go to the operating room. You will get medicine so you do not feel pain. You may be fully asleep, or you may be awake but very relaxed.

The doctor will make a small cut in your groin or chest. They will put a thin tube into a large blood vessel. A folded new valve is pushed through the tube to your heart.

The doctor will also place a temporary pacemaker. This keeps your heartbeat steady. You may also get a small filter put into a blood vessel in your neck. The doctor puts this in through a small cut in your wrist. The filter catches any bits of plaque that break loose. This keeps them from reaching your brain and helps prevent a stroke.

When the new valve is in the right spot, it is opened up. It pushes the flaps of your old valve out of the way. The new valve takes over to pump blood. It is made from cow or pig tissue, which is safe for humans.

After the valve is in place, the doctor removes the filter and the pacemaker. If your heart rate is slow, the pacemaker might stay in until the next day. The whole process takes about two to three hours. This is half the time of open-heart surgery. You will have another ultrasound right away to check the new valve.

After the procedure, you will wake up in a recovery room. Then, you will move to a hospital room. You will be asked to get up and walk within 24 hours. Most people stay in the hospital for one or two nights. Your care team will watch you closely. They will do tests like an X-ray, EKG, and ultrasound to make sure your heart is working well.

Before you go home, your team will teach you how to care for yourself. They will talk about diet, exercise, your cuts, and new medicines.

After you go home, you must see your doctors for regular checkups. This helps them make sure your new valve is working right. Pay close attention to how you feel. If you have strange headaches, dizziness, or other worrying signs, call your doctor or 911 right away.

Do not drive for at least 72 hours. This gives your cuts time to heal. Do not do hard work or heavy exercise for 10 days. Most people can get back to normal activities in two weeks. A full recovery takes one to two months.

If you need extra help at home, a social worker can help you. You might need a home nurse or a cardiac rehab program. Cardiac rehab is a special program with exercise and education to help your heart get stronger. Ask your doctor if this is right for you.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Original page

Transcatheter Aortic Valve Replacement (TAVR)

On this page

What is transcatheter aortic valve replacement (TAVR)?

Transcatheter aortic valve replacement (TAVR), sometimes called transcatheter aortic valve implantation (TAVI), is a minimally invasive procedure for treating aortic valve stenosis. As blood exits the heart, it passes through the aortic valve. In patients with aortic stenosis, the valve is stiff and narrow and doesn't open and close properly. This makes it harder for the heart to pump blood to the body.

As a result, patients feel short of breath and fatigued, and are at high risk for a heart attack.

The only treatment for aortic valve stenosis is to replace the faulty valve, which traditionally required open-heart surgery. For many patients, however, there is an equally effective, much less invasive alternative - TAVR.

How is the TAVR procedure done?

Instead of opening the chest to remove and replace the damaged valve, the doctor threads a catheter, a thin, flexible tube, through a blood vessel to reach the heart. The catheter is usually threaded through the groin or the chest. Then, the doctor inserts a new valve using the catheter, positioning it inside of the faulty valve to restore healthy blood flow. The TAVR procedure takes much less time than open-heart surgery, and patients recover more quickly, with less pain and scarring.

UCSF has a comprehensive valve program that offers catheter-based treatment options for all four valves in the heart.

Evaluation for TAVR

Initially, TAVR was intended for patients who were at too high a risk to undergo open-heart surgery. Now, it is approved for anyone with severe aortic stenosis. To determine whether it's a good option for you, our team of heart specialists will evaluate your heart health as well as your overall health. You may have all or some of the following tests as part of the evaluation:

- Blood tests. These tests help doctors assess your health. They will likely include a red blood cell count, which indicates how much oxygen is being delivered to your body's cells; electrolyte levels, which can be disturbed by an underlying illness; and tests to measure substances in the bloodstream that reflect how well your kidneys, liver and thyroid are working.

- Chest X-ray. This exam can show the size of your heart and reveal whether you have fluid buildup in your lungs.

- Electrocardiogram (EKG). This is a painless test to assess the heart's electrical activity. Electrodes are taped to your chest, arms and legs to record and measure the signals that cause your heart to beat.

- Transesophageal echocardiogram (TEE). During a TEE, an ultrasound probe is inserted into your esophagus to capture clear images of your heart's chambers and valves. These images are used to assess how blood flows through your heart.

- Cardiac MRI or CT scan. These exams provide detailed images of your blood vessels, the blood flow through your heart, and your heart's structure and motion.

- Right heart catheterization. This exam may be performed to collect more information about how well your heart is pumping. A catheter is inserted through a blood vessel and threaded into the heart, where it measures the blood pressure in your heart and in the main arteries of your lungs. The catheter can also measure cardiac output and blood oxygen levels.

If your doctor suspects that you have coronary artery disease (CAD) in addition to aortic stenosis, you may also have these tests:

- Coronary angiogram. This provides images of the heart's blood vessels to detect blockages. During this test, a catheter is inserted into a blood vessel (usually in the groin area) and threaded up to the heart. A contrast dye (a harmless substance that shows up well on X-rays) is injected through the catheter, and X-ray images track the blood flow to your heart muscle.

- Exercise stress test. This shows your heart's response to exertion. You'll either be given medication that makes your heart work extra hard, or asked to exercise (on a treadmill or stationary bike), while we track your vital signs and monitor your heart's response.

If your tests indicate that you would benefit from TAVR, our cardiac care team will develop a treatment plan for you. Each patient's team includes a cardiologist, cardiac surgeon, interventional cardiologist, nurse practitioner and imaging specialist.

Preparing for a TAVR

Our team will set you up for an appointment with the UCSF Prepare Program. The program staff are dedicated to preparing you for your scheduled procedure, including ensuring you complete all required testing. They will review your medications to be certain you have the correct instructions regarding their use and will discuss your pain control options. They will share all their communication and information with the TAVR team.

How you can prepare for your TAVR procedure:

- Talk to your doctor about your medications. Find out which ones you should or shouldn't take on the days leading up to your procedure, on the day itself, and on the following days.

- Get a dental checkup. Oral bacteria can infect the heart valve.

- Make a recovery plan. It's important to work out details, such as who will take you home, stay with you at home, and do things like help you prepare meals.

TAVR Procedure

We perform TAVR procedures in a state-of-the-art operating room at the UCSF Helen Diller Medical Center at Parnassus Heights. Before the procedure, you will have an echocardiogram, then we will bring you to the operating room.

Our team will have already determined the type of anesthesia they'll use for you, as the best choice depends on certain health factors. You may be given general anesthesia where you are fully asleep, or a sedative that leaves you awake but relaxed and not feeling pain.

Once you are under anesthesia, the doctor will make a small incision in either your groin or chest and insert a catheter into the major artery there. The doctor will use this to convey a new valve to your heart. The new valve is collapsed so it can pass easily through the catheter to your heart.

As part of the procedure, we will place a temporary pacemaker to ensure your heartbeat stays regular. In many patients, we also place a filtration device inside a blood vessel in your neck. This device lowers the risk of stroke by capturing any tiny bits of plaque that come loose during the procedure, preventing them from reaching the brain. The device is inserted through a small incision in your wrist.

Once the replacement valve is inside your damaged valve, the new valve is expanded to full size. It then pushes aside the "leaflets" of the old valve - the flaps that open and close - and takes over the job of regulating blood flow from the heart. The new valve is typically made of bovine (cow) or porcine (pig) tissue; both are compatible with human tissue.

After we place the valve, we remove the temporary pacemaker. If we detect a slow heart rate, however, we may wait until the following day to remove it. The filtration device in your neck is also removed. The whole operation takes about two to three hours - half as long as open-heart surgery.

Another echocardiogram is performed immediately after the procedure to assess how the new valve is functioning. Once that's completed, you'll be taken to the recovery area.

Recovery from TAVR

Immediately after the procedure, you'll recover in the post-anesthesia care unit. You'll then be transferred to either our transitional care or intensive care unit, depending on the level of care you need.

Patients are encouraged to be up and walking within 24 hours. Most patients spend one to two nights in the hospital, where they are closely monitored. During your hospital stay, your cardiac care team will conduct several follow-up tests, including a chest X-ray, EKG and echocardiogram, to ensure your heart is functioning properly.

Before you go home, your team will talk with you about caring for yourself. They'll cover topics such as diet, exercise, care of your incision sites, and any new medications.

After discharge, you'll need to have regular checkups with your primary care doctor, local cardiologist and the heart team, as necessary. These checkups are important to ensure all is well with your new valve and to detect any potential problems early.

Particularly during the first weeks of recovery at home, you'll need to be mindful of how you're feeling. If you experience unusual headaches, dizziness or other concerning symptoms, immediately call the doctor who performed your procedure or dial 911.

We advise patients not to drive for at least 72 hours after the procedure (to allow the incisions to heal) and to avoid strenuous physical activity for 10 days. TAVR recovery is much faster than recovery from open-heart surgery, and most patients can resume normal activities within two weeks. However, it may take one to two months for a full recovery.

You can talk to one of our social workers about the help and care you may need at home as you heal. Some patients may require a homecare nurse or a cardiac rehabilitation program after a TAVR.

This program includes a team of health care professionals who provide an individualized exercise plan, education and support to help you improve your heart health and quality of life. Ask your doctor whether cardiac rehab is right for you.

Awards & recognition

One of the nation's best for heart & vascular care

Rated high-performing hospital for aortic valve surgery

Rated high-performing hospital for transcatheter aortic valve replacement

Related services & conditions

Specialties

Conditions

- Aortic Valve Disease

UCSF Health medical specialists have reviewed this information. It is for educational purposes only and is not intended to replace the advice of your doctor or other health care provider. We encourage you to discuss any questions or concerns you may have with your provider.

AI rewrite

Transcatheter Aortic Valve Replacement (TAVR)

On this page

What is transcatheter aortic valve replacement (TAVR)?

Transcatheter aortic valve replacement is called TAVR. It is sometimes called TAVI. TAVR is a less invasive procedure to treat aortic valve stenosis.

Blood leaves the heart through the aortic valve. In aortic stenosis, this valve becomes stiff and narrow. It does not open and close the right way. This makes it harder for the heart to pump blood to the body.

Because of this, patients may feel short of breath and tired. They also have a high risk for a heart attack.

The only treatment for aortic valve stenosis is to replace the bad valve. In the past, this was done with open-heart surgery. For many patients, TAVR works just as well. It is also much less invasive.

How is the TAVR procedure done?

The doctor does not open the chest to remove and replace the bad valve. Instead, the doctor uses a catheter. A catheter is a thin, bendable tube. The doctor puts the catheter into a blood vessel and moves it to the heart. The catheter is usually put in through the groin or the chest.

The doctor uses the catheter to place a new valve inside the bad valve. This helps blood flow the right way again.

TAVR takes much less time than open-heart surgery. Patients often heal faster. They usually have less pain and less scarring.

UCSF has a full valve program. It offers catheter-based treatment options for all four heart valves.

Evaluation for TAVR

At first, TAVR was for patients who were too high risk for open-heart surgery. Now, it is approved for anyone with severe aortic stenosis.

To see if TAVR is a good choice for you, our heart team will check your heart health and your overall health. You may have some or all of these tests:

- Blood tests. These tests help doctors check your health. They may include a red blood cell count. This shows how much oxygen is being carried to your body’s cells. They may also check electrolytes. Electrolytes can be changed by an illness. Other tests may check substances in your blood. These show how well your kidneys, liver and thyroid are working.

- Chest X-ray. This test can show the size of your heart. It can also show if fluid has built up in your lungs.

- Electrocardiogram (EKG). This is a painless test. It checks the heart’s electrical activity. Small sticky pads called electrodes are placed on your chest, arms and legs. They record and measure the signals that make your heart beat.

- Transesophageal echocardiogram (TEE). During a TEE, an ultrasound probe is put into your esophagus. The esophagus is the tube that carries food from your mouth to your stomach. This test takes clear pictures of your heart’s chambers and valves.

The pictures show how blood flows through your heart.

- Cardiac MRI or CT scan. These tests take detailed pictures. They show your blood vessels, blood flow through your heart, and the shape and movement of your heart.

- Right heart catheterization. This test may be done to learn more about how well your heart pumps. A catheter is put into a blood vessel and moved into the heart. It measures blood pressure in your heart and in the main arteries of your lungs. The catheter can also measure how much blood your heart pumps and the oxygen level in your blood.

If your doctor thinks you may have coronary artery disease (CAD) along with aortic stenosis, you may also have these tests:

- Coronary angiogram. This test takes pictures of the heart's blood vessels to look for blockages. During the test, a catheter is put into a blood vessel, usually in the groin. It is moved up to the heart. A contrast dye is put through the catheter. This dye is harmless and shows up well on X-rays. X-ray pictures show blood flow to your heart muscle.

- Exercise stress test. This test shows how your heart responds to hard work. You may get medicine that makes your heart work harder. Or you may exercise on a treadmill or stationary bike. We will check your vital signs and watch how your heart responds.

If your tests show that TAVR may help you, our heart care team will make a treatment plan for you. Each patient's team includes a cardiologist, heart surgeon, interventional cardiologist, nurse practitioner and imaging specialist.

Preparing for TAVR

Our team will set up an appointment for you with the UCSF Prepare Program. The program staff will help you get ready for your procedure. They will make sure you complete all needed tests.

They will review your medicines. They will make sure you know which medicines to take and when to take them. They will also talk with you about pain control options. They will share all information with the TAVR team.

How you can prepare for your TAVR procedure:

- Talk to your doctor about your medicines. Ask which ones you should or should not take before the procedure, on the day of the procedure, and after the procedure.

- Get a dental checkup. Bacteria in the mouth can infect the heart valve.

- Make a recovery plan. Plan who will take you home. Plan who will stay with you at home. Plan who can help with things like making meals.

TAVR Procedure

We do TAVR procedures in a modern operating room at the UCSF Helen Diller Medical Center at Parnassus Heights. Before the procedure, you will have an echocardiogram. Then we will bring you to the operating room.

Our team will already have chosen the type of anesthesia for you. The best choice depends on your health. You may get general anesthesia, which means you are fully asleep. Or you may get a sedative, which keeps you awake but relaxed and not feeling pain.

Once you have anesthesia, the doctor will make a small cut in your groin or chest. The doctor will put a catheter into the major artery there. The doctor uses the catheter to move the new valve to your heart. The new valve is folded down so it can pass through the catheter easily.

As part of the procedure, we will place a temporary pacemaker. This helps keep your heartbeat regular. In many patients, we also place a filter device inside a blood vessel in the neck. This device lowers the risk of stroke. It catches tiny bits of plaque that may break loose during the procedure. This helps keep them from reaching the brain. The device is put in through a small cut in your wrist.

Once the new valve is inside your damaged valve, it is opened to full size. It pushes aside the leaflets of the old valve. Leaflets are the flaps that open and close. The new valve then takes over the job of controlling blood flow from the heart.

The new valve is usually made from cow or pig tissue. Both can work with human tissue.

After we place the valve, we remove the temporary pacemaker. If we find that your heart rate is slow, we may wait until the next day to remove it. The filter device in your neck is also removed.

The whole operation takes about two to three hours. This is about half as long as open-heart surgery.

Another echocardiogram is done right after the procedure. This checks how the new valve is working. After that, you will be taken to the recovery area.

Recovery from TAVR

Right after the procedure, you will recover in the post-anesthesia care unit. Then you will move to either our transitional care unit or intensive care unit. This depends on the level of care you need.

Patients are encouraged to get up and walk within 24 hours. Most patients stay in the hospital for one to two nights. You will be watched closely.

During your hospital stay, your heart care team will do follow-up tests. These may include a chest X-ray, EKG and echocardiogram. These tests make sure your heart is working properly.

Before you go home, your team will talk with you about how to care for yourself. They will talk about diet, exercise, care for your incision sites, and any new medicines.

After you leave the hospital, you will need regular checkups. These may be with your primary care doctor, local cardiologist and the heart team, as needed. These visits are important. They help make sure your new valve is working well. They also help find any problems early.

Pay close attention to how you feel, especially during the first weeks at home. If you have unusual headaches, dizziness or other symptoms that worry you, call the doctor who did your procedure right away or call 911.

We advise patients not to drive for at least 72 hours after the procedure. This gives the cuts time to heal. We also advise patients to avoid hard physical activity for 10 days.

Recovery from TAVR is much faster than recovery from open-heart surgery. Most patients can return to normal activities within two weeks. But full recovery may take one to two months.

You can talk to one of our social workers about the help and care you may need at home while you heal. Some patients may need a homecare nurse or a cardiac rehabilitation program after TAVR.

Cardiac rehabilitation is also called cardiac rehab. This program has a team of health care professionals. They give you an exercise plan made for you. They also give education and support. The goal is to help improve your heart health and quality of life. Ask your doctor if cardiac rehab is right for you.

Awards & recognition

One of the nation’s best for heart and vascular care

Rated high-performing hospital for aortic valve surgery

Rated high-performing hospital for transcatheter aortic valve replacement

Related services & conditions

Specialties

Conditions

- Aortic Valve Disease

UCSF Health medical specialists have reviewed this information. It is for education only. It is not meant to replace advice from your doctor or other health care provider. Please talk with your provider about any questions or concerns you have.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

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Original page

What is TAVR? (TAVI)

During this minimally invasive procedure, a new valve is inserted without removing the old, damaged valve. The new valve is placed inside the diseased valve. The surgery may be called a transcatheter aortic valve replacement (TAVR) or transcatheter aortic valve implantation (TAVI).

Valve-within-valve - How does it work?

Somewhat similar to placing a stent in an artery, the TAVR approach delivers a fully collapsible replacement valve to the valve site through a catheter.

Once the new valve is expanded, it pushes the old valve leaflets out of the way, and the tissue in the replacement valve takes over the job of regulating blood flow.

This procedure is available for people with symptomatic severe aortic stenosis at low, intermediate or high risk for standard valve replacement surgery.

What is involved in a TAVR procedure?

Valve replacement usually requires an open-heart procedure with a "sternotomy," in which the chest is surgically opened for the procedure. The TAVR or TAVI procedure can be done through very small openings that leave all the chest bones in place.

While TAVR is not without risks, it provides beneficial treatment options to people who may not have been considered for valve replacement. A person's experience with a TAVR procedure may be similar to a coronary angiogram in terms of recovery. You will likely spend less time in the hospital after TAVR compared to surgical valve replacement.

The TAVR procedure is performed using one of two approaches, allowing the cardiologist or surgeon to choose which one provides the best and safest way to access the valve:

- Transfemoral approach: Entering through the femoral artery (large artery in the groin), which does not require a surgical incision in the chest.

- Transapical approach: Using a minimally invasive surgical approach with a small incision in the chest and entering through a large artery in the chest or through the tip of the left ventricle (the apex).

Who's a good candidate for this type of valve surgery?

The procedure is available to people in all risk categories.

TAVR can be an effective option to improve the quality of life in people who otherwise have limited choices for the repair of their aortic valve.

View Aortic Stenosis: Considerations for Replacing Your Aortic Valve (PDF)

Is there financial assistance available for people in need of TAVR or TAVI valve replacement?

TAVR is approved and available for qualifying patients receiving Medicare and Medicaid. More information is available on the Centers for Medicare & Medicaid Services (CMS) website. Learn more about health care laws and government programs seeking to provide affordable coverage at HealthCare.gov(link opens in new window).

Additional resources:

- Answers by Heart sheets: What is TAVR? (PDF) | Spanish (PDF)

- Pre-surgery Checklist (PDF) | Spanish (PDF)

- Recovery Milestones Checklist (PDF) | Spanish (PDF)

Support that lifts you up

Our online community of patients, survivors and caregivers is here to keep you going no matter the obstacles. We've been there, and we won't let you do it alone.

AI rewrite

What is TAVR? (TAVI)

TAVR is a way to fix a sick heart valve. The doctor does not take out the old valve. Instead, the doctor puts a new valve inside the old one. This surgery has two names. One is transcatheter aortic valve replacement (TAVR). The other is transcatheter aortic valve implantation (TAVI).

Valve-within-valve – How does it work?

This is a little like putting a stent in an artery. The doctor uses a thin tube called a catheter. The new valve is folded up small. The doctor moves it through the tube to your heart.

Then the new valve opens up. It pushes the old valve flaps out of the way. Now the new valve does the job. It helps your blood flow the right way.

This procedure can help people who have a problem called severe aortic stenosis. It can help people who have low, medium, or high risk for normal valve surgery.

What is involved in a TAVR procedure?

Most valve surgery is open-heart surgery. The doctor cuts open the chest. This is called a "sternotomy."

TAVR is not like that. The doctor uses very small openings. Your chest bones stay in place.

TAVR has some risks. But it can help people who could not have normal valve surgery. Getting better after TAVR may feel a lot like getting better after a heart test called a coronary angiogram. You will likely spend less time in the hospital than with open-heart valve surgery.

The doctor can do TAVR in one of two ways. The doctor picks the way that is best and safest for you:

- Transfemoral way: The doctor goes in through a big artery in your groin. This does not need a cut in your chest.

- Transapical way: The doctor makes a small cut in your chest. Then the doctor goes in through a big artery in the chest or through the tip of the lower left part of the heart. This tip is called the apex.

Who's a good candidate for this type of valve surgery?

TAVR can help people in all risk groups.

TAVR can be a good choice to help you feel better. It may help people who have few other ways to fix their aortic valve.

View Aortic Stenosis: Considerations for Replacing Your Aortic Valve (PDF)

Is there financial help for people who need TAVR or TAVI valve replacement?

TAVR is approved for people who qualify and have Medicare and Medicaid. You can learn more on the Centers for Medicare & Medicaid Services (CMS) website. You can also learn about health care laws and government programs that help with cost at HealthCare.gov (link opens in new window).

More resources:

- Answers by Heart sheets: What is TAVR? (PDF) | Spanish (PDF)

- Pre-surgery Checklist (PDF) | Spanish (PDF)

- Recovery Milestones Checklist (PDF) | Spanish (PDF)

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We have an online community of patients, survivors, and caregivers. We are here to help you keep going, no matter what gets in your way. We have been there. We will not let you do it alone.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

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Original page

What is TAVR? (TAVI)

During this minimally invasive procedure, a new valve is inserted without removing the old, damaged valve. The new valve is placed inside the diseased valve. The surgery may be called a transcatheter aortic valve replacement (TAVR) or transcatheter aortic valve implantation (TAVI).

Valve-within-valve - How does it work?

Somewhat similar to placing a stent in an artery, the TAVR approach delivers a fully collapsible replacement valve to the valve site through a catheter.

Once the new valve is expanded, it pushes the old valve leaflets out of the way, and the tissue in the replacement valve takes over the job of regulating blood flow.

This procedure is available for people with symptomatic severe aortic stenosis at low, intermediate or high risk for standard valve replacement surgery.

What is involved in a TAVR procedure?

Valve replacement usually requires an open-heart procedure with a "sternotomy," in which the chest is surgically opened for the procedure. The TAVR or TAVI procedure can be done through very small openings that leave all the chest bones in place.

While TAVR is not without risks, it provides beneficial treatment options to people who may not have been considered for valve replacement. A person's experience with a TAVR procedure may be similar to a coronary angiogram in terms of recovery. You will likely spend less time in the hospital after TAVR compared to surgical valve replacement.

The TAVR procedure is performed using one of two approaches, allowing the cardiologist or surgeon to choose which one provides the best and safest way to access the valve:

- Transfemoral approach: Entering through the femoral artery (large artery in the groin), which does not require a surgical incision in the chest.

- Transapical approach: Using a minimally invasive surgical approach with a small incision in the chest and entering through a large artery in the chest or through the tip of the left ventricle (the apex).

Who's a good candidate for this type of valve surgery?

The procedure is available to people in all risk categories.

TAVR can be an effective option to improve the quality of life in people who otherwise have limited choices for the repair of their aortic valve.

View Aortic Stenosis: Considerations for Replacing Your Aortic Valve (PDF)

Is there financial assistance available for people in need of TAVR or TAVI valve replacement?

TAVR is approved and available for qualifying patients receiving Medicare and Medicaid. More information is available on the Centers for Medicare & Medicaid Services (CMS) website. Learn more about health care laws and government programs seeking to provide affordable coverage at HealthCare.gov(link opens in new window).

Additional resources:

- Answers by Heart sheets: What is TAVR? (PDF) | Spanish (PDF)

- Pre-surgery Checklist (PDF) | Spanish (PDF)

- Recovery Milestones Checklist (PDF) | Spanish (PDF)

Support that lifts you up

Our online community of patients, survivors and caregivers is here to keep you going no matter the obstacles. We've been there, and we won't let you do it alone.

AI rewrite

What is TAVR or TAVI?

TAVR is a way to fix a bad heart valve. The doctor puts a new valve inside your old, damaged valve. They do not take the old valve out. This is called transcatheter aortic valve replacement (TAVR). It is also called TAVI.

How does it work?

The doctor uses a long, thin tube called a catheter. They use the tube to move a folded-up new valve into your heart. When the new valve opens, it pushes the old valve out of the way. The new valve takes over to help your blood flow the right way. You can get this if you have severe aortic stenosis. This is a stiff heart valve that causes symptoms. TAVR is for people at low, medium, or high risk for regular open-heart surgery.

What happens during TAVR?

Regular heart surgery requires a large cut to open your chest bone. TAVR uses very small cuts. Your chest bones stay in place. TAVR still has some risks. But it helps people who might not be able to have regular surgery. Your recovery may be like the recovery from a heart test called a coronary angiogram. You will likely spend less time in the hospital than you would for regular surgery.

The doctor will pick the best and safest way to put in the new valve. There are two main ways to do this:

- Transfemoral approach: The doctor goes through a large blood vessel in your groin. This does not need a cut in your chest.

- Transapical approach: The doctor makes a small cut in your chest. They go through a large blood vessel in your chest or through the bottom tip of your heart.

Who can get this surgery?

People in all risk groups can get TAVR. It is a good choice to help you live a better life. It is very helpful if you have few choices to fix your valve. You can read more in the PDF guide called "Aortic Stenosis: Considerations for Replacing Your Aortic Valve."

Is there help to pay for TAVR?

Medicare and Medicaid will pay for TAVR if you qualify. You can find more facts on the Centers for Medicare & Medicaid Services (CMS) website. You can also learn about health care laws and coverage you can afford at HealthCare.gov.

More helpful guides:

- Answers by Heart sheets: What is TAVR? (PDFs in English and Spanish)

- Pre-surgery Checklist (PDFs in English and Spanish)

- Recovery Milestones Checklist (PDFs in English and Spanish)

Support to help you:

We have an online group of patients, survivors, and caregivers. We are here to help you keep going through hard times. We have been there, and you do not have to do this alone.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

Item 50 of 77 · Item ID B-2ESPRVU8FI

Original page

What is TAVR? (TAVI)

During this minimally invasive procedure, a new valve is inserted without removing the old, damaged valve. The new valve is placed inside the diseased valve. The surgery may be called a transcatheter aortic valve replacement (TAVR) or transcatheter aortic valve implantation (TAVI).

Valve-within-valve - How does it work?

Somewhat similar to placing a stent in an artery, the TAVR approach delivers a fully collapsible replacement valve to the valve site through a catheter.

Once the new valve is expanded, it pushes the old valve leaflets out of the way, and the tissue in the replacement valve takes over the job of regulating blood flow.

This procedure is available for people with symptomatic severe aortic stenosis at low, intermediate or high risk for standard valve replacement surgery.

What is involved in a TAVR procedure?

Valve replacement usually requires an open-heart procedure with a "sternotomy," in which the chest is surgically opened for the procedure. The TAVR or TAVI procedure can be done through very small openings that leave all the chest bones in place.

While TAVR is not without risks, it provides beneficial treatment options to people who may not have been considered for valve replacement. A person's experience with a TAVR procedure may be similar to a coronary angiogram in terms of recovery. You will likely spend less time in the hospital after TAVR compared to surgical valve replacement.

The TAVR procedure is performed using one of two approaches, allowing the cardiologist or surgeon to choose which one provides the best and safest way to access the valve:

- Transfemoral approach: Entering through the femoral artery (large artery in the groin), which does not require a surgical incision in the chest.

- Transapical approach: Using a minimally invasive surgical approach with a small incision in the chest and entering through a large artery in the chest or through the tip of the left ventricle (the apex).

Who's a good candidate for this type of valve surgery?

The procedure is available to people in all risk categories.

TAVR can be an effective option to improve the quality of life in people who otherwise have limited choices for the repair of their aortic valve.

View Aortic Stenosis: Considerations for Replacing Your Aortic Valve (PDF)

Is there financial assistance available for people in need of TAVR or TAVI valve replacement?

TAVR is approved and available for qualifying patients receiving Medicare and Medicaid. More information is available on the Centers for Medicare & Medicaid Services (CMS) website. Learn more about health care laws and government programs seeking to provide affordable coverage at HealthCare.gov(link opens in new window).

Additional resources:

- Answers by Heart sheets: What is TAVR? (PDF) | Spanish (PDF)

- Pre-surgery Checklist (PDF) | Spanish (PDF)

- Recovery Milestones Checklist (PDF) | Spanish (PDF)

Support that lifts you up

Our online community of patients, survivors and caregivers is here to keep you going no matter the obstacles. We've been there, and we won't let you do it alone.

AI rewrite

What is TAVR? (TAVI)

TAVR is a procedure to put in a new heart valve. The old, damaged valve is not removed. The new valve is placed inside the old valve.

TAVR stands for transcatheter aortic valve replacement. It may also be called TAVI, which stands for transcatheter aortic valve implantation.

How does “valve-within-valve” work?

TAVR is a little like placing a stent in an artery. The new valve can fold down very small. It is moved to the heart through a thin tube called a catheter.

When the new valve opens up, it pushes the old valve flaps out of the way. The new valve tissue then takes over the job of controlling blood flow.

This procedure is for people who have severe aortic stenosis with symptoms. It is available for people at low, medium, or high risk for standard valve replacement surgery.

What happens during a TAVR procedure?

Valve replacement often needs open-heart surgery. This may include a sternotomy. A sternotomy means the chest is opened during surgery.

TAVR can be done through very small openings. The chest bones stay in place.

TAVR has risks. But it can help people who may not have had other choices for valve replacement. Recovery from TAVR may feel similar to recovery from a coronary angiogram. You will likely spend less time in the hospital after TAVR than after surgical valve replacement.

TAVR is done in one of two ways. The cardiologist or surgeon chooses the best and safest way to reach the valve.

- Transfemoral approach: The doctor enters through the femoral artery. This is a large artery in the groin. This approach does not need a cut in the chest.

- Transapical approach: The doctor uses a small cut in the chest. The doctor enters through a large artery in the chest or through the tip of the left ventricle. This tip is called the apex.

Who may be a good candidate for this valve surgery?

This procedure is available to people in all risk groups.

TAVR can help improve quality of life for people who have limited choices to repair their aortic valve.

View Aortic Stenosis: Considerations for Replacing Your Aortic Valve (PDF)

Is financial help available for people who need TAVR or TAVI?

TAVR is approved and available for qualifying patients who have Medicare or Medicaid. More information is on the Centers for Medicare & Medicaid Services (CMS) website.

Learn more about health care laws and government programs that help provide affordable coverage at HealthCare.gov (link opens in new window).

Additional resources:

- Answers by Heart sheets: What is TAVR? (PDF) | Spanish (PDF)

- Pre-surgery Checklist (PDF) | Spanish (PDF)

- Recovery Milestones Checklist (PDF) | Spanish (PDF)

Support that lifts you up

Our online community of patients, survivors, and caregivers is here to help you keep going, no matter what gets in the way. We have been there, and we will not let you do it alone.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

Item 51 of 77 · Item ID B-Y088O5OR15

Original page

Aortic stenosis is a tightening of the aortic valve in the heart. If you have been diagnosed with severe aortic stenosis, you may have to decide whether to fix your valve.

Treatments may include transcatheter aortic valve replacement (TAVR), surgical aortic valve replacement (SAVR) or symptom management. TAVR is also referred to as transcatheter aortic valve implantation, or TAVI.

Depending on what you are considering, booklets in the sections below can help you understand your condition and learn about your options.

You, your family, and your clinicians can begin to talk about what decision is best for you.

If you have been told that TAVR (also called TAVI) or SAVR are options for you or your loved one, use the decision aids below to help decide which treatment might work best for you.

Booklet: This

booklet (in PDF format) will help you understand what aortic stenosis is and what treatment options are available to you.

Download in English |

Spanish | French

Video: This video is for patients with severe aortic stenosis who are deciding between TAVR and open-heart surgery.

If you have been told that TAVR (also called TAVI) or symptom management are options for you or your loved one, use the decision aids below to help decide which treatment might work best for you.

Booklet:This booklet (in PDF format) will help you understand what aortic stenosis is and what treatment options are available to you.

Download in English | Spanish |

French

Video: This video is

for patients with severe aortic stenosis who are deciding between TAVR and symptom management.

AI rewrite

Aortic stenosis means the aortic valve in your heart has become tight. If you have severe aortic stenosis, you may need to decide whether to have your valve fixed.

Treatment choices may include:

- Transcatheter aortic valve replacement (TAVR)

- Surgical aortic valve replacement (SAVR)

- Managing your symptoms

TAVR is also called transcatheter aortic valve implantation, or TAVI.

The booklets below can help you learn about your condition and your treatment choices.

You, your family, and your doctors can talk about which choice may be best for you.

If you have been told that TAVR, also called TAVI, or SAVR are choices for you or your loved one, use the decision aids below. They can help you decide which treatment may work best for you.

Booklet: This booklet, in PDF format, will help you understand aortic stenosis and your treatment choices.

Download in English | Spanish | French

Video: This video is for patients with severe aortic stenosis who are deciding between TAVR and open-heart surgery.

If you have been told that TAVR, also called TAVI, or symptom management are choices for you or your loved one, use the decision aids below. They can help you decide which treatment may work best for you.

Booklet: This booklet, in PDF format, will help you understand aortic stenosis and your treatment choices.

Download in English | Spanish | French

Video: This video is for patients with severe aortic stenosis who are deciding between TAVR and symptom management.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 52 of 77 · Item ID B-DN0YBY4AWT

Original page

Aortic stenosis is a tightening of the aortic valve in the heart. If you have been diagnosed with severe aortic stenosis, you may have to decide whether to fix your valve.

Treatments may include transcatheter aortic valve replacement (TAVR), surgical aortic valve replacement (SAVR) or symptom management. TAVR is also referred to as transcatheter aortic valve implantation, or TAVI.

Depending on what you are considering, booklets in the sections below can help you understand your condition and learn about your options.

You, your family, and your clinicians can begin to talk about what decision is best for you.

If you have been told that TAVR (also called TAVI) or SAVR are options for you or your loved one, use the decision aids below to help decide which treatment might work best for you.

Booklet: This

booklet (in PDF format) will help you understand what aortic stenosis is and what treatment options are available to you.

Download in English |

Spanish | French

Video: This video is for patients with severe aortic stenosis who are deciding between TAVR and open-heart surgery.

If you have been told that TAVR (also called TAVI) or symptom management are options for you or your loved one, use the decision aids below to help decide which treatment might work best for you.

Booklet:This booklet (in PDF format) will help you understand what aortic stenosis is and what treatment options are available to you.

Download in English | Spanish |

French

Video: This video is

for patients with severe aortic stenosis who are deciding between TAVR and symptom management.

AI rewrite

Aortic stenosis is when the aortic valve in your heart gets too tight. The aortic valve is one of the valves that helps blood move through your heart. If a doctor told you that you have severe aortic stenosis, you may need to decide if you want to fix your valve.

There are a few ways to treat it. One way is called TAVR. This stands for transcatheter aortic valve replacement. It is also called TAVI. Another way is called SAVR. This stands for surgical aortic valve replacement. A third way is to manage your symptoms instead of fixing the valve.

The booklets in the sections below can help you learn about your condition. They can help you learn about your choices, too.

You, your family, and your care team can start to talk about what is best for you.

Maybe a doctor told you that TAVR (also called TAVI) or SAVR could work for you or someone you love. If so, the tools below can help you pick the best treatment.

Booklet: This booklet is a PDF. It will help you learn what aortic stenosis is. It will also help you learn about your treatment choices.

Download in English | Spanish | French

Video: This video is for people with severe aortic stenosis. It is for people who are choosing between TAVR and open-heart surgery.

Maybe a doctor told you that TAVR (also called TAVI) or managing your symptoms could work for you or someone you love. If so, the tools below can help you pick the best treatment.

Booklet: This booklet is a PDF. It will help you learn what aortic stenosis is. It will also help you learn about your treatment choices.

Download in English | Spanish | French

Video: This video is for people with severe aortic stenosis. It is for people who are choosing between TAVR and managing their symptoms.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 53 of 77 · Item ID B-BYVZK3I8BZ

Original page

Aortic stenosis is a tightening of the aortic valve in the heart. If you have been diagnosed with severe aortic stenosis, you may have to decide whether to fix your valve.

Treatments may include transcatheter aortic valve replacement (TAVR), surgical aortic valve replacement (SAVR) or symptom management. TAVR is also referred to as transcatheter aortic valve implantation, or TAVI.

Depending on what you are considering, booklets in the sections below can help you understand your condition and learn about your options.

You, your family, and your clinicians can begin to talk about what decision is best for you.

If you have been told that TAVR (also called TAVI) or SAVR are options for you or your loved one, use the decision aids below to help decide which treatment might work best for you.

Booklet: This

booklet (in PDF format) will help you understand what aortic stenosis is and what treatment options are available to you.

Download in English |

Spanish | French

Video: This video is for patients with severe aortic stenosis who are deciding between TAVR and open-heart surgery.

If you have been told that TAVR (also called TAVI) or symptom management are options for you or your loved one, use the decision aids below to help decide which treatment might work best for you.

Booklet:This booklet (in PDF format) will help you understand what aortic stenosis is and what treatment options are available to you.

Download in English | Spanish |

French

Video: This video is

for patients with severe aortic stenosis who are deciding between TAVR and symptom management.

AI rewrite

Aortic stenosis means the aortic valve in your heart is too tight. If your valve is very tight, you may need to decide if you want to fix it.

Treatments include transcatheter aortic valve replacement (TAVR), surgical aortic valve replacement (SAVR), or treating your symptoms. TAVR is also called TAVI.

The booklets below can help you learn about your heart problem and your choices. You, your family, and your doctors can talk about what is best for you.

If you are told that TAVR or SAVR are choices for you or your loved one, use the tools below to help you decide.

Booklet: This PDF booklet helps you understand aortic stenosis and your treatment choices.

Download in English | Spanish | French

Video: This video is for patients with severe aortic stenosis. Watch it if you are deciding between TAVR and open-heart surgery.

If you are told that TAVR or treating your symptoms are choices for you or your loved one, use the tools below to help you decide.

Booklet: This PDF booklet helps you understand aortic stenosis and your treatment choices.

Download in English | Spanish | French

Video: This video is for patients with severe aortic stenosis. Watch it if you are deciding between TAVR and treating your symptoms.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 54 of 77 · Item ID B-3UVYR0QF4B

Original page

Transcatheter aortic valve replacement (TAVR) is a minimally invasive procedure to replace the aortic valve of the heart for high-risk patients with severe aortic valve stenosis who are not candidates for traditional open heart surgery. TAVR is performed on a beating heart and does not require a large surgical incision or the support of a heart-lung machine.

At the Brigham and Women's Hospital (BWH) Heart & Vascular Center, one of the nation's most experienced centers in transcatheter valve replacement, cardiac surgeons and interventional cardiologists work alongside imaging specialists and anesthesiologists to perform this advanced technique. Brigham and Women's Hospital was one of two hospitals in New England and one of 24 nationwide to examine the benefits of TAVR as part of the PARTNER (Placement of AoRTic TraNscathetER Valve) clinical trial. TAVR was approved by the FDA in 2011. Brigham and Women's Hospital successfully completed its first TAVR procedure in 2009 and has performed over 1000 since then.

This deep experience and their collaboration with a multidisciplinary team of cardiac specialists through the Interventional Cardiology Program enables our cardiac surgeons to handle the most complicated cases, with a range of treatment options that improve the lives of patients experiencing aortic stenosis.

With 47,000 outpatient visits each year, the Heart & Vascular Center is one of the largest in the United States, treating over 7,000 inpatients and performing more than 8,000 procedures annually at our state-of-the-art Shapiro Cardiovascular Center.

During transcatheter aortic valve replacement, a cardiac surgeon and interventional cardiologist use imaging equipment to evaluate the arteries. Your cardiac specialist then decides which entry point on the body is the best and safest way to access the aortic valve.

A catheter with a stented valve on the end is guided into the heart through a small incision in one of the following locations:

The valve is then deployed. The catheter is then removed. Imaging equipment helps confirm that the new valve is working properly before the catheter is removed and the access sites are closed.

Because TAVR is a minimally invasive technique, patients experience a quicker recovery than they would from a traditional, open-heart valve replacement. The entire procedure, performed under local and/or general anesthesia, takes 90 minutes while open heart surgery typically takes four to six hours, followed by a two to three month recovery period. With TAVR, patients resume daily living quickly with a high quality of life.

Transcatheter aortic valve replacement is a relatively new procedure reserved for people who:

Learn more about aortic valve repair and replacement options.

The Heart & Vascular Center is located in the Shapiro Cardiovascular Center, across the street from BWH's main 75 Francis Street entrance. The Heart & Vascular Center brings together the full range of services in one location, fostering seamless and coordinated care for all cardiovascular patients.

Prior to surgery, you will be scheduled for a visit to the Watkins Clinic in the Shapiro Cardiovascular Center for preoperative information and tests.

The day of surgery, your care will be provided by surgeons, anesthesiologists and nurses who specialize in aortic valve disease. The Heart & Vascular Center is home to one of the most advanced hybrid operating rooms in the country. After surgery, you will recover in the post-surgical care unit where you will receive comprehensive care by an experienced surgical and nursing staff.

During your surgery, family and friends can wait in the Shapiro Family Center where staff members will provide surgery updates.

In addition to our cardiac surgeons and interventional cardiologists, patients also benefit from the teamwork of medical cardiologists, cardiovascular imaging experts and radiologists, and anesthesiologists, all experts in cardiovascular disease. They work alongside nurses, physician assistants, physical therapists, dietitians and social workers to achieve outstanding outcomes for our patients.

Learn more about heart valve repair and replacement surgery in our health library.

Visit the Kessler Health Education Library in the Bretholtz Center for Patients and Families.

Access a complete directory of patient and family services.

Learn about the Watkins Clinic in the Shapiro Center.

Download Cardiac Surgery: A Guide for Patients in English or Spanish.

For over a century, a leader in patient care, medical education and research, with expertise in virtually every specialty of medicine and surgery.

About BWH

AI rewrite

Transcatheter aortic valve replacement is also called TAVR. It is a less invasive procedure to replace the aortic valve in the heart. It is used for high-risk patients who have severe aortic valve stenosis. These patients cannot have regular open heart surgery.

TAVR is done while the heart is still beating. It does not need a large cut. It does not need a heart-lung machine.

At the Brigham and Women’s Hospital Heart & Vascular Center, heart surgeons and heart doctors work together to do TAVR. They also work with imaging specialists and anesthesia doctors.

Brigham and Women’s Hospital was one of two hospitals in New England and one of 24 hospitals in the United States that studied TAVR in the PARTNER clinical trial. TAVR was approved by the FDA in 2011. Brigham and Women’s Hospital did its first TAVR procedure in 2009. Since then, it has done more than 1,000 TAVR procedures.

This experience helps the heart team care for very complex cases. The team offers different treatment options to help people with aortic stenosis.

The Heart & Vascular Center has 47,000 outpatient visits each year. It is one of the largest centers in the United States. It treats more than 7,000 patients in the hospital each year. It also does more than 8,000 procedures each year at the Shapiro Cardiovascular Center.

During TAVR, a heart surgeon and an interventional cardiologist use imaging equipment to look at the arteries. Your heart specialist then chooses the best and safest place to enter the body and reach the aortic valve.

A catheter is a thin tube. A new valve is attached to the end of the catheter. The catheter is guided into the heart through a small cut.

The new valve is then opened and placed. The catheter is removed. Imaging equipment helps the team make sure the new valve is working well. Then the catheter is removed and the entry sites are closed.

TAVR is less invasive than regular open-heart valve replacement. Because of this, patients often recover faster.

The procedure takes about 90 minutes. It is done with local anesthesia, general anesthesia, or both. Open heart surgery usually takes 4 to 6 hours. Recovery after open heart surgery can take 2 to 3 months. With TAVR, patients often return to daily life quickly and have a high quality of life.

TAVR is a fairly new procedure. It is saved for certain people.

You can learn more about aortic valve repair and replacement options.

The Heart & Vascular Center is in the Shapiro Cardiovascular Center. It is across the street from the main Brigham and Women’s Hospital entrance at 75 Francis Street. The Heart & Vascular Center brings many services together in one place. This helps patients get smooth and well-planned heart and blood vessel care.

Before surgery, you will have a visit at the Watkins Clinic in the Shapiro Cardiovascular Center. You will get information before the procedure. You will also have tests.

On the day of surgery, your care will be given by surgeons, anesthesia doctors, and nurses who specialize in aortic valve disease. The Heart & Vascular Center has one of the most advanced hybrid operating rooms in the country.

After surgery, you will recover in the post-surgical care unit. There, an experienced surgery and nursing team will care for you.

During your surgery, family and friends can wait in the Shapiro Family Center. Staff members will give them updates about your surgery.

Patients also benefit from a large care team. This team includes heart surgeons, interventional cardiologists, medical cardiologists, heart imaging experts, radiologists, and anesthesia doctors. They are all experts in heart and blood vessel disease.

They work with nurses, physician assistants, physical therapists, dietitians, and social workers. Together, they help care for patients.

You can learn more about heart valve repair and replacement surgery in the health library.

You can visit the Kessler Health Education Library in the Bretholtz Center for Patients and Families.

You can access a full list of patient and family services.

You can learn about the Watkins Clinic in the Shapiro Center.

You can download “Cardiac Surgery: A Guide for Patients” in English or Spanish.

For more than 100 years, Brigham and Women’s Hospital has been a leader in patient care, medical education, and research. It has experts in almost every area of medicine and surgery.

About BWH

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 55 of 77 · Item ID B-F1I3LDQYUN

Original page

Transcatheter aortic valve replacement (TAVR) is a minimally invasive procedure to replace the aortic valve of the heart for high-risk patients with severe aortic valve stenosis who are not candidates for traditional open heart surgery. TAVR is performed on a beating heart and does not require a large surgical incision or the support of a heart-lung machine.

At the Brigham and Women's Hospital (BWH) Heart & Vascular Center, one of the nation's most experienced centers in transcatheter valve replacement, cardiac surgeons and interventional cardiologists work alongside imaging specialists and anesthesiologists to perform this advanced technique. Brigham and Women's Hospital was one of two hospitals in New England and one of 24 nationwide to examine the benefits of TAVR as part of the PARTNER (Placement of AoRTic TraNscathetER Valve) clinical trial. TAVR was approved by the FDA in 2011. Brigham and Women's Hospital successfully completed its first TAVR procedure in 2009 and has performed over 1000 since then.

This deep experience and their collaboration with a multidisciplinary team of cardiac specialists through the Interventional Cardiology Program enables our cardiac surgeons to handle the most complicated cases, with a range of treatment options that improve the lives of patients experiencing aortic stenosis.

With 47,000 outpatient visits each year, the Heart & Vascular Center is one of the largest in the United States, treating over 7,000 inpatients and performing more than 8,000 procedures annually at our state-of-the-art Shapiro Cardiovascular Center.

During transcatheter aortic valve replacement, a cardiac surgeon and interventional cardiologist use imaging equipment to evaluate the arteries. Your cardiac specialist then decides which entry point on the body is the best and safest way to access the aortic valve.

A catheter with a stented valve on the end is guided into the heart through a small incision in one of the following locations:

The valve is then deployed. The catheter is then removed. Imaging equipment helps confirm that the new valve is working properly before the catheter is removed and the access sites are closed.

Because TAVR is a minimally invasive technique, patients experience a quicker recovery than they would from a traditional, open-heart valve replacement. The entire procedure, performed under local and/or general anesthesia, takes 90 minutes while open heart surgery typically takes four to six hours, followed by a two to three month recovery period. With TAVR, patients resume daily living quickly with a high quality of life.

Transcatheter aortic valve replacement is a relatively new procedure reserved for people who:

Learn more about aortic valve repair and replacement options.

The Heart & Vascular Center is located in the Shapiro Cardiovascular Center, across the street from BWH's main 75 Francis Street entrance. The Heart & Vascular Center brings together the full range of services in one location, fostering seamless and coordinated care for all cardiovascular patients.

Prior to surgery, you will be scheduled for a visit to the Watkins Clinic in the Shapiro Cardiovascular Center for preoperative information and tests.

The day of surgery, your care will be provided by surgeons, anesthesiologists and nurses who specialize in aortic valve disease. The Heart & Vascular Center is home to one of the most advanced hybrid operating rooms in the country. After surgery, you will recover in the post-surgical care unit where you will receive comprehensive care by an experienced surgical and nursing staff.

During your surgery, family and friends can wait in the Shapiro Family Center where staff members will provide surgery updates.

In addition to our cardiac surgeons and interventional cardiologists, patients also benefit from the teamwork of medical cardiologists, cardiovascular imaging experts and radiologists, and anesthesiologists, all experts in cardiovascular disease. They work alongside nurses, physician assistants, physical therapists, dietitians and social workers to achieve outstanding outcomes for our patients.

Learn more about heart valve repair and replacement surgery in our health library.

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Access a complete directory of patient and family services.

Learn about the Watkins Clinic in the Shapiro Center.

Download Cardiac Surgery: A Guide for Patients in English or Spanish.

For over a century, a leader in patient care, medical education and research, with expertise in virtually every specialty of medicine and surgery.

About BWH

AI rewrite

TAVR stands for transcatheter aortic valve replacement. It is a way to replace the aortic valve in your heart. The aortic valve helps blood flow the right way. Sometimes this valve gets too narrow. This is called severe aortic valve stenosis. TAVR is for people who are very sick and cannot have open heart surgery.

TAVR is a less invasive procedure. This means it does not need a large cut. The heart keeps beating during TAVR. You do not need a heart-lung machine.

The Brigham and Women's Hospital (BWH) Heart & Vascular Center is one of the most experienced centers in the country for this kind of valve replacement. Heart surgeons and heart doctors work together with imaging experts and anesthesia doctors to do this procedure.

BWH was one of two hospitals in New England, and one of 24 in the whole country, to study how TAVR helps people. This study was called the PARTNER trial. The FDA approved TAVR in 2011. BWH did its first TAVR in 2009. Since then, it has done over 1,000 of them.

This deep experience helps our heart surgeons handle even the hardest cases. They work with a team of heart experts through the Interventional Cardiology Program. They offer many treatment choices to help people with aortic stenosis.

The Heart & Vascular Center is one of the largest in the United States. It has 47,000 outpatient visits each year. It cares for over 7,000 inpatients. It does more than 8,000 procedures each year. This all happens at our modern Shapiro Cardiovascular Center.

Here is how TAVR works. First, a heart surgeon and a heart doctor use imaging tools to look at your arteries. Then your heart doctor picks the best and safest spot on your body to reach the aortic valve.

A thin tube is called a catheter. It has a new valve on the end. The catheter goes into your heart through a small cut. The new valve is put in place. Imaging tools check that the new valve works well. Then the catheter is taken out. The doctors close the spots where the catheter went in.

Because TAVR is less invasive, you get better faster than with open heart surgery. The whole procedure takes about 90 minutes. You may get local or general anesthesia. Open heart surgery usually takes four to six hours. After open heart surgery, it can take two to three months to get better. With TAVR, you can go back to your daily life quickly and feel good.

TAVR is a fairly new procedure. It is only for certain people.

Learn more about ways to repair and replace the aortic valve.

The Heart & Vascular Center is in the Shapiro Cardiovascular Center. It is across the street from the main BWH entrance at 75 Francis Street. The Heart & Vascular Center has all its services in one place. This makes care smooth and well planned for all heart patients.

Before your surgery, you will have a visit at the Watkins Clinic in the Shapiro Cardiovascular Center. You will get information and tests before your surgery.

On the day of your surgery, your care team will include surgeons, anesthesia doctors, and nurses. They all focus on aortic valve disease. The Heart & Vascular Center has one of the most advanced hybrid operating rooms in the country. After surgery, you will recover in the post-surgical care unit. There, a skilled surgical and nursing staff will take good care of you.

During your surgery, your family and friends can wait in the Shapiro Family Center. Staff there will give them updates about your surgery.

Many experts help care for you. These include heart surgeons, heart doctors, medical heart doctors, heart imaging experts, radiologists, and anesthesia doctors. They all know a lot about heart disease. They work with nurses, physician assistants, physical therapists, dietitians, and social workers. Together, they help you get the best results.

Learn more about heart valve repair and replacement surgery in our health library.

Visit the Kessler Health Education Library in the Bretholtz Center for Patients and Families.

Find a full list of services for patients and families.

Learn about the Watkins Clinic in the Shapiro Center.

Download Cardiac Surgery: A Guide for Patients in English or Spanish.

For over a hundred years, we have been a leader in patient care, medical teaching, and research. We have experts in almost every kind of medicine and surgery.

About BWH

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 56 of 77 · Item ID B-8DWOECBPMB

Original page

Transcatheter aortic valve replacement (TAVR) is a minimally invasive procedure to replace the aortic valve of the heart for high-risk patients with severe aortic valve stenosis who are not candidates for traditional open heart surgery. TAVR is performed on a beating heart and does not require a large surgical incision or the support of a heart-lung machine.

At the Brigham and Women's Hospital (BWH) Heart & Vascular Center, one of the nation's most experienced centers in transcatheter valve replacement, cardiac surgeons and interventional cardiologists work alongside imaging specialists and anesthesiologists to perform this advanced technique. Brigham and Women's Hospital was one of two hospitals in New England and one of 24 nationwide to examine the benefits of TAVR as part of the PARTNER (Placement of AoRTic TraNscathetER Valve) clinical trial. TAVR was approved by the FDA in 2011. Brigham and Women's Hospital successfully completed its first TAVR procedure in 2009 and has performed over 1000 since then.

This deep experience and their collaboration with a multidisciplinary team of cardiac specialists through the Interventional Cardiology Program enables our cardiac surgeons to handle the most complicated cases, with a range of treatment options that improve the lives of patients experiencing aortic stenosis.

With 47,000 outpatient visits each year, the Heart & Vascular Center is one of the largest in the United States, treating over 7,000 inpatients and performing more than 8,000 procedures annually at our state-of-the-art Shapiro Cardiovascular Center.

During transcatheter aortic valve replacement, a cardiac surgeon and interventional cardiologist use imaging equipment to evaluate the arteries. Your cardiac specialist then decides which entry point on the body is the best and safest way to access the aortic valve.

A catheter with a stented valve on the end is guided into the heart through a small incision in one of the following locations:

The valve is then deployed. The catheter is then removed. Imaging equipment helps confirm that the new valve is working properly before the catheter is removed and the access sites are closed.

Because TAVR is a minimally invasive technique, patients experience a quicker recovery than they would from a traditional, open-heart valve replacement. The entire procedure, performed under local and/or general anesthesia, takes 90 minutes while open heart surgery typically takes four to six hours, followed by a two to three month recovery period. With TAVR, patients resume daily living quickly with a high quality of life.

Transcatheter aortic valve replacement is a relatively new procedure reserved for people who:

Learn more about aortic valve repair and replacement options.

The Heart & Vascular Center is located in the Shapiro Cardiovascular Center, across the street from BWH's main 75 Francis Street entrance. The Heart & Vascular Center brings together the full range of services in one location, fostering seamless and coordinated care for all cardiovascular patients.

Prior to surgery, you will be scheduled for a visit to the Watkins Clinic in the Shapiro Cardiovascular Center for preoperative information and tests.

The day of surgery, your care will be provided by surgeons, anesthesiologists and nurses who specialize in aortic valve disease. The Heart & Vascular Center is home to one of the most advanced hybrid operating rooms in the country. After surgery, you will recover in the post-surgical care unit where you will receive comprehensive care by an experienced surgical and nursing staff.

During your surgery, family and friends can wait in the Shapiro Family Center where staff members will provide surgery updates.

In addition to our cardiac surgeons and interventional cardiologists, patients also benefit from the teamwork of medical cardiologists, cardiovascular imaging experts and radiologists, and anesthesiologists, all experts in cardiovascular disease. They work alongside nurses, physician assistants, physical therapists, dietitians and social workers to achieve outstanding outcomes for our patients.

Learn more about heart valve repair and replacement surgery in our health library.

Visit the Kessler Health Education Library in the Bretholtz Center for Patients and Families.

Access a complete directory of patient and family services.

Learn about the Watkins Clinic in the Shapiro Center.

Download Cardiac Surgery: A Guide for Patients in English or Spanish.

For over a century, a leader in patient care, medical education and research, with expertise in virtually every specialty of medicine and surgery.

About BWH

AI rewrite

Transcatheter aortic valve replacement (TAVR) is a way to replace the aortic valve in your heart. It is for high-risk patients who have severe aortic valve stenosis. TAVR is for people who cannot have regular open-heart surgery. The doctor does not make a large cut. Your heart keeps beating during the procedure. You do not need a heart-lung machine.

At the Brigham and Women's Hospital (BWH) Heart & Vascular Center, our team has a lot of experience with TAVR. Heart doctors, picture experts, and anesthesiologists (doctors who give sleep medicine) work together. Our hospital was one of 24 in the country to study TAVR in a big test called the PARTNER trial. The FDA approved TAVR in 2011. Our hospital did its first TAVR in 2009. We have done over 1,000 of them since then. Our team can handle the hardest cases to help you feel better.

Our Heart & Vascular Center is one of the largest in the country. We have 47,000 clinic visits each year. We treat over 7,000 hospital patients and do more than 8,000 procedures every year at our modern Shapiro Cardiovascular Center.

During TAVR, your doctors use special pictures to look at your arteries. They decide the safest place to make a small cut on your body. A thin tube called a catheter is put through this small cut. A new valve with a wire frame (stent) is on the end of this tube. The doctor guides the tube into your heart. The new valve is then put into place. Doctors take pictures to make sure the new valve works right. Then, they take the tube out and close the cut.

Because the cut is small, you will heal faster than you would from open-heart surgery. You will get medicine to numb the area or medicine to make you sleep (anesthesia). The TAVR procedure takes about 90 minutes. Regular open-heart surgery takes four to six hours and takes two to three months to heal. With TAVR, you can get back to your normal life quickly and feel good.

TAVR is a newer choice for high-risk people who cannot have open-heart surgery. You can learn more about your choices for valve repair and replacement.

The Heart & Vascular Center is in the Shapiro Cardiovascular Center. It is across the street from the main hospital entrance at 75 Francis Street. Before your surgery, you will visit the Watkins Clinic in the Shapiro Center. You will get information and have tests done there.

On the day of your surgery, experts in heart valve disease will care for you. We have very modern operating rooms. After the surgery, you will wake up in a special recovery room. Expert nurses and staff will take great care of you. Your family and friends can wait in the Shapiro Family Center. Our staff will give them updates about your surgery.

Many experts work together to give you the best care. Your team includes heart doctors, sleep doctors, and doctors who read X-rays. You will also see nurses, physician assistants, physical therapists, dietitians, and social workers.

You can learn more about heart surgery in our health library. You can also visit the Kessler Health Education Library in the Bretholtz Center for Patients and Families. We have a full list of services for patients and families. You can learn more about the Watkins Clinic. You can also download our guide called "Cardiac Surgery: A Guide for Patients" in English or Spanish. Brigham and Women's Hospital has been a leader in patient care and research for over 100 years.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 57 of 77 · Item ID B-PI4HN3QXKH

Original page

TAVI (transcatheter aortic valve implantation) is a procedure to improve the blood flow in your heart by replacing an aortic valve that does not open fully.

Page updated:

Next review:

Page updated:

Next review:

TAVI explained

What is a TAVI?

TAVI is a procedure to replace the aortic valve in your heart. It helps improve the blood flow through your heart and into the rest of your body.

TAVI stands for transcatheter aortic valve implantation. It's sometimes called transcatheter aortic valve replacement (TAVR).

Why do you need a TAVI?

You may need a TAVI if the aortic valve in your heart is narrow and does not open properly. This is called aortic stenosis.

Your heart valves play an important role in keeping blood flowing through your heart. The aortic valve opens to let blood flow from your heart to your body.

Benefits and risks

What are the benefits and risks of a TAVI procedure?

Your treatment team will give you medicine to help reduce your risk of blood clots or a stroke.

The benefits and risks of TAVI are different for everyone and depend on how serious your condition is, your age, and your overall health.

Will I need open heart surgery?

A small number of people may need open heart surgery to correct complications that happen during the TAVI procedure. Your surgeon will explain this to you before you have surgery.

Speak to your doctor if you're worried or have more questions.

During a TAVI procedure

What happens during a TAVI?

TAVI can be done under sedation (you're awake, but sleepy and relaxed) or general anaesthetic (you're asleep). You should not feel any pain with either option. Speak to the hospital team about which option is best for you.

A thin tube is put into an artery near your collarbone or the top of your leg (groin).

The tube is guided to your heart until it reaches your aortic valve. A new valve, made of metal and animal tissue, is put inside your aortic valve.

The valve either expands by itself or a balloon at the end of the tube may be blown up to make room for the new valve.

The tube is then taken out and a dressing put over the cut. There may be a small amount of bleeding when it is taken out.

Immediately after your procedure, you'll be taken to a recovery room and a few hours later you'll be moved to a ward.

Some people feel pressure in their chest during a TAVI. Tell the person doing the procedure if you feel unwell or have any chest pain.

Watch our video to find out what happens during a TAVI procedure

Recovery

How long will my recovery take?

Most people go home 1 to 2 days after a TAVI procedure, but some people may need to stay a little longer. Usually, you will fully recover after 6 to 10 weeks. However, it takes people different periods of time to recover after a TAVI procedure.

Are there any activities I will need to avoid?

Before you leave hospital, someone will have a chat about what you can and cannot do. You may also be offered the support of a TAVI nurse while you're recovering.

You may be told to avoid certain activities while recovering, such as:

vacuuming

carrying heavy bags

lifting heavy pans when cooking

mowing

picking up children, grandchildren or heavy pets.

Speak to your hospital team about safe movement during your recovery, like walking. Your doctor can give you more information on getting active.

You'll need to take some time off from work after your procedure and stop driving for a period of time. Speak to your doctor and the DVLA to find out more.

What follow-up appointments will I have?

You should be invited to cardiac rehab. This is a programme of exercise and educational support that will help you recover.

You will also have a follow-up appointment with a cardiologist. If you have any questions after this appointment, speak to your GP or your regular doctor.

What side effects will I have?

You may have bruising or tenderness in the area where you had the catheter tube put in. This should wear off in a few days.

When you get home, check the area where the catheter tube was put in. You should contact your doctor if you notice:

redness or swelling

pain gets worse

bruising gets worse

you have a temperature (over 38C).

When you leave hospital, you'll be given medicine to help reduce your risk of blood clots or a stroke. Ask the hospital team how long you need to take it for and if there are any changes to other medicines you take.

How long does a TAVI heart valve last?

More research is needed before we can say for certain how long a TAVI valve will last. For most people, their TAVI valve will still be in good condition 6 years after their procedure. This will likely improve as techniques and technology evolve.

Get support

It's normal to feel worried about your procedure. There are places you can go to for support and to talk things through:

Speak to one of our cardiac nurses by calling the Heart Helpline.

Order our free booklet on heart surgery. It can help you and your loved ones understand what's going to happen, how to prepare and what your recovery looks like.

To find out more, or to support British Heart Foundation's work, please visit www.bhf.org.uk. You can speak to one of our cardiac nurses by calling our helpline on 0808 802 1234 (freephone), Monday to Friday, 9am to 5pm. For general customer service enquiries, please call 0300 330 3322, Monday to Friday, 9am to 5pm.

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AI rewrite

TAVI is a procedure to help blood flow through your heart. It does this by replacing an aortic valve that does not open all the way.

Page updated:

Next review:

Page updated:

Next review:

TAVI explained

What is a TAVI?

A TAVI replaces the aortic valve in your heart. This helps blood flow through your heart and out to the rest of your body.

TAVI is short for transcatheter aortic valve implantation. It is sometimes called transcatheter aortic valve replacement, or TAVR.

Why do you need a TAVI?

You may need a TAVI if the aortic valve in your heart is too narrow and does not open right. This is called aortic stenosis.

Your heart valves help keep blood moving through your heart. The aortic valve opens to let blood flow from your heart to your body.

Benefits and risks

What are the benefits and risks of a TAVI?

Your care team will give you medicine to lower your risk of blood clots or a stroke.

The benefits and risks of TAVI are not the same for everyone. They depend on how serious your problem is, your age, and your overall health.

Will I need open heart surgery?

A small number of people may need open heart surgery to fix problems that happen during the TAVI. Your surgeon will explain this to you before your surgery.

Talk to your doctor if you are worried or have more questions.

During a TAVI

What happens during a TAVI?

You can have a TAVI under sedation (you are awake, but sleepy and relaxed) or general anaesthetic (you are asleep). You should not feel any pain either way. Talk to the hospital team about which is best for you.

A thin tube is put into a blood vessel (artery) near your collarbone or at the top of your leg (groin).

The tube is moved up to your heart until it reaches your aortic valve. A new valve is put inside your old aortic valve. The new valve is made of metal and animal tissue.

The new valve opens up on its own. Or a small balloon at the end of the tube may be blown up to make room for it.

Then the tube is taken out and a dressing is put over the cut. There may be a small amount of bleeding when the tube comes out.

Right after the procedure, you go to a recovery room. A few hours later, you move to a ward.

Some people feel pressure in their chest during a TAVI. Tell the person doing the procedure if you feel unwell or have any chest pain.

Watch our video to find out what happens during a TAVI.

Recovery

How long will my recovery take?

Most people go home 1 to 2 days after a TAVI. Some people may need to stay a bit longer. You will usually feel fully better after 6 to 10 weeks. But recovery takes a different amount of time for each person.

Are there any activities I should avoid?

Before you leave hospital, someone will talk with you about what you can and cannot do. You may also get help from a TAVI nurse while you recover.

You may be told to avoid some activities while you recover, such as:

- vacuuming

- carrying heavy bags

- lifting heavy pans when cooking

- mowing the lawn

- picking up children, grandchildren, or heavy pets.

Talk to your hospital team about safe ways to move during your recovery, like walking. Your doctor can tell you more about getting active.

You will need to take some time off work after your procedure. You will also need to stop driving for a while. Talk to your doctor and the DVLA to find out more.

What follow-up appointments will I have?

You should be invited to cardiac rehab. This is a program of exercise and learning that helps you recover.

You will also have a follow-up appointment with a heart doctor (cardiologist). If you have questions after this appointment, talk to your GP or your usual doctor.

What side effects will I have?

You may have bruising or sore skin where the tube went in. This should go away in a few days.

When you get home, check the area where the tube went in. Call your doctor if you notice:

- redness or swelling

- pain that gets worse

- bruising that gets worse

- a temperature over 38C.

When you leave hospital, you will be given medicine to lower your risk of blood clots or a stroke. Ask the hospital team how long you need to take it. Also ask if it changes any other medicines you take.

How long does a TAVI heart valve last?

More research is needed before we can say for sure how long a TAVI valve will last. For most people, the valve will still be in good shape 6 years after the procedure. This will likely get better as the tools and methods improve.

Get support

It is normal to feel worried about your procedure. There are places you can go for support and to talk things through:

Talk to one of our cardiac nurses by calling the Heart Helpline.

Order our free booklet on heart surgery. It can help you and your loved ones learn what will happen, how to get ready, and what your recovery will look like.

To find out more, or to support the British Heart Foundation's work, please visit www.bhf.org.uk. You can talk to one of our cardiac nurses by calling our helpline on 0808 802 1234 (freephone), Monday to Friday, 9am to 5pm. For general customer service questions, please call 0300 330 3322, Monday to Friday, 9am to 5pm.

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Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 58 of 77 · Item ID B-KTGBTYZMEP

Original page

TAVI (transcatheter aortic valve implantation) is a procedure to improve the blood flow in your heart by replacing an aortic valve that does not open fully.

Page updated:

Next review:

Page updated:

Next review:

TAVI explained

What is a TAVI?

TAVI is a procedure to replace the aortic valve in your heart. It helps improve the blood flow through your heart and into the rest of your body.

TAVI stands for transcatheter aortic valve implantation. It's sometimes called transcatheter aortic valve replacement (TAVR).

Why do you need a TAVI?

You may need a TAVI if the aortic valve in your heart is narrow and does not open properly. This is called aortic stenosis.

Your heart valves play an important role in keeping blood flowing through your heart. The aortic valve opens to let blood flow from your heart to your body.

Benefits and risks

What are the benefits and risks of a TAVI procedure?

Your treatment team will give you medicine to help reduce your risk of blood clots or a stroke.

The benefits and risks of TAVI are different for everyone and depend on how serious your condition is, your age, and your overall health.

Will I need open heart surgery?

A small number of people may need open heart surgery to correct complications that happen during the TAVI procedure. Your surgeon will explain this to you before you have surgery.

Speak to your doctor if you're worried or have more questions.

During a TAVI procedure

What happens during a TAVI?

TAVI can be done under sedation (you're awake, but sleepy and relaxed) or general anaesthetic (you're asleep). You should not feel any pain with either option. Speak to the hospital team about which option is best for you.

A thin tube is put into an artery near your collarbone or the top of your leg (groin).

The tube is guided to your heart until it reaches your aortic valve. A new valve, made of metal and animal tissue, is put inside your aortic valve.

The valve either expands by itself or a balloon at the end of the tube may be blown up to make room for the new valve.

The tube is then taken out and a dressing put over the cut. There may be a small amount of bleeding when it is taken out.

Immediately after your procedure, you'll be taken to a recovery room and a few hours later you'll be moved to a ward.

Some people feel pressure in their chest during a TAVI. Tell the person doing the procedure if you feel unwell or have any chest pain.

Watch our video to find out what happens during a TAVI procedure

Recovery

How long will my recovery take?

Most people go home 1 to 2 days after a TAVI procedure, but some people may need to stay a little longer. Usually, you will fully recover after 6 to 10 weeks. However, it takes people different periods of time to recover after a TAVI procedure.

Are there any activities I will need to avoid?

Before you leave hospital, someone will have a chat about what you can and cannot do. You may also be offered the support of a TAVI nurse while you're recovering.

You may be told to avoid certain activities while recovering, such as:

vacuuming

carrying heavy bags

lifting heavy pans when cooking

mowing

picking up children, grandchildren or heavy pets.

Speak to your hospital team about safe movement during your recovery, like walking. Your doctor can give you more information on getting active.

You'll need to take some time off from work after your procedure and stop driving for a period of time. Speak to your doctor and the DVLA to find out more.

What follow-up appointments will I have?

You should be invited to cardiac rehab. This is a programme of exercise and educational support that will help you recover.

You will also have a follow-up appointment with a cardiologist. If you have any questions after this appointment, speak to your GP or your regular doctor.

What side effects will I have?

You may have bruising or tenderness in the area where you had the catheter tube put in. This should wear off in a few days.

When you get home, check the area where the catheter tube was put in. You should contact your doctor if you notice:

redness or swelling

pain gets worse

bruising gets worse

you have a temperature (over 38C).

When you leave hospital, you'll be given medicine to help reduce your risk of blood clots or a stroke. Ask the hospital team how long you need to take it for and if there are any changes to other medicines you take.

How long does a TAVI heart valve last?

More research is needed before we can say for certain how long a TAVI valve will last. For most people, their TAVI valve will still be in good condition 6 years after their procedure. This will likely improve as techniques and technology evolve.

Get support

It's normal to feel worried about your procedure. There are places you can go to for support and to talk things through:

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AI rewrite

TAVI, also called TAVR, is a procedure to replace the aortic valve in your heart. This valve may be too narrow and may not open fully. TAVI helps blood flow better through your heart and to the rest of your body.

TAVI stands for transcatheter aortic valve implantation. TAVR stands for transcatheter aortic valve replacement.

You may need a TAVI if your aortic valve is narrow and does not open the right way. This is called aortic stenosis.

Your heart valves help keep blood moving through your heart. The aortic valve opens to let blood flow from your heart to your body.

Your treatment team will give you medicine to help lower your risk of blood clots or a stroke.

The benefits and risks of TAVI are different for each person. They depend on how serious your condition is, your age, and your overall health.

A small number of people may need open heart surgery. This may be needed to fix problems that happen during the TAVI procedure. Your surgeon will explain this before you have surgery.

Talk to your doctor if you are worried or have more questions.

TAVI can be done with sedation or with a general anaesthetic.

With sedation, you are awake, but sleepy and relaxed.

With a general anaesthetic, you are asleep.

You should not feel pain with either choice. Talk to the hospital team about which choice is best for you.

A thin tube is put into an artery near your collarbone or at the top of your leg, called the groin.

The tube is moved to your heart until it reaches your aortic valve. A new valve is put inside your aortic valve. The new valve is made of metal and animal tissue.

The valve may open by itself. Or a balloon at the end of the tube may be blown up to make space for the new valve.

The tube is then taken out. A dressing is put over the cut. There may be a small amount of bleeding when the tube is taken out.

Right after the procedure, you will go to a recovery room. A few hours later, you will be moved to a ward.

Some people feel pressure in their chest during a TAVI. Tell the person doing the procedure if you feel unwell or have any chest pain.

Most people go home 1 to 2 days after a TAVI. Some people need to stay in hospital a little longer.

Most people fully recover after 6 to 10 weeks. But recovery time is different for each person.

Before you leave hospital, someone will talk with you about what you can and cannot do. You may also be offered help from a TAVI nurse while you recover.

You may be told to avoid some activities while you recover, such as:

vacuuming

carrying heavy bags

lifting heavy pans when cooking

mowing

picking up children, grandchildren, or heavy pets

Talk to your hospital team about safe movement during recovery, such as walking. Your doctor can give you more information about getting active.

You will need to take some time off work after your procedure. You will also need to stop driving for a period of time. Talk to your doctor and the DVLA to find out more.

You should be invited to cardiac rehab. This is a programme with exercise and learning support to help you recover.

You will also have a follow-up appointment with a cardiologist. If you have questions after this appointment, talk to your GP or your regular doctor.

You may have bruising or soreness where the catheter tube was put in. This should get better in a few days.

When you get home, check the area where the catheter tube was put in. Contact your doctor if you notice:

redness or swelling

pain gets worse

bruising gets worse

you have a temperature over 38C

When you leave hospital, you will be given medicine to help lower your risk of blood clots or a stroke. Ask the hospital team how long you need to take it. Also ask if there are any changes to your other medicines.

More research is needed before we can say for sure how long a TAVI valve will last. For most people, their TAVI valve will still be in good condition 6 years after the procedure. This will likely get better as methods and technology improve.

It is normal to feel worried about your procedure. There are places you can go for support and to talk things through.

You can speak to one of our cardiac nurses by calling the Heart Helpline.

You can order our free booklet on heart surgery. It can help you and your loved ones understand what will happen, how to prepare, and what recovery may be like.

To find out more, or to support the British Heart Foundation’s work, visit www.bhf.org.uk.

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It is registered as a charity in Scotland No. SC039426.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

Original page

TAVI (transcatheter aortic valve implantation) is a procedure to improve the blood flow in your heart by replacing an aortic valve that does not open fully.

Page updated:

Next review:

Page updated:

Next review:

TAVI explained

What is a TAVI?

TAVI is a procedure to replace the aortic valve in your heart. It helps improve the blood flow through your heart and into the rest of your body.

TAVI stands for transcatheter aortic valve implantation. It's sometimes called transcatheter aortic valve replacement (TAVR).

Why do you need a TAVI?

You may need a TAVI if the aortic valve in your heart is narrow and does not open properly. This is called aortic stenosis.

Your heart valves play an important role in keeping blood flowing through your heart. The aortic valve opens to let blood flow from your heart to your body.

Benefits and risks

What are the benefits and risks of a TAVI procedure?

Your treatment team will give you medicine to help reduce your risk of blood clots or a stroke.

The benefits and risks of TAVI are different for everyone and depend on how serious your condition is, your age, and your overall health.

Will I need open heart surgery?

A small number of people may need open heart surgery to correct complications that happen during the TAVI procedure. Your surgeon will explain this to you before you have surgery.

Speak to your doctor if you're worried or have more questions.

During a TAVI procedure

What happens during a TAVI?

TAVI can be done under sedation (you're awake, but sleepy and relaxed) or general anaesthetic (you're asleep). You should not feel any pain with either option. Speak to the hospital team about which option is best for you.

A thin tube is put into an artery near your collarbone or the top of your leg (groin).

The tube is guided to your heart until it reaches your aortic valve. A new valve, made of metal and animal tissue, is put inside your aortic valve.

The valve either expands by itself or a balloon at the end of the tube may be blown up to make room for the new valve.

The tube is then taken out and a dressing put over the cut. There may be a small amount of bleeding when it is taken out.

Immediately after your procedure, you'll be taken to a recovery room and a few hours later you'll be moved to a ward.

Some people feel pressure in their chest during a TAVI. Tell the person doing the procedure if you feel unwell or have any chest pain.

Watch our video to find out what happens during a TAVI procedure

Recovery

How long will my recovery take?

Most people go home 1 to 2 days after a TAVI procedure, but some people may need to stay a little longer. Usually, you will fully recover after 6 to 10 weeks. However, it takes people different periods of time to recover after a TAVI procedure.

Are there any activities I will need to avoid?

Before you leave hospital, someone will have a chat about what you can and cannot do. You may also be offered the support of a TAVI nurse while you're recovering.

You may be told to avoid certain activities while recovering, such as:

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carrying heavy bags

lifting heavy pans when cooking

mowing

picking up children, grandchildren or heavy pets.

Speak to your hospital team about safe movement during your recovery, like walking. Your doctor can give you more information on getting active.

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pain gets worse

bruising gets worse

you have a temperature (over 38C).

When you leave hospital, you'll be given medicine to help reduce your risk of blood clots or a stroke. Ask the hospital team how long you need to take it for and if there are any changes to other medicines you take.

How long does a TAVI heart valve last?

More research is needed before we can say for certain how long a TAVI valve will last. For most people, their TAVI valve will still be in good condition 6 years after their procedure. This will likely improve as techniques and technology evolve.

Get support

It's normal to feel worried about your procedure. There are places you can go to for support and to talk things through:

Speak to one of our cardiac nurses by calling the Heart Helpline.

Order our free booklet on heart surgery. It can help you and your loved ones understand what's going to happen, how to prepare and what your recovery looks like.

To find out more, or to support British Heart Foundation's work, please visit www.bhf.org.uk. You can speak to one of our cardiac nurses by calling our helpline on 0808 802 1234 (freephone), Monday to Friday, 9am to 5pm. For general customer service enquiries, please call 0300 330 3322, Monday to Friday, 9am to 5pm.

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AI rewrite

TAVI is a procedure to fix your heart. It stands for transcatheter aortic valve implantation. Sometimes it is called TAVR. It replaces a heart valve that does not open all the way. This helps blood flow better through your heart and to your body.

You may need a TAVI if your aortic valve is too narrow. This problem is called aortic stenosis. Your heart valves keep blood moving the right way. The aortic valve opens to let blood flow from your heart to the rest of your body.

The good and bad points of a TAVI depend on your age and health. They also depend on how sick you are. Your care team will give you medicine. This medicine lowers your chance of getting blood clots or having a stroke. A small number of people might need open-heart surgery if problems happen during the TAVI. Your doctor will talk to you about this before your procedure. Tell your doctor if you are worried or have questions.

You can have medicine to make you sleepy and relaxed. Or, you can have medicine to make you fully asleep. You should not feel any pain. Talk to your doctors about which choice is best for you.

A doctor will put a thin tube into an artery. This will be near your collarbone or at the top of your leg. The doctor guides the tube to your heart. A new valve is put inside your old valve. The new valve is made of metal and animal tissue. It might open up on its own. Or, the doctor might use a small balloon to open it.

Then, the doctor takes the tube out and covers the cut with a bandage. It might bleed a little bit. You will go to a recovery room right after. A few hours later, you will move to a regular hospital room. Some people feel pressure in their chest during the TAVI. Tell the person doing the procedure right away if you feel sick or have chest pain.

Most people go home 1 or 2 days after a TAVI. Some people need to stay longer. It usually takes 6 to 10 weeks to fully heal. Everyone heals at their own speed. Before you leave, your team will tell you what you can and cannot do. You might get help from a special TAVI nurse.

While you heal, you should not do heavy work. Do not vacuum or mow the lawn. Do not carry heavy bags or lift heavy pots. Do not pick up children or heavy pets. Ask your team about safe ways to move, like walking. You will need to take time off from work. You also must stop driving for a while. Ask your doctor and the DVLA to find out more.

You will be asked to join a heart rehab program. This program uses exercise and teaching to help you heal. You will also have a check-up with your heart doctor. If you have questions, ask your GP or regular doctor.

You might have bruises or soreness where the tube went in. This should go away in a few days. Check this spot when you get home. Call your doctor if it gets red or swollen. Call if the pain or bruising gets worse. Call if you have a temperature over 38C.

When you go home, you will take medicine to stop blood clots and strokes. Ask your team how long you need to take it. Also ask if you need to change any of your other medicines.

Doctors are still studying how long a TAVI valve lasts. For most people, it works well for at least 6 years. This time will likely get longer as technology gets better.

It is normal to feel worried. You can get support and talk about your feelings. You can order a free booklet on heart surgery to help you and your family prepare. To learn more, visit www.bhf.org.uk. You can talk to a heart nurse by calling 0808 802 1234 for free. This line is open Monday to Friday, 9 am to 5 pm. For general questions, call 0300 330 3322 during the same hours.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

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Original page

Last reviewed on April 23, 2024

Preprocedural Planning for Transcatheter Aortic Valve Replacement

Transcatheter aortic valve replacement (TAVR) is a less invasive way to replace an improperly functioning aortic valve as opposed to open-heart surgery. The TAVR procedure consists of inserting a catheter into a blood vessel via small incisions in the groin or chest area. The valve is then moved to the correct location using imaging for guidance. Accurate measurements are important in choosing the correct replacement valve size as well as planning the path for appropriate valve placement.

To assess the root of the aorta (the portion of the aorta leading out of the valve) prior to the TAVR procedure, ultrasound echocardiography transesophageal, MRI heart function and morphology without and with intravenous (IV) contrast, and CT heart function and morphology with IV contrast are usually appropriate. MR angiography (MRA) chest with IV contrast, MRA chest without and with IV contrast, CT angiography (CTA) chest with IV contrast, and CTA coronary arteries with IV contrast may also be appropriate.

To assess the supravalvular aorta (narrowest section of the aorta just above the valve) and to assess the access to the valve through the blood vessels prior to the TAVR procedure, CTA chest with IV contrast, CTA abdomen/pelvis with IV contrast, and CTA chest, abdomen/pelvis with IV contrast are usually appropriate. MRA abdomen and pelvis without and with IV contrast, MRA chest abdomen pelvis with IV contrast, and MRA chest without and with IV contrast may also be appropriate.

AI rewrite

This text was last reviewed on April 23, 2024.

Transcatheter aortic valve replacement (TAVR) fixes a bad heart valve without open-heart surgery. Doctors make a small cut in your groin or chest. Then, they put a thin tube called a catheter into a blood vessel. They use special pictures to guide the new valve to the right spot. Doctors need exact measurements to pick the right valve size and plan its path.

Before your TAVR, doctors check the aorta root. This is the part of your main blood vessel just outside the valve. To do this, they often use a heart ultrasound done through your throat. They may also use a heart MRI scan. This is done with and without IV contrast (a special dye put in your vein). Another common test is a heart CT scan with IV contrast. Doctors might also use an MRA scan of the chest. This can be done with IV contrast, or with and without it. They might also use a CTA scan of the chest or heart blood vessels with IV contrast.

Doctors also check the narrow part just above the valve. They check your blood vessels to see how the tube will reach your heart. For this, they usually use a CTA scan of the chest, the belly and pelvis, or all three areas. These scans use IV contrast. Sometimes, doctors use MRA scans instead. They might do an MRA scan of the belly and pelvis with and without IV contrast. They might do an MRA scan of the chest, belly, and pelvis with IV contrast. Another choice is a chest MRA scan with and without IV contrast.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Last reviewed on April 23, 2024

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AI rewrite

Last reviewed on April 23, 2024

Getting Ready for Transcatheter Aortic Valve Replacement

Transcatheter aortic valve replacement (TAVR) is a way to replace a heart valve that is not working right. The aortic valve is one of the valves in your heart. TAVR is not open-heart surgery. It uses smaller cuts, so it is less hard on your body.

In TAVR, the doctor puts a thin tube called a catheter into a blood vessel. The tube goes in through small cuts in your groin or chest. The doctor then moves the new valve to the right spot. They use special pictures to help guide it.

Good measurements matter a lot. They help the doctor pick the right size for the new valve. They also help the doctor plan the best path to put the valve in place.

Before TAVR, the doctor needs to look at the root of your aorta. The aorta is the big blood vessel that carries blood out of your heart. The root is the part right next to the valve. These tests are usually a good way to check it:

- An ultrasound of the heart done through the throat (transesophageal echocardiography)

- An MRI of the heart, done both without and with dye in your vein (IV contrast)

- A CT of the heart, done with IV contrast

These tests may also work to check the aortic root:

- MR angiography (MRA) of the chest with IV contrast

- MRA of the chest without and with IV contrast

- CT angiography (CTA) of the chest with IV contrast

- CTA of the heart arteries with IV contrast

Before TAVR, the doctor also needs to look at two more things. One is the part of the aorta just above the valve. This is the most narrow part. The other is the path through your blood vessels to reach the valve. These tests are usually a good way to check them:

- CTA of the chest with IV contrast

- CTA of the belly and pelvis with IV contrast

- CTA of the chest, belly, and pelvis with IV contrast

These tests may also work:

- MRA of the belly and pelvis without and with IV contrast

- MRA of the chest, belly, and pelvis with IV contrast
- MRA of the chest without and with IV contrast

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Last reviewed on April 23, 2024

Preprocedural Planning for Transcatheter Aortic Valve Replacement

Transcatheter aortic valve replacement (TAVR) is a less invasive way to replace an improperly functioning aortic valve as opposed to open-heart surgery. The TAVR procedure consists of inserting a catheter into a blood vessel via small incisions in the groin or chest area. The valve is then moved to the correct location using imaging for guidance. Accurate measurements are important in choosing the correct replacement valve size as well as planning the path for appropriate valve placement.

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AI rewrite

Last reviewed on April 23, 2024

Planning Before Transcatheter Aortic Valve Replacement

Transcatheter aortic valve replacement is also called TAVR. It is a less invasive way to replace an aortic valve that is not working right. It is used instead of open-heart surgery.

During TAVR, the doctor puts a thin tube called a catheter into a blood vessel. This is done through small cuts in the groin or chest. The new valve is moved through the catheter to the right spot. Doctors use imaging tests to guide it.

Doctors need exact measurements before TAVR. These measurements help them choose the right size valve. They also help doctors plan the best path to place the valve.

Before TAVR, doctors may need to check the root of the aorta. This is the part of the aorta that comes out of the valve. The tests that are usually appropriate include:

- Transesophageal echocardiography, an ultrasound test done through the food pipe

- MRI of the heart, without and with IV contrast

- CT of the heart, with IV contrast

Other tests that may also be appropriate include:

- MRA of the chest, with IV contrast

- MRA of the chest, without and with IV contrast

- CTA of the chest, with IV contrast

- CTA of the heart arteries, with IV contrast

Before TAVR, doctors may also need to check the part of the aorta just above the valve. This is the narrowest part above the valve. Doctors also check the blood vessels used to reach the valve.

The tests that are usually appropriate include:

- CTA of the chest, with IV contrast

- CTA of the belly and pelvis, with IV contrast

- CTA of the chest, belly, and pelvis, with IV contrast

Other tests that may also be appropriate include:

- MRA of the belly and pelvis, without and with IV contrast

- MRA of the chest, belly, and pelvis, with IV contrast
- MRA of the chest, without and with IV contrast

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Left Atrial Appendage (LAA) Closure With Watchman or Lariat

On this page

Left Atrial Appendage (LAA) Closure With Watchman or Lariat

More than 2.6 million Americans have atrial fibrillation (sometimes called A-fib), a fast, irregular heartbeat rhythm caused by chaotic electrical activity in the heart's two upper chambers (atria). Because the heart isn't pumping in a steady, coordinated way, the blood flow out to the body is reduced.

A-fib can cause uncomfortable symptoms, such as fatigue, palpitations and shortness of breath. More serious, it can cause blood clots that may lead to a stroke. Up to 90 percent of these stroke-causing clots originate in the left atrial appendage, a small, hollow sac attached to the left atrium where blood can collect. To prevent clots from forming, patients with A-fib are usually prescribed blood-thinning medications. But some patients can't tolerate these drugs. For A-fib patients who aren't on blood thinners, the risk of stroke increases with age: A-fib causes 15 percent of all strokes, but in patients 70 years and older, it accounts for up to 25 percent of strokes.

An effective way to reduce stroke risk from A-fib is to close off the left atrial appendage. For decades, this was done through open-heart surgery. But recently developed devices have made it possible to close off the appendage through minimally invasive procedures, in which thin, flexible tubes (catheters) are guided through blood vessels to access the heart. If you are considering these procedures, understand that closing the appendage lowers stroke risk but doesn't cure A-fib.

The Watchman is an implantable device that acts as a plug to seal off the left atrial appendage. This parachute-shaped device is about the size of a quarter and made of light, compact materials used in many other medical implants.

UCSF was among the first medical centers to perform a procedure that uses a lasso-like device called the Lariat to close the appendage. The Lariat is a pre-tied loop of suture (stitching material) that the doctor places around the outside of the appendage to tie it off from the heart. UCSF cardiologist Dr. Randall Lee helped develop the Lariat and has since trained cardiologists from more than 100 medical centers to perform the procedure.

UCSF is one of the few hospitals in the country to offer both treatments.

Evaluation

Potential candidates for either procedure are evaluated in our electrophysiology clinic. To be eligible for the Watchman device, you must be able to take anticoagulation and antiplatelet medications for a period of time after the procedure. Having the Lariat procedure doesn't require this. Some patients can't undergo either procedure because of prior heart surgery or the particular anatomy of their left atrial appendage.

To determine the size and shape of your atrial appendage, your doctor will order one or more imaging tests. Options include:

- A cardiac MRI uses powerful magnets and radio waves to create pictures of the heart. A harmless dye that highlights blood vessels and heart structures may be injected into the circulatory system.

- A cardiac CT scan takes detailed X-ray images of the heart and blood vessels. When patients may be having the Lariat procedure, this test is used to evaluate the size and shape of the left atrial appendage.

- The transesophageal echocardiogram (TEE) is an ultrasound test in which a thin tube conveys a transducer (a device emitting sound waves) down the throat to obtain clear heart images from inside the body. TEE is the preferred imaging test for the Watchman procedure.

Procedure

For both the Watchman and Lariat procedures, patients stay in the hospital for at least one night.

These minimally invasive procedures are performed in a similar manner. A catheter carrying the device is inserted into the femoral artery (in the groin) and threaded through blood vessels to the heart. A combination of X-rays and TEE imaging is used to guide implantation. Once the device is in place, the catheter is withdrawn. Each procedure takes two to three hours, and patients are under general anesthesia (completely asleep).

In the Watchman procedure, the left atrial appendage is accessed from inside the heart. The device is placed across the opening of the appendage, sealing off the space so that blood can't collect in it. Over time, scar tissue grows around the device, securing it in place.

In the Lariat procedure, the loop is placed around the outside of the left atrial appendage and pulled tight, sealing it off from the rest of the heart. This requires accessing the heart through both the groin and chest wall. The chest access is obtained by passing a long needle under the rib cage, rather than cutting through the chest wall as in open-heart surgery.

Recovery

Following either procedure, you have to stay in bed for at least four hours to prevent bleeding. During that time, we closely monitor your heart rate, blood pressure, oxygen levels and breathing.

Most patients can go home the day after their procedure. Your doctor may advise you to avoid driving for several days to a few weeks and to avoid strenuous exercise for a week. You may be instructed to take aspirin for about a month to reduce the risk of clots forming in your heart and blood vessels.

Remember, closing the left atrial appendage doesn't cure atrial fibrillation. These procedures reduce stroke risk in people with A-fib. Patients who undergo one of these procedures still have A-fib and may experience palpitations or other symptoms related to the condition.

Related services & conditions

Specialties

Conditions

- Atrial Fibrillation

- Cerebrovascular Accident (CVA)

- Ischemic Stroke

UCSF Health medical specialists have reviewed this information. It is for educational purposes only and is not intended to replace the advice of your doctor or other health care provider. We encourage you to discuss any questions or concerns you may have with your provider.

AI rewrite

Left Atrial Appendage (LAA) Closure With Watchman or Lariat

On this page

Left Atrial Appendage (LAA) Closure With Watchman or Lariat

More than 2.6 million Americans have atrial fibrillation. People often call it A-fib. A-fib is a fast, uneven heartbeat. It happens when the electrical signals in the heart's two top spaces (the atria) get mixed up. When this happens, the heart does not pump in a smooth, steady way. So less blood flows out to the body.

A-fib can make you feel bad. You may feel tired, feel your heart flutter, or feel short of breath. A-fib can also be more serious. It can cause blood clots. A clot can lead to a stroke. Up to 90 out of 100 of these clots start in the left atrial appendage. This is a small, hollow sac on the left atrium. Blood can pool there.

To stop clots, most people with A-fib take blood-thinning medicine. But some people cannot take these drugs. For people with A-fib who do not take blood thinners, the stroke risk goes up with age. A-fib causes 15 out of 100 of all strokes. But in people 70 years and older, it causes up to 25 out of 100 of strokes.

One good way to lower stroke risk from A-fib is to close off the left atrial appendage. For many years, doctors did this with open-heart surgery. But new devices now let doctors close the sac in an easier way. They use thin, bendable tubes called catheters. They guide these tubes through your blood vessels to reach your heart. If you think about having these procedures, know this: closing the sac lowers your stroke risk. But it does not cure A-fib.

The Watchman is a small device that is put inside you. It works like a plug to seal off the left atrial appendage. It is shaped like a parachute. It is about the size of a quarter. It is made of light, small materials. These same materials are used in many other medical implants.

UCSF was one of the first medical centers to use a lasso-like device called the Lariat to close the sac. The Lariat is a loop of stitching material that is already tied. The doctor puts it around the outside of the sac to tie it off from the heart. UCSF heart doctor Dr. Randall Lee helped make the Lariat. Since then, he has trained heart doctors from more than 100 medical centers to do this procedure.

UCSF is one of the few hospitals in the country that offers both treatments.

Evaluation

We check people who may want either procedure in our heart rhythm (electrophysiology) clinic. To get the Watchman device, you must be able to take blood-thinning and anti-clotting medicine for a while after the procedure. The Lariat procedure does not need this. Some people cannot have either procedure. This may be because of past heart surgery or the shape of their left atrial appendage.

Your doctor will order one or more imaging tests. These tests show the size and shape of your sac. The tests may include:

- A cardiac MRI. This uses strong magnets and radio waves to make pictures of the heart. A safe dye may be put into your blood. The dye helps show your blood vessels and heart parts.

- A cardiac CT scan. This takes clear X-ray pictures of the heart and blood vessels. If you may have the Lariat procedure, this test checks the size and shape of the left atrial appendage.

- A transesophageal echocardiogram (TEE). This is an ultrasound test. A thin tube carries a small device down your throat. This device sends out sound waves. It makes clear pictures of the heart from inside the body. TEE is the best imaging test for the Watchman procedure.

Procedure

For both the Watchman and Lariat procedures, you stay in the hospital for at least one night.

These two procedures are done in a similar way. A catheter holds the device. The doctor puts the catheter into the femoral artery (in your groin). The doctor then guides it through blood vessels to your heart. The doctor uses X-rays and TEE pictures to help place the device. Once the device is in place, the doctor pulls out the catheter. Each procedure takes two to three hours. You are fully asleep under general anesthesia.

In the Watchman procedure, the doctor reaches the left atrial appendage from inside the heart. The doctor places the device across the opening of the sac. This seals off the space so blood cannot pool in it. Over time, scar tissue grows around the device. This holds it in place.

In the Lariat procedure, the doctor places the loop around the outside of the left atrial appendage. The doctor pulls it tight to seal it off from the rest of the heart. To do this, the doctor reaches the heart through both the groin and the chest wall. To reach the chest, the doctor passes a long needle under the rib cage. The doctor does not cut through the chest wall like in open-heart surgery.

Recovery

After either procedure, you must stay in bed for at least four hours. This helps stop bleeding. During this time, we watch your heart rate, blood pressure, oxygen levels, and breathing closely.

Most people can go home the day after their procedure. Your doctor may tell you not to drive for several days to a few weeks. Your doctor may also tell you not to do hard exercise for a week. You may be told to take aspirin for about a month. This lowers the risk of clots in your heart and blood vessels.

Remember, closing the left atrial appendage does not cure atrial fibrillation. These procedures lower stroke risk in people with A-fib. People who have one of these procedures still have A-fib. They may still feel their heart flutter or have other symptoms from A-fib.

Related services & conditions

Specialties

Conditions

- Atrial Fibrillation

- Cerebrovascular Accident (CVA)

- Ischemic Stroke

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Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Left Atrial Appendage (LAA) Closure With Watchman or Lariat

On this page

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A-fib can cause uncomfortable symptoms, such as fatigue, palpitations and shortness of breath. More serious, it can cause blood clots that may lead to a stroke. Up to 90 percent of these stroke-causing clots originate in the left atrial appendage, a small, hollow sac attached to the left atrium where blood can collect. To prevent clots from forming, patients with A-fib are usually prescribed blood-thinning medications. But some patients can't tolerate these drugs. For A-fib patients who aren't on blood thinners, the risk of stroke increases with age: A-fib causes 15 percent of all strokes, but in patients 70 years and older, it accounts for up to 25 percent of strokes.

An effective way to reduce stroke risk from A-fib is to close off the left atrial appendage. For decades, this was done through open-heart surgery. But recently developed devices have made it possible to close off the appendage through minimally invasive procedures, in which thin, flexible tubes (catheters) are guided through blood vessels to access the heart. If you are considering these procedures, understand that closing the appendage lowers stroke risk but doesn't cure A-fib.

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For both the Watchman and Lariat procedures, patients stay in the hospital for at least one night.

These minimally invasive procedures are performed in a similar manner. A catheter carrying the device is inserted into the femoral artery (in the groin) and threaded through blood vessels to the heart. A combination of X-rays and TEE imaging is used to guide implantation. Once the device is in place, the catheter is withdrawn. Each procedure takes two to three hours, and patients are under general anesthesia (completely asleep).

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AI rewrite

More than 2.6 million Americans have atrial fibrillation, or A-fib. This is a fast, uneven heartbeat. It happens when the top parts of the heart do not beat normally. Because the heart does not pump well, less blood flows to the body. A-fib can make you feel tired or short of breath. It can also make your heart feel like it is fluttering.

A-fib can cause blood clots. These clots can lead to a stroke. Up to 90 percent of these clots form in a small pouch in the heart. This pouch is called the left atrial appendage. Doctors often give blood-thinning medicine to stop clots. But some people cannot take these drugs. If you have A-fib and do not take blood thinners, your risk of a stroke goes up as you get older. A-fib causes 15 percent of all strokes. For people 70 and older, it causes up to 25 percent of strokes.

One way to lower your stroke risk is to close off this heart pouch. In the past, doctors did open-heart surgery to do this. Now, doctors can use thin, flexible tubes called catheters. They guide these tubes through your blood vessels to your heart. Closing the pouch lowers your risk of a stroke, but it does not cure A-fib.

There are two main ways to close the pouch. One uses a device called the Watchman. The other uses a device called the Lariat. The Watchman is a small plug that seals the pouch. It looks like a tiny parachute and is about the size of a quarter. It is made of light materials used in many other medical implants. The Lariat is a loop of string that acts like a lasso. The doctor puts it around the outside of the pouch to tie it off. UCSF was one of the first hospitals to use the Lariat. A UCSF heart doctor named Dr. Randall Lee helped create it. He has taught doctors from over 100 hospitals how to do this procedure. UCSF is one of the few hospitals that offers both the Watchman and the Lariat.

Doctors in our electrophysiology clinic will check to see if these treatments are right for you. To get the Watchman, you must be able to take medicines that thin your blood and prevent clots for a short time after the procedure. The Lariat does not require this. Some people cannot have either treatment. This might be because of past heart surgery or the shape of their heart pouch.

Your doctor will order tests to check the size and shape of your heart pouch. A cardiac MRI uses strong magnets and radio waves to take pictures of your heart. You might get a safe dye in your blood to help the doctor see your heart better. A cardiac CT scan takes detailed X-ray pictures of your heart and blood vessels. Doctors use this test if you might get the Lariat. A transesophageal echocardiogram (TEE) is an ultrasound test. A thin tube goes down your throat to take clear pictures of your heart from the inside. Doctors prefer this test for the Watchman.

For both treatments, you will stay in the hospital for at least one night. The procedures take two to three hours. You will be completely asleep. The doctor will put a thin tube into a blood vessel in your groin. The tube carries the device to your heart. The doctor uses X-rays and ultrasound pictures to guide the device. When the device is in place, the tube is taken out.

For the Watchman, the doctor works from inside the heart. The plug is placed over the opening of the pouch. This stops blood from pooling inside. Over time, scar tissue grows over the plug to hold it in place.

For the Lariat, the doctor puts the loop around the outside of the pouch and pulls it tight. To do this, the doctor needs to reach the heart from the groin and the chest. A long needle is passed under your rib cage. There are no large cuts like in open-heart surgery.

After the procedure, you must stay in bed for at least four hours. This stops bleeding. Nurses will watch your heart rate, blood pressure, oxygen, and breathing. Most people go home the next day. Your doctor may tell you not to drive for a few days or weeks. You should not do hard exercise for a week. You may need to take aspirin for about a month. This helps stop clots from forming.

Remember, closing the pouch does not cure A-fib. It only lowers your risk of a stroke, which is also called an ischemic stroke or cerebrovascular accident. You will still have A-fib. You might still feel a fluttering heart or other symptoms.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

Item 65 of 77 · Item ID B-P059P452BF

Original page

Left Atrial Appendage (LAA) Closure With Watchman or Lariat

On this page

Left Atrial Appendage (LAA) Closure With Watchman or Lariat

More than 2.6 million Americans have atrial fibrillation (sometimes called A-fib), a fast, irregular heartbeat rhythm caused by chaotic electrical activity in the heart's two upper chambers (atria). Because the heart isn't pumping in a steady, coordinated way, the blood flow out to the body is reduced.

A-fib can cause uncomfortable symptoms, such as fatigue, palpitations and shortness of breath. More serious, it can cause blood clots that may lead to a stroke. Up to 90 percent of these stroke-causing clots originate in the left atrial appendage, a small, hollow sac attached to the left atrium where blood can collect. To prevent clots from forming, patients with A-fib are usually prescribed blood-thinning medications. But some patients can't tolerate these drugs. For A-fib patients who aren't on blood thinners, the risk of stroke increases with age: A-fib causes 15 percent of all strokes, but in patients 70 years and older, it accounts for up to 25 percent of strokes.

An effective way to reduce stroke risk from A-fib is to close off the left atrial appendage. For decades, this was done through open-heart surgery. But recently developed devices have made it possible to close off the appendage through minimally invasive procedures, in which thin, flexible tubes (catheters) are guided through blood vessels to access the heart. If you are considering these procedures, understand that closing the appendage lowers stroke risk but doesn't cure A-fib.

The Watchman is an implantable device that acts as a plug to seal off the left atrial appendage. This parachute-shaped device is about the size of a quarter and made of light, compact materials used in many other medical implants.

UCSF was among the first medical centers to perform a procedure that uses a lasso-like device called the Lariat to close the appendage. The Lariat is a pre-tied loop of suture (stitching material) that the doctor places around the outside of the appendage to tie it off from the heart. UCSF cardiologist Dr. Randall Lee helped develop the Lariat and has since trained cardiologists from more than 100 medical centers to perform the procedure.

UCSF is one of the few hospitals in the country to offer both treatments.

Evaluation

Potential candidates for either procedure are evaluated in our electrophysiology clinic. To be eligible for the Watchman device, you must be able to take anticoagulation and antiplatelet medications for a period of time after the procedure. Having the Lariat procedure doesn't require this. Some patients can't undergo either procedure because of prior heart surgery or the particular anatomy of their left atrial appendage.

To determine the size and shape of your atrial appendage, your doctor will order one or more imaging tests. Options include:

- A cardiac MRI uses powerful magnets and radio waves to create pictures of the heart. A harmless dye that highlights blood vessels and heart structures may be injected into the circulatory system.

- A cardiac CT scan takes detailed X-ray images of the heart and blood vessels. When patients may be having the Lariat procedure, this test is used to evaluate the size and shape of the left atrial appendage.

- The transesophageal echocardiogram (TEE) is an ultrasound test in which a thin tube conveys a transducer (a device emitting sound waves) down the throat to obtain clear heart images from inside the body. TEE is the preferred imaging test for the Watchman procedure.

Procedure

For both the Watchman and Lariat procedures, patients stay in the hospital for at least one night.

These minimally invasive procedures are performed in a similar manner. A catheter carrying the device is inserted into the femoral artery (in the groin) and threaded through blood vessels to the heart. A combination of X-rays and TEE imaging is used to guide implantation. Once the device is in place, the catheter is withdrawn. Each procedure takes two to three hours, and patients are under general anesthesia (completely asleep).

In the Watchman procedure, the left atrial appendage is accessed from inside the heart. The device is placed across the opening of the appendage, sealing off the space so that blood can't collect in it. Over time, scar tissue grows around the device, securing it in place.

In the Lariat procedure, the loop is placed around the outside of the left atrial appendage and pulled tight, sealing it off from the rest of the heart. This requires accessing the heart through both the groin and chest wall. The chest access is obtained by passing a long needle under the rib cage, rather than cutting through the chest wall as in open-heart surgery.

Recovery

Following either procedure, you have to stay in bed for at least four hours to prevent bleeding. During that time, we closely monitor your heart rate, blood pressure, oxygen levels and breathing.

Most patients can go home the day after their procedure. Your doctor may advise you to avoid driving for several days to a few weeks and to avoid strenuous exercise for a week. You may be instructed to take aspirin for about a month to reduce the risk of clots forming in your heart and blood vessels.

Remember, closing the left atrial appendage doesn't cure atrial fibrillation. These procedures reduce stroke risk in people with A-fib. Patients who undergo one of these procedures still have A-fib and may experience palpitations or other symptoms related to the condition.

Related services & conditions

Specialties

Conditions

- Atrial Fibrillation

- Cerebrovascular Accident (CVA)

- Ischemic Stroke

UCSF Health medical specialists have reviewed this information. It is for educational purposes only and is not intended to replace the advice of your doctor or other health care provider. We encourage you to discuss any questions or concerns you may have with your provider.

AI rewrite

Left Atrial Appendage (LAA) Closure With Watchman or Lariat

More than 2.6 million Americans have atrial fibrillation. It is often called A-fib. A-fib is a fast, uneven heartbeat. It happens because of mixed-up electrical signals in the heart’s two upper chambers, called the atria. When the heart does not pump in a steady way, less blood flows out to the body.

A-fib can cause symptoms that feel uncomfortable. These may include feeling very tired, feeling your heart race or flutter, and shortness of breath. A-fib can also cause blood clots. These clots can lead to a stroke.

Up to 90% of clots that cause strokes in people with A-fib start in the left atrial appendage. This is a small, hollow pouch attached to the left atrium. Blood can collect there. To help stop clots from forming, people with A-fib are usually given blood-thinning medicines. But some people cannot take these medicines.

For people with A-fib who are not taking blood thinners, the risk of stroke goes up with age. A-fib causes 15% of all strokes. In people age 70 and older, A-fib causes up to 25% of strokes.

One way to lower stroke risk from A-fib is to close off the left atrial appendage. For many years, this was done with open-heart surgery. New devices now make it possible to close the pouch with less invasive procedures. In these procedures, thin, flexible tubes called catheters are guided through blood vessels to the heart.

If you are thinking about these procedures, it is important to know that closing the appendage lowers stroke risk. It does not cure A-fib.

The Watchman is a device that is placed in the body. It works like a plug to seal off the left atrial appendage. It is shaped like a parachute. It is about the size of a quarter. It is made of light, small materials used in many other medical implants.

UCSF was one of the first medical centers to use a device called the Lariat to close the appendage. The Lariat works like a lasso. It is a pre-tied loop of suture, or stitching material. The doctor places it around the outside of the appendage to tie it off from the heart. UCSF cardiologist Dr. Randall Lee helped develop the Lariat. He has trained cardiologists from more than 100 medical centers to do the procedure.

UCSF is one of the few hospitals in the country that offers both treatments.

Evaluation

People who may be able to have either procedure are checked in our electrophysiology clinic. To be able to get the Watchman device, you must be able to take blood-thinning and antiplatelet medicines for a time after the procedure. The Lariat procedure does not require this.

Some people cannot have either procedure. This may be because of past heart surgery. It may also be because of the size or shape of their left atrial appendage.

To find out the size and shape of your atrial appendage, your doctor will order one or more imaging tests. These may include:

- Cardiac MRI. This test uses strong magnets and radio waves to make pictures of the heart. A harmless dye may be injected into your blood. The dye helps show blood vessels and heart structures.

- Cardiac CT scan. This test takes detailed X-ray pictures of the heart and blood vessels. For people who may have the Lariat procedure, this test is used to check the size and shape of the left atrial appendage.

- Transesophageal echocardiogram, or TEE. This is an ultrasound test. A thin tube carries a small device down the throat. The device sends out sound waves. This gives clear pictures of the heart from inside the body. TEE is the preferred imaging test for the Watchman procedure.

Procedure

For both the Watchman and Lariat procedures, you will stay in the hospital for at least one night.

These procedures are done in similar ways. A catheter carrying the device is put into the femoral artery in the groin. It is then moved through blood vessels to the heart. Doctors use X-rays and TEE pictures to guide the device into place. Once the device is in place, the catheter is removed.

Each procedure takes 2 to 3 hours. You will have general anesthesia. This means you will be completely asleep.

In the Watchman procedure, the doctor reaches the left atrial appendage from inside the heart. The device is placed across the opening of the appendage. This seals off the space so blood cannot collect in it. Over time, scar tissue grows around the device. This helps hold it in place.

In the Lariat procedure, the loop is placed around the outside of the left atrial appendage. It is then pulled tight. This seals the appendage off from the rest of the heart. This procedure requires access through both the groin and the chest wall. To reach the chest area, the doctor passes a long needle under the rib cage. The doctor does not cut through the chest wall as in open-heart surgery.

Recovery

After either procedure, you must stay in bed for at least 4 hours. This helps prevent bleeding. During this time, we closely watch your heart rate, blood pressure, oxygen levels, and breathing.

Most people can go home the day after the procedure. Your doctor may tell you not to drive for several days to a few weeks. Your doctor may also tell you not to do hard exercise for 1 week. You may be told to take aspirin for about 1 month. This helps lower the risk of clots forming in your heart and blood vessels.

Remember, closing the left atrial appendage does not cure atrial fibrillation. These procedures lower stroke risk in people with A-fib. People who have one of these procedures still have A-fib. They may still feel heart racing, fluttering, or other symptoms from A-fib.

Related services and conditions

Specialties

Conditions

- Atrial Fibrillation

- Cerebrovascular Accident, or CVA

- Ischemic Stroke

UCSF Health medical specialists have reviewed this information. It is for education only. It is not meant to replace advice from your doctor or other health care provider. Please talk with your provider about any questions or concerns you have.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Left Atrial Appendage Closure

Our Approach

If you have atrial fibrillation (AFib), left atrial appendage closure may reduce your risk of stroke. This procedure safely seals off a small part of your heart to help prevent blood clots. Left atrial appendage closure is an alternative to blood thinners for eligible patients.

At Stanford Health Care, we work together to find the best way to reduce your risk of stroke. We offer left atrial appendage closure using the Watchman FLXTM and the AmplatzerTM AmuletTM. These devices are FDA-approved to lower the risk of AFib-related strokes.

Stanford Health Care is one of the few centers in the country that performs left atrial appendage closure without general anesthesia and transesophageal echocardiogram. Our approach is less invasive, and you go home the same day.

WHAT WE OFFER YOU FOR LEFT ATRIAL APPENDAGE CLOSURE

- Extensive experience and promising results with the Watchman FLX and Amplatzer Amulet devices, offering both devices commercially before any other program in Northern California after FDA approval.

- Team approach to reducing AFib-related stroke risk, bringing together top electrophysiologists, interventional cardiologists, surgeons, and imaging specialists.

- Tailored care for your needs, with our team taking the time to explain your options and the potential benefits and risks of each option.

- Active clinical research program to develop future left atrial appendage closure techniques and devices.

To schedule an appointment, please call: 650-723-6459

Left Atrial Appendage Closure

Our doctors have extensive expertise in closing left atrial appendages with WatchmanTM technology, reducing the risk for atrial fibrillation-related stroke.

left atrial appendage closure

stroke

blood clot

atrial fibrillation

watchman

warfarin

blood thinner

LAA

device

Lariat

AI rewrite

If you have an irregular heartbeat called AFib, you have a higher risk of a stroke. We can do a procedure to close off a small part of your heart. This part is called the left atrial appendage, or LAA. Closing it helps stop blood clots from forming. This is a good choice for some people who cannot take blood thinners like warfarin.

At Stanford Health Care, our team works with you to lower your stroke risk. We use special devices to close the LAA. These include the Watchman FLX and the Amplatzer Amulet. The FDA has approved these devices to help prevent strokes caused by AFib. Our doctors are experts at using Watchman technology and other devices to keep you safe.

We are one of the few hospitals that can do this procedure without putting you completely to sleep. We also do not need to put an ultrasound tube down your throat. Because of this, the treatment is easier on your body, and you can go home the exact same day.

When you come to us, you get care from a team of top heart rhythm doctors, heart surgeons, and imaging experts. We have a lot of experience. We were the first hospital in Northern California to offer these devices after they were approved. We take the time to explain all your choices. We will make sure you understand the benefits and the risks of each option. We also do research to find even better ways to close the LAA in the future.

To make an appointment, please call 650-723-6459.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Left Atrial Appendage Closure

Our Approach

If you have atrial fibrillation (AFib), left atrial appendage closure may reduce your risk of stroke. This procedure safely seals off a small part of your heart to help prevent blood clots. Left atrial appendage closure is an alternative to blood thinners for eligible patients.

At Stanford Health Care, we work together to find the best way to reduce your risk of stroke. We offer left atrial appendage closure using the Watchman FLXTM and the AmplatzerTM AmuletTM. These devices are FDA-approved to lower the risk of AFib-related strokes.

Stanford Health Care is one of the few centers in the country that performs left atrial appendage closure without general anesthesia and transesophageal echocardiogram. Our approach is less invasive, and you go home the same day.

WHAT WE OFFER YOU FOR LEFT ATRIAL APPENDAGE CLOSURE

- Extensive experience and promising results with the Watchman FLX and Amplatzer Amulet devices, offering both devices commercially before any other program in Northern California after FDA approval.

- Team approach to reducing AFib-related stroke risk, bringing together top electrophysiologists, interventional cardiologists, surgeons, and imaging specialists.

- Tailored care for your needs, with our team taking the time to explain your options and the potential benefits and risks of each option.

- Active clinical research program to develop future left atrial appendage closure techniques and devices.

To schedule an appointment, please call: 650-723-6459

Left Atrial Appendage Closure

Our doctors have extensive expertise in closing left atrial appendages with WatchmanTM technology, reducing the risk for atrial fibrillation-related stroke.

left atrial appendage closure

stroke

blood clot

atrial fibrillation

watchman

warfarin

blood thinner

LAA

device

Lariat

AI rewrite

Left Atrial Appendage Closure

Our Approach

If you have atrial fibrillation, also called AFib, left atrial appendage closure may lower your risk of stroke.

This procedure seals off a small part of your heart. This helps stop blood clots from forming there. For some patients, this procedure can be an option instead of taking blood thinners.

At Stanford Health Care, we work as a team to find the best way to lower your risk of stroke. We offer left atrial appendage closure using the Watchman FLX™ and the Amplatzer™ Amulet™. These devices are approved by the FDA to lower the risk of strokes related to AFib.

Stanford Health Care is one of the few centers in the country that does left atrial appendage closure without general anesthesia and without a transesophageal echocardiogram. This means our approach is less invasive. You go home the same day.

What We Offer You for Left Atrial Appendage Closure

- A lot of experience and promising results with the Watchman FLX and Amplatzer Amulet devices. We offered both devices before any other program in Northern California after FDA approval.

- A team approach to lowering stroke risk from AFib. Our team includes top electrophysiologists, interventional cardiologists, surgeons, and imaging specialists.

- Care that fits your needs. Our team takes time to explain your choices. We also explain the possible benefits and risks of each choice.

- An active research program to help create future left atrial appendage closure methods and devices.

To schedule an appointment, please call: 650-723-6459

Left Atrial Appendage Closure

Our doctors have a lot of experience using Watchman™ technology to close the left atrial appendage. This helps lower the risk of stroke related to atrial fibrillation.

left atrial appendage closure

stroke

blood clot

atrial fibrillation

Watchman

warfarin

blood thinner

LAA

device

Lariat

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Left Atrial Appendage Closure

Our Approach

If you have atrial fibrillation (AFib), left atrial appendage closure may reduce your risk of stroke. This procedure safely seals off a small part of your heart to help prevent blood clots. Left atrial appendage closure is an alternative to blood thinners for eligible patients.

At Stanford Health Care, we work together to find the best way to reduce your risk of stroke. We offer left atrial appendage closure using the Watchman FLXTM and the AmplatzerTM AmuletTM. These devices are FDA-approved to lower the risk of AFib-related strokes.

Stanford Health Care is one of the few centers in the country that performs left atrial appendage closure without general anesthesia and transesophageal echocardiogram. Our approach is less invasive, and you go home the same day.

WHAT WE OFFER YOU FOR LEFT ATRIAL APPENDAGE CLOSURE

- Extensive experience and promising results with the Watchman FLX and Amplatzer Amulet devices, offering both devices commercially before any other program in Northern California after FDA approval.

- Team approach to reducing AFib-related stroke risk, bringing together top electrophysiologists, interventional cardiologists, surgeons, and imaging specialists.

- Tailored care for your needs, with our team taking the time to explain your options and the potential benefits and risks of each option.

- Active clinical research program to develop future left atrial appendage closure techniques and devices.

To schedule an appointment, please call: 650-723-6459

Left Atrial Appendage Closure

Our doctors have extensive expertise in closing left atrial appendages with WatchmanTM technology, reducing the risk for atrial fibrillation-related stroke.

left atrial appendage closure

stroke

blood clot

atrial fibrillation

watchman

warfarin

blood thinner

LAA

device

Lariat

AI rewrite

Left Atrial Appendage Closure

Our Approach

Do you have atrial fibrillation (AFib)? AFib is a kind of irregular heartbeat. If you have AFib, a procedure called left atrial appendage closure may lower your risk of stroke. This procedure seals off a small part of your heart. This helps stop blood clots from forming. It is one choice instead of taking blood thinners. Not everyone can have it, but some people can.

At Stanford Health Care, we work as a team. We help find the best way to lower your risk of stroke. We use two devices to close the appendage. They are called the Watchman FLX™ and the Amplatzer™ Amulet™. The FDA has approved both. They help lower the risk of strokes caused by AFib.

Stanford Health Care is one of the few places in the country that can do this procedure without general anesthesia. We also do it without a test called a transesophageal echocardiogram. This means our way is easier on your body. Most people go home the same day.

WHAT WE OFFER YOU FOR LEFT ATRIAL APPENDAGE CLOSURE

- We have lots of experience and good results with the Watchman FLX and Amplatzer Amulet devices. We were the first program in Northern California to offer both devices after the FDA approved them.

- We use a team to lower your risk of AFib-related stroke. Our team includes top heart doctors, heart surgeons, and imaging experts.

- We give care that fits your needs. Our team takes time to explain your choices. We talk about the good points and the risks of each choice.

- We do active research. This helps us build better ways and devices for the future.

To make an appointment, please call: 650-723-6459

Left Atrial Appendage Closure

Our doctors know how to close left atrial appendages with Watchman™ technology. This lowers the risk of stroke caused by atrial fibrillation.

left atrial appendage closure

stroke

blood clot

atrial fibrillation

watchman

warfarin

blood thinner

LAA

device

Lariat

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

New to MyHealth?

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Access your health information from any device with MyHealth. You can message your clinic, view lab results, schedule an appointment, and pay your bill.

ALREADY HAVE AN ACCESS CODE?

DON'T HAVE AN ACCESS CODE?

NEED MORE DETAILS?

MyHealth for Mobile

Watchman FLX

Our Approach

If you have atrial fibrillation (AFib), left atrial appendage closure may reduce your risk of stroke. This procedure safely seals off a small part of your heart to help prevent blood clots. Left atrial appendage closure is an alternative to blood thinners for eligible patients.

At Stanford Health Care, we work together to find the best way to reduce your risk of stroke. We offer left atrial appendage closure using the Watchman FLXTM and the AmplatzerTM AmuletTM. These devices are FDA-approved to lower the risk of AFib-related strokes.

Stanford Health Care is one of the few centers in the country that performs left atrial appendage closure without general anesthesia and transesophageal echocardiogram. Our approach is less invasive, and you go home the same day.

WHAT WE OFFER YOU FOR LEFT ATRIAL APPENDAGE CLOSURE

- Extensive experience and promising results with the Watchman FLX and Amplatzer Amulet devices, offering both devices commercially before any other program in Northern California after FDA approval.

- Team approach to reducing AFib-related stroke risk, bringing together top electrophysiologists, interventional cardiologists, surgeons, and imaging specialists.

- Tailored care for your needs, with our team taking the time to explain your options and the potential benefits and risks of each option.

- Active clinical research program to develop future left atrial appendage closure techniques and devices.

To schedule an appointment, please call: 650-723-6459

Watchman FLX

Watchman FLX is a metal and fabric implant the size of a quarter that blocks the opening to your left atrial appendage. It comes in different sizes, and our team chooses the size that fits you.

With this implant in place, blood clots cannot form in your left atrial appendage, so your risk of stroke is reduced.

In clinical trials, more than 90% of people who received the Watchman FLX could stop taking warfarin once the treatment fully started working. During the procedure, our team:

- Gives you local anesthesia and sedation, so you will be comfortable

- Makes a small incision in your femoral vein in your upper leg

- Guides a catheter through your femoral vein to your heart

- Inserts the Watchman FLX in the catheter and places it in your left atrial appendage, with tiny hooks holding it in place

- Uses advanced imaging to make sure the device firmly attaches and completely covers the opening of your left atrial appendage

- Removes the catheter and closes the incision in your leg

It takes several weeks for your heart tissue to completely close around the Watchman FLX. During these first few weeks, keep taking your blood thinners. Don't stop until your doctor says it's safe to do so.

Learn more about Watchman FLX and the implant technique here.

Interested in an Online Second Opinion?

The Stanford Medicine Online Second Opinion program offers you easy access to our world-class doctors. It's all done remotely, and you don't have to visit our hospital or one of our clinics for this service. You don't even need to leave home!

Visit our online second opinion page to learn more.

Our Clinics

Our Structural Heart Program offers expertise in leading-edge, minimally invasive procedures, and open-heart surgery to treat structural heart disease. Together, our interventional cardiologists, heart surgeons, heart failure specialists, and imaging experts provide the most advanced and personalized care.

AI rewrite

New to MyHealth?

Take Care of Your Health From Anywhere.

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ALREADY HAVE AN ACCESS CODE?

DON'T HAVE AN ACCESS CODE?

NEED MORE DETAILS?

MyHealth for Mobile

Watchman FLX

Our Approach

Do you have atrial fibrillation (AFib)? If so, a procedure called left atrial appendage closure may lower your risk of stroke. This procedure safely seals off a small part of your heart. This helps stop blood clots from forming. It is another choice instead of blood thinners for people who qualify.

At Stanford Health Care, we work as a team. We find the best way to lower your risk of stroke. We use two devices for this procedure. They are the Watchman FLX™ and the Amplatzer™ Amulet™. The FDA has approved both devices to lower the risk of strokes from AFib.

Stanford Health Care is one of the few centers in the country that does this procedure without general anesthesia. We also do not use a transesophageal echocardiogram. Our way is easier on your body. You can go home the same day.

WHAT WE OFFER YOU FOR LEFT ATRIAL APPENDAGE CLOSURE

- We have a lot of experience with the Watchman FLX and Amplatzer Amulet devices. We see good results. We were the first program in Northern California to offer both devices after the FDA approved them.

- We use a team to lower your risk of stroke from AFib. Our team brings together top heart doctors, surgeons, and imaging experts.

- We give you care that fits your needs. Our team takes time to explain your choices. We tell you the good points and the risks of each choice.

- We have an active research program. We work to make new tools and ways to do this procedure.

To set up an appointment, please call: 650-723-6459

Watchman FLX

The Watchman FLX is a small implant made of metal and fabric. It is about the size of a quarter. It blocks the opening to your left atrial appendage. It comes in different sizes. Our team picks the size that fits you.

With this implant in place, blood clots cannot form in your left atrial appendage. This lowers your risk of stroke.

In studies, more than 9 out of 10 people who got the Watchman FLX could stop taking warfarin once the treatment fully worked. During the procedure, our team:

- Gives you local anesthesia and medicine to help you relax, so you feel comfortable

- Makes a small cut in the femoral vein in your upper leg

- Guides a thin tube, called a catheter, through this vein to your heart

- Puts the Watchman FLX in the tube and places it in your left atrial appendage. Tiny hooks hold it in place.

- Uses special imaging to make sure the device is firmly in place. It checks that the device fully covers the opening of your left atrial appendage.

- Takes out the tube and closes the cut in your leg

It takes a few weeks for your heart tissue to fully close around the Watchman FLX. During these first weeks, keep taking your blood thinners. Do not stop until your doctor says it is safe.

Learn more about Watchman FLX and how the implant works here.

Interested in an Online Second Opinion?

The Stanford Medicine Online Second Opinion program gives you easy access to our top doctors. It is all done online. You do not have to come to our hospital or clinic. You do not even need to leave home!

Visit our online second opinion page to learn more.

Our Clinics

Our Structural Heart Program has experts in new procedures that are easier on your body. We also do open-heart surgery to treat heart problems. Our team works together. It includes heart doctors, heart surgeons, heart failure experts, and imaging experts. Together, they give you the most advanced care that fits your needs.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**

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Original page

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MyHealth for Mobile

Watchman FLX

Our Approach

If you have atrial fibrillation (AFib), left atrial appendage closure may reduce your risk of stroke. This procedure safely seals off a small part of your heart to help prevent blood clots. Left atrial appendage closure is an alternative to blood thinners for eligible patients.

At Stanford Health Care, we work together to find the best way to reduce your risk of stroke. We offer left atrial appendage closure using the Watchman FLXTM and the AmplatzerTM AmuletTM. These devices are FDA-approved to lower the risk of AFib-related strokes.

Stanford Health Care is one of the few centers in the country that performs left atrial appendage closure without general anesthesia and transesophageal echocardiogram. Our approach is less invasive, and you go home the same day.

WHAT WE OFFER YOU FOR LEFT ATRIAL APPENDAGE CLOSURE

- Extensive experience and promising results with the Watchman FLX and Amplatzer Amulet devices, offering both devices commercially before any other program in Northern California after FDA approval.

- Team approach to reducing AFib-related stroke risk, bringing together top electrophysiologists, interventional cardiologists, surgeons, and imaging specialists.

- Tailored care for your needs, with our team taking the time to explain your options and the potential benefits and risks of each option.

- Active clinical research program to develop future left atrial appendage closure techniques and devices.

To schedule an appointment, please call: 650-723-6459

Watchman FLX

Watchman FLX is a metal and fabric implant the size of a quarter that blocks the opening to your left atrial appendage. It comes in different sizes, and our team chooses the size that fits you.

With this implant in place, blood clots cannot form in your left atrial appendage, so your risk of stroke is reduced.

In clinical trials, more than 90% of people who received the Watchman FLX could stop taking warfarin once the treatment fully started working. During the procedure, our team:

- Gives you local anesthesia and sedation, so you will be comfortable

- Makes a small incision in your femoral vein in your upper leg

- Guides a catheter through your femoral vein to your heart

- Inserts the Watchman FLX in the catheter and places it in your left atrial appendage, with tiny hooks holding it in place

- Uses advanced imaging to make sure the device firmly attaches and completely covers the opening of your left atrial appendage

- Removes the catheter and closes the incision in your leg

It takes several weeks for your heart tissue to completely close around the Watchman FLX. During these first few weeks, keep taking your blood thinners. Don't stop until your doctor says it's safe to do so.

Learn more about Watchman FLX and the implant technique here.

Interested in an Online Second Opinion?

The Stanford Medicine Online Second Opinion program offers you easy access to our world-class doctors. It's all done remotely, and you don't have to visit our hospital or one of our clinics for this service. You don't even need to leave home!

Visit our online second opinion page to learn more.

Our Clinics

Our Structural Heart Program offers expertise in leading-edge, minimally invasive procedures, and open-heart surgery to treat structural heart disease. Together, our interventional cardiologists, heart surgeons, heart failure specialists, and imaging experts provide the most advanced and personalized care.

AI rewrite

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Watchman FLX

Our Approach

If you have atrial fibrillation (AFib), closing a small part of your heart can lower your chance of a stroke. This procedure safely seals off a small pouch in your heart to help stop blood clots. Closing this pouch is a choice instead of taking blood thinners for some patients.

At Stanford Health Care, we work together to lower your chance of a stroke. We use devices called the Watchman FLX and the Amplatzer Amulet. The FDA approves these devices to lower the chance of strokes caused by AFib.

We are one of the few hospitals in the country that do this without putting you completely to sleep. We also do not need to put an ultrasound tube down your throat to look at your heart. Our way is gentler on your body, and you can go home the same day.

What we offer you for this procedure:

- We have a lot of experience with the Watchman FLX and Amplatzer Amulet. We were the first program in Northern California to offer both devices after FDA approval.

- A team of top heart doctors, surgeons, and imaging experts works together to lower your stroke risk.

- We plan care just for you. We take time to explain your choices and the good and bad points of each one.

- We do active research to find new ways and tools to treat the heart.

To make an appointment, please call: 650-723-6459

Watchman FLX

The Watchman FLX is an implant made of metal and cloth. It is the size of a quarter. It blocks the opening to the small pouch in your heart. It comes in different sizes, and our team picks the best size for you.

When this implant is in place, blood clots cannot form in that pouch. This lowers your chance of a stroke.

In studies, more than 90 out of 100 people who got the Watchman FLX could stop taking their blood thinner medicine (warfarin). They stopped once the treatment fully worked. During the procedure, our team will:

- Give you medicine to numb the area and help you relax, so you will be comfortable.

- Make a small cut in a large blood vessel in your upper leg.

- Guide a thin tube through this blood vessel up to your heart.

- Put the Watchman FLX through the tube and into the pouch in your heart. Tiny hooks hold it in place.

- Use special pictures to make sure the device is attached tightly and covers the opening.

- Take out the tube and close the cut in your leg.

It takes a few weeks for your heart tissue to grow completely over the Watchman FLX. You must keep taking your blood thinners during these first few weeks. Do not stop taking them until your doctor says it is safe to do so.

Learn more about Watchman FLX and how it is put in here.

Interested in an online second opinion?

Our online second opinion program lets you talk to our top doctors easily. It is all done over the computer or phone. You do not have to visit our hospital or clinics for this service. You do not even need to leave your home!

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Our Clinics

Our Structural Heart Program treats heart disease. We use the newest procedures that use small cuts, as well as open-heart surgery. Our team of heart doctors, surgeons, and imaging experts works together. They give you the most advanced care made just for you.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

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At Stanford Health Care, we work together to find the best way to reduce your risk of stroke. We offer left atrial appendage closure using the Watchman FLXTM and the AmplatzerTM AmuletTM. These devices are FDA-approved to lower the risk of AFib-related strokes.

Stanford Health Care is one of the few centers in the country that performs left atrial appendage closure without general anesthesia and transesophageal echocardiogram. Our approach is less invasive, and you go home the same day.

WHAT WE OFFER YOU FOR LEFT ATRIAL APPENDAGE CLOSURE

- Extensive experience and promising results with the Watchman FLX and Amplatzer Amulet devices, offering both devices commercially before any other program in Northern California after FDA approval.

- Team approach to reducing AFib-related stroke risk, bringing together top electrophysiologists, interventional cardiologists, surgeons, and imaging specialists.

- Tailored care for your needs, with our team taking the time to explain your options and the potential benefits and risks of each option.

- Active clinical research program to develop future left atrial appendage closure techniques and devices.

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Watchman FLX

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In clinical trials, more than 90% of people who received the Watchman FLX could stop taking warfarin once the treatment fully started working. During the procedure, our team:

- Gives you local anesthesia and sedation, so you will be comfortable

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- Guides a catheter through your femoral vein to your heart

- Inserts the Watchman FLX in the catheter and places it in your left atrial appendage, with tiny hooks holding it in place

- Uses advanced imaging to make sure the device firmly attaches and completely covers the opening of your left atrial appendage

- Removes the catheter and closes the incision in your leg

It takes several weeks for your heart tissue to completely close around the Watchman FLX. During these first few weeks, keep taking your blood thinners. Don't stop until your doctor says it's safe to do so.

Learn more about Watchman FLX and the implant technique here.

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AI rewrite

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Do you need an access code?

Need more details?

MyHealth for Mobile

Watchman FLX

Our Approach

If you have atrial fibrillation (AFib), closing the left atrial appendage may lower your risk of stroke. The left atrial appendage is a small part of your heart. This procedure seals it off to help stop blood clots from forming. This is an option instead of blood thinners for some patients.

At Stanford Health Care, we work with you to find the best way to lower your risk of stroke. We offer left atrial appendage closure with the Watchman FLXTM and the AmplatzerTM AmuletTM. These devices are approved by the FDA to lower the risk of strokes related to AFib.

Stanford Health Care is one of the few centers in the country that can do left atrial appendage closure without general anesthesia and without a transesophageal echocardiogram. This way is less invasive. You can go home the same day.

What we offer for left atrial appendage closure:

- A lot of experience and promising results with the Watchman FLX and Amplatzer Amulet devices. We offered both devices before any other program in Northern California after FDA approval.

- A team approach to lowering stroke risk related to AFib. Our team includes top heart rhythm doctors, heart procedure doctors, surgeons, and imaging specialists.

- Care that fits your needs. Our team takes time to explain your choices. We also explain the possible benefits and risks of each choice.

- An active research program to help develop future left atrial appendage closure methods and devices.

To make an appointment, please call: 650-723-6459

Watchman FLX

Watchman FLX is an implant made of metal and fabric. It is about the size of a quarter. It blocks the opening to your left atrial appendage. It comes in different sizes. Our team chooses the size that fits you.

With this implant in place, blood clots cannot form in your left atrial appendage. This lowers your risk of stroke.

In clinical trials, more than 90% of people who got the Watchman FLX were able to stop taking warfarin after the treatment was fully working.

During the procedure, our team:

- Gives you local anesthesia and sedation so you feel comfortable
- Makes a small cut in your femoral vein in your upper leg
- Guides a thin tube called a catheter through your femoral vein to your heart
- Puts the Watchman FLX into the catheter and places it in your left atrial appendage
- Uses tiny hooks to hold it in place
- Uses advanced imaging to make sure the device is attached well and fully covers the opening of your left atrial appendage
- Removes the catheter and closes the cut in your leg

It takes several weeks for your heart tissue to fully close around the Watchman FLX. During these first few weeks, keep taking your blood thinners. Do not stop until your doctor says it is safe.

Learn more about Watchman FLX and how the implant is placed here.

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Our Clinics

Our Structural Heart Program offers expert care in advanced, less invasive procedures and open-heart surgery. We treat structural heart disease. Our heart procedure doctors, heart surgeons, heart failure specialists, and imaging experts work together. They provide advanced care that is made for you.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Left atrial appendage (LAA) closure is a procedure that blocks or closes the opening to your LAA to keep blood clots from leaving there and going into your bloodstream. This prevents strokes in people with atrial fibrillation, but without blood thinners. This appendage in one of your heart's upper chambers doesn't have a function.

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Left atrial appendage closure is a surgical or minimally invasive procedure to seal off your left atrial appendage (LAA). This is a small sac in the muscle wall of your left atrium (top left chamber of your heart). Removing it or closing it off can reduce your risk of stroke and end the need to take blood thinners.

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Among people with atrial fibrillation (Afib) who don't have valve disease, most of the blood clots that happen in the left atrium start in the LAA.

Normal hearts contract with each heartbeat, squeezing the blood in your left atrium and LAA out into your left ventricle (bottom left chamber of the heart). When you have Afib, the electrical impulses that control your heartbeat don't travel in an orderly way. They can be fast and chaotic, which doesn't give your atria time to squeeze blood into your ventricles.

Because the LAA is a pouch, blood collects there and can form clots. When your heart pumps out blood clots, they can cause a stroke. People with Afib are three to five times more likely to have a stroke than the general population. While this is unsettling, your provider can help you protect yourself with a solution that works for you. This can involve medications that make it more difficult for these blood clots to form or closing off the left atrial appendage.

Your heart can keep doing its job with a closed left atrial appendage.

Thousands of people worldwide have had an LAA closure. Providers use different methods and devices, like:

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LAA closure reduces the stroke risk associated with Afib. But it doesn't treat the Afib itself. Left atrial appendage closure can benefit people who have an increased risk of stroke due to Afib. If you're at risk of getting blood clots in your left atrium/left atrial appendage, your healthcare provider may recommend sealing off your LAA to reduce your risk of stroke from Afib. They can do this instead of prescribing a blood thinner like warfarin (Coumadin® or Jantoven®), apixaban (Eliquis®), rivaroxaban (Xarelto®) or dabigatran (Pradaxa®).

Many people have concerns about, or dislike, taking warfarin. Some of the reasons for this are:

People with Afib who don't have heart valve disease can take other medications - apixaban, dabigatran and rivaroxaban. But some people have concerns and problems with these medications:

Before an LAA closure, you'll get a transesophageal echocardiogram (TEE) and/or cardiac CT. This gives your provider dimensions of your left atrial appendage. This matters because left atrial appendages vary from person to person. Researchers have compared the left atrial appendage's shape to a chicken wing, cactus, windsock or cauliflower.

Your provider may also do three-dimensional printing to help choose the correct device and the right size.

You'll need to fast (stop eating and drinking) at a certain time before your procedure. Your provider will tell you when to fast.

You'll receive anesthesia of some type, depending on which procedure you're having.

If you're having heart surgery for other reasons, like coronary artery disease or valve disease, your healthcare provider frequently can remove your left atrial appendage at the same time. They may sew or staple the area closed. This is a surgical left atrial appendage closure.

In a minimally invasive approach, they can use a catheter to insert a special device to close or block your left atrial appendage. These devices may act as a plug in your left atrial appendage's opening to your left atrium. Other devices tie or clamp your left atrial appendage closed.

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For a device your provider uses a catheter to implant, they will:

Several types of imaging guide your provider while they work inside your heart. Types of imaging they use include:

After your surgery or minimally invasive procedure, you may:

People who get an LAA closure can:

Left atrial appendage closure complications may include:

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If you had your LAA closure as part of a heart surgery for another issue, your recovery time will vary depending on which other surgery you had.

If you had a minimally invasive procedure using a catheter to put in a left atrial appendage closure device, your recovery will be quicker. You may be able to go home the next day.

After getting a device, you'll need to take an anticoagulant (blood thinner) and aspirin for 45 days. This is how long it takes your body's tissue to form around the device so there aren't any gaps around it. Then, you'll stop taking an anticoagulant and switch to clopidogrel (Plavix®) and aspirin for six months.

You may be on aspirin for the long term.

You'll have another transesophageal echo (TEE) at a follow-up visit 45 days after your procedure. This helps your provider see if your left atrial appendage is fully blocked.

If the TEE shows that your left atrial appendage isn't blocked, you'll keep taking the anticoagulant and have another TEE and follow-up visit after six months.

Once your left atrial appendage is blocked, you'll have an annual follow-up visit with your provider.

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Researchers have linked LAA closure with suture to a 40% decrease in the risk of a blood clot, and over 90% with clip or surgical excision. They found an 80% decrease in strokes in device studies. Some devices have rates above 95% for successful placement.

While there's no surefire way to prevent strokes, removing or blocking your left atrial appendage reduces your risk significantly. This is because 90% of strokes that start in an upper heart chamber start in your LAA.

In addition to going to your scheduled visits, you should contact your provider if you have:

It can be hard living with the threat of stroke from Afib, especially if you have trouble with blood thinners. Deciding on a left atrial appendage closure is a personal decision. After talking with your healthcare provider and discussing your options, you can make an informed decision. The procedure isn't right for everyone. But if you have Afib, it's an option you can consider. Be sure to ask your provider any questions you have about the procedure.

If you have a left atrial appendage closure procedure, keep taking your medicines until your provider tells you to stop. Follow-up visits are important, too. These ensure you're healing and your device is fitting well.

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When your heart rhythm is out of sync, the experts at Cleveland Clinic can find out why. We offer personalized care for all types of arrhythmias.

AI rewrite

Left atrial appendage closure is a procedure that blocks or closes the opening to your left atrial appendage, or LAA. This helps keep blood clots from leaving the LAA and going into your blood. It can help prevent strokes in people with atrial fibrillation, or Afib, without using blood thinners. The LAA is a small pouch in one of the top chambers of your heart. It does not have a known function.

The LAA is in the wall of your left atrium. The left atrium is the top left chamber of your heart. Removing or closing the LAA can lower your risk of stroke. It may also mean you do not need to take blood thinners.

In people with Afib who do not have heart valve disease, most blood clots in the left atrium start in the LAA.

A normal heart squeezes with each heartbeat. This pushes blood from the left atrium and LAA into the left ventricle. The left ventricle is the bottom left chamber of the heart.

When you have Afib, the electrical signals that control your heartbeat do not move in a normal way. They can be fast and mixed up. This can keep your atria from squeezing blood well into your ventricles.

The LAA is like a pouch. Blood can collect there and form clots. If your heart pumps out these clots, they can cause a stroke. People with Afib are three to five times more likely to have a stroke than people in general.

Your provider can help you find a way to lower your risk. This may include medicines that make clots harder to form. It may also include closing the LAA.

Your heart can still work with a closed LAA.

Thousands of people around the world have had LAA closure. Providers use different methods and devices.

LAA closure lowers the stroke risk linked to Afib. But it does not treat Afib itself.

LAA closure may help people who have a higher risk of stroke because of Afib. If you are at risk for blood clots in your left atrium or LAA, your provider may suggest closing your LAA. This can lower your risk of stroke from Afib.

Your provider may suggest this instead of a blood thinner. Blood thinners include warfarin, also called Coumadin® or Jantoven®. They also include apixaban, also called Eliquis®, rivaroxaban, also called Xarelto®, and dabigatran, also called Pradaxa®.

Many people worry about taking warfarin or do not like taking it. Some people with Afib who do not have heart valve disease can take other medicines, such as apixaban, dabigatran or rivaroxaban. But some people have worries or problems with these medicines, too.

Before LAA closure, you will have a test called a transesophageal echocardiogram, or TEE. You may also have a cardiac CT scan. These tests show the size and shape of your LAA. This is important because each person’s LAA is different. Researchers have said the LAA can look like a chicken wing, cactus, windsock or cauliflower.

Your provider may also use 3D printing to help choose the right device and size.

You will need to fast before your procedure. This means you must stop eating and drinking at a certain time. Your provider will tell you when to stop.

You will get some type of anesthesia. The type depends on which procedure you have.

If you are having heart surgery for another reason, your provider may be able to remove your LAA at the same time. This may happen during surgery for coronary artery disease or valve disease. Your provider may sew or staple the area closed. This is surgical LAA closure.

In a less invasive procedure, your provider can use a catheter to place a special device. A catheter is a thin tube. The device closes or blocks your LAA. Some devices act like a plug at the opening between the LAA and the left atrium. Other devices tie or clamp the LAA closed.

Your provider uses imaging while working inside your heart. These images help guide the procedure.

After surgery or a less invasive procedure, you may need time to recover.

People who get LAA closure may be able to lower their risk of stroke from Afib. Some people may be able to stop taking long-term blood thinners.

LAA closure can have complications.

If your LAA closure was part of heart surgery for another problem, your recovery time will depend on the other surgery you had.

If you had a less invasive procedure with a catheter and a closure device, you may recover faster. You may be able to go home the next day.

After getting a device, you will need to take an anticoagulant and aspirin for 45 days. An anticoagulant is a blood thinner. This gives your body time to grow tissue around the device so there are no gaps around it.

After that, you will stop taking the anticoagulant. You will switch to clopidogrel, also called Plavix®, and aspirin for six months.

You may need to take aspirin for a long time.

You will have another TEE at a follow-up visit 45 days after your procedure. This helps your provider see if your LAA is fully blocked.

If the TEE shows that your LAA is not blocked, you will keep taking the anticoagulant. You will have another TEE and follow-up visit after six months.

Once your LAA is blocked, you will have a follow-up visit with your provider every year.

Researchers have linked LAA closure with suture to a 40% lower risk of a blood clot. They found a risk drop of more than 90% with a clip or surgical removal. Device studies found an 80% drop in strokes. Some devices have success rates above 95% for being placed correctly.

There is no sure way to prevent strokes. But removing or blocking your LAA can greatly lower your risk. This is because 90% of strokes that start in an upper heart chamber start in the LAA.

Go to all of your scheduled visits. You should also contact your provider if you have symptoms or concerns after the procedure.

It can be hard to live with the risk of stroke from Afib. This may be even harder if you have trouble with blood thinners. Choosing LAA closure is a personal choice. Talk with your provider about your options. Then you can make a choice with the facts you need.

This procedure is not right for everyone. But if you have Afib, it may be an option to think about. Ask your provider any questions you have.

If you have LAA closure, keep taking your medicines until your provider tells you to stop. Follow-up visits are important. They help make sure you are healing and that your device fits well.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 73 of 77 · Item ID B-COSS3EOQM1

Original page

Left atrial appendage (LAA) closure is a procedure that blocks or closes the opening to your LAA to keep blood clots from leaving there and going into your bloodstream. This prevents strokes in people with atrial fibrillation, but without blood thinners. This appendage in one of your heart's upper chambers doesn't have a function.

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Left atrial appendage closure is a surgical or minimally invasive procedure to seal off your left atrial appendage (LAA). This is a small sac in the muscle wall of your left atrium (top left chamber of your heart). Removing it or closing it off can reduce your risk of stroke and end the need to take blood thinners.

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Among people with atrial fibrillation (Afib) who don't have valve disease, most of the blood clots that happen in the left atrium start in the LAA.

Normal hearts contract with each heartbeat, squeezing the blood in your left atrium and LAA out into your left ventricle (bottom left chamber of the heart). When you have Afib, the electrical impulses that control your heartbeat don't travel in an orderly way. They can be fast and chaotic, which doesn't give your atria time to squeeze blood into your ventricles.

Because the LAA is a pouch, blood collects there and can form clots. When your heart pumps out blood clots, they can cause a stroke. People with Afib are three to five times more likely to have a stroke than the general population. While this is unsettling, your provider can help you protect yourself with a solution that works for you. This can involve medications that make it more difficult for these blood clots to form or closing off the left atrial appendage.

Your heart can keep doing its job with a closed left atrial appendage.

Thousands of people worldwide have had an LAA closure. Providers use different methods and devices, like:

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LAA closure reduces the stroke risk associated with Afib. But it doesn't treat the Afib itself. Left atrial appendage closure can benefit people who have an increased risk of stroke due to Afib. If you're at risk of getting blood clots in your left atrium/left atrial appendage, your healthcare provider may recommend sealing off your LAA to reduce your risk of stroke from Afib. They can do this instead of prescribing a blood thinner like warfarin (Coumadin® or Jantoven®), apixaban (Eliquis®), rivaroxaban (Xarelto®) or dabigatran (Pradaxa®).

Many people have concerns about, or dislike, taking warfarin. Some of the reasons for this are:

People with Afib who don't have heart valve disease can take other medications - apixaban, dabigatran and rivaroxaban. But some people have concerns and problems with these medications:

Before an LAA closure, you'll get a transesophageal echocardiogram (TEE) and/or cardiac CT. This gives your provider dimensions of your left atrial appendage. This matters because left atrial appendages vary from person to person. Researchers have compared the left atrial appendage's shape to a chicken wing, cactus, windsock or cauliflower.

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You'll receive anesthesia of some type, depending on which procedure you're having.

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In a minimally invasive approach, they can use a catheter to insert a special device to close or block your left atrial appendage. These devices may act as a plug in your left atrial appendage's opening to your left atrium. Other devices tie or clamp your left atrial appendage closed.

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For a device your provider uses a catheter to implant, they will:

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People who get an LAA closure can:

Left atrial appendage closure complications may include:

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If you had your LAA closure as part of a heart surgery for another issue, your recovery time will vary depending on which other surgery you had.

If you had a minimally invasive procedure using a catheter to put in a left atrial appendage closure device, your recovery will be quicker. You may be able to go home the next day.

After getting a device, you'll need to take an anticoagulant (blood thinner) and aspirin for 45 days. This is how long it takes your body's tissue to form around the device so there aren't any gaps around it. Then, you'll stop taking an anticoagulant and switch to clopidogrel (Plavix®) and aspirin for six months.

You may be on aspirin for the long term.

You'll have another transesophageal echo (TEE) at a follow-up visit 45 days after your procedure. This helps your provider see if your left atrial appendage is fully blocked.

If the TEE shows that your left atrial appendage isn't blocked, you'll keep taking the anticoagulant and have another TEE and follow-up visit after six months.

Once your left atrial appendage is blocked, you'll have an annual follow-up visit with your provider.

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While there's no surefire way to prevent strokes, removing or blocking your left atrial appendage reduces your risk significantly. This is because 90% of strokes that start in an upper heart chamber start in your LAA.

In addition to going to your scheduled visits, you should contact your provider if you have:

It can be hard living with the threat of stroke from Afib, especially if you have trouble with blood thinners. Deciding on a left atrial appendage closure is a personal decision. After talking with your healthcare provider and discussing your options, you can make an informed decision. The procedure isn't right for everyone. But if you have Afib, it's an option you can consider. Be sure to ask your provider any questions you have about the procedure.

If you have a left atrial appendage closure procedure, keep taking your medicines until your provider tells you to stop. Follow-up visits are important, too. These ensure you're healing and your device is fitting well.

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Cleveland Clinic's health articles are based on evidence-backed information and review by medical professionals to ensure accuracy, reliability and up-to-date clinical standards.

When your heart rhythm is out of sync, the experts at Cleveland Clinic can find out why. We offer personalized care for all types of arrhythmias.

AI rewrite

Your heart has a small pouch in its top left chamber. This pouch is called the left atrial appendage, or LAA. It does not have a special job, and your heart works fine without it. Sometimes, doctors need to close or remove this pouch. This procedure is called LAA closure. It helps stop blood clots from leaving the pouch and going into your body. This lowers your risk of having a stroke. It can also help you stop taking blood thinner medicines.

A normal heart squeezes blood out with each beat. If you have atrial fibrillation, or Afib, your heartbeat is fast and uneven. Your heart does not have time to squeeze the blood out. Because the LAA is shaped like a pouch, blood can pool there and turn into clots. If your heart pumps a clot out, it can cause a stroke. People with Afib are three to five times more likely to have a stroke. Most clots in the top left part of the heart start in the LAA. Closing the LAA lowers your stroke risk, but it does not cure Afib.

This procedure is for people with Afib who do not have heart valve disease. It is a good choice if you have a high risk of stroke but have problems taking blood thinners. Blood thinners include medicines like warfarin, apixaban, rivaroxaban, and dabigatran.

Before the procedure, your doctor will take pictures of your heart. They may use a special ultrasound called a TEE or a CT scan. This helps them measure your LAA, because everyone's LAA is a different shape. Your doctor might even print a 3D model of your heart to pick the right size device. You will need to stop eating and drinking before your procedure. Your doctor will tell you exactly when to stop. You will also get medicine called anesthesia so you do not feel pain.

There are two main ways to close the LAA. If you are already having heart surgery, the doctor can close the LAA at the same time. They will sew or staple the pouch shut. The second way uses a long, thin tube called a catheter. The doctor puts the tube into a blood vessel and guides it to your heart using X-rays and ultrasounds. Then, they place a special device through the tube to plug, tie, or clamp the LAA closed.

If you had the tube procedure, you will likely go home the next day. If you had heart surgery, your recovery will take longer. After you get a device, you must take a blood thinner and aspirin for 45 days. This gives your body time to grow tissue over the device to seal it. After 45 days, you will stop the blood thinner. You will then take aspirin and a medicine called clopidogrel for six months. You might need to take aspirin for the rest of your life. Do not stop taking your medicines until your doctor tells you to.

You will see your doctor 45 days after the procedure. They will do another TEE ultrasound to make sure the pouch is fully blocked. If it is not blocked yet, you will keep taking your blood thinner. You will have another check-up in six months. Once the pouch is fully blocked, you will see your doctor once a year.

Closing the LAA works very well. It can lower your risk of a stroke or blood clot by 40 to over 90 percent. Like any procedure, there are risks. Problems can include bleeding, fluid around the heart, or a stroke. The device could also move, or a clot could form on the device itself.

Call your doctor right away if you have chest pain or trouble breathing. You should also call if you feel dizzy or if you are bleeding where the tube went into your body.

Choosing to have this procedure is a personal choice. Talk to your doctor to see if it is right for you. Going to your follow-up visits is very important to make sure you are healing well.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Left atrial appendage (LAA) closure is a procedure that blocks or closes the opening to your LAA to keep blood clots from leaving there and going into your bloodstream. This prevents strokes in people with atrial fibrillation, but without blood thinners. This appendage in one of your heart's upper chambers doesn't have a function.

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Left atrial appendage closure is a surgical or minimally invasive procedure to seal off your left atrial appendage (LAA). This is a small sac in the muscle wall of your left atrium (top left chamber of your heart). Removing it or closing it off can reduce your risk of stroke and end the need to take blood thinners.

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Among people with atrial fibrillation (Afib) who don't have valve disease, most of the blood clots that happen in the left atrium start in the LAA.

Normal hearts contract with each heartbeat, squeezing the blood in your left atrium and LAA out into your left ventricle (bottom left chamber of the heart). When you have Afib, the electrical impulses that control your heartbeat don't travel in an orderly way. They can be fast and chaotic, which doesn't give your atria time to squeeze blood into your ventricles.

Because the LAA is a pouch, blood collects there and can form clots. When your heart pumps out blood clots, they can cause a stroke. People with Afib are three to five times more likely to have a stroke than the general population. While this is unsettling, your provider can help you protect yourself with a solution that works for you. This can involve medications that make it more difficult for these blood clots to form or closing off the left atrial appendage.

Your heart can keep doing its job with a closed left atrial appendage.

Thousands of people worldwide have had an LAA closure. Providers use different methods and devices, like:

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LAA closure reduces the stroke risk associated with Afib. But it doesn't treat the Afib itself. Left atrial appendage closure can benefit people who have an increased risk of stroke due to Afib. If you're at risk of getting blood clots in your left atrium/left atrial appendage, your healthcare provider may recommend sealing off your LAA to reduce your risk of stroke from Afib. They can do this instead of prescribing a blood thinner like warfarin (Coumadin® or Jantoven®), apixaban (Eliquis®), rivaroxaban (Xarelto®) or dabigatran (Pradaxa®).

Many people have concerns about, or dislike, taking warfarin. Some of the reasons for this are:

People with Afib who don't have heart valve disease can take other medications - apixaban, dabigatran and rivaroxaban. But some people have concerns and problems with these medications:

Before an LAA closure, you'll get a transesophageal echocardiogram (TEE) and/or cardiac CT. This gives your provider dimensions of your left atrial appendage. This matters because left atrial appendages vary from person to person. Researchers have compared the left atrial appendage's shape to a chicken wing, cactus, windsock or cauliflower.

Your provider may also do three-dimensional printing to help choose the correct device and the right size.

You'll need to fast (stop eating and drinking) at a certain time before your procedure. Your provider will tell you when to fast.

You'll receive anesthesia of some type, depending on which procedure you're having.

If you're having heart surgery for other reasons, like coronary artery disease or valve disease, your healthcare provider frequently can remove your left atrial appendage at the same time. They may sew or staple the area closed. This is a surgical left atrial appendage closure.

In a minimally invasive approach, they can use a catheter to insert a special device to close or block your left atrial appendage. These devices may act as a plug in your left atrial appendage's opening to your left atrium. Other devices tie or clamp your left atrial appendage closed.

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For a device your provider uses a catheter to implant, they will:

Several types of imaging guide your provider while they work inside your heart. Types of imaging they use include:

After your surgery or minimally invasive procedure, you may:

People who get an LAA closure can:

Left atrial appendage closure complications may include:

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If you had your LAA closure as part of a heart surgery for another issue, your recovery time will vary depending on which other surgery you had.

If you had a minimally invasive procedure using a catheter to put in a left atrial appendage closure device, your recovery will be quicker. You may be able to go home the next day.

After getting a device, you'll need to take an anticoagulant (blood thinner) and aspirin for 45 days. This is how long it takes your body's tissue to form around the device so there aren't any gaps around it. Then, you'll stop taking an anticoagulant and switch to clopidogrel (Plavix®) and aspirin for six months.

You may be on aspirin for the long term.

You'll have another transesophageal echo (TEE) at a follow-up visit 45 days after your procedure. This helps your provider see if your left atrial appendage is fully blocked.

If the TEE shows that your left atrial appendage isn't blocked, you'll keep taking the anticoagulant and have another TEE and follow-up visit after six months.

Once your left atrial appendage is blocked, you'll have an annual follow-up visit with your provider.

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Researchers have linked LAA closure with suture to a 40% decrease in the risk of a blood clot, and over 90% with clip or surgical excision. They found an 80% decrease in strokes in device studies. Some devices have rates above 95% for successful placement.

While there's no surefire way to prevent strokes, removing or blocking your left atrial appendage reduces your risk significantly. This is because 90% of strokes that start in an upper heart chamber start in your LAA.

In addition to going to your scheduled visits, you should contact your provider if you have:

It can be hard living with the threat of stroke from Afib, especially if you have trouble with blood thinners. Deciding on a left atrial appendage closure is a personal decision. After talking with your healthcare provider and discussing your options, you can make an informed decision. The procedure isn't right for everyone. But if you have Afib, it's an option you can consider. Be sure to ask your provider any questions you have about the procedure.

If you have a left atrial appendage closure procedure, keep taking your medicines until your provider tells you to stop. Follow-up visits are important, too. These ensure you're healing and your device is fitting well.

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When your heart rhythm is out of sync, the experts at Cleveland Clinic can find out why. We offer personalized care for all types of arrhythmias.

AI rewrite

Left atrial appendage (LAA) closure is a procedure that blocks or closes the opening to your LAA. This keeps blood clots from leaving there and going into your blood. It helps stop strokes in people with atrial fibrillation, but without blood thinners. The LAA is a small pouch in one of your heart's upper chambers. It does not have a job to do.

LAA closure can be a surgery or a smaller procedure. It seals off your left atrial appendage. The LAA is a small sac in the muscle wall of your left atrium. This is the top left part of your heart. Closing it off or taking it out can lower your risk of stroke. It can also end the need to take blood thinners.

Some people have atrial fibrillation (Afib) but no valve disease. In these people, most blood clots in the left atrium start in the LAA.

A normal heart squeezes with each heartbeat. This pushes blood from your left atrium and LAA into your left ventricle. This is the bottom left part of the heart. When you have Afib, the signals that control your heartbeat do not move in a smooth way. They can be fast and mixed up. This does not give your atria time to push blood into your ventricles.

The LAA is a pouch, so blood can pool there and form clots. When your heart pumps out a clot, it can cause a stroke. People with Afib are three to five times more likely to have a stroke than other people. This is scary, but your provider can help you stay safe. They can find a plan that works for you. This may be medicine that makes it harder for clots to form. Or it may be closing off the left atrial appendage.

Your heart can keep doing its job with a closed left atrial appendage.

Thousands of people around the world have had an LAA closure. Providers use different ways and tools to do it.

LAA closure lowers the stroke risk that comes with Afib. But it does not treat the Afib itself. LAA closure can help people who have a higher risk of stroke from Afib. If you are at risk of getting blood clots in your left atrium or LAA, your provider may suggest sealing off your LAA. This can lower your risk of stroke from Afib. They can do this instead of giving you a blood thinner. Common blood thinners are warfarin (Coumadin® or Jantoven®), apixaban (Eliquis®), rivaroxaban (Xarelto®) and dabigatran (Pradaxa®).

Many people worry about, or do not like, taking warfarin. Here are some reasons why:

People with Afib who do not have heart valve disease can take other medicines. These are apixaban, dabigatran and rivaroxaban. But some people worry about, or have problems with, these medicines too:

Before an LAA closure, you will get a test called a transesophageal echocardiogram (TEE). You may also get a heart CT scan. These show your provider the size and shape of your LAA. This matters because the LAA is different in each person. Researchers say the LAA can look like a chicken wing, a cactus, a windsock or a cauliflower.

Your provider may also use 3D printing. This helps them pick the right tool and the right size.

You will need to fast before your procedure. This means you stop eating and drinking at a certain time. Your provider will tell you when to start fasting.

You will get some type of anesthesia. The kind depends on which procedure you have.

You may be having heart surgery for other reasons, like blocked heart arteries or valve disease. If so, your provider can often take out your LAA at the same time. They may sew or staple the area closed. This is a surgical LAA closure.

In a smaller procedure, they use a thin tube called a catheter. They put a special tool through it to close or block your LAA. Some tools act like a plug in the LAA opening. Other tools tie or clamp the LAA closed.

For a tool that goes in through a catheter, your provider will:

Several types of imaging help your provider see inside your heart while they work. The types of imaging they use include:

After your surgery or smaller procedure, you may:

People who get an LAA closure can:

LAA closure problems may include:

If you had your LAA closure as part of another heart surgery, your recovery time will depend on that other surgery.

If you had a smaller procedure with a catheter and a closure tool, your recovery will be faster. You may be able to go home the next day.

After you get a tool, you will need to take a blood thinner and aspirin for 45 days. This is how long it takes your body to grow tissue around the tool. This makes sure there are no gaps around it. Then you will stop the blood thinner. You will switch to clopidogrel (Plavix®) and aspirin for six months.

You may need to take aspirin for a long time.

You will have another TEE test at a follow-up visit 45 days after your procedure. This helps your provider see if your LAA is fully blocked.

If the TEE shows your LAA is not blocked, you will keep taking the blood thinner. You will have another TEE and follow-up visit after six months.

Once your LAA is blocked, you will have a follow-up visit with your provider once a year.

Researchers have studied how well LAA closure works. Closing it with stitches lowered the risk of a blood clot by 40%. Closing it with a clip or by taking it out lowered the risk by more than 90%. Studies of tools showed an 80% drop in strokes. Some tools are placed well more than 95% of the time.

There is no sure way to stop strokes. But taking out or blocking your LAA lowers your risk a lot. This is because 90% of strokes that start in an upper heart chamber start in the LAA.

Go to all your scheduled visits. Also, call your provider if you have:

It can be hard to live with the fear of stroke from Afib. This is even harder if you have trouble with blood thinners. Choosing to have an LAA closure is a personal choice. You can talk with your provider and go over your options. Then you can make a choice that is right for you. This procedure is not right for everyone. But if you have Afib, it is an option you can think about. Be sure to ask your provider any questions you have about the procedure.

If you have an LAA closure procedure, keep taking your medicines until your provider tells you to stop. Follow-up visits are important too. They make sure you are healing and your tool fits well.

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Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

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Original page

Surgical Procedures for Atrial Fibrillation

Quick Facts

- Some people with atrial fibrillation (AFib) need surgery to help their heart beat normally.

- Surgery might include a device placed in the body to help control heart rhythm and speed.

- Maze heart surgery creates scar tissue to help stop AFib.

For most people, medications may be the best option for AFib treatment. But there are also nonsurgical and surgical methods to treat AFib. Discuss your options with your health care professional.

Pacemakers

A pacemaker is a small electrical device that is placed under the skin near the collarbone during a surgical procedure. Wires from the pacemaker go to your heart. The device sends electric al signals to help your heart beat at a steady pace. Some pacemakers can tell when your heart is beating too fast or too slow. If it is, the pacemaker sends signals that help the heart return to the right rhythm and speed.

Watch an animation of a pacemaker.

Learn more about living with your pacemaker. Read about devices that may interfere with your pacemaker.

Printable patient information sheet: What is a pacemaker? (PDF)

Implantable cardioverter defibrillators (ICDs)

An ICD is another device placed in your chest through surgery. It can give small electric shocks if your heart beats in an unsafe way. This helps your heart beat normally. An ICD can also work as a pacemaker or defibrillator.

Learn more about living with your ICD and questions to ask your health care professional.

Left atrial appendage occlusion

Left atrial appendage occlusion (LAAO), or left atrial appendage closure, is a procedure that blocks or closes the opening of your left atrial appendage (LAA). The LAA is a pouch in the left atrium that can pool with blood if someone has AFib. This allows blood clots to form. If these blood clots enter the bloodstream, it increases the risk of stroke. Closing the LAA reduces the risk of stroke. It also can allow the person to stop taking anticoagulants (blood thinners) to prevent clots. The heart can continue to work normally with the LAA closed.

There are several types of procedures and devices that are used to close the LAA. These include those that:

- Block the LAA opening

- Clamp the base of the LAA to close it off

- Use a band or suture loop to close it off

If you have AFib and blood thinners aren't a good fit for you, ask your health care professional if LAA closure is a treatment option for you.

Open-heart maze procedure

Maze heart surgery is a complex procedure. A surgeon typically makes small cuts in the top part of the heart. Stitching those cuts together forms scar tissue. The scar tissue blocks the signals that cause AFib. This can help restore a normal heartbeat.

AI rewrite

Surgery for Atrial Fibrillation

Quick Facts

- Some people with atrial fibrillation (AFib) need surgery to help their heart beat the right way.

- Surgery might put a small device in the body. This device helps control how fast the heart beats.

- A surgery called the maze surgery makes scar tissue to help stop AFib.

For most people, medicine may be the best way to treat AFib. But there are other ways too. Some need surgery. Some do not. Talk with your doctor about what is best for you.

Pacemakers

A pacemaker is a small device that runs on electricity. A doctor puts it under your skin near your collarbone. This is done during surgery. Wires go from the pacemaker to your heart. The device sends out signals to help your heart beat at a steady pace. Some pacemakers can tell when your heart beats too fast or too slow. When this happens, the pacemaker sends signals to help your heart go back to the right pace.

Watch a video of how a pacemaker works.

Learn more about living with your pacemaker. Read about things that may bother your pacemaker.

You can print this fact sheet: What is a pacemaker? (PDF)

Implantable Cardioverter Defibrillators (ICDs)

An ICD is another device. A doctor puts it in your chest during surgery. It can give small shocks if your heart beats in an unsafe way. This helps your heart beat the right way. An ICD can also work like a pacemaker.

Learn more about living with your ICD. Also learn what questions to ask your doctor.

Left Atrial Appendage Closure

The left atrial appendage (LAA) is a small pouch in your heart. If you have AFib, blood can pool in this pouch. When blood pools, it can form clots. If a clot moves into your blood, it can cause a stroke. Closing the LAA lowers the chance of a stroke. It may also let you stop taking blood thinners. Blood thinners are medicine that stops clots. Your heart can still work fine with the LAA closed.

There are a few ways to close the LAA. Doctors may use a device or tool that:

- Blocks the opening of the LAA

- Clamps the base of the LAA to close it

- Uses a band or loop to close it

Do you have AFib? Are blood thinners not a good fit for you? If so, ask your doctor if closing the LAA is right for you.

Open-Heart Maze Surgery

The maze surgery is hard and takes a lot of skill. The doctor makes small cuts in the top part of the heart. Then the doctor stitches the cuts back together. This makes scar tissue. The scar tissue blocks the signals that cause AFib. This can help your heart beat the right way again.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 76 of 77 · Item ID B-1UQBFOBRCL

Original page

Surgical Procedures for Atrial Fibrillation

Quick Facts

- Some people with atrial fibrillation (AFib) need surgery to help their heart beat normally.

- Surgery might include a device placed in the body to help control heart rhythm and speed.

- Maze heart surgery creates scar tissue to help stop AFib.

For most people, medications may be the best option for AFib treatment. But there are also nonsurgical and surgical methods to treat AFib. Discuss your options with your health care professional.

Pacemakers

A pacemaker is a small electrical device that is placed under the skin near the collarbone during a surgical procedure. Wires from the pacemaker go to your heart. The device sends electric al signals to help your heart beat at a steady pace. Some pacemakers can tell when your heart is beating too fast or too slow. If it is, the pacemaker sends signals that help the heart return to the right rhythm and speed.

Watch an animation of a pacemaker.

Learn more about living with your pacemaker. Read about devices that may interfere with your pacemaker.

Printable patient information sheet: What is a pacemaker? (PDF)

Implantable cardioverter defibrillators (ICDs)

An ICD is another device placed in your chest through surgery. It can give small electric shocks if your heart beats in an unsafe way. This helps your heart beat normally. An ICD can also work as a pacemaker or defibrillator.

Learn more about living with your ICD and questions to ask your health care professional.

Left atrial appendage occlusion

Left atrial appendage occlusion (LAAO), or left atrial appendage closure, is a procedure that blocks or closes the opening of your left atrial appendage (LAA). The LAA is a pouch in the left atrium that can pool with blood if someone has AFib. This allows blood clots to form. If these blood clots enter the bloodstream, it increases the risk of stroke. Closing the LAA reduces the risk of stroke. It also can allow the person to stop taking anticoagulants (blood thinners) to prevent clots. The heart can continue to work normally with the LAA closed.

There are several types of procedures and devices that are used to close the LAA. These include those that:

- Block the LAA opening

- Clamp the base of the LAA to close it off

- Use a band or suture loop to close it off

If you have AFib and blood thinners aren't a good fit for you, ask your health care professional if LAA closure is a treatment option for you.

Open-heart maze procedure

Maze heart surgery is a complex procedure. A surgeon typically makes small cuts in the top part of the heart. Stitching those cuts together forms scar tissue. The scar tissue blocks the signals that cause AFib. This can help restore a normal heartbeat.

AI rewrite

Surgery for Atrial Fibrillation (AFib)

Quick Facts

- Some people with atrial fibrillation (AFib) need surgery to help their heart beat normally.

- Surgery might put a small device inside your body. This helps control how fast and steady your heart beats.

- Maze heart surgery makes scar tissue to help stop AFib.

For most people, medicine is the best way to treat AFib. But there are also ways to treat it with or without surgery. Talk with your doctor about your choices.

Pacemakers

A pacemaker is a small device. A doctor puts it under the skin near your collarbone during surgery. Wires go from the pacemaker to your heart. The device sends small electric signals to help your heart beat at a steady pace. Some pacemakers know if your heart is beating too fast or too slow. If it is, the pacemaker sends signals to fix your heart rate.

Watch a video of a pacemaker.

Learn more about living with your pacemaker. Read about things that can affect your pacemaker.

Printable patient sheet: What is a pacemaker? (PDF)

Implantable Cardioverter Defibrillators (ICDs)

An ICD is another device put in your chest during surgery. It can give small electric shocks if your heart beats in a dangerous way. This helps your heart beat normally again. An ICD can also work as a pacemaker or defibrillator.

Learn more about living with your ICD and questions to ask your doctor.

Closing the Left Atrial Appendage

Left atrial appendage closure is a surgery that blocks a small pouch in your heart. This pouch is called the LAA. If you have AFib, blood can pool in this pouch. This can cause blood clots. If these clots get into your blood, they can cause a stroke. Closing the pouch lowers your chance of having a stroke. It may also let you stop taking blood thinner medicine. Your heart will still work normally with the pouch closed.

Doctors can close the pouch in a few ways. They might:

- Block the opening of the pouch.

- Clamp the bottom of the pouch to close it.

- Use a band or a loop of thread to tie it off.

If you have AFib and cannot take blood thinners, ask your doctor if this surgery is right for you.

Open-Heart Maze Surgery

Maze heart surgery is a major operation. A doctor makes small cuts in the top part of your heart. The doctor then stitches the cuts together. This makes scar tissue. The scar tissue blocks the bad signals that cause AFib. This helps your heart beat normally again.

Factual accuracy (1–5)	Completeness (1–5)	Added errors (1–5)	Comments

Item 77 of 77 · Item ID B-3TZR6852FC

Original page

Surgical Procedures for Atrial Fibrillation

Quick Facts

- Some people with atrial fibrillation (AFib) need surgery to help their heart beat normally.

- Surgery might include a device placed in the body to help control heart rhythm and speed.

- Maze heart surgery creates scar tissue to help stop AFib.

For most people, medications may be the best option for AFib treatment. But there are also nonsurgical and surgical methods to treat AFib. Discuss your options with your health care professional.

Pacemakers

A pacemaker is a small electrical device that is placed under the skin near the collarbone during a surgical procedure. Wires from the pacemaker go to your heart. The device sends electric al signals to help your heart beat at a steady pace. Some pacemakers can tell when your heart is beating too fast or too slow. If it is, the pacemaker sends signals that help the heart return to the right rhythm and speed.

Watch an animation of a pacemaker.

Learn more about living with your pacemaker. Read about devices that may interfere with your pacemaker.

Printable patient information sheet: What is a pacemaker? (PDF)

Implantable cardioverter defibrillators (ICDs)

An ICD is another device placed in your chest through surgery. It can give small electric shocks if your heart beats in an unsafe way. This helps your heart beat normally. An ICD can also work as a pacemaker or defibrillator.

Learn more about living with your ICD and questions to ask your health care professional.

Left atrial appendage occlusion

Left atrial appendage occlusion (LAAO), or left atrial appendage closure, is a procedure that blocks or closes the opening of your left atrial appendage (LAA). The LAA is a pouch in the left atrium that can pool with blood if someone has AFib. This allows blood clots to form. If these blood clots enter the bloodstream, it increases the risk of stroke. Closing the LAA reduces the risk of stroke. It also can allow the person to stop taking anticoagulants (blood thinners) to prevent clots. The heart can continue to work normally with the LAA closed.

There are several types of procedures and devices that are used to close the LAA. These include those that:

- Block the LAA opening

- Clamp the base of the LAA to close it off

- Use a band or suture loop to close it off

If you have AFib and blood thinners aren't a good fit for you, ask your health care professional if LAA closure is a treatment option for you.

Open-heart maze procedure

Maze heart surgery is a complex procedure. A surgeon typically makes small cuts in the top part of the heart. Stitching those cuts together forms scar tissue. The scar tissue blocks the signals that cause AFib. This can help restore a normal heartbeat.

AI rewrite

Surgery for Atrial Fibrillation

Quick Facts

- Some people with atrial fibrillation, or AFib, need surgery to help their heart beat in a normal way.

- Surgery may include putting a device in the body. The device helps control the heart’s rhythm and speed.

- Maze heart surgery makes scar tissue. This can help stop AFib.

For most people, medicine may be the best treatment for AFib. But there are also treatments that do not use surgery and treatments that do use surgery. Talk with your health care professional about your choices.

Pacemakers

A pacemaker is a small electric device. It is placed under the skin near the collarbone during surgery. Wires from the pacemaker go to your heart.

The pacemaker sends electric signals. These signals help your heart beat at a steady pace.

Some pacemakers can tell when your heart is beating too fast or too slow. If this happens, the pacemaker sends signals to help your heart return to the right rhythm and speed.

Watch an animation of a pacemaker.

Learn more about living with your pacemaker. Read about devices that may get in the way of your pacemaker.

Printable patient information sheet: What is a pacemaker? (PDF)

Implantable Cardioverter Defibrillators, or ICDs

An ICD is another device placed in your chest during surgery. It can give small electric shocks if your heart beats in an unsafe way. This helps your heart beat normally.

An ICD can also work as a pacemaker or defibrillator.

Learn more about living with your ICD. Learn what questions to ask your health care professional.

Left Atrial Appendage Occlusion

Left atrial appendage occlusion, or LAAO, is also called left atrial appendage closure. It is a procedure that blocks or closes the opening of your left atrial appendage, or LAA.

The LAA is a small pouch in the left atrium of the heart. If a person has AFib, blood can pool in this pouch. This can allow blood clots to form.

If these blood clots enter the bloodstream, the risk of stroke goes up. Closing the LAA lowers the risk of stroke.

Closing the LAA may also let a person stop taking anticoagulants, also called blood thinners, to prevent clots. The heart can still work normally when the LAA is closed.

There are several types of procedures and devices used to close the LAA. They can:

- Block the LAA opening
- Clamp the base of the LAA to close it
- Use a band or suture loop to close it

If you have AFib and blood thinners are not a good fit for you, ask your health care professional if LAA closure may be a treatment option for you.

Open-Heart Maze Procedure

Maze heart surgery is a complex procedure. A surgeon usually makes small cuts in the top part of the heart.

The surgeon stitches the cuts together. This forms scar tissue. The scar tissue blocks the signals that cause AFib. This can help bring back a normal heartbeat.

Factual accuracy (1–5)	**Completeness (1–5)**	**Added errors (1–5)**	**Comments**